PROBABILITY & STATISTICS · COMMON CORE HIGH SCHOOL MATH

# Question Collection

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

54 question sets • 1435 questions

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# Represent data with dot plots

| | | |
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| **BACKWARD 2** | Locate values on a number-line scale | **6.NS.C.6.c** |
| **BACKWARD 3** | Read counts from a dot plot | **6.SP.B.4** |
| **FORWARD 1** | Find the median and mean from a dot plot | **HSS-ID.A.2** |
| **FORWARD 2** | Choose the display that keeps the information you need | **HSS-ID.A.1** |

SET 1 • HSS-ID.A.1

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

The numbers of books 15 students read over the summer are 2, 4, 3, 2, 5, 3, 2, 6, 3, 4, 2, 3, 1, 3, 4. A dot plot shows one dot for each student above the number of books. How many dots are above 2, above 3, and above 4?

Dots above 2: _____

Dots above 3: _____

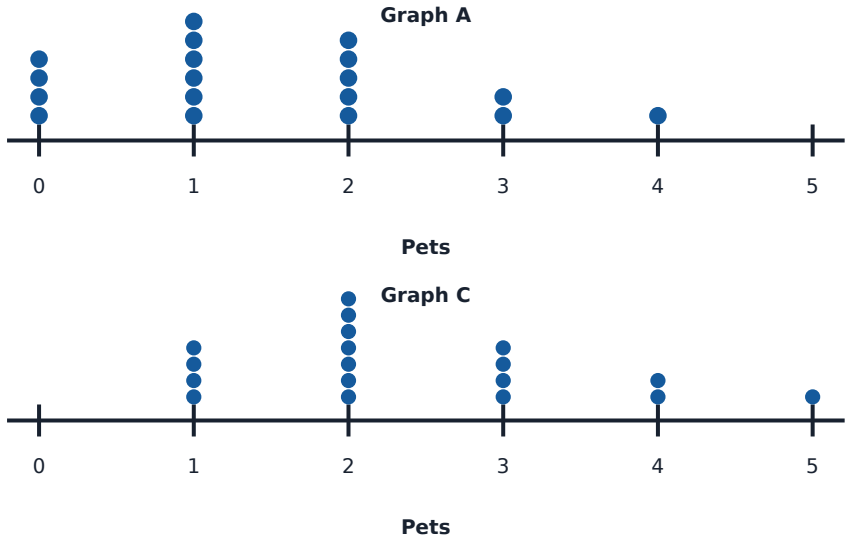
Dots above 4: _____

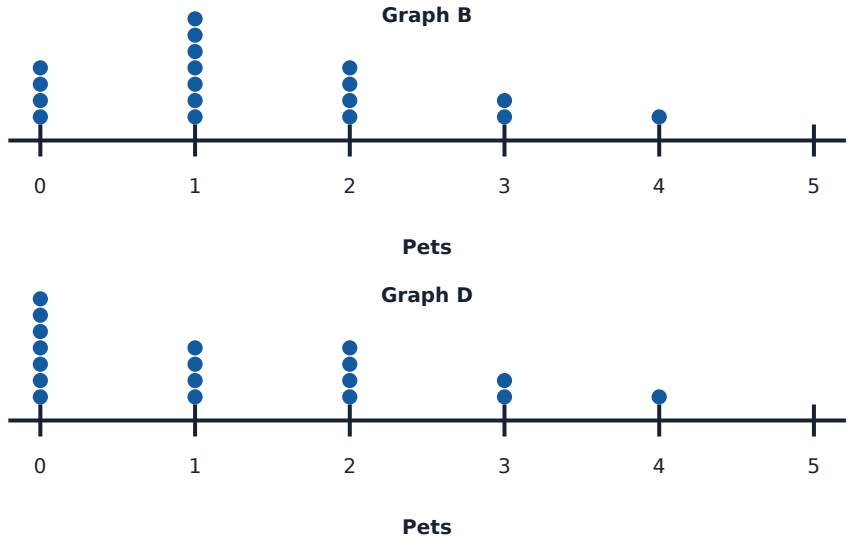
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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S1-M-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 1 • MAIN • Q1/5**

The numbers of pets in 18 households are 0, 1, 1, 2, 0, 3, 1, 2, 1, 0, 4, 1, 2, 1, 0, 2, 1, 3. Which dot plot shows the data?










**A.** Graph A

**B.** Graph B

**C.** Graph C

**D.** Graph D

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S1-M-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 1 • MAIN • Q2/5**

Fourteen quiz scores (out of 10) are 7, 9, 8, 10, 7, 8, 8, 6, 9, 8, 7, 10, 8, 9. In a dot plot of the scores, which score has the tallest stack of dots, and how many dots are in that stack?

Score: _____

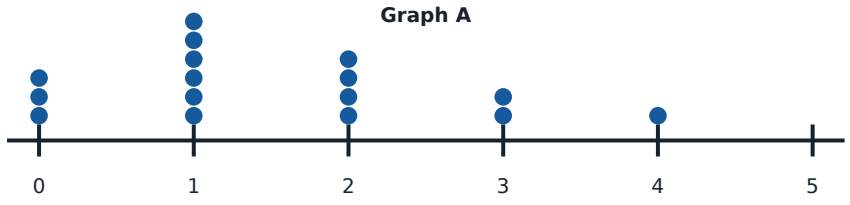
Dots in that stack: _____

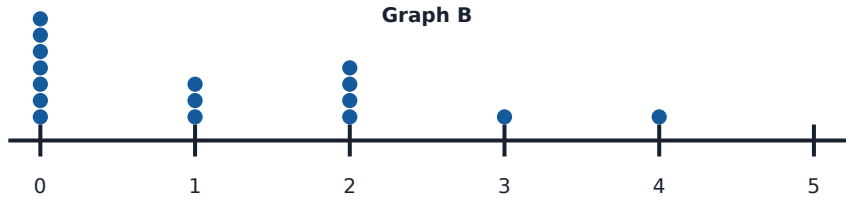
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

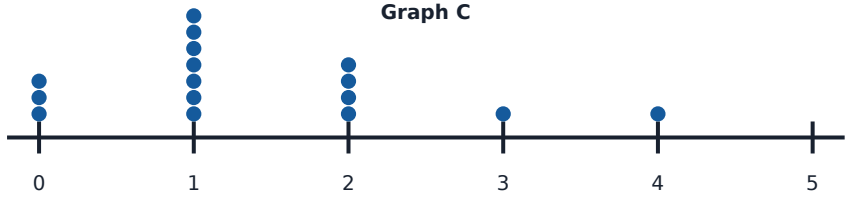
ID S1-M-Q3

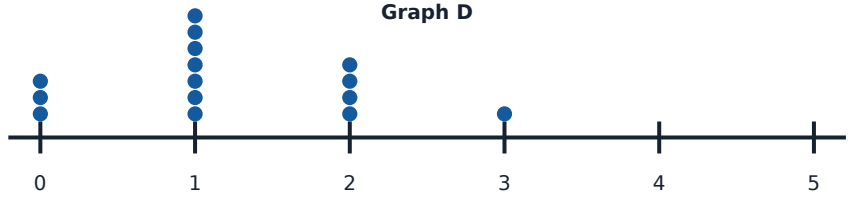
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 1 • MAIN • Q3/5

A team scored these numbers of goals in 16 games: 1, 0, 2, 1, 3, 1, 0, 2, 2, 1, 4, 1, 0, 1, 2, 1. Which dot plot shows the data?




**A.** Graph A

**B.** Graph B

**C.** Graph C

**D.** Graph D

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S1-M-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 1 • MAIN • Q4/5**

Twenty students reported their numbers of siblings: 1, 2, 0, 1, 3, 2, 1, 1, 0, 2, 5, 1, 2, 0, 1, 2, 3, 1, 1, 2. In a dot plot of the data, how many dots are at values of 2 or more, and how many dots are there in all?

Dots at 2 or more: _____

Dots in all: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSS-ID.A.1

# BACKWARD 1

Count how often each value occurs • Grade 6 • Statistics & Probability • 6.SP.B.5.a

In the list 3, 5, 3, 4, 3, 5, how many times does 3 appear?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q1/5**

In the list 1, 2, 2, 4, 2, 1, 2, how many times does 2 appear?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q2/5**

## True or false?

The list 6, 6, 7, 8, 6 has three values equal to 6.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q3/5**

How many values are in the list 2, 4, 4, 5, 2, 4, 4?

**A.** 3

**B.** 5

**C.** 4

**D.** 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q4/5**

In the list 0, 1, 0, 2, 1, 0, how many values are 0 or 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q5/5**

SET 1 • HSS-ID.A.1

# BACKWARD 2

Locate values on a number-line scale • Grade 6 • The Number System • 6.NS.C.6.c

A number line has marks labeled 0, 1, 2, 3, 4, 5, 6. Where does a dot for the value 4 go?

**A.** at the right end of the number line

**B.** above the mark labeled 4

**C.** above the fourth mark from the left, labeled 3

**D.** above the mark labeled 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

**Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q1/5**

## True or false?

On a number line with marks at 0, 2, 4, 6, 8, the value 5 is halfway between the marks 4 and 6.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

**Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q2/5**

A number line has marks at 10, 15, 20, 25, 30. What value is halfway between the marks 15 and 20?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

**Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q3/5**

A number line runs from 0 to 10 with a mark at every whole number. How many marks are there?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

**Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q4/5**

## True or false?

On a horizontal number line, a larger value is placed to the left of a smaller value.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

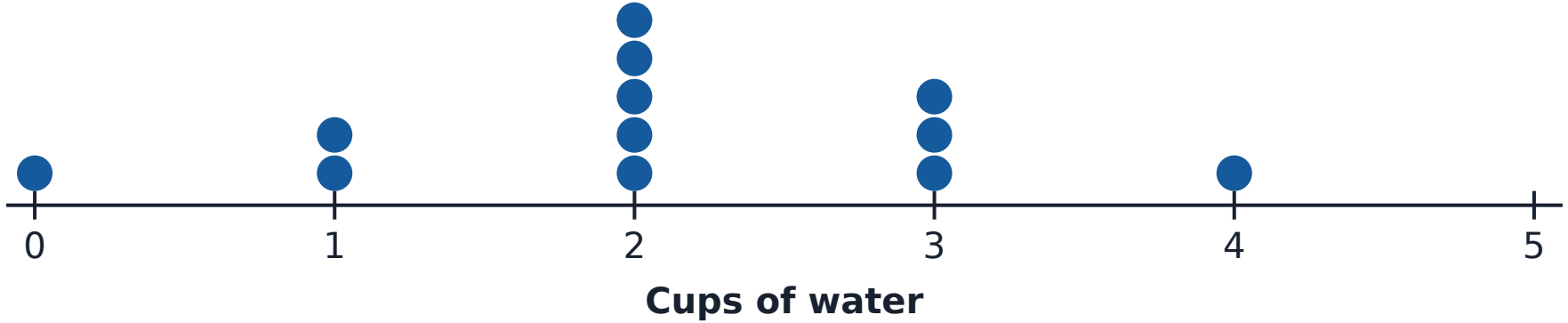
**Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q5/5**

SET 1 • HSS-ID.A.1

# BACKWARD 3

Read counts from a dot plot • Grade 6 • Statistics & Probability • 6.SP.B.4

The dot plot shows the cups of water 12 workers drank before noon. How many workers drank 3 cups?



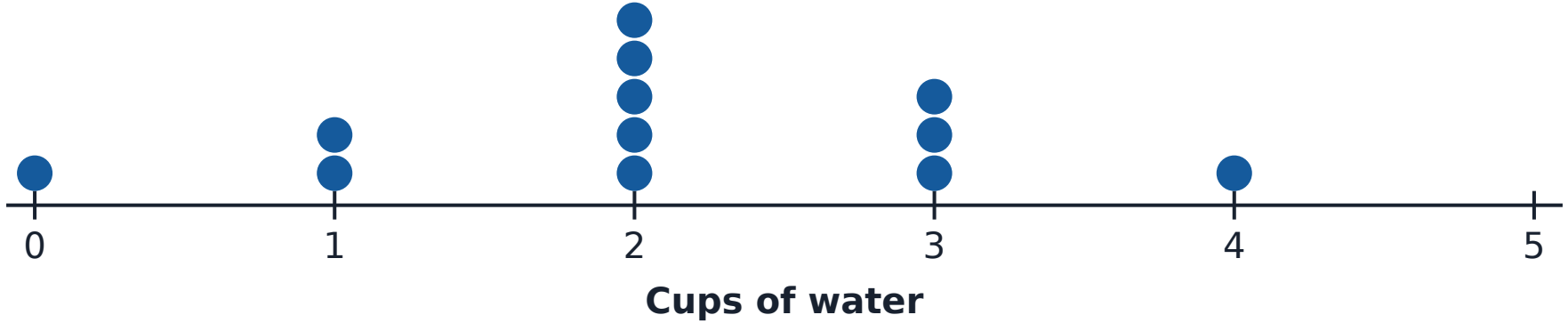

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q1/5**

The dot plot shows the cups of water some workers drank before noon. How many workers are shown in all?




Answer: _____

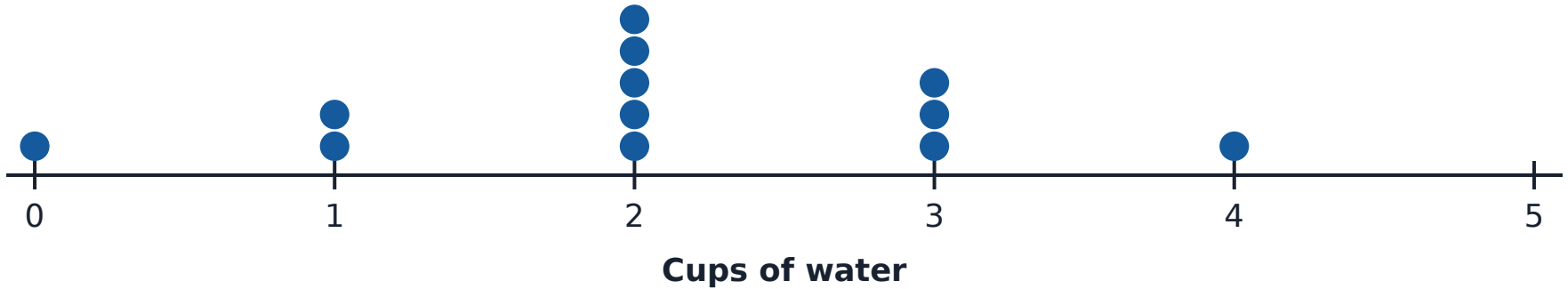
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q2/5**

## True or false?

The dot plot shows the cups of water 12 workers drank before noon. No worker drank 5 cups.




**A.** True

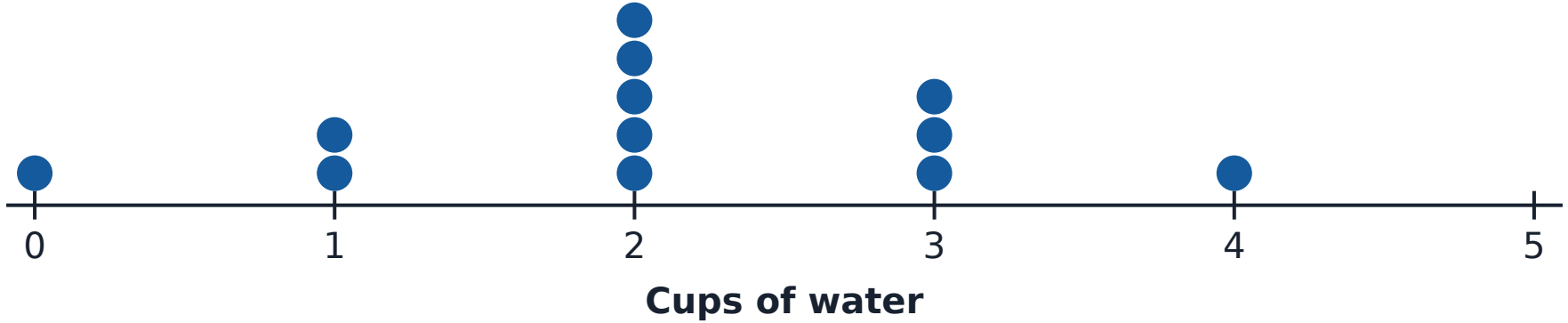
**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q3/5**

The dot plot shows the cups of water 12 workers drank before noon. Which number of cups occurs most often?




**A.** 3 **B.** 12 **C.** 2 **D.** 5

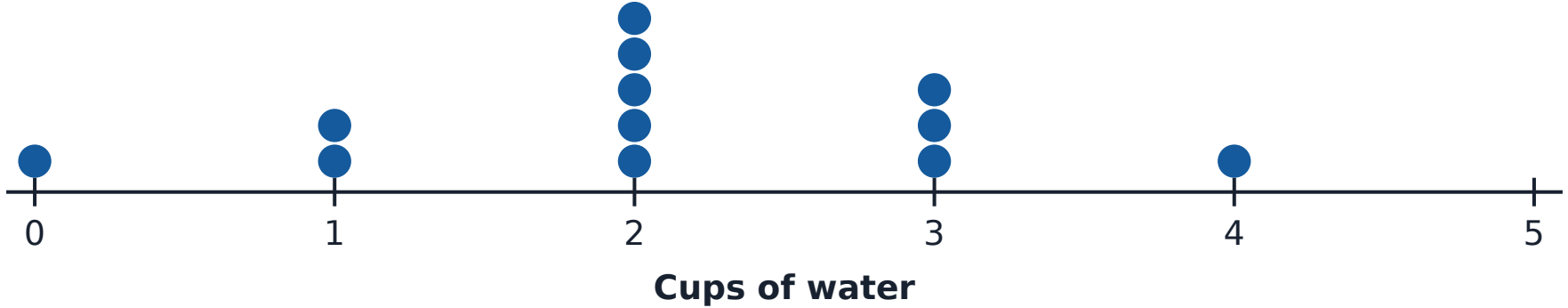
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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q4/5**

The dot plot shows the cups of water 12 workers drank before noon. How many workers drank fewer than 2 cups?




Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q5/5**

SET 1 • HSS-ID.A.1

# FORWARD 1

Find the median and mean from a dot plot • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

The dot plot shows rainy days in 10 two-week periods. Find the median number of rainy days.




Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 1 • FORWARD 1 • Q1/5

The dot plot shows rainy days in 10 two-week periods. Find the mean number of rainy days. Give a decimal.




Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 1 • FORWARD 1 • Q2/5

## True or false?

A dot plot shows the values 3, 3, 4, 4, 4, 4, 5, 5, 6, 7. Adding one more dot at 20 changes the median more than it changes the mean.

**A.** True

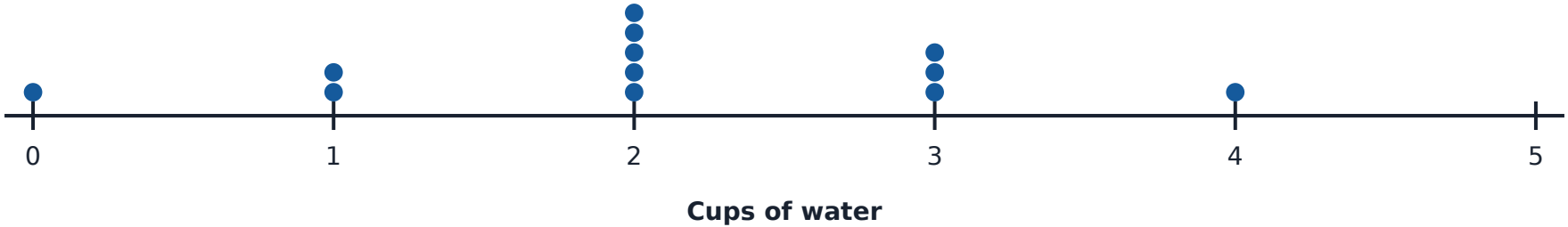
**B.** False

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S1-F1-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 1 • FORWARD 1 • Q3/5**

The dot plot shows the cups of water 12 workers drank before noon. Which statement is correct?




**A.** The mean is 12 cups, because 12 workers are shown in the plot.

**B.** The median is 2.5 cups, halfway between the smallest and largest values.

**C.** The median is 5 cups, the height of the tallest stack in the plot.

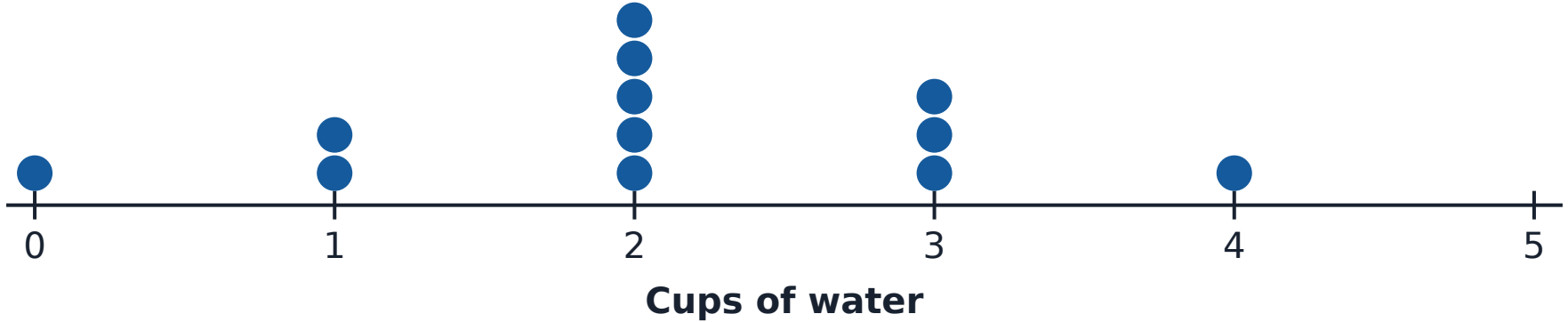
**D.** The median is 2 cups and the mean is 2.08 cups (to the nearest hundredth).

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S1-F1-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 1 • FORWARD 1 • Q4/5**

The dot plot shows the cups of water 12 workers drank before noon. Find the total number of cups drunk.




Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSS-ID.A.1

# FORWARD 2

Choose the display that keeps the information you need • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Which display always shows every individual value of a small data set?

**A.** a box plot

**B.** a histogram with bins of width 10

**C.** a dot plot

**D.** a box plot with outliers marked

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S1-F2-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 1 • FORWARD 2 • Q1/5**

From a box plot alone, which of these can always be found?

**A.** the median

**B.** the number of values

**C.** the most common value

**D.** the mean

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S1-F2-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 1 • FORWARD 2 • Q2/5**

## True or false?

A histogram with bins 0-10, 10-20, 20-30 shows how many values are exactly 15.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 1 • FORWARD 2 • Q3/5

A school records the number of people in each of 400 students' households (whole numbers from 1 to 9). Which display shows how many households have each size and stays readable?

**A.** a histogram with one bin for each whole number

**B.** a list of all 400 household sizes

**C.** a box plot of the household sizes

**D.** a dot plot with one dot per household

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S1-F2-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 1 • FORWARD 2 • Q4/5**

## True or false?

Two different data sets can have the same box plot but different dot plots.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 1 • FORWARD 2 • Q5/5

# Represent data with histograms

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.A.1** |
| **BACKWARD 1** | Assign values to bins with an endpoint rule | **6.SP.B.4** |
| **BACKWARD 2** | Count the values in an interval | **6.SP.B.5.a** |
| **BACKWARD 3** | Read a histogram | **6.SP.B.4** |
| **FORWARD 1** | Regroup a histogram with wider bins | **HSS-ID.A.1** |
| **FORWARD 2** | Locate the median in a histogram | **HSS-ID.A.2** |

SET 2 • HSS-ID.A.1

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Commute times (minutes) of 20 employees are 12, 25, 8, 31, 17, 22, 15, 40, 19, 27, 10, 33, 21, 14, 26, 18, 35, 23, 29, 16. A histogram uses the bins 0-10, 10-20, 20-30, 30-40 and 40-50. Each bin includes its left endpoint and excludes its right endpoint. How many values are in the bins 10-20, 30-40 and 40-50?

10-20: _____

30-40: _____

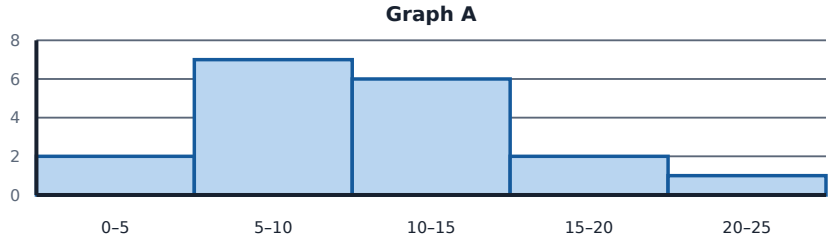
40-50: _____

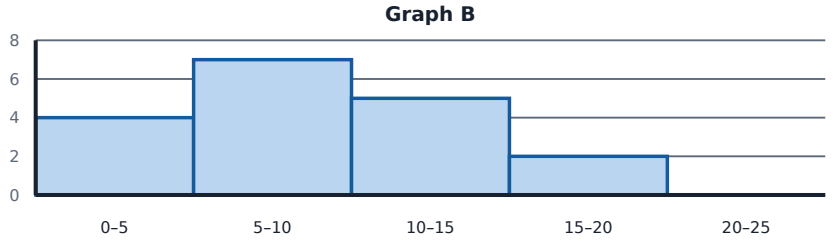
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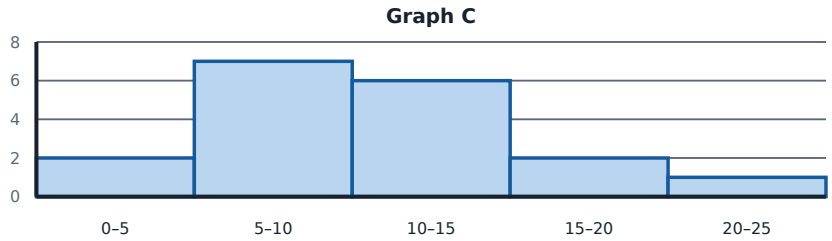
**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S2-M-Q1**

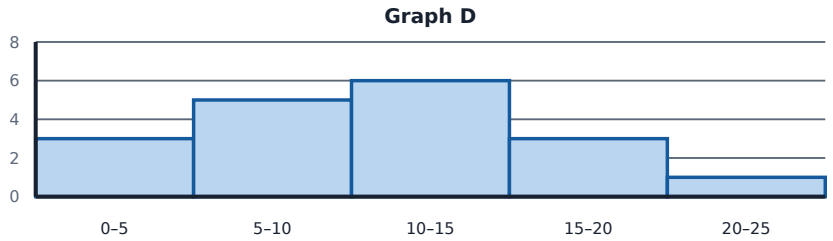
**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 2 • MAIN • Q1/5**

Waiting times (minutes) of 18 patients are 4, 11, 7, 15, 9, 5, 12, 20, 10, 6, 14, 8, 17, 10, 3, 13, 5, 9. Which histogram shows the data with bins 0-5, 5-10, 10-15, 15-20, 20-25? Each bin includes its left endpoint and excludes its right endpoint. In each histogram, the horizontal axis shows minutes and the vertical axis shows frequency.




**A.** Graph A

**B.** Graph B

**C.** Graph C

**D.** Graph D

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S2-M-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 2 • MAIN • Q2/5**

Standing long-jump distances (cm) of 16 students are 152, 168, 145, 175, 160, 158, 171, 149, 163, 180, 155, 166, 150, 172, 161, 157. A histogram uses the bins 140-150, 150-160, 160-170, 170-180 and 180-190. Each bin includes its left endpoint and excludes its right endpoint. How many values are in the bins 150-160 and 170-180?

150-160: _____

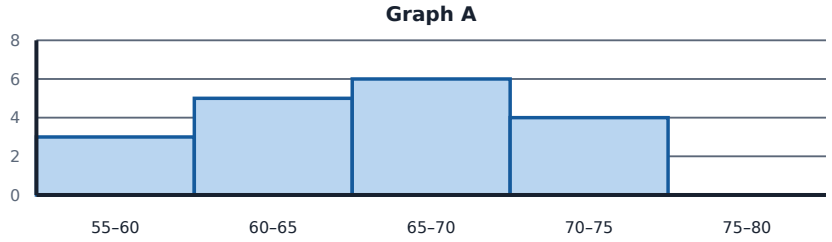
170-180: _____

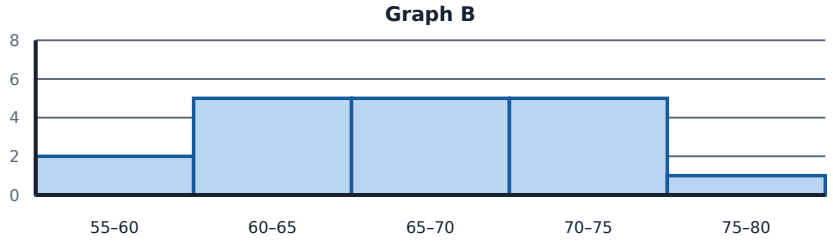
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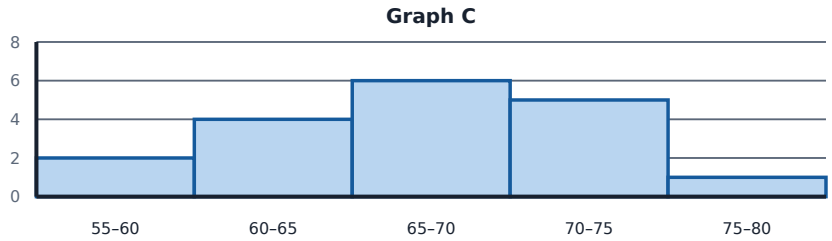
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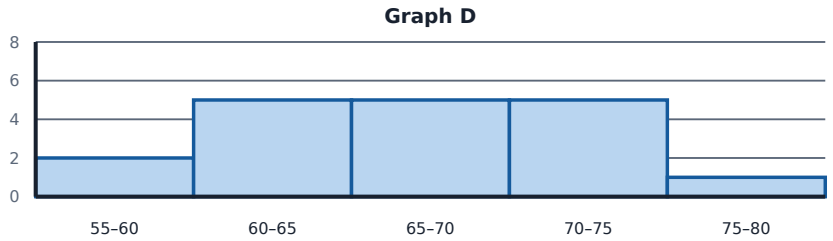
**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 2 • MAIN • Q3/5**

Daily high temperatures (°F) over 18 days are 61, 58, 70, 66, 72, 59, 64, 75, 68, 63, 71, 60, 67, 65, 74, 69, 62, 70. Which histogram shows the data with bins 55-60, 60-65, 65-70, 70-75, 75-80? Each bin includes its left endpoint and excludes its right endpoint. In each histogram, the horizontal axis shows °f and the vertical axis shows frequency.




**A.** Graph A

**B.** Graph B

**C.** Graph C

**D.** Graph D

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S2-M-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 2 • MAIN • Q4/5**

Page counts of 14 novels are 120, 255, 188, 310, 96, 205, 150, 275, 230, 182, 140, 200, 260, 175. A histogram uses bins of width 50: 50-100, 100-150, ..., 300-350. Each bin includes its left endpoint and excludes its right endpoint. Which bin has the most values (give its left endpoint), and how many values does it have?

Left endpoint of the tallest bin: _____

Values in that bin: _____

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S2-M-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 2 • MAIN • Q5/5**

SET 2 • HSS-ID.A.1

# BACKWARD 1

Assign values to bins with an endpoint rule • Grade 6 • Statistics & Probability • 6.SP.B.4

Bins 10-20 and 20-30 each include their left endpoint and exclude their right endpoint. Which bin holds 20?

**A.** 20-30

**B.** 10-20

**C.** neither bin

**D.** both bins

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q1/5**

## True or false?

With bins that include the left endpoint and exclude the right endpoint, the value 30 goes in the bin 20-30.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q2/5**

Bins 0-5, 5-10 and 10-15 include their left endpoint and exclude their right endpoint. Which bin holds 9.9? Give its left endpoint.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q3/5**

How many of the values 4, 5, 9, 10, 12 fall in the bin 5-10, which includes 5 and excludes 10?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q4/5**

**True or false?**

If every bin included both of its endpoints, a value equal to an endpoint shared by two bins would be counted twice.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q5/5**

SET 2 • HSS-ID.A.1

# BACKWARD 2

Count the values in an interval • Grade 6 • Statistics & Probability • 6.SP.B.5.a

How many of the values 12, 18, 25, 31, 19, 22 are at least 15 and less than 25?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q1/5**

How many of the values 3.5, 4, 6.2, 7, 8 are at least 4 and less than 7?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q2/5**

## True or false?

Exactly two of the values 40, 45, 50, 55 are at least 45 and less than 55.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q3/5**

How many of the values 2, 9, 11, 14, 20, 21 are at least 10 and less than 20?

**A.** 2

**B.** 3

**C.** 4

**D.** 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q4/5**

How many of the values 100, 150, 199, 200, 250 are at least 100 and less than 200?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q5/5**

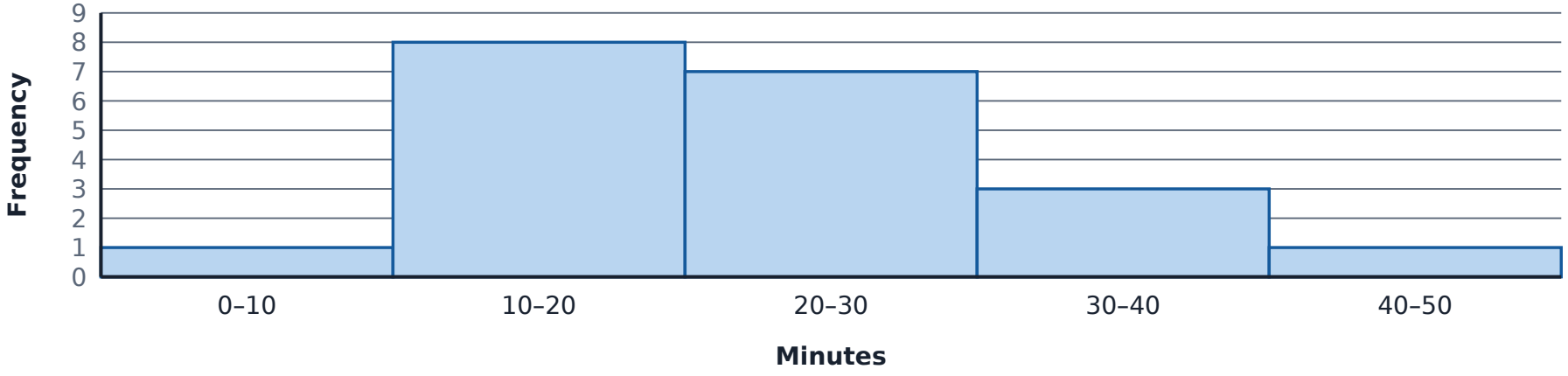
SET 2 • HSS-ID.A.1

# BACKWARD 3

Read a histogram • Grade 6 • Statistics & Probability • 6.SP.B.4

The histogram shows the commute times of some employees.

How many employees have commutes in the bin 20-30?




Answer: _____

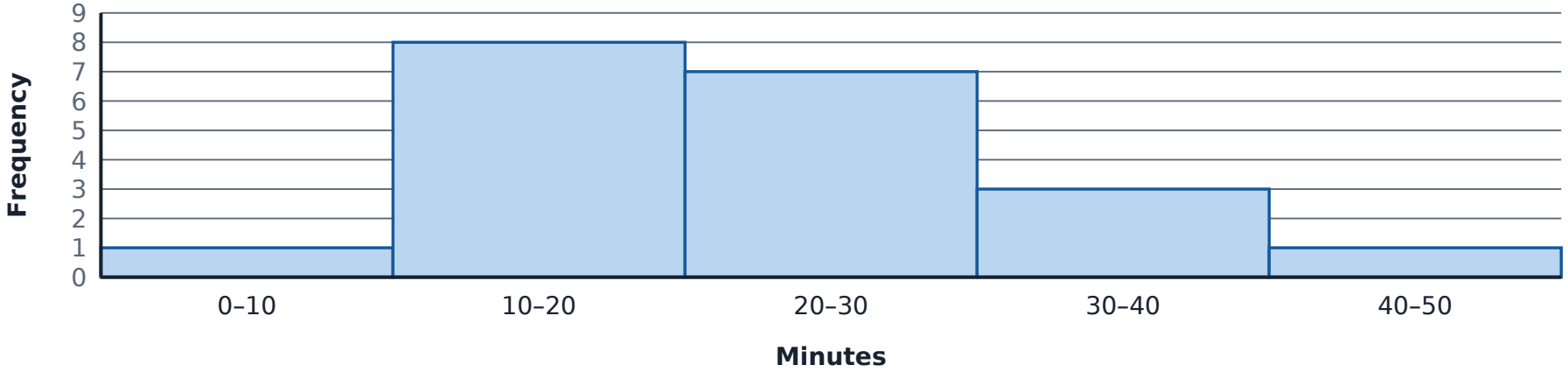
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q1/5**

The histogram shows the commute times of some employees.

How many employees are shown in all?




Answer: _____

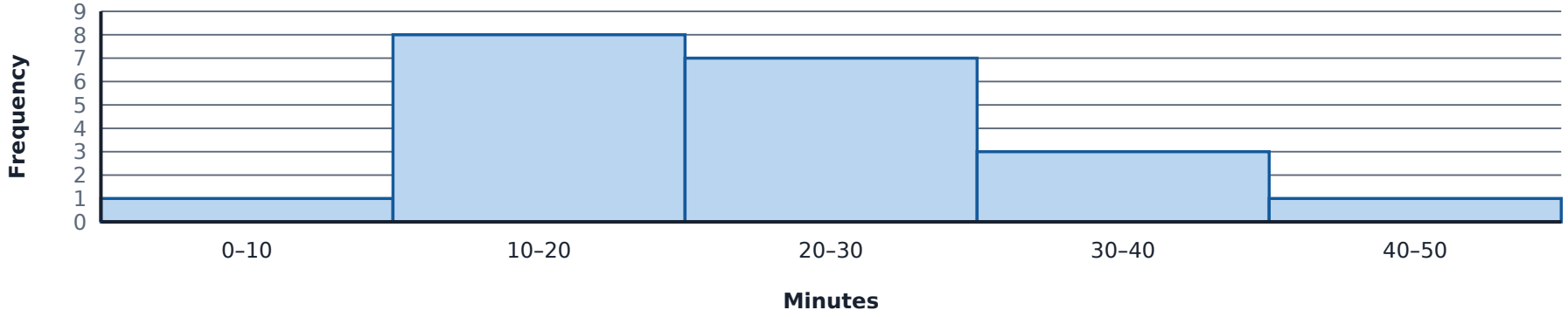
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q2/5**

## True or false?

The histogram shows the commute times of some employees. It shows that one employee's commute was exactly 15 minutes.




**A.** True

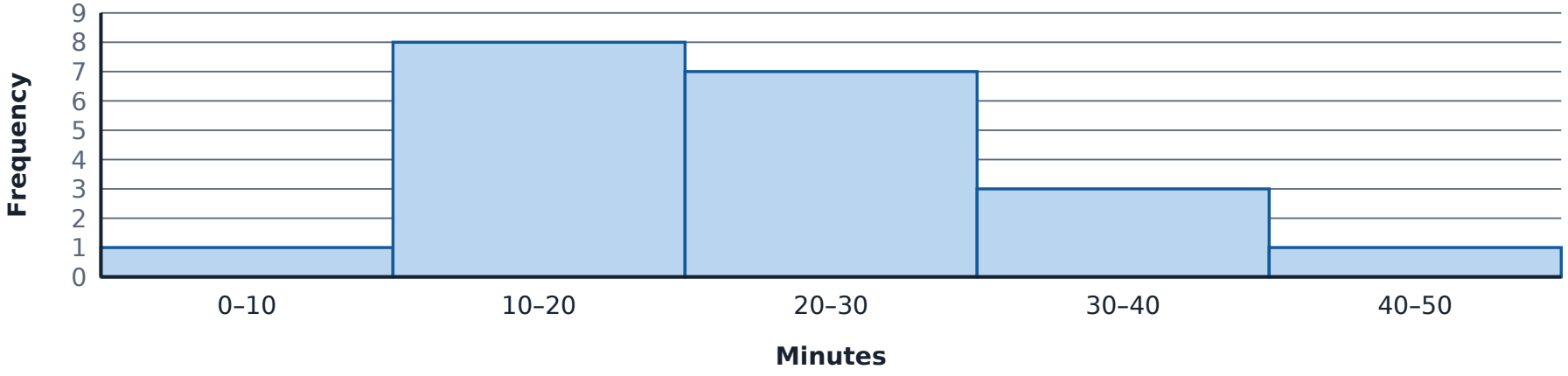
**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q3/5**

The histogram shows the commute times of some employees. Which bin has the most values?




**A.** 0-10 **B.** 20-30 **C.** 30-40 **D.** 10-20

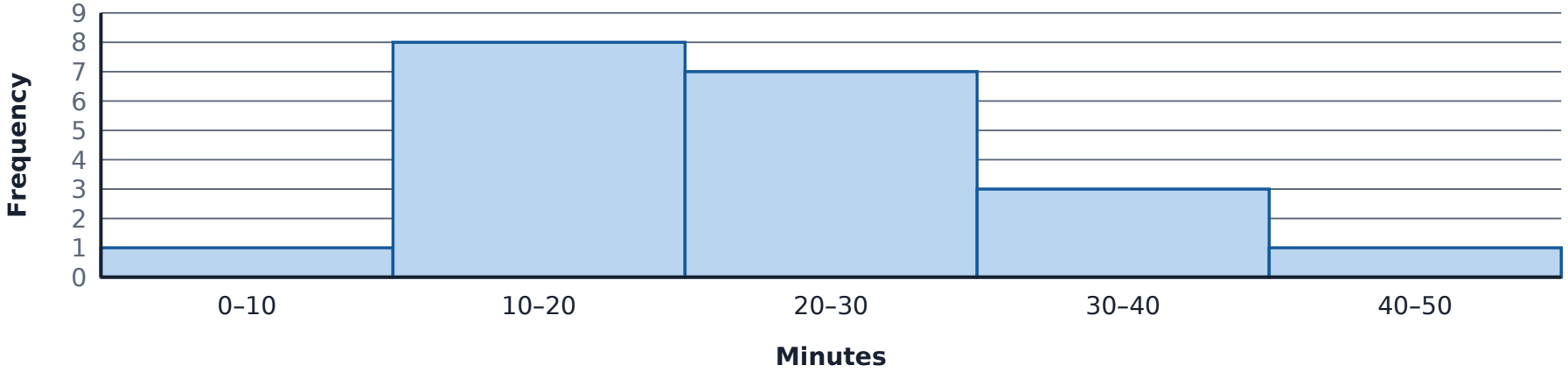
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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q4/5**

The histogram shows the commute times of some employees.
How many employees have commutes of at least 30 minutes?




Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q5/5**

SET 2 • HSS-ID.A.1

# FORWARD 1

Regroup a histogram with wider bins • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

A histogram with bins 0-5, 5-10, 10-15, 15-20 has counts 3, 6, 8, 2. What are the counts for the wider bins 0-10 and 10-20?

0-10: _____

10-20: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 2 • FORWARD 1 • Q1/5

A histogram with bins 60-65, 65-70, 70-75, 75-80 has counts 2, 5, 6, 4. What are the counts for 60-70 and 70-80?

60-70: _____

70-80: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 2 • FORWARD 1 • Q2/5

## True or false?

Combining each pair of neighboring bins into one bin never changes the total number of values shown.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 2 • FORWARD 1 • Q3/5

A histogram with bins of width 1 has counts 1, 0, 2, 0, 1, 0, 3, 0, 2, so it looks jagged. Why might bins of width 2 help?

**A.** Wider bins always make a distribution look symmetric.

**B.** Wider bins change the data values, so the shape becomes smoother.

**C.** Wider bins group nearby values, so the overall shape is easier to see.

**D.** Wider bins add values to the empty bins, so no bin is 0.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S2-F1-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 2 • FORWARD 1 • Q4/5**

A histogram with bins of width 10 has counts 4, 9, 5. Can the counts for bins of width 5 be found from it?

**A.** Yes: halve each count, because each narrower bin is half as wide.

**B.** Yes: keep the same counts, because the data have not changed.

**C.** No: each count could be split between the two narrower bins in many ways.

**D.** No: a histogram with bins of width 10 cannot be redrawn at all.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S2-F1-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 2 • FORWARD 1 • Q5/5**

SET 2 • HSS-ID.A.1

# FORWARD 2

Locate the median in a histogram • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

A histogram of 21 values has counts 0-10: 4, 10-20: 6, 20-30: 7, 30-40: 4. The median is the 11th value in order. Which bin holds it? Give its left endpoint.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 2 • FORWARD 2 • Q1/5

A histogram of 20 values has counts 50-60: 3, 60-70: 5, 70-80: 8, 80-90: 4. The median is the mean of the 10th and 11th values. Which bin holds both? Give its left endpoint.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 2 • FORWARD 2 • Q2/5

## True or false?

The bin with the tallest bar always holds the median.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 2 • FORWARD 2 • Q3/5

A histogram of 15 values has counts 0-5: 10, 5-10: 2, 10-15: 2, 15-20: 1. Which bin holds the median (the 8th value)?

**A.** 15-20

**B.** 10-15

**C.** 5-10

**D.** 0-5

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S2-F2-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 2 • FORWARD 2 • Q4/5**

A histogram of 15 values has counts 0-10: 2, 10-20: 3, 20-30: 3, 30-40: 7. Which bin holds the median (the 8th value)? Give its left endpoint.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 2 • FORWARD 2 • Q5/5

# Represent data with box plots

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.A.1** |
| **BACKWARD 1** | Order data and find the median | **6.SP.B.5.c** |
| **BACKWARD 2** | Find quartiles as medians of the halves | **6.SP.B.5.c** |
| **BACKWARD 3** | Read a box plot | **6.SP.B.4** |
| **FORWARD 1** | Mark outliers on a box plot | **HSS-ID.A.3** |
| **FORWARD 2** | Compare two box plots | **HSS-ID.A.2** |

SET 3 • HSS-ID.A.1

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Laps swum by 11 swimmers: 12, 15, 9, 18, 14, 11, 16, 20, 13, 10, 17. Give the five-number summary for a box plot. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

Minimum: _____

Q1: _____

Median: _____

Q3: _____

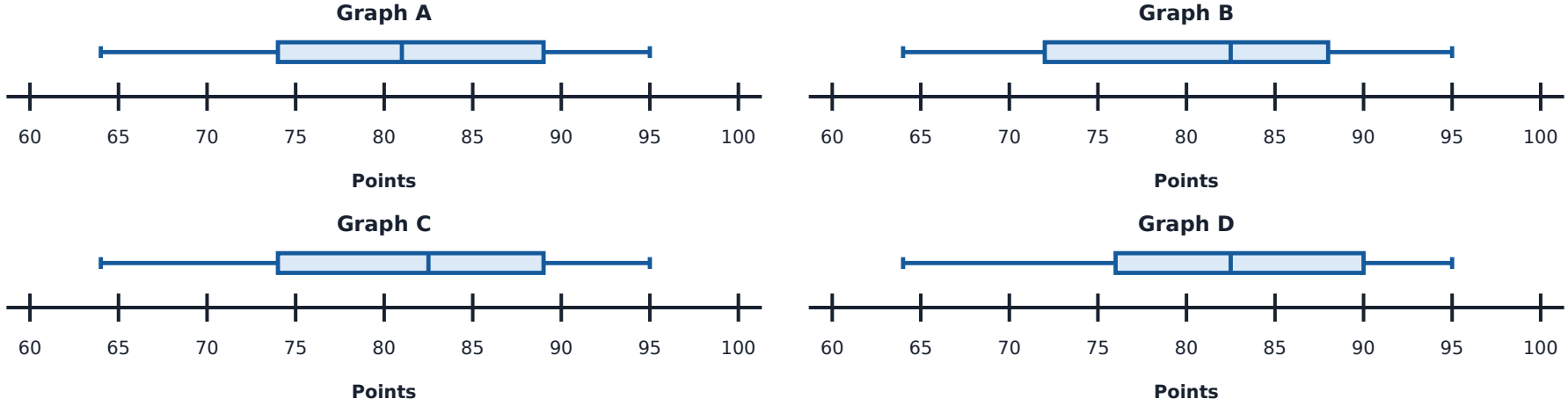
Maximum: _____

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S3-M-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 3 • MAIN • Q1/5**

Test scores of 12 students: 72, 85, 90, 64, 78, 88, 95, 70, 81, 76, 84, 92. Which box plot shows the data? Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.




**A.** Graph A **B.** Graph B **C.** Graph C **D.** Graph D

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S3-M-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 3 • MAIN • Q2/5**

Hours of sleep of 14 teenagers: 7.5, 6, 8, 6.5, 9, 7, 5.5, 8.5, 7, 6.5, 7.5, 8, 6, 7. Give the five-number summary for a box plot. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

Minimum: _____

Q1: _____

Median: _____

Q3: _____

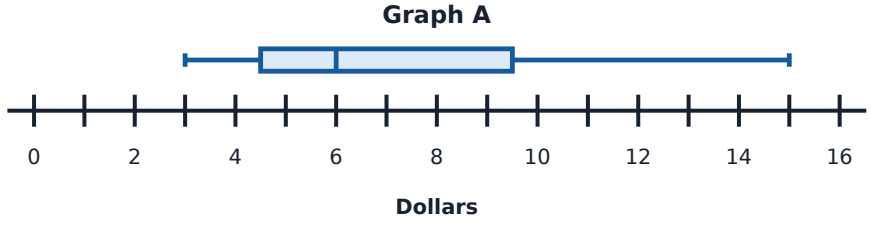
Maximum: _____

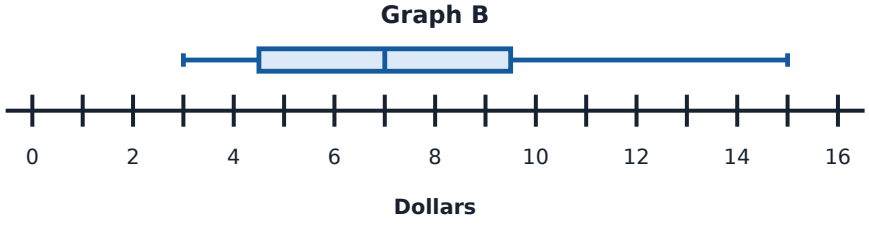
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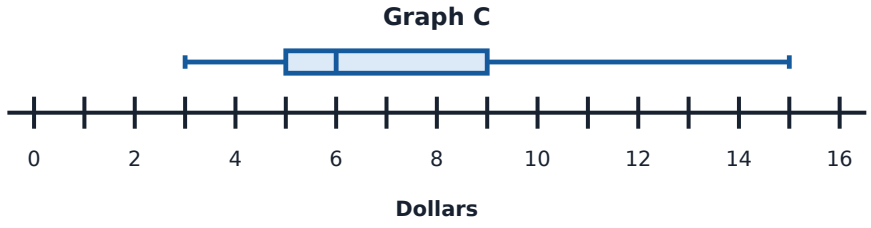
**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S3-M-Q3**

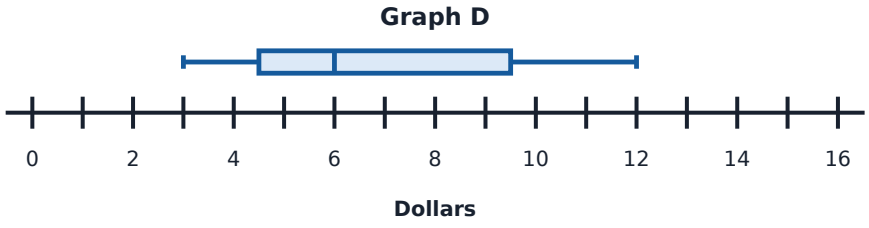
**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 3 • MAIN • Q3/5**

Prices (dollars) of 13 menu items: 3, 8, 5, 12, 6, 4, 9, 7, 10, 5, 6, 15, 4. Which box plot shows the data? Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.




**A.** Graph A **B.** Graph B **C.** Graph C **D.** Graph D

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S3-M-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 3 • MAIN • Q4/5**

Distances (km) 10 students live from school: 2.1, 3.4, 1.8, 5, 2.7, 4.2, 3, 2.4, 3.9, 1.5. Give the five-number summary for a box plot. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

Minimum: _____

Q1: _____

Median: _____

Q3: _____

Maximum: _____

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S3-M-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 3 • MAIN • Q5/5**

SET 3 • HSS-ID.A.1

# BACKWARD 1

Order data and find the median • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the median of 9, 3, 7, 5, 11.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q1/5**

Find the median of 4, 10, 6, 8.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q2/5**

## True or false?

The median of 2, 9, 4 is 9, the middle value of the list as written.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q3/5**

What is the median of 1, 3, 3, 6, 7, 8?

**A.** 4.5

**B.** 3

**C.** 6

**D.** 4.67

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q4/5**

Find the median of 12, 15, 11, 14, 13, 16, 10.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q5/5**

SET 3 • HSS-ID.A.1

# BACKWARD 2

Find quartiles as medians of the halves • Grade 6 • Statistics & Probability • 6.SP.B.5.c

For 2, 4, 6, 8, 10, 12, 14, 16, find Q1 and Q3 (each the median of a half).

Q1: _____

Q3: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q1/5**

For 1, 3, 5, 7, 9, 11, 13 (an odd count), leave out the median and find Q1 and Q3.

Q1: _____

Q3: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q2/5**

## True or false?

For 1, 2, 3, 4, 5, 6, 7, 8, 9, Q1 is 2.5 when the median 5 is left out of both halves.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q3/5**

For 10, 20, 30, 40, 50, 60, what is the interquartile range Q3 − Q1?

**A.** 50

**B.** 20

**C.** 35

**D.** 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q4/5**

For 5, 6, 8, 9, 12, 15, find Q1.

Q1: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

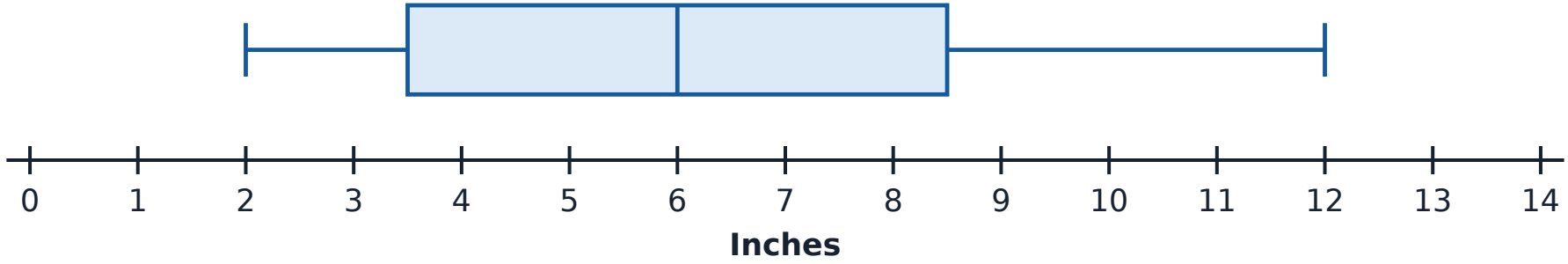
**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q5/5**

SET 3 • HSS-ID.A.1

# BACKWARD 3

Read a box plot • Grade 6 • Statistics & Probability • 6.SP.B.4

The box plot shows the snowfall in some storms. What is the median snowfall?



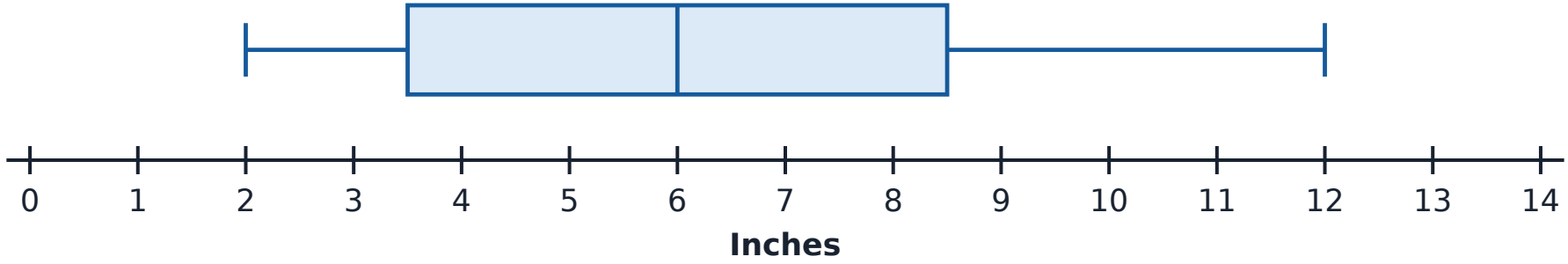

Median (inches): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q1/5**

The box plot shows the snowfall in some storms. What is the interquartile range?




IQR (inches): _____

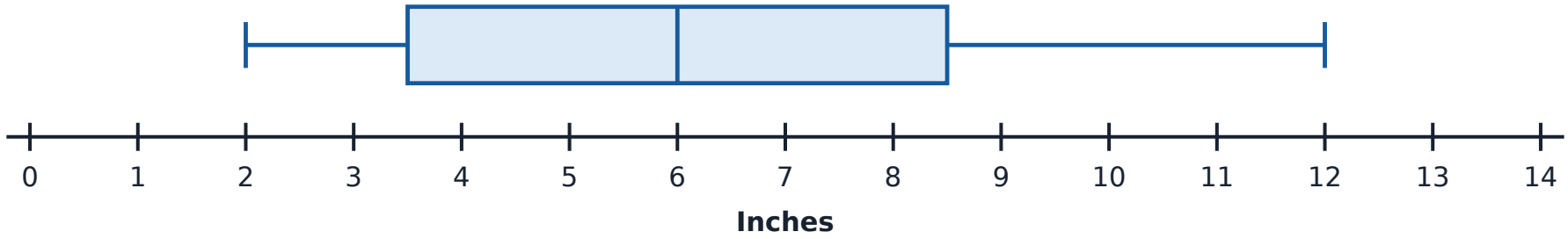
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q2/5**

**True or false?**

The box plot shows the snowfall in some storms. About half of the storms had at least 6 inches.




**A.** True

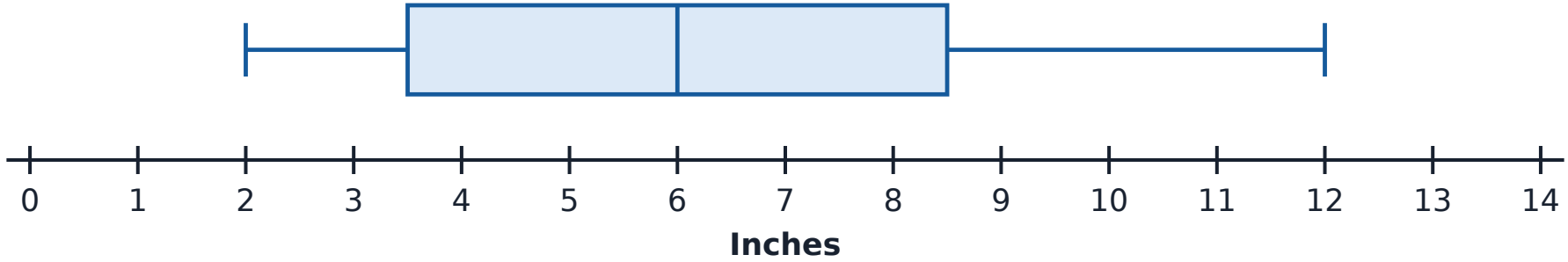
**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q3/5**

The box plot shows the snowfall in some storms. What is the range?



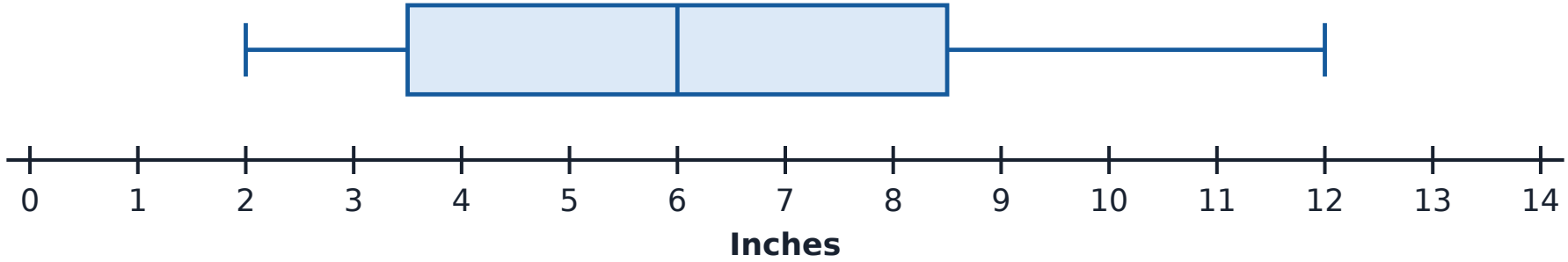

**A.** 6 inches **B.** 10 inches **C.** 12 inches **D.** 5 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q4/5**

The box plot shows the snowfall in some storms. What is Q1?




Q1 (inches): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q5/5**

SET 3 • HSS-ID.A.1

# FORWARD 1

Mark outliers on a box plot • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

A data set has Q1 = 20 and Q3 = 30. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. What are the lower and upper fences?

Lower fence: _____

Upper fence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 3 • FORWARD 1 • Q1/5

Prices of 10 used bikes (dollars): 180, 210, 195, 205, 190, 600, 200, 185, 215, 220. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. How many outliers are there, and where does the upper whisker end?

Number of outliers: _____

Upper whisker ends at: _____

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S3-F1-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 3 • FORWARD 1 • Q2/5**

## True or false?

In a box plot that marks outliers, a whisker ends at the fence value.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 3 • FORWARD 1 • Q3/5

Race times (seconds) of 9 swimmers: 52, 55, 54, 31, 56, 53, 57, 55, 54. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. Which value is an outlier?

**A.** 31

**B.** 57

**C.** both 31 and 57

**D.** none

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S3-F1-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 3 • FORWARD 1 • Q4/5**

A box plot that marks outliers has Q1 = 14 and Q3 = 17, and the largest data values are 18 and 41. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. Where does the upper whisker end?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

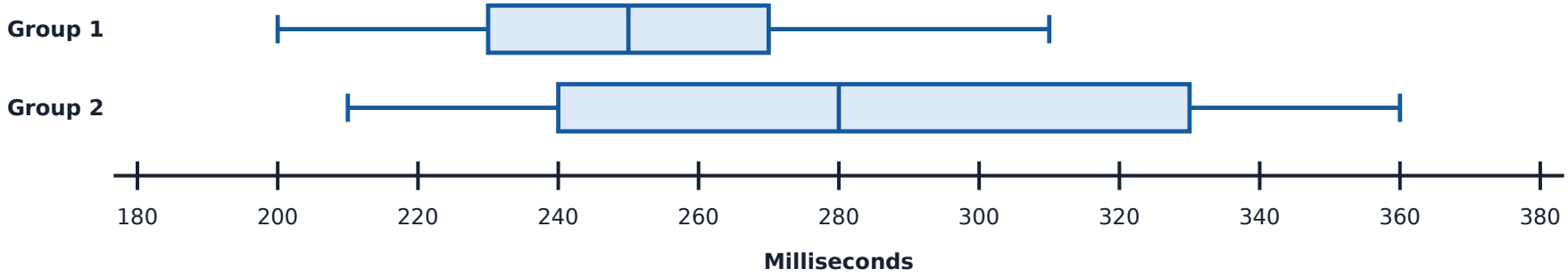
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSS-ID.A.1

# FORWARD 2

Compare two box plots • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

The box plots show the reaction times of two groups. Which comparison is correct?




**A.** Group 1 has the greater median and the greater IQR.

**B.** Group 1 has the greater range and the greater median.

**C.** The groups have the same median but different IQRs.

**D.** Group 2 has the greater median and the greater IQR.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S3-F2-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 3 • FORWARD 2 • Q1/5**

Box plot 1 has median 40 and IQR 12; box plot 2 has median 46 and IQR 8. By how much do the medians differ, and which has the smaller IQR (enter 1 or 2)?

Difference of medians: _____

Smaller IQR (1 or 2): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 3 • FORWARD 2 • Q2/5

## True or false?

If two box plots have the same median, the two data sets have the same mean.

**A.** True

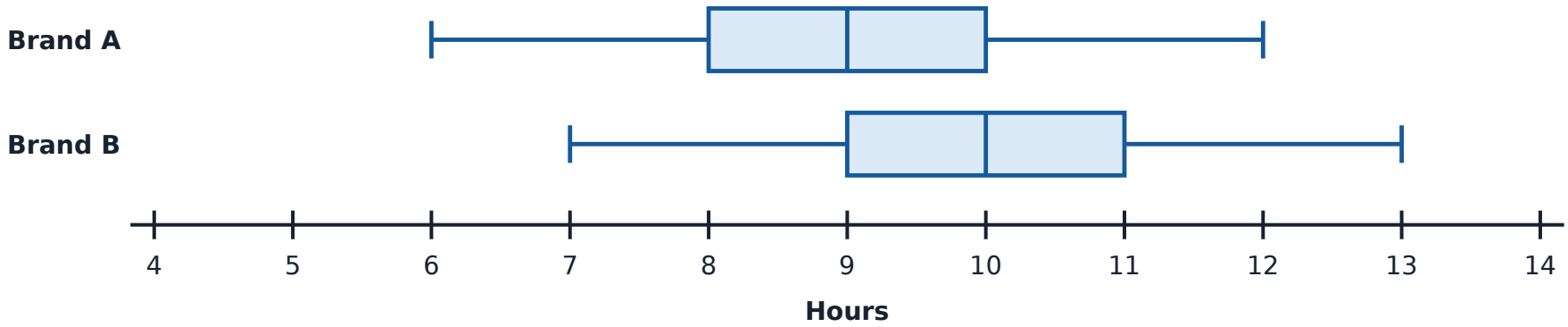
**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 3 • FORWARD 2 • Q3/5

The box plots show the battery lives of two brands. About what fraction of brand B batteries last longer than the median life of brand A?




**A.** about 1/2 **B.** about 3/4 **C.** none **D.** about 1/4

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S3-F2-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 3 • FORWARD 2 • Q4/5**

Two box plots: class A has Q1 = 70 and Q3 = 85; class B has Q1 = 72 and Q3 = 80. Give each IQR.

IQR of A: _____

IQR of B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 3 • FORWARD 2 • Q5/5

SET 4 • HSS-ID.A.2 • INTERPRETING CATEGORICAL AND QUANTITATIVE DATA

# Compare centers with the measure suited to the shape

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.A.2** |
| **BACKWARD 1** | Compute a mean | **6.SP.B.5.c** |
| **BACKWARD 2** | Compute a median | **6.SP.B.5.c** |
| **BACKWARD 3** | Recognize skew and extreme values | **6.SP.A.2** |
| **FORWARD 1** | Find a missing value from a target mean or median | **HSS-ID.A.2** |
| **FORWARD 2** | Predict how changes in data move the mean and median | **HSS-ID.A.3** |

SET 4 • HSS-ID.A.2

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Sale prices (thousands of dollars) of 8 homes in each of two towns: Town A: 210, 225, 230, 240, 245, 250, 260, 900; Town B: 190, 220, 235, 255, 270, 280, 300, 310. Compare the typical prices with one measure of center suited to the shapes of both data sets. Give that measure for each town.

Town A: _____

Town B: _____

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S4-M-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 4 • MAIN • Q1/5**

Scores of 7 students in each of two classes: class P: 70, 74, 76, 78, 80, 82, 86; class Q: 66, 70, 72, 74, 76, 78, 82. Which comparison of centers is appropriate and correct?

**A.** Compare medians: P's median 76 is 2 points above Q's median 74, so P is 2 points higher.

**B.** Compare the highest scores: P's 86 is 4 points above Q's 82, so P is 4 points higher.

**C.** Both are roughly symmetric with no outliers, so compare means: P's mean 78 is 4 points above Q's mean 74.

**D.** Both are roughly symmetric, so compare means: P's mean 76 is 2 points above Q's mean 74.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S4-M-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 4 • MAIN • Q2/5**

Delivery times (days) of 9 orders from each of two stores: store X: 2, 2, 3, 3, 3, 4, 4, 5, 14; store Y: 2, 3, 3, 4, 4, 5, 5, 6, 6. Compare the typical delivery times with one measure of center suited to the shapes of both data sets. Give that measure for each store.

Store X: _____

Store Y: _____

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S4-M-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 4 • MAIN • Q3/5**

Heights (cm) of 7 seedlings of each of two plants: plant A: 5, 6, 6, 7, 7, 8, 30; plant B: 8, 8, 9, 9, 9, 10, 10. Which comparison of typical heights is appropriate?

**A.** A is skewed by the value 30, so compare medians: B's median 9 cm is greater than A's median 7 cm.

**B.** Compare means: A's mean (about 9.9 cm) is greater, so A's seedlings are typically taller.

**C.** Compare means: both are about 9 cm, so the seedlings are typically the same height.

**D.** Compare maximums: A's tallest seedling (30 cm) is taller, so A's seedlings are typically taller.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S4-M-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 4 • MAIN • Q4/5**

Masses (grams) of 6 bags filled by each of two machines: machine 1: 498, 499, 500, 500, 501, 502; machine 2: 503, 504, 505, 505, 506, 507. Compare the typical masses with one measure of center suited to the shapes of both data sets. Give that measure for each machine.

Machine 1: _____

Machine 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 4 • MAIN • Q5/5

SET 4 • HSS-ID.A.2

# BACKWARD 1

Compute a mean • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean of 4, 7, 10.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q1/5**

Find the mean of 2, 3, 5, 6. Give a decimal.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q2/5**

## True or false?

The mean of 1, 2, 3, 10 is 4.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q3/5**

What is the mean of 5, 5, 6, 8?

**A.** 24

**B.** 5

**C.** 6

**D.** 5.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q4/5**

Find the mean of 1.5, 2.5, 3.5. Give a decimal.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q5/5**

SET 4 • HSS-ID.A.2

# BACKWARD 2

Compute a median • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the median of 8, 2, 6.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q1/5**

Find the median of 3, 9, 5, 7.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q2/5**

## True or false?

The median of 1, 2, 3, 100 is 2.5.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q3/5**

What is the median of 10, 12, 12, 15, 30?

**A.** 15.8

**B.** 12

**C.** 30

**D.** 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q4/5**

Find the median of 4.5, 6, 5, 7.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q5/5**

SET 4 • HSS-ID.A.2

# BACKWARD 3

Recognize skew and extreme values • Grade 6 • Statistics & Probability • 6.SP.A.2

Which data set is skewed to the right?

**A.** 5, 5, 5, 5, 5

**B.** 1, 2, 2, 3, 3, 3, 4, 15

**C.** 1, 12, 13, 13, 14, 14, 14, 15

**D.** 1, 2, 3, 4, 5, 6, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q1

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q1/5**

## True or false?

A data set with a single value far below all the others is skewed to the left.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q2

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q2/5**

Which value is extreme in 20, 22, 21, 23, 22, 60?

**A.** 23

**B.** none

**C.** 20

**D.** 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q3

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q3/5**

## True or false?

In 4, 5, 5, 6, 6, 6, 7, 7, 8 the values are spread evenly on both sides of 6, so the data are roughly symmetric.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q4

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q4/5**

Incomes in a town include a few very large incomes. The distribution is most likely

**A.** skewed to the right

**B.** skewed to the left

**C.** uniform

**D.** symmetric

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q5

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q5/5**

SET 4 • HSS-ID.A.2

# FORWARD 1

Find a missing value from a target mean or median • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Four quiz scores are 80, 85, 90, 75. What fifth score makes the mean of all five scores 84?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 4 • FORWARD 1 • Q1/5

The values 3, 5, 8 and x have mean 7. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 4 • FORWARD 1 • Q2/5

## True or false?

In a data set of 7 values, replacing the largest value with a larger number can change the median.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • FORWARD 1 • Q3/5

The values 4, 6, 9, 10, 12 and one more value x have a median of 8. What is x?

**A.** 16

**B.** 7

**C.** 8

**D.** 6

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S4-F1-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 4 • FORWARD 1 • Q4/5**

A team's mean score over 9 games is 70. What score in a 10th game makes the mean of all 10 games 72?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSS-ID.A.2

# FORWARD 2

Predict how changes in data move the mean and median • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

The values are 10, 12, 13, 15, 40. If 40 is replaced by 20, what happens?

**A.** The mean decreases by 20 and the median stays 13.

**B.** The mean decreases by 4 and the median stays 13.

**C.** The mean and the median both decrease by 4.

**D.** The mean stays the same and the median decreases.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S4-F2-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 4 • FORWARD 2 • Q1/5**

## True or false?

Adding 5 to every value in a data set increases both the mean and the median by 5.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 4 • FORWARD 2 • Q2/5

Find the mean and the median of 2, 4, 6, 8, 100.

Mean: _____

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 4 • FORWARD 2 • Q3/5

Salaries at a company include a few very high salaries. Which statement is most likely true?

**A.** The mean salary is greater than the median salary.

**B.** The median is pulled up more than the mean.

**C.** The median salary is greater than the mean salary.

**D.** The mean and the median are equal.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S4-F2-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 4 • FORWARD 2 • Q4/5**

## True or false?

Doubling every value in a data set doubles the median but leaves the mean unchanged.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 4 • FORWARD 2 • Q5/5

# Compare spreads with the IQR or the standard deviation

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.A.2** |
| **BACKWARD 1** | Find the range and the interquartile range | **6.SP.B.5.c** |
| **BACKWARD 2** | Measure distances from the mean | **6.SP.B.5.c** |
| **BACKWARD 3** | Square, average, and take square roots | **8.EE.A.2** |
| **FORWARD 1** | Compare population and sample standard deviations | **HSS-ID.A.2** |
| **FORWARD 2** | Predict how adding or multiplying changes center and spread | **HSS-ID.A.2** |

SET 5 • HSS-ID.A.2

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Ages of 9 players on each of two teams: team A: 14, 15, 15, 16, 16, 17, 18, 18, 19; team B: 13, 14, 16, 16, 17, 18, 20, 22, 25. Give the interquartile range of each team. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

IQR of team A: _____

IQR of team B: _____

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S5-M-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 5 • MAIN • Q1/5**

Bus delays (minutes) on 5 days: route 1: 4, 6, 8, 10, 12; route 2: 7, 8, 8, 8, 9. Both are symmetric. Give the standard deviation of each route, to the nearest tenth. Use the population standard deviation (divide by n). A calculator or spreadsheet may be used.

Route 1: _____

Route 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.2 • Set 5 • MAIN • Q2/5

Monthly rents (dollars) of 7 apartments in each of two buildings: A: 800, 850, 900, 950, 1000, 1050, 2400; B: 700, 800, 900, 1000, 1100, 1200, 1300. Which comparison of spread is appropriate and correct?

**A.** A has an outlier, so compare IQRs: B's IQR ($400) is twice A's ($200), so B's typical rents vary more.

**B.** Compare standard deviations: A's is greater, so A's typical rents vary more than B's.

**C.** Compare IQRs: A's IQR ($400) is twice B's ($200), so A's typical rents vary more.

**D.** Compare ranges: A's range ($1,600) is greater than B's ($600), so A's typical rents vary more.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S5-M-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 5 • MAIN • Q3/5**

Lengths (cm) of 8 fish from each of two lakes: lake A: 20, 22, 23, 25, 26, 28, 30, 31; lake B: 18, 24, 24, 25, 25, 26, 26, 40. Lake B has outliers. Give the interquartile range of each lake. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

IQR of lake A: _____

IQR of lake B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 5 • MAIN • Q4/5

Weekly savings (dollars) of 5 students in each of two groups: group 1: 10, 20, 30, 40, 50; group 2: 25, 28, 30, 32, 35. Both are symmetric. Give the standard deviation of each group, to the nearest tenth. Use the population standard deviation (divide by n). A calculator or spreadsheet may be used.

Group 1: _____

Group 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 5 • MAIN • Q5/5

SET 5 • HSS-ID.A.2

# BACKWARD 1

Find the range and the interquartile range • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the range of 7, 3, 12, 5, 9.

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q1/5**

Q1 = 18 and Q3 = 27. Find the IQR.

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q2/5**

## True or false?

The IQR is the spread of the middle half of the data.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q3/5**

For 2, 4, 6, 8, 10, 12, the IQR (with Q1 and Q3 the medians of the halves) is

**A.** 10

**B.** 6

**C.** 4

**D.** 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q4/5**

A box plot has minimum 5, Q1 9, median 12, Q3 16, maximum 25. Give the IQR and the range.

IQR: _____

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q5/5**

SET 5 • HSS-ID.A.2

# BACKWARD 2

Measure distances from the mean • Grade 6 • Statistics & Probability • 6.SP.B.5.c

The mean of 2, 6, 10 is 6. What are the distances of the values from the mean?

Distance of 2: _____

Distance of 10: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q1/5**

Find the mean absolute deviation of 1, 3, 5, 7.

MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q2/5**

**True or false?**

A data set whose values are all far from the mean has a large spread.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q3/5**

Which set has the larger spread around its mean?

**A.** 0, 10, 20

**B.** cannot tell

**C.** 9, 10, 11

**D.** they are the same

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q4/5**

The values 5, 5, 5, 5 have mean 5. What is their mean absolute deviation?

MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q5/5**

SET 5 • HSS-ID.A.2

# BACKWARD 3

Square, average, and take square roots • Grade 8 • Expressions & Equations • 8.EE.A.2

Find $\sqrt{16} + \sqrt{9}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q1

**Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q1/5**

The squared distances from a mean are 4, 0, 4. Find their average.

Average: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q2

**Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q2/5**

**True or false?**

$\sqrt{(9 + 16)} = \sqrt{9} + \sqrt{16}$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

**Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q3/5**

Which is √0.25?

**A.** 0.125

**B.** 0.0625

**C.** 0.05

**D.** 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q4

**Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q4/5**

Find $\sqrt{8}$ to the nearest tenth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q5

**Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q5/5**

SET 5 • HSS-ID.A.2

# FORWARD 1

Compare population and sample standard deviations • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

For the data 4, 6, 8, 10, 12, give the sample standard deviation (divide by $n - 1$), to the nearest hundredth. A calculator or spreadsheet may be used.

$s =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 5 • FORWARD 1 • Q1/5

## True or false?

For the same data, the sample standard deviation (divide by $n - 1$) is never smaller than the population standard deviation (divide by $n$).

**A.** True

**B.** False

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 1 • Q2/5

For 2, 4, 6, give the population and the sample standard deviations, to the nearest hundredth. A calculator or spreadsheet may be used.

Population: _____

Sample: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 5 • FORWARD 1 • Q3/5

A spreadsheet gives 5.2 for STDEV.S and 4.9 for STDEV.P of the same data. Which statement is correct?

**A.** STDEV.S divides by n and treats the data as the whole population; STDEV.P divides by n − 1 for a sample.

**B.** The two functions measure different spreads: STDEV.S the range of the data and STDEV.P their IQR.

**C.** One of the two values must be an error, because a data set has only one standard deviation.

**D.** STDEV.S divides by n − 1 and treats the data as a sample; STDEV.P divides by n and treats them as the population.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S5-F1-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 5 • FORWARD 1 • Q4/5**

Ten values have sum of squared deviations from their mean equal to 90. Give the population and sample variances.

Population variance: _____

Sample variance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSS-ID.A.2

# FORWARD 2

Predict how adding or multiplying changes center and spread • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Every value in a data set is increased by 10. What happens to the standard deviation and the IQR?

**A.** Both stay the same.

**B.** The SD stays the same and the IQR increases by 10.

**C.** Both increase by 10.

**D.** Both are multiplied by 10.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S5-F2-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 5 • FORWARD 2 • Q1/5**

A data set has mean 20 and standard deviation 4. Every value is multiplied by 3. Give the new mean and standard deviation.

Mean: _____

Standard deviation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 5 • FORWARD 2 • Q2/5

## True or false?

Converting temperatures from degrees to tenths of a degree (multiplying by 10) multiplies the IQR by 10.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 2 • Q3/5

A data set has median 50 and IQR 8. Each value is changed to 2·value + 5. Give the new median and IQR.

Median: _____

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 5 • FORWARD 2 • Q4/5

Test scores have SD 6. The teacher adds 3 bonus points to every score. The new SD is

**A.** 6

**B.** 18

**C.** 9

**D.** 3

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S5-F2-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 5 • FORWARD 2 • Q5/5**

# Interpret differences in shape, center and spread in context

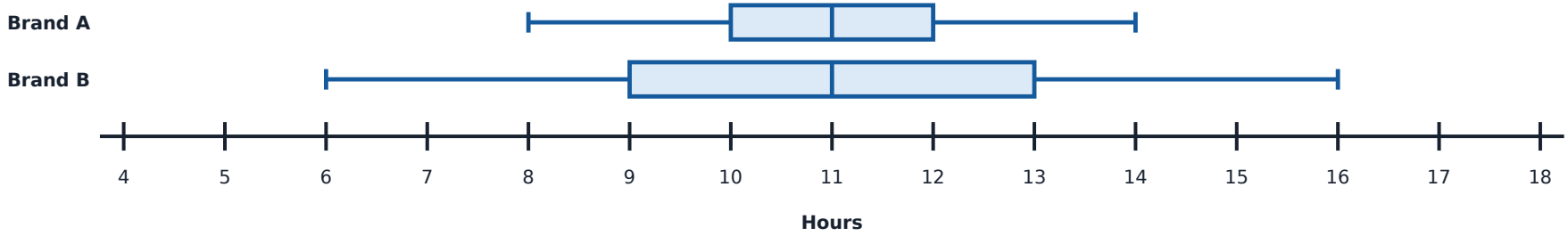
| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.A.3** |
| **BACKWARD 1** | Describe the shape of a distribution | **6.SP.A.2** |
| **BACKWARD 2** | What center and spread tell about data | **6.SP.A.3** |
| **BACKWARD 3** | Compare two distributions informally | **7.SP.B.4** |
| **FORWARD 1** | Measure separation in units of spread | **HSS-ID.A.3** |
| **FORWARD 2** | Judge whether a difference supports a claim | **HSS-IC.B.6** |

SET 6 • HSS-ID.A.3

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

The box plots show the battery lives (hours) of two brands. Which interpretation is correct?




**A.** Typical lives are the same (both medians 11 hours), but brand B's lives vary more (IQR 4 versus 2 hours).

**B.** Brand A's lives vary more, because its box is narrower than brand B's box is.

**C.** Brand B lasts longer, because its longest-lasting battery runs longer than brand A's does.

**D.** Brand B is better in every way, because its whiskers and its box are both longer.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S6-M-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 6 • MAIN • Q1/5**

Waiting times at two stores: store 1 is roughly symmetric with mean 6 minutes and standard deviation 1.5 minutes; store 2 is skewed right with median 5 minutes, IQR 6 minutes, and a few waits over 20 minutes. Which interpretation is correct?

**A.** Store 2 often has a shorter wait, but its waits are much less predictable, with occasional very long waits.

**B.** Store 2 always has shorter waits than store 1, because its median is 1 minute lower than the mean of store 1.

**C.** The stores cannot be compared at all, because they are summarized with different measures.

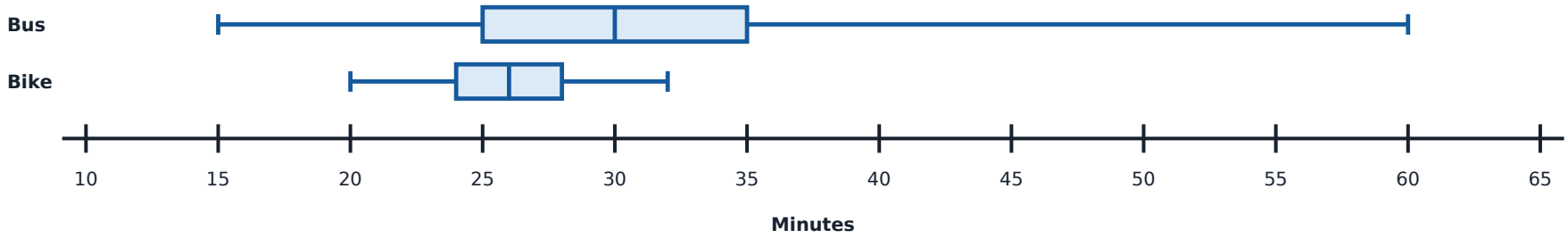
**D.** Store 1's waits vary more, because its waits are described with a mean and a standard deviation.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S6-M-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 6 • MAIN • Q2/5**

The box plots show commute times (minutes) by bus and by bike for the same group of workers. Which interpretation is correct?




**A.** Bus trips are shorter in general, because the bus box plot starts at a lower time than the bike box plot.

**B.** Bike trips vary more, because the bike box plot is shorter than the bus box plot on the scale.

**C.** Bike trips are more consistent; bus trips take a little longer in the middle and vary much more, including long trips.

**D.** Both ways take the same time, because both box plots contain the times from 25 to 28 minutes.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S6-M-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 6 • MAIN • Q3/5**

Two classes both have mean score 70 on a test. Class X's scores form two clusters, around 55 and around 85; class Y's form one mound around 70. Which interpretation is correct?

**A.** Class X did better, because some of its students scored around 85, which is above every score in class Y.

**B.** Class Y has more spread, because its scores are all close to the middle of the range of scores.

**C.** The equal means hide a difference: class X seems to contain two groups doing very differently; class Y is more alike.

**D.** The classes performed the same way, because their means are equal and the mean summarizes the whole class.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S6-M-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 6 • MAIN • Q4/5**

Apples from two farms both have mean weight 150 g; farm A's weights have standard deviation 5 g and farm B's have standard deviation 20 g. Both distributions are roughly symmetric. Which interpretation is correct?

**A.** Farm B's apples are heavier, because a standard deviation of 20 g is larger than one of 5 g.

**B.** Farm A's apples are more uniform in size; on average, apples from both farms weigh about the same.

**C.** Farm A's apples are lighter, because a smaller standard deviation means a smaller mean weight.

**D.** The farms' apples are the same in every way, because their mean weights are both 150 g.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S6-M-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 6 • MAIN • Q5/5**

SET 6 • HSS-ID.A.3

# BACKWARD 1

Describe the shape of a distribution • Grade 6 • Statistics & Probability • 6.SP.A.2

A histogram has one peak in the middle and falls off evenly on both sides. Its shape is

**A.** symmetric

**B.** bimodal

**C.** skewed left

**D.** skewed right

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q1/5**

## True or false?

A histogram with two separate peaks is called bimodal.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q2/5**

A histogram has a long tail toward small values. Its shape is

**A.** uniform

**B.** symmetric

**C.** skewed right

**D.** skewed left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q3/5**

## True or false?

A distribution whose bars are all about the same height is skewed right.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q4/5**

Waiting times at a busy clinic are mostly short with a few very long waits. The shape is most likely

**A.** skewed left

**B.** symmetric

**C.** bimodal

**D.** skewed right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q5/5**

SET 6 • HSS-ID.A.3

# BACKWARD 2

What center and spread tell about data • Grade 6 • Statistics & Probability • 6.SP.A.3

## True or false?

A measure of center summarizes all the values of a data set with a single number.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

**Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q1/5**

## True or false?

A measure of spread tells how much the values vary.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

**Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q2/5**

Two data sets have the same median but different IQRs. Which is true?

**A.** Their typical values are similar, but one varies more.

**B.** One of them has a larger typical value than the other.

**C.** They are identical data sets with the same values.

**D.** They have the same range and the same maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

**Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q3/5**

Which describes spread?

**A.** the median

**B.** the IQR

**C.** the mean

**D.** the mode

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

**Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q4/5**

## True or false?

A data set with a larger mean must have a larger standard deviation.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

**Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q5/5**

SET 6 • HSS-ID.A.3

# BACKWARD 3

Compare two distributions informally • Grade 7 • Statistics & Probability • 7.SP.B.4

Group 1 has median 12 and IQR 3; group 2 has median 15 and IQR 3. Which is true?

**A.** Group 2 varies more, since its median is higher.

**B.** The groups are the same, since the IQRs are equal.

**C.** Group 2's typical value is 3 higher, with similar spread.

**D.** Group 1's values are higher, since its IQR is the same.

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q1

**Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q1/5**

## True or false?

If two groups have very different medians and small IQRs, their distributions overlap very little.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q2

**Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q2/5**

Group A has mean 40 and group B has mean 46. By how much are the means different?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q3

**Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q3/5**

Which pair of groups overlaps the most?

**A.** medians 20 and 30, both IQRs 1

**B.** medians 20 and 21, both IQRs 10

**C.** medians 20 and 40, both IQRs 2

**D.** medians 10 and 50, both IQRs 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q4

**Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q4/5**

## True or false?

Two groups with the same center must have the same spread.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q5

**Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q5/5**

SET 6 • HSS-ID.A.3

# FORWARD 1

Measure separation in units of spread • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

Two roughly symmetric groups have means 70 and 76; both have standard deviation 4. How many standard deviations apart are the means?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 6 • FORWARD 1 • Q1/5

Group medians are 30 and 45; both IQRs are 10. How many IQRs apart are the medians?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 6 • FORWARD 1 • Q2/5

Pair 1: means 50 and 55, both SDs 10. Pair 2: means 50 and 53, both SDs 1. Which pair is more clearly separated?

**A.** pair 1, 5 points apart

**B.** pair 1, 0.5 SDs apart

**C.** pair 2, 3 SDs apart

**D.** they are equally separated

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S6-F1-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 6 • FORWARD 1 • Q3/5**

## True or false?

Two groups whose means are 0.2 standard deviations apart overlap a great deal.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • FORWARD 1 • Q4/5

Why measure a difference in units of spread?

**A.** Units of spread show which group has more values in it.

**B.** A gap matters only relative to how much values vary within each group.

**C.** Units of spread remove the need to compare the centers of the groups.

**D.** Units of spread make every difference look larger than it really is.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S6-F1-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 6 • FORWARD 1 • Q5/5**

SET 6 • HSS-ID.A.3

# FORWARD 2

Judge whether a difference supports a claim • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A report says that brand X bulbs "last much longer" than brand Y: means 1,010 and 1,000 hours, both with standard deviation 80 hours. Evaluate.

**A.** Weak: the means differ by about 0.13 standard deviations, so the lives overlap almost completely.

**B.** Weak: the claim would be supported only if the standard deviations were different.

**C.** Strong: both brands have the same standard deviation, so the difference in means must be large.

**D.** Strong: 1,010 is greater than 1,000, so every brand X bulb lasts longer than any brand Y bulb.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S6-F2-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q1/5**

## True or false?

A difference of 5 points between group means is more convincing when each group's standard deviation is 2 than when it is 20.

**A.** True

**B.** False

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q2/5

A coach claims the new training "improved race times a lot": median time dropped from 52 to 50 seconds; IQRs 8 seconds. Which evaluation is best?

**A.** The claim cannot be judged, because the IQRs of the two groups are equal.

**B.** The claim is wrong, because a drop in the median time means the athletes slowed down.

**C.** The drop is large, because two seconds is a long time in a race of 52 seconds.

**D.** The drop is small relative to the spread (a quarter of an IQR), so "a lot" overstates it.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S6-F2-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q3/5**

Two groups have means 120 and 135 and the same standard deviation 10. How many standard deviations apart are they?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q4/5

## True or false?

If two groups' means differ by less than one standard deviation, the groups cannot differ in any important way.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q5/5

# Identify outliers and account for their effects

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.A.3** |
| **BACKWARD 1** | Find the interquartile range | **6.SP.B.5.c** |
| **BACKWARD 2** | Compute the fences | **6.NS.B.3** |
| **BACKWARD 3** | How an extreme value affects the mean and the median | **6.SP.B.5.d** |
| **FORWARD 1** | Decide whether an outlier is an error or a real value | **HSS-ID.A.3** |
| **FORWARD 2** | Report resistant summaries after checking outliers | **HSS-ID.A.3** |

SET 7 • HSS-ID.A.3

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

Points a player scored in 11 games: 12, 15, 14, 18, 16, 13, 17, 15, 41, 14, 16. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. Give the lower fence, the upper fence, and the number of outliers.

Lower fence: _____

Upper fence: _____

Number of outliers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q1/5

Points a player scored in 11 games: 12, 15, 14, 18, 16, 13, 17, 15, 41, 14, 16. The value 41 is an outlier. Find the mean with and without it, each to the nearest tenth.

Mean with 41: _____

Mean without 41: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q2/5

Hours of screen time 11 students reported for one weekend: 6, 8, 7, 9, 7, 8, 30, 6, 7, 8, 1. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. Give the lower fence, the upper fence, and the number of outliers.

Lower fence: _____

Upper fence: _____

Number of outliers: _____

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S7-M-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 7 • MAIN • Q3/5**

Prices (dollars) of 10 used bikes: 180, 210, 195, 205, 190, 600, 200, 185, 215, 220. The $600 bike is an outlier. Which describes the effect of removing it?

**A.** The mean and the median both drop by $40.

**B.** The median drops by $40, while the mean drops by only $2.50.

**C.** The mean drops by $40, while the median drops by only $2.50.

**D.** Neither the mean nor the median changes.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S7-M-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 7 • MAIN • Q4/5**

Support calls received in each of 10 hours: 3, 5, 4, 6, 5, 4, 22, 5, 3, 4. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. Give the upper fence and the number of outliers.

Upper fence: _____

Number of outliers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 7 • MAIN • Q5/5

SET 7 • HSS-ID.A.3

# BACKWARD 1

Find the interquartile range • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Q1 = 12 and Q3 = 20. Find the IQR.

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q1/5**

For 1, 3, 4, 6, 8, 9, find the IQR (Q1 and Q3 are the medians of the halves).

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q2/5**

## True or false?

The IQR is the difference between the maximum and the minimum.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q3/5**

A box plot has Q1 = 30 and Q3 = 42. The IQR is

**A.** 12

**B.** 36

**C.** 42

**D.** 72

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q4/5**

For 2, 4, 5, 7, 9, 10, 12 (odd count; leave out the median), find the IQR.

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q5/5**

SET 7 • HSS-ID.A.3

# BACKWARD 2

Compute the fences • Grade 6 • The Number System • 6.NS.B.3

Find 1.5 × 6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

**Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q1/5**

Find 1.5 × 2.5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

**Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q2/5**

**True or false?**

If Q1 = 10 and the IQR is 4, the lower fence is $10 - 6 = 4$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

**Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q3/5**

If Q3 = 25 and the IQR is 10, the upper fence is

**A.** 15

**B.** 35

**C.** 40

**D.** 37.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

**Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q4/5**

Q1 = 7.5 and the IQR is 3. Find the lower fence.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

**Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q5/5**

SET 7 • HSS-ID.A.3

# BACKWARD 3

How an extreme value affects the mean and the median • Grade 6 • Statistics & Probability • 6.SP.B.5.d

In 3, 4, 5, 6, 50, which is pulled more by the value 50?

**A.** neither

**B.** the median

**C.** the mean

**D.** both equally

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q1/5**

## True or false?

The median of 2, 3, 4, 5, 100 is 4.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q2/5**

Find the mean of 2, 3, 4, 5, 100.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q3/5**

## True or false?

Removing an extreme value always changes the median more than the mean.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q4/5**

For skewed data with outliers, the better summary of a typical value is usually

**A.** the maximum

**B.** the median

**C.** the range

**D.** the mean

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q5/5**

SET 7 • HSS-ID.A.3

# FORWARD 1

Decide whether an outlier is an error or a real value • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

Heights of 9 adults in a study are listed in centimeters; one value is 17.5. Which is the most reasonable judgment?

**A.** Outliers are always errors, so every outlier in the data should be removed.

**B.** It is a real height, so it must be kept exactly as it is in the data.

**C.** It is probably a recording error (likely 175 cm), so check the record before using it.

**D.** It is an outlier, so it should be deleted without any further checking.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S7-F1-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 7 • FORWARD 1 • Q1/5**

## True or false?

A value that the 1.5·IQR rule flags as an outlier must be removed from the data.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q2/5

Among 30 home prices, one mansion sells for 10 times the median. Which is correct?

**A.** It must be removed, because the outlier rule requires it to be deleted.

**B.** It is a legitimate unusual value; report resistant summaries such as the median and the IQR.

**C.** It is a recording error, because no real home could cost ten times the median price.

**D.** It should replace the median, because it is the most informative value in the data.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S7-F1-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 7 • FORWARD 1 • Q3/5**

## True or false?

An outlier can be a correct measurement of an unusual case.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q4/5

A thermometer log shows 18, 19, 21, 20, 210, 19 °C in a classroom. What is the best next step?

**A.** Report the mean including 210, because every reading is data and must be used as recorded.

**B.** Delete the 19s, because they are the most common values and do not add information.

**C.** Report the 210 reading as the median, because it is the most unusual value in the list.

**D.** Check the 210 reading, since a decimal point was likely lost (21.0), before computing summaries.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S7-F1-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 7 • FORWARD 1 • Q5/5**

SET 7 • HSS-ID.A.3

# FORWARD 2

Report resistant summaries after checking outliers • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

After checking, the $600 bike price is real. Which summaries best describe a typical used bike and the spread?

**A.** the median and the IQR

**B.** the mean and the range

**C.** the maximum and the minimum

**D.** the mean and the standard deviation

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S7-F2-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 7 • FORWARD 2 • Q1/5**

After a recording error is corrected, the data are 12, 15, 14, 18, 16, 13, 17, 15, 14, 14, 16 (the 41 was really 14). Find the new mean to the nearest tenth and the new median.

Mean: _____

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 7 • FORWARD 2 • Q2/5

## True or false?

The IQR barely changes when a single extreme value is removed from a large data set.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 2 • Q3/5

Which summary is most changed by one extreme value?

**A.** Q1

**B.** the standard deviation

**C.** the median

**D.** the IQR

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S7-F2-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 7 • FORWARD 2 • Q4/5**

Data 5, 6, 6, 7, 8, 30: find the median and the IQR (Q1 and Q3 are the medians of the halves).

Median: _____

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 7 • FORWARD 2 • Q5/5

# Estimate percentages with the 68-95-99.7 rule

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.A.4** |
| **BACKWARD 1** | Count standard deviations from the mean | **HSS-ID.A.2** |
| **BACKWARD 2** | Find a percent of a population | **6.RP.A.3.c** |
| **BACKWARD 3** | Use the symmetry of the normal curve | **6.SP.A.2** |
| **FORWARD 1** | Find a cutoff from a percent | **HSS-ID.A.4** |
| **FORWARD 2** | Compare positions in two populations with standardized values | **HSS-ID.A.4** |

SET 8 • HSS-ID.A.4

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Heights of adult women in a population are approximately normal with mean 64 inches and standard deviation 2.5 inches. Use the 68-95-99.7 rule. What percent are between 61.5 and 66.5 inches, and what percent are between 59 and 69 inches?

61.5 to 66.5: _____

59 to 69: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q1/5

Battery lives are approximately normal with mean 40 hours and standard deviation 4 hours. Use the 68-95-99.7 rule. What percent of batteries last more than 48 hours?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q2/5

Commute times are approximately normal with mean 30 minutes and standard deviation 5 minutes. Use the 68-95-99.7 rule. What percent are between 25 and 40 minutes?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q3/5

Egg masses are approximately normal with mean 58 g and standard deviation 3 g. Use the 68-95-99.7 rule. What percent are less than 52 g, and what percent are more than 61 g?

Less than 52 g: _____

More than 61 g: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q4/5

Scores on a test are approximately normal with mean 500 and standard deviation 100. Use the 68-95-99.7 rule. What percent are between 300 and 600, and what percent are below 400?

300 to 600: _____

Below 400: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q5/5

SET 8 • HSS-ID.A.4

# BACKWARD 1

Count standard deviations from the mean • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

The mean is 50 and the standard deviation is 5. How many standard deviations above the mean is 60?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q1/5

The mean is 12 and the standard deviation is 3. How many standard deviations below the mean is 9?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q2/5

## True or false?

With mean 100 and standard deviation 15, the value 70 is 2 standard deviations below the mean.

**A.** True

**B.** False

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q3/5

With mean 64 and standard deviation 2.5, which value is 1 standard deviation above the mean?

**A.** 66.5

**B.** 62.5

**C.** 65

**D.** 67

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S8-B1-Q4**

**High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q4/5**

The mean is 30 and the standard deviation is 5. What value is 2 standard deviations below the mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • HSS-ID.A.4

# BACKWARD 2

Find a percent of a population • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 68% of 500?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

**Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q1/5**

Half of 5% is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

**Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q2/5**

## True or false?

34% + 13.5% = 47.5%.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

**Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q3/5**

If 95% of values are inside an interval, what percent are outside it?

**A.** 5%

**B.** 95%

**C.** 0%

**D.** 2.5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

**Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q4/5**

What is 100% minus 68%, divided by 2?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

**Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q5/5**

SET 8 • HSS-ID.A.4

# BACKWARD 3

Use the symmetry of the normal curve • Grade 6 • Statistics & Probability • 6.SP.A.2

## True or false?

In a normal distribution, half of the values are above the mean.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q1

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q1/5**

**True or false?**

In a normal distribution, more values are between the mean and 1 SD above it than between the mean and 1 SD below it.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q2

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q2/5**

What percent of a normal distribution lies between the mean and 2 standard deviations above it?

**A.** 95%

**B.** 50%

**C.** 47.5%

**D.** 34%

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q3

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q3/5**

What percent of a normal distribution lies between 1 SD below the mean and the mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q4

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q4/5**

What percent of a normal distribution is more than 1 standard deviation below the mean?

**A.** 16%

**B.** 34%

**C.** 2.5%

**D.** 32%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q5

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q5/5**

SET 8 • HSS-ID.A.4

# FORWARD 1

Find a cutoff from a percent • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Heights are approximately normal with mean 64 in and standard deviation 2.5 in. Use the 68-95-99.7 rule. Above what height are the tallest 2.5%?

Height (in): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q1/5

Times are approximately normal with mean 30 min and standard deviation 5 min. Use the 68-95-99.7 rule. Below what time are the fastest 16%?

Time (min): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q2/5

## True or false?

In a normal distribution with mean 200 and standard deviation 20, the middle 95% is from 160 to 240.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q3/5

Test scores are approximately normal with mean 70 and SD 8. Which score separates the top 16%?

**A.** 74

**B.** 62

**C.** 86

**D.** 78

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S8-F1-Q4**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q4/5**

Lengths are approximately normal with mean 12 cm and standard deviation 0.5 cm. Use the 68-95-99.7 rule. Give the interval that holds the middle 68%.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q5/5

SET 8 • HSS-ID.A.4

# FORWARD 2

Compare positions in two populations with standardized values • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Ana scored 85 on a test with mean 75 and SD 5; Ben scored 90 on a test with mean 80 and SD 10. How many standard deviations above the mean is each score?

Ana: _____

Ben: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q1/5

Using the scores above (Ana 2 SDs above, Ben 1 SD above, both tests normal), which is correct?

**A.** Ana's score is relatively higher: about 2.5% scored above her, but about 16% scored above Ben.

**B.** Ben's score is relatively higher, because 90 is a larger number than 85.

**C.** The scores are equally good, because both are 10 points above their means.

**D.** They cannot be compared, because the two tests had different means.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S8-F2-Q2**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q2/5**

## True or false?

A value 1 SD below the mean in one normal population and a value 1 SD below the mean in another are at the same percentile.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q3/5

A 100-m time of 12.0 s comes from a population with mean 13.0 s and SD 0.5 s. How many standard deviations below the mean is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q4/5

Two normal populations: in A, mean 50, SD 5; in B, mean 50, SD 10. Which value is more unusual?

**A.** they are equally unusual

**B.** 60 in population A

**C.** 55 in population B

**D.** 60 in population B

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S8-F2-Q5**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q5/5**

# Estimate normal areas with a table or calculator

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.A.4** |
| **BACKWARD 1** | Compute a standardized value | **7.NS.A.3** |
| **BACKWARD 2** | Read areas as proportions | **7.SP.C.5** |
| **BACKWARD 3** | Find right tails and areas between from left areas | **HSS-ID.A.4** |
| **FORWARD 1** | Find the cutoff for a percentile | **HSS-ID.A.4** |
| **FORWARD 2** | Make decisions with normal areas | **HSS-ID.A.4** |

SET 9 • HSS-ID.A.4

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

IQ scores are approximately normal with mean 100 and standard deviation 15. What proportion of people score below 118? Use the table values printed below (areas to the left of z). Give four decimal places.

| z | 1.0 | 1.2 | 1.5 | 2.0 |
|---|---|---|---|---|
| Area to the left | 0.8413 | 0.8849 | 0.9332 | 0.9772 |

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q1/5

Lengths of calls are approximately normal with mean 50 seconds and standard deviation 8 seconds. What proportion of calls last more than 60 seconds? Use the table values printed below (areas to the left of z). Give four decimal places.

| z | 1.0 | 1.25 | 1.5 | 2.0 |
|---|---|---|---|---|
| Area to the left | 0.8413 | 0.8944 | 0.9332 | 0.9772 |

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q2/5

Daily water use is approximately normal with mean 20 gallons and standard deviation 2.5 gallons. What proportion of days use between 18 and 23 gallons? Use the table values printed below (areas to the left of z). Give four decimal places.

| z | −1.0 | −0.8 | 0.8 | 1.2 |
|---|---|---|---|---|
| Area to the left | 0.1587 | 0.2119 | 0.7881 | 0.8849 |

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q3/5

Grade-point averages at a school are approximately normal with mean 3.0 and standard deviation 0.4. What proportion are below 2.5? Use the table values printed below (areas to the left of z). Give four decimal places.

| z | −1.5 | −1.25 | −1.0 | −0.5 |
|---|---|---|---|---|
| Area to the left | 0.0668 | 0.1056 | 0.1587 | 0.3085 |

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q4/5

Heights of adult men are approximately normal with mean 175 cm and standard deviation 7 cm. What proportion are between 168 and 189 cm? Use the table values printed below (areas to the left of $z$). Give four decimal places.

| $z$ | $-2.0$ | $-1.0$ | $1.0$ | $2.0$ |
|---|---|---|---|---|
| Area to the left | 0.0228 | 0.1587 | 0.8413 | 0.9772 |

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q5/5

SET 9 • HSS-ID.A.4

# BACKWARD 1

Compute a standardized value • Grade 7 • The Number System • 7.NS.A.3

Compute (130 − 100)/15.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

**Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q1/5**

Compute $(42 - 50)/4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

**Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q2/5**

**True or false?**

$(9.5 - 10)/0.5 = -1$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

**Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q3/5**

Compute $(61 - 58)/2$.

**A.** 0.67

**B.** 3

**C.** 1.5

**D.** $-1.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

**Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q4/5**

Compute $(2.2 - 3.0)/0.4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

**Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q5/5**

SET 9 • HSS-ID.A.4

# BACKWARD 2

Read areas as proportions • Grade 7 • Statistics & Probability • 7.SP.C.5

**True or false?**

An area of 0.8413 to the left of $z = 1$ means about 84% of the population is below 1 SD above the mean.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

**Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q1/5**

The table gives area 0.0228 to the left of $z = -2$. What does this mean?

**A.** Exactly 2 values are below the mean.

**B.** About 22.8% of values are below the mean.

**C.** About 2.28% of values are above the mean.

**D.** About 2.28% of values are below 2 SDs below the mean.

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

**Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q2/5**

Write 0.1587 as a percent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q3

**Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q3/5**

## True or false?

A proportion of a population can be greater than 1.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

**Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q4/5**

Which is the largest proportion?

**A.** 0.5

**B.** 0.8413

**C.** 0.9772

**D.** 0.0228

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

**Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q5/5**

SET 9 • HSS-ID.A.4

# BACKWARD 3

Find right tails and areas between from left areas • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

The area to the left of $z = 1.5$ is 0.9332. What is the area to the right of $z = 1.5$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q1/5

Areas to the left: $z = -1$ gives 0.1587; $z = 1$ gives 0.8413. Find the area between $z = -1$ and $z = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q2/5

## True or false?

The area to the right of $z = 0$ is 0.5.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q3/5

The area to the left of $z = 0.5$ is 0.6915. The area between $z = 0$ and $z = 0.5$ is

**A.** 0.3085

**B.** 0.1915

**C.** 0.6915

**D.** 0.5

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S9-B3-Q4**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q4/5**

The area to the left of $z = 2$ is 0.9772. What is the area to the left of $z = -2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q5/5

SET 9 • HSS-ID.A.4

# FORWARD 1

Find the cutoff for a percentile • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

IQ scores are normal with mean 100 and SD 15. The table gives area 0.9332 to the left of $z = 1.5$. What score is at the 93.32nd percentile?

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q1/5

Call lengths are normal with mean 50 s and SD 8 s. The table gives area 0.1056 to the left of $z = -1.25$. What length is at the 10.56th percentile?

Seconds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q2/5

## True or false?

A value with $z = 0$ is at the 50th percentile of a normal distribution.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q3/5

A factory wants only the longest 2.28% of rods to be rejected. Rod lengths are normal with mean 10 cm and SD 0.2 cm; the table gives area 0.9772 to the left of $z = 2$. What cutoff should it use?

**A.** 10.2 cm

**B.** 12 cm

**C.** 10.4 cm

**D.** 9.6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q4/5

Heights are normal with mean 175 cm and SD 7 cm; area 0.8413 is to the left of $z = 1$. What height separates the tallest 15.87%?

Height (cm): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q5/5

SET 9 • HSS-ID.A.4

# FORWARD 2

Make decisions with normal areas • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

A bolt fits if its diameter is between 9.8 and 10.2 mm. Diameters are normal with mean 10.0 mm and SD 0.1 mm. Table: area 0.9772 left of $z = 2$ and 0.0228 left of $z = -2$. What proportion of bolts fit?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q1/5

Machine A fills bottles with mean 500 mL, SD 4 mL; machine B with mean 502 mL, SD 1 mL. A bottle is underfilled below 498 mL. Table: area 0.3085 to the left of $z = -0.5$; the area to the left of $z = -4$ is less than 0.0001. Which machine underfills fewer bottles?

**A.** A, because its standard deviation is larger

**B.** B: under 0.01% versus about 31% for A

**C.** A, because its mean is closer to 498 mL

**D.** They underfill the same proportion

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q2/5

## True or false?

Reducing a process's standard deviation, with the mean unchanged, reduces the proportion of items far from the mean.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q3/5

Scores are normal with mean 70 and SD 10. A prize goes to scores of 85 or more. Table: area 0.9332 left of z = 1.5. What proportion of students win?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q4/5

Two normal processes have the same mean. Process 1 has SD 2; process 2 has SD 5. Which has more items within 4 units of the mean?

**A.** process 2

**B.** the same

**C.** cannot tell

**D.** process 1

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S9-F2-Q5**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q5/5**

# Recognize when a normal model is not appropriate

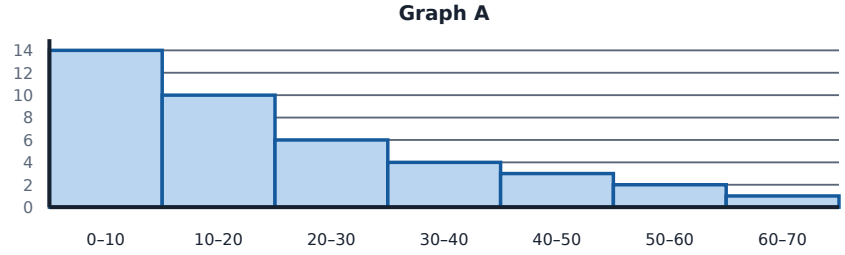
| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.A.4** |
| **BACKWARD 1** | Describe the shape of a distribution | **6.SP.A.2** |
| **BACKWARD 2** | Relate the mean and median to the shape | **6.SP.B.5.d** |
| **BACKWARD 3** | Test a normal model against the data | **HSS-ID.A.4** |
| **FORWARD 1** | Compare actual percents with the 68-95-99.7 rule | **HSS-ID.A.4** |
| **FORWARD 2** | Choose summaries for data that are not normal | **HSS-ID.A.2** |

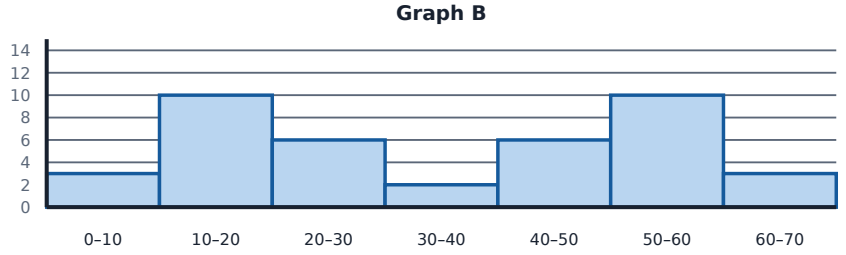
SET 10 • HSS-ID.A.4

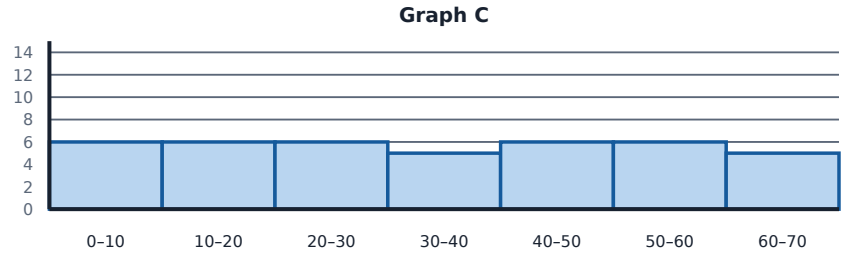
# MAIN

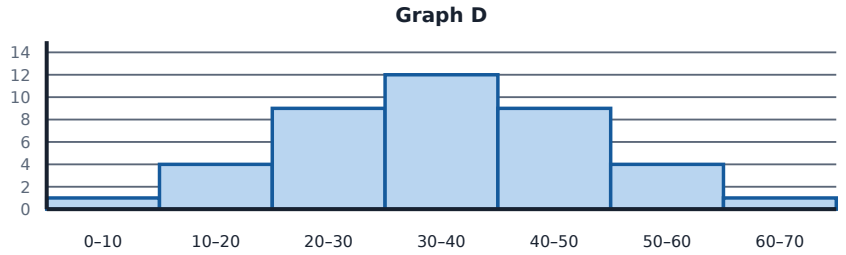
Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Each histogram shows 40 measurements on the same scale. For which data set is a normal model most reasonable? In each histogram, the horizontal axis shows value and the vertical axis shows frequency.




**A.** Graph A

**B.** Graph B

**C.** Graph C

**D.** Graph D

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S10-M-Q1**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q1/5**

Waiting times at a clinic can never be negative. Their mean is 3 minutes and their standard deviation is 4 minutes. If a normal model with this mean and standard deviation were used, what proportion of waiting times would it place below 0 minutes? Use the table values printed below (areas to the left of z). Give four decimal places.

| z | −1.0 | −0.75 | −0.5 | 0.5 |
|---|---|---|---|---|
| Area to the left | 0.1587 | 0.2266 | 0.3085 | 0.6915 |

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q2/5

Waiting times can never be negative, yet a normal model with their mean (3 minutes) and standard deviation (4 minutes) places about 23% of times below 0. What does this show?

**A.** The times are normal, but 23% of them were recorded wrong.

**B.** The standard deviation must have been computed incorrectly.

**C.** A normal model fits, because the mean is a positive number.

**D.** A normal model is not appropriate for these waiting times.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S10-M-Q3**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q3/5**

Which data set is least likely to be approximately normal?

**A.** the masses of eggs from one large farm

**B.** repeated careful measurements of one object's length

**C.** the yearly incomes of all households in a large city

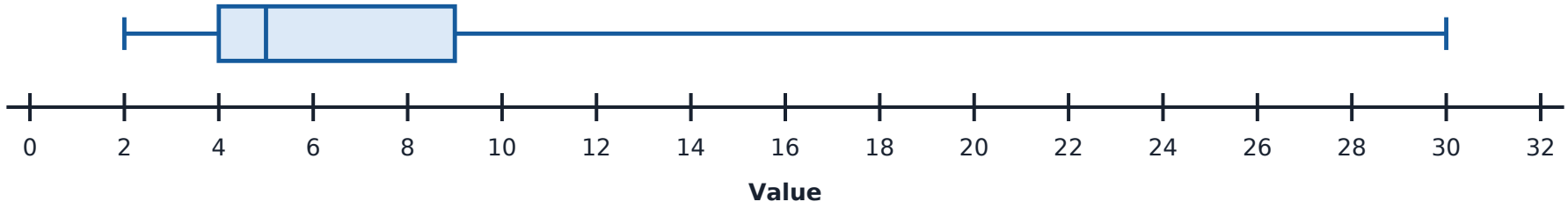
**D.** the heights of adult women in one country

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S10-M-Q4**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q4/5**

A box plot of 60 values shows minimum 2, Q1 4, median 5, Q3 9 and maximum 30. Which conclusion is correct?




**A.** The data are roughly symmetric, so a normal model fits well.

**B.** The data are skewed left, so a normal model is not a good fit.

**C.** A median of 5 shows that the data must be normal.

**D.** The data are skewed right, so a normal model is not a good fit.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S10-M-Q5**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q5/5**

SET 10 • HSS-ID.A.4

# BACKWARD 1

Describe the shape of a distribution • Grade 6 • Statistics & Probability • 6.SP.A.2

## True or false?

A histogram with a long tail to the right is called skewed right.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q1

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q1/5**

What is the shape of the data 1, 1, 2, 2, 2, 3, 3, 4, 6, 9?

**A.** skewed right

**B.** skewed left

**C.** symmetric

**D.** uniform

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q2/5**

## True or false?

A distribution with two separate peaks is called unimodal.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q3

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q3/5**

Which list of values is roughly symmetric?

**A.** 1, 10, 18, 19, 19, 20

**B.** 3, 3, 3, 3, 3, 30

**C.** 1, 1, 1, 2, 9, 20

**D.** 2, 4, 5, 5, 6, 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q4/5**

For the data 4, 5, 5, 6, 6, 6, 7, 7, 8, find the mean and the median.

Mean: _____

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q5/5**

SET 10 • HSS-ID.A.4

# BACKWARD 2

Relate the mean and median to the shape • Grade 6 • Statistics & Probability • 6.SP.B.5.d

**True or false?**

In a distribution skewed right, the mean is usually greater than the median.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q1/5**

A data set has mean 52 and median 40. Which shape is most likely?

**A.** skewed left

**B.** uniform

**C.** symmetric

**D.** skewed right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q2/5**

For the data 1, 2, 2, 3, 22, find the mean and the median.

Mean: _____

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q3/5**

**True or false?**

For strongly skewed data, the mean and standard deviation are the best summaries of center and spread.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q4/5**

For data skewed right, which measure best describes a typical value?

**A.** mean

**B.** midrange

**C.** maximum

**D.** median

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q5/5**

SET 10 • HSS-ID.A.4

# BACKWARD 3

Test a normal model against the data • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Counts of errors per page (never negative) have mean 8 and standard deviation 4. How many standard deviations below the mean is 0, and what percent would a normal model place below 0 by the 68-95-99.7 rule?

Standard deviations: _____

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q1/5

## True or false?

A data set with any mean and any standard deviation is approximately normal.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q2/5

Of 20 values with mean 50 and standard deviation 10, 19 lie between 40 and 60. Is this consistent with a normal model?

**A.** Yes: a normal model predicts about 95% within 1 SD.

**B.** Yes: every value is within 3 SDs of the mean.

**C.** No: a normal model predicts about 68% within 1 SD, but 95% are.

**D.** No: a normal model predicts all values within 1 SD.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S10-B3-Q3**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q3/5**

A normal model is used for 200 values. About how many values does it predict within 2 standard deviations of the mean?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q4/5

## True or false?

A normal model with mean 3 and SD 4 places about 16% of its values below −1.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q5/5

SET 10 • HSS-ID.A.4

# FORWARD 1

Compare actual percents with the 68-95-99.7 rule • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Of 40 values, 27 lie within 1 standard deviation of the mean and 38 lie within 2. What percent is each?

Within 1 SD: _____

Within 2 SD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q1/5

Of 40 values, 67.5% are within 1 SD of the mean and 95% within 2 SDs, and a histogram is one symmetric mound. Which conclusion is reasonable?

**A.** A normal model is not reasonable, because 67.5% is not 68%.

**B.** The data must be skewed right.

**C.** A normal model is reasonable for these data.

**D.** No conclusion is possible without the raw values.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S10-F1-Q2**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q2/5**

## True or false?

If 92% of a data set lies within 1 SD of its mean, the data fit a normal model well.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q3/5

Of 50 values, 46 lie within 1 standard deviation of the mean. What percent is that?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q4/5

Which result most strongly suggests a data set is not normal?

**A.** About half of the values lie above the mean.

**B.** 92% of the values lie within 1 SD of the mean.

**C.** 96% of the values lie within 2 SDs of the mean.

**D.** 70% of the values lie within 1 SD of the mean.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S10-F1-Q5**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q5/5**

SET 10 • HSS-ID.A.4

# FORWARD 2

Choose summaries for data that are not normal • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Home prices in a town are strongly skewed right. Which pair of summaries is most appropriate?

**A.** mode and maximum

**B.** mean and range

**C.** mean and standard deviation

**D.** median and IQR

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S10-F2-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 10 • FORWARD 2 • Q1/5**

Home prices (thousands of dollars) are 150, 160, 170, 180, 190, 200, 900. Find the median and the IQR. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

Median: _____

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.2 • Set 10 • FORWARD 2 • Q2/5

## True or false?

Using a normal model for strongly skewed data can give misleading percentages.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q3/5

For the prices 150, 160, 170, 180, 190, 200, 900 (thousands), which statement is correct?

**A.** The value 900 pulls the median above the mean.

**B.** The mean and the median are equal.

**C.** The value 900 pulls the mean above the median.

**D.** The value 900 changes the IQR more than the mean.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S10-F2-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 10 • FORWARD 2 • Q4/5**

## True or false?

The standard deviation is resistant to one extreme value.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q5/5

# Joint and marginal frequencies in two-way tables

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.B.5** |
| **BACKWARD 1** | Complete and read a two-way frequency table | **8.SP.A.4** |
| **BACKWARD 2** | Write a part of a whole as a fraction, decimal and percent | **7.NS.A.2.d** |
| **FORWARD 1** | Build a joint relative-frequency table | **HSS-ID.B.5** |
| **FORWARD 2** | Look for an association by comparing rows | **HSS-ID.B.5** |

SET 11 • HSS-ID.B.5

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

160 students were asked how they prefer to take a class. What fraction of all students are 10th graders who chose online? Give a fraction in lowest terms.

| | Online | In person | Total |
|---|---|---|---|
| 9th grade | 30 | 50 | 80 |
| 10th grade | 45 | 35 | 80 |
| Total | 75 | 85 | 160 |

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q1/5

160 students were asked how they prefer to take a class. What is the marginal relative frequency of choosing online? Give a fraction in lowest terms.

| | Online | In person | Total |
|---|---|---|---|
| 9th grade | 30 | 50 | 80 |
| 10th grade | 45 | 35 | 80 |
| Total | 75 | 85 | 160 |

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.B.5 • Set 11 • MAIN • Q2/5

A survey of 120 students: 50 are 7th graders and 70 are 8th graders. 18 of the 7th graders play an instrument, and 20 of the 8th graders do not. How many 7th graders do not play, and how many 8th graders do play?

7th graders who do not play: _____

8th graders who play: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q3/5

160 students were surveyed (table). Which statement gives a marginal relative frequency?

| | Online | In person | Total |
|---|---|---|---|
| 9th grade | 30 | 50 | 80 |
| 10th grade | 45 | 35 | 80 |
| Total | 75 | 85 | 160 |

**A.** 53.125% of all students chose in person.

**B.** 28.125% of all students are 10th graders who chose online.

**C.** 56.25% of the 10th graders chose online.

**D.** 37.5% of the 9th graders chose online.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S11-M-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.5 • Set 11 • MAIN • Q4/5**

A dealer has 100 cars in two lots (table). What percent of all the cars are gray, and what percent of all the cars are red cars in lot B?

| | White | Red | Gray | Total |
|---|---|---|---|---|
| Lot A | 18 | 12 | 10 | 40 |
| Lot B | 22 | 8 | 30 | 60 |
| Total | 40 | 20 | 40 | 100 |

Gray: _____

Red in lot B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q5/5

SET 11 • HSS-ID.B.5

# BACKWARD 1

Complete and read a two-way frequency table • Grade 8 • Statistics & Probability • 8.SP.A.4

120 adults were asked whether they own a dog and whether they walk daily (table). How many walk daily?

| | Walks daily | Does not | Total |
|---|---|---|---|
| Has a dog | 36 | 24 | 60 |
| No dog | 15 | 45 | 60 |
| Total | 51 | 69 | 120 |

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q1/5**

From the dog-and-walking table, how many adults have no dog and do not walk daily?

| | Walks daily | Does not | Total |
|---|---|---|---|
| Has a dog | 36 | 24 | 60 |
| No dog | 15 | 45 | 60 |
| Total | 51 | 69 | 120 |

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q2/5**

Of 50 students, 20 are in band and 12 of the band members are 9th graders. 30 of the 50 students are 9th graders. How many 9th graders are not in band?

**A.** 30

**B.** 8

**C.** 18

**D.** 12

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q3

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q3/5**

## True or false?

In a two-way table, the row totals add up to the grand total.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q4/5**

## True or false?

In a two-way table, a cell count can be greater than its row total.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q5/5**

SET 11 • HSS-ID.B.5

# BACKWARD 2

Write a part of a whole as a fraction, decimal and percent • Grade 7 • The Number System • 7.NS.A.2.d

Write 45/160 as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q1

**Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q1/5**

Write 24/120 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

**Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q2/5**

## True or false?

3/8 = 0.375.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

**Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q3/5**

Write 15/32 as a percent.

**A.** 15.32%

**B.** 47%

**C.** 2.13%

**D.** 46.875%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

**Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q4/5**

Write 30/160 in lowest terms.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

**Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q5/5**

SET 11 • HSS-ID.B.5

# FORWARD 1

Build a joint relative-frequency table • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

Convert the online-class table to relative frequencies of the grand total (160). What are the joint relative frequencies for 9th grade online and 9th grade in person? Give decimals.

| | Online | In person | Total |
|---|---|---|---|
| 9th grade | 30 | 50 | 80 |
| 10th grade | 45 | 35 | 80 |
| Total | 75 | 85 | 160 |

9th online: _____

9th in person: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q1/5

## True or false?

In a table of joint relative frequencies of the grand total, all the cells add to 1.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q2/5

In a joint relative-frequency table, the 9th-grade cells are 0.1875 and 0.3125. What is the 9th-grade marginal relative frequency?

**A.** 0.5

**B.** 0.1875

**C.** 0.6

**D.** 0.125

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S11-F1-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.5 • Set 11 • FORWARD 1 • Q3/5**

In the car table (100 cars), what are the joint relative frequencies for white cars in lot A and gray cars in lot B? Give decimals.

| | White | Red | Gray | Total |
|---|---|---|---|---|
| Lot A | 18 | 12 | 10 | 40 |
| Lot B | 22 | 8 | 30 | 60 |
| Total | 40 | 20 | 40 | 100 |

White, lot A: _____

Gray, lot B: _____

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S11-F1-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.5 • Set 11 • FORWARD 1 • Q4/5**

## True or false?

A marginal relative frequency is always smaller than every joint relative frequency in its row.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q5/5

SET 11 • HSS-ID.B.5

# FORWARD 2

Look for an association by comparing rows • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

In the online-class table, what percent of 9th graders and what percent of 10th graders chose online?

| | Online | In person | Total |
|---|---|---|---|
| 9th grade | 30 | 50 | 80 |
| 10th grade | 45 | 35 | 80 |
| Total | 75 | 85 | 160 |

9th: _____

10th: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q1/5

If grade and choice of class were not associated, what would be true?

**A.** Every cell of the table would contain the same count.

**B.** The percent choosing online would be about the same in each grade.

**C.** The number choosing online would be 0 in one grade.

**D.** The grade totals would have to be unequal.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S11-F2-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.5 • Set 11 • FORWARD 2 • Q2/5**

## True or false?

Because more 10th graders (45) than 9th graders (30) chose online, the counts alone prove an association, even if the grades had very different sizes.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q3/5

In the car table, what percent of lot A cars and what percent of lot B cars are gray?

| | White | Red | Gray | Total |
|---|---|---|---|---|
| Lot A | 18 | 12 | 10 | 40 |
| Lot B | 22 | 8 | 30 | 60 |
| Total | 40 | 20 | 40 | 100 |

Lot A: _____

Lot B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.B.5 • Set 11 • FORWARD 2 • Q4/5

Gray makes up 25% of lot A and 50% of lot B. Which is correct?

**A.** Lot B has exactly twice as many cars as lot A.

**B.** Lot and color are associated: gray is more common in lot B.

**C.** Gray is the most common color in lot A.

**D.** Lot and color are not associated.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S11-F2-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.5 • Set 11 • FORWARD 2 • Q5/5**

# Conditional relative frequencies and association

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.B.5** |
| **BACKWARD 1** | Choose the denominator for a conditional relative frequency | **8.SP.A.4** |
| **BACKWARD 2** | Find a percent of a group | **6.RP.A.3.c** |
| **BACKWARD 3** | Compare two fractions | **4.NF.A.2** |
| **FORWARD 1** | Distinguish the two directions of a conditional rate | **HSS-ID.B.5** |
| **FORWARD 2** | Recognize when combining groups reverses a comparison | **HSS-ID.B.5** |

SET 12 • HSS-ID.B.5

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

160 students were surveyed (table). What fraction of the 9th graders chose online, and what fraction of the 10th graders chose online? Give fractions in lowest terms.

| | Online | In person | Total |
|---|---|---|---|
| 9th grade | 30 | 50 | 80 |
| 10th grade | 45 | 35 | 80 |
| Total | 75 | 85 | 160 |

9th graders: _____

10th graders: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q1/5

In the survey, 37.5% of 9th graders and 56.25% of 10th graders chose online. Which statement is correct?

| | Online | In person | Total |
|---|---|---|---|
| 9th grade | 30 | 50 | 80 |
| 10th grade | 45 | 35 | 80 |
| Total | 75 | 85 | 160 |

**A.** 9th graders chose online at a higher rate, so grade and choice are associated.

**B.** The rates are equal, so grade and choice are not associated.

**C.** 10th graders chose online at a higher rate, so grade and choice are associated.

**D.** Each grade has 80 students, so grade and choice cannot be associated.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S12-M-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.5 • Set 12 • MAIN • Q2/5**

160 students were surveyed (table). Of the students who chose online, what percent were 10th graders?

| | Online | In person | Total |
|---|---|---|---|
| 9th grade | 30 | 50 | 80 |
| 10th grade | 45 | 35 | 80 |
| Total | 75 | 85 | 160 |

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q3/5

120 adults were surveyed (table). What percent of dog owners walk daily, and what percent of adults without a dog walk daily?

| | Walks daily | Does not | Total |
|---|---|---|---|
| Has a dog | 36 | 24 | 60 |
| No dog | 15 | 45 | 60 |
| Total | 51 | 69 | 120 |

Dog owners: _____

No dog: _____

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S12-M-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.5 • Set 12 • MAIN • Q4/5**

In the survey, 60% of dog owners and 25% of adults without a dog walk daily. Which conclusion is supported?

| | Walks daily | Does not | Total |
|---|---|---|---|
| Has a dog | 36 | 24 | 60 |
| No dog | 15 | 45 | 60 |
| Total | 51 | 69 | 120 |

**A.** Dog ownership and walking daily are not associated.

**B.** Owning a dog causes people to walk daily.

**C.** Owning a dog and walking daily are associated in this sample.

**D.** Adults without a dog walk daily more often.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S12-M-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.5 • Set 12 • MAIN • Q5/5**

SET 12 • HSS-ID.B.5

# BACKWARD 1

Choose the denominator for a conditional relative frequency • Grade 8 • Statistics & Probability • 8.SP.A.4

Of the 60 dog owners, 36 walk daily. What fraction of dog owners walk daily? Give lowest terms.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q1/5**

**True or false?**

To find the percent of 9th graders who chose online, divide the 9th-grade online count by the grand total of all students.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q2/5**

In the online-class table, which number is the denominator for the percent of online-choosers who are 9th graders?

| | Online | In person | Total |
|---|---|---|---|
| 9th grade | 30 | 50 | 80 |
| 10th grade | 45 | 35 | 80 |
| Total | 75 | 85 | 160 |

**A.** 75, the online total

**B.** 80, the 9th-grade total

**C.** 30, the cell count

**D.** 160, the grand total

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q3/5**

Write 30 out of 80 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q4/5**

## True or false?

A row relative frequency uses that row's total as the denominator.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q5/5**

SET 12 • HSS-ID.B.5

# BACKWARD 2

Find a percent of a group • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

45 is what percent of 80?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q1

**Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q1/5**

What is 25% of 60?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q2

**Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q2/5**

18 out of 24 is what percent?

**A.** 75%

**B.** 18%

**C.** 133%

**D.** 24%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q3

**Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q3/5**

## True or false?

9/16 = 56.25%.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

**Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q4/5**

## True or false?

45 is 60% of 80.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

**Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q5/5**

SET 12 • HSS-ID.B.5

# BACKWARD 3

Compare two fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.2

Which is larger, 3/8 or 9/16?

**A.** 3/8

**B.** they cannot be compared

**C.** 9/16

**D.** they are equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q1

**Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q1/5**

## True or false?

3/5 > 1/4.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q2

**Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q2/5**

Write 3/8 as an equivalent fraction with denominator 16. What is the numerator?

Numerator: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q3

**Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q3/5**

## True or false?

2/5 > 3/4.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q4

**Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q4/5**

Order 1/4, 3/5 and 3/8 from least to greatest.

**A.** 1/4, 3/5, 3/8

**B.** 1/4, 3/8, 3/5

**C.** 3/8, 1/4, 3/5

**D.** 3/5, 3/8, 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q5

**Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q5/5**

SET 12 • HSS-ID.B.5

# FORWARD 1

Distinguish the two directions of a conditional rate • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

In the car table, what percent of the gray cars are in lot B, and what percent of the lot B cars are gray?

| | White | Red | Gray | Total |
|---|---|---|---|---|
| Lot A | 18 | 12 | 10 | 40 |
| Lot B | 22 | 8 | 30 | 60 |
| Total | 40 | 20 | 40 | 100 |

Gray cars in lot B: _____

Lot B cars that are gray: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q1/5

## True or false?

The percent of lot B cars that are gray equals the percent of gray cars that are in lot B.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q2/5

In the car table, what percent of the white cars are in lot A?

| | White | Red | Gray | Total |
|---|---|---|---|---|
| Lot A | 18 | 12 | 10 | 40 |
| Lot B | 22 | 8 | 30 | 60 |
| Total | 40 | 20 | 40 | 100 |

**A.** 30%

**B.** 45%

**C.** 18%

**D.** 40%

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S12-F1-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.5 • Set 12 • FORWARD 1 • Q3/5**

In the car table, what percent of the lot A cars are red?

| | White | Red | Gray | Total |
|---|---|---|---|---|
| Lot A | 18 | 12 | 10 | 40 |
| Lot B | 22 | 8 | 30 | 60 |
| Total | 40 | 20 | 40 | 100 |

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.B.5 • Set 12 • FORWARD 1 • Q4/5

Which statement uses a column total of the car table as its denominator?

| | White | Red | Gray | Total |
|---|---|---|---|---|
| Lot A | 18 | 12 | 10 | 40 |
| Lot B | 22 | 8 | 30 | 60 |
| Total | 40 | 20 | 40 | 100 |

**A.** 12% of all cars are red cars in lot A.

**B.** Of the lot A cars, 30% are red.

**C.** Of the red cars, 60% are in lot A.

**D.** 40% of all cars are in lot A.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S12-F1-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.5 • Set 12 • FORWARD 1 • Q5/5**

SET 12 • HSS-ID.B.5

# FORWARD 2

Recognize when combining groups reverses a comparison • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

Use the hospital table. What is the survival rate for mild cases at hospital A and at hospital B?

| Hospital | Mild: survived / patients | Severe: survived / patients |
|---|---|---|
| A | 95 / 100 | 120 / 400 |
| B | 360 / 400 | 25 / 100 |

A: _____

B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.B.5 • Set 12 • FORWARD 2 • Q1/5

Use the hospital table. What is the survival rate for severe cases at hospital A and at hospital B?

| Hospital | Mild: survived / patients | Severe: survived / patients |
|---|---|---|
| A | 95 / 100 | 120 / 400 |
| B | 360 / 400 | 25 / 100 |

A: _____

B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.B.5 • Set 12 • FORWARD 2 • Q2/5

Use the hospital table. What is the overall survival rate (all 500 patients) at hospital A and at hospital B?

| Hospital | Mild: survived / patients | Severe: survived / patients |
|---|---|---|
| A | 95 / 100 | 120 / 400 |
| B | 360 / 400 | 25 / 100 |

A: _____

B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.B.5 • Set 12 • FORWARD 2 • Q3/5

Hospital A has the higher survival rate for mild cases and for severe cases, but the lower rate overall. Why?

| Hospital | Mild: survived / patients | Severe: survived / patients |
|---|---|---|
| A | 95 / 100 | 120 / 400 |
| B | 360 / 400 | 25 / 100 |

**A.** The two hospitals treated different total numbers of patients.

**B.** The overall rates were computed with the wrong denominators.

**C.** A treats many more severe cases, and severe cases survive less often.

**D.** A treats many more mild cases, and mild cases survive less often.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S12-F2-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.5 • Set 12 • FORWARD 2 • Q4/5**

## True or false?

If one group has the higher rate within every category, it must have the higher rate overall.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 2 • Q5/5

# Represent two quantitative variables on a scatter plot

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.B.6** |
| **BACKWARD 1** | Plot and read points in the first quadrant | **5.G.A.2** |
| **BACKWARD 2** | Describe patterns of association | **8.SP.A.1** |
| **FORWARD 1** | Compare the direction, form and strength of associations | **HSS-ID.B.6** |
| **FORWARD 2** | Distinguish association from causation | **HSS-ID.C.9** |

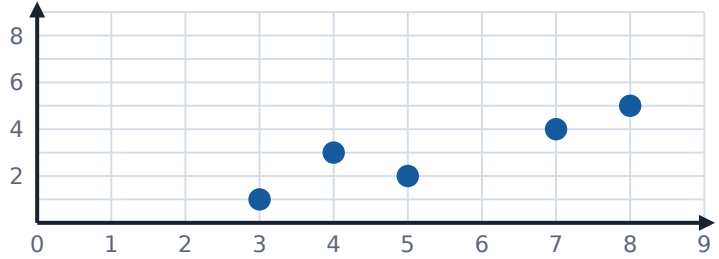
SET 13 • HSS-ID.B.6

# MAIN

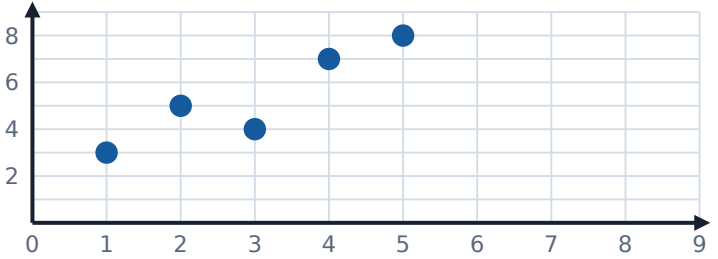
Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6

A seedling was measured weekly: (week, height in cm) = (1, 3), (2, 5), (3, 4), (4, 7), (5, 8). Which scatter plot shows the data?

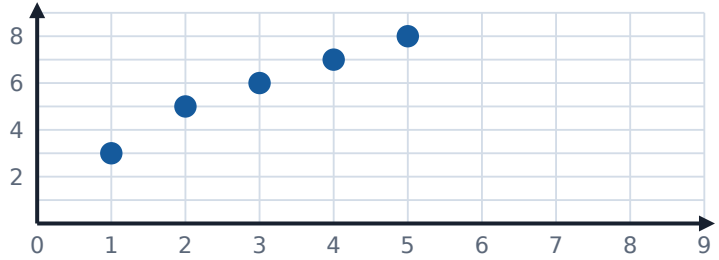
**A**



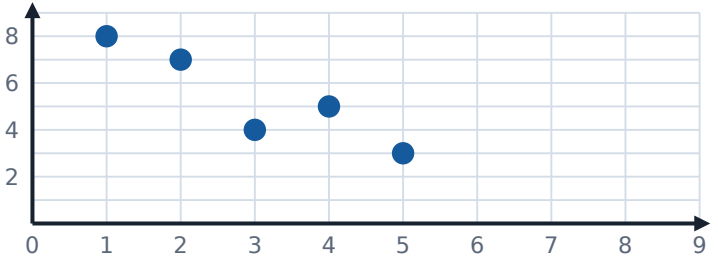

**B**




**C**




**D**




**A.** Graph A

**B.** Graph B

**C.** Graph C

**D.** Graph D

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S13-M-Q1**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q1/5**

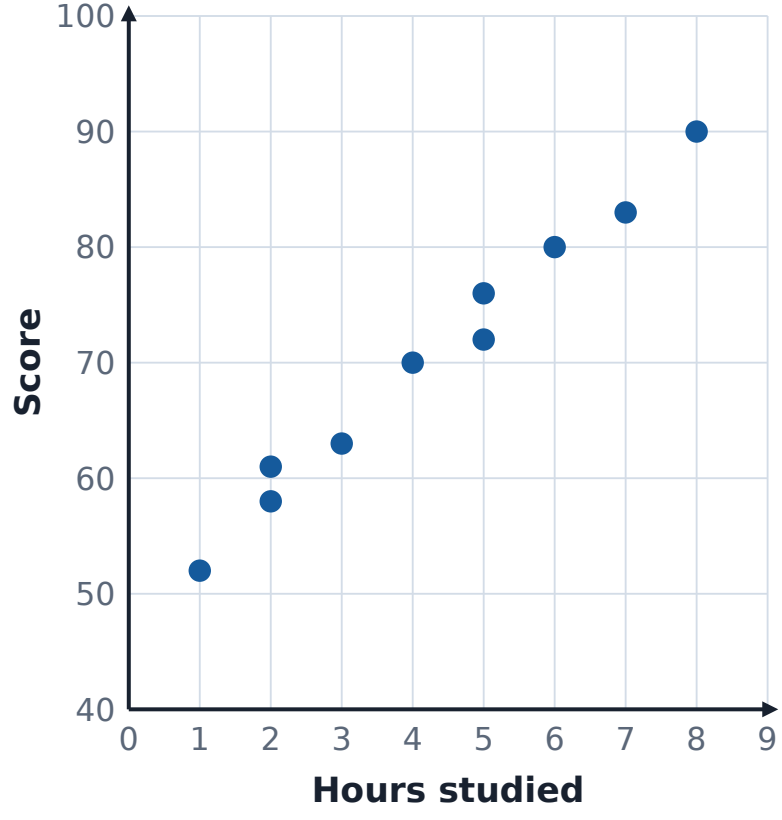
The scatter plot shows hours studied and test scores. Which describes the association?

**A.** curved, not linear

**B.** no clear association

**C.** strong, negative, linear

**D.** strong, positive, linear




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q2/5

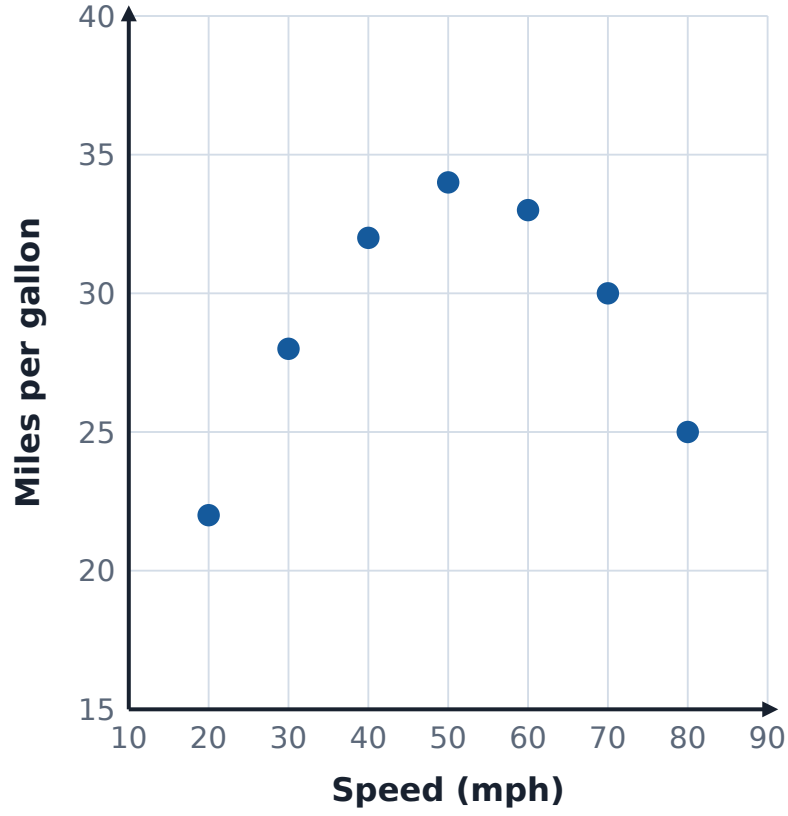
The scatter plot shows a car's speed and fuel efficiency. Which describes the relationship?

**A.** no association at all

**B.** strong, negative, linear

**C.** strong, positive, linear

**D.** curved: rises, then falls




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

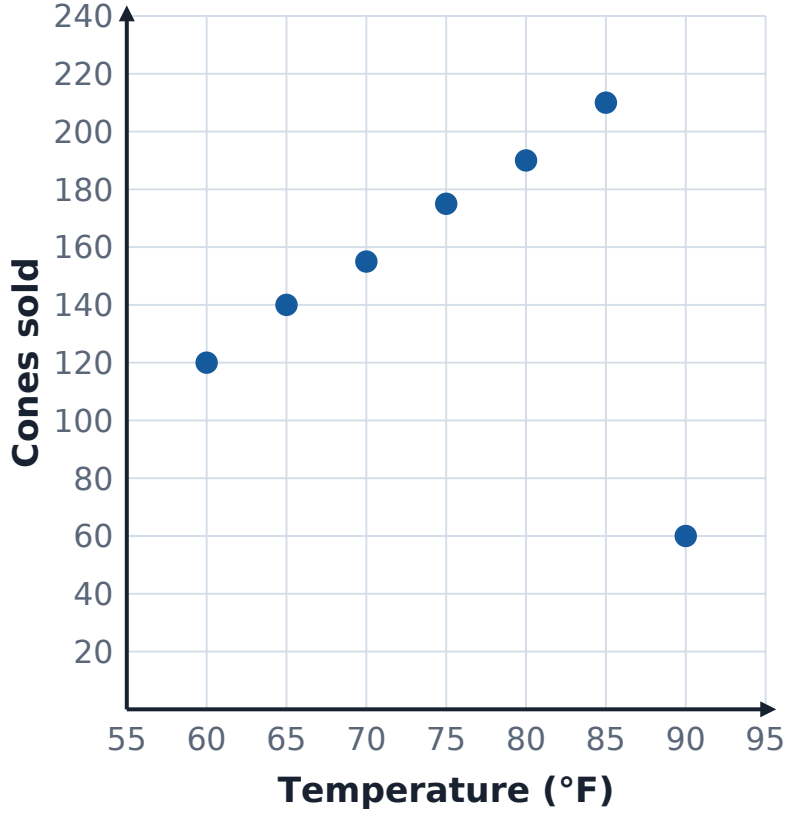
ID S13-M-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q3/5

The scatter plot shows daily high temperature and ice-cream cones sold. One day does not follow the pattern. Give its temperature and its cones sold.

Temperature (°F): _____

Cones sold: _____




**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S13-M-Q4**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q4/5**

Which pair of variables would most likely show a negative association?

**A.** the height and arm span of adults

**B.** hours studied and test score

**C.** shoe size and quiz score of adults

**D.** the age of a car and its resale value

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S13-M-Q5**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q5/5**

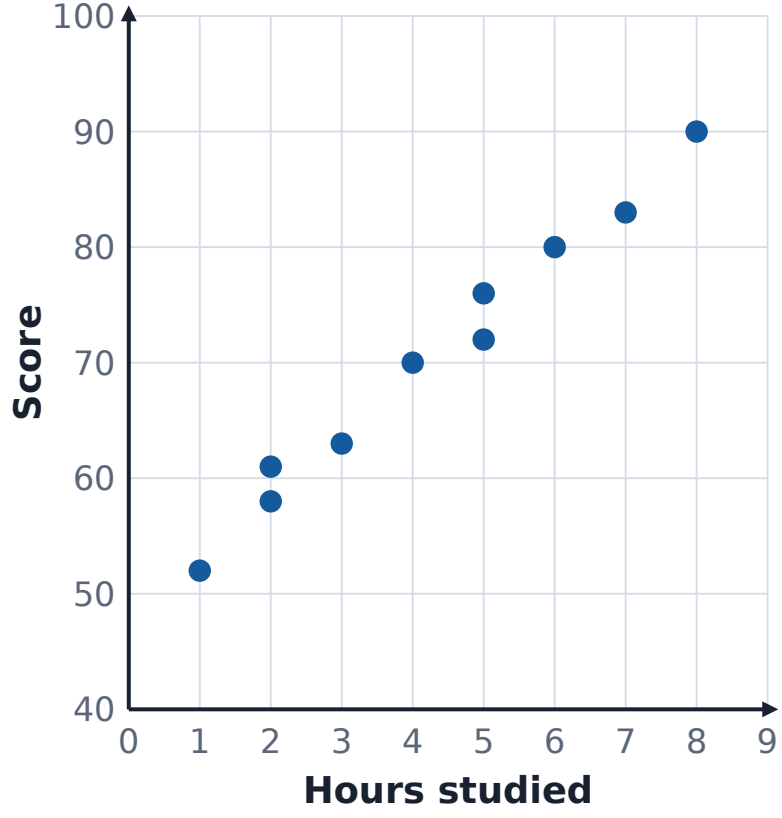
SET 13 • HSS-ID.B.6

# BACKWARD 1

Plot and read points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

In the study scatter plot, what score did the student who studied 7 hours get?

Score: _____



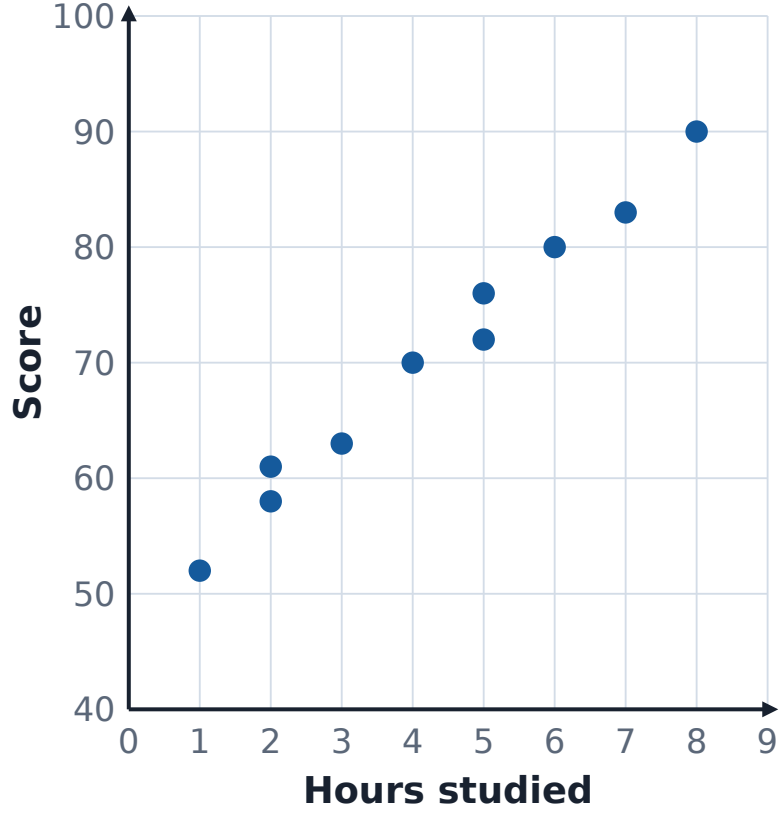

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q1

**Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q1/5**

In the study scatter plot, how many students studied 2 hours?

Number: _____




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

**Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q2/5**

In the study data, what does the point (4, 70) mean?

**A.** a student studied 4 hours and scored 70

**B.** a student studied 70 hours and scored 4

**C.** 4 students scored 70

**D.** the average score was 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

**Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q3/5**

**True or false?**

In the ordered pair (3, 63), 63 is the x-coordinate.

**A.** True

**B.** False

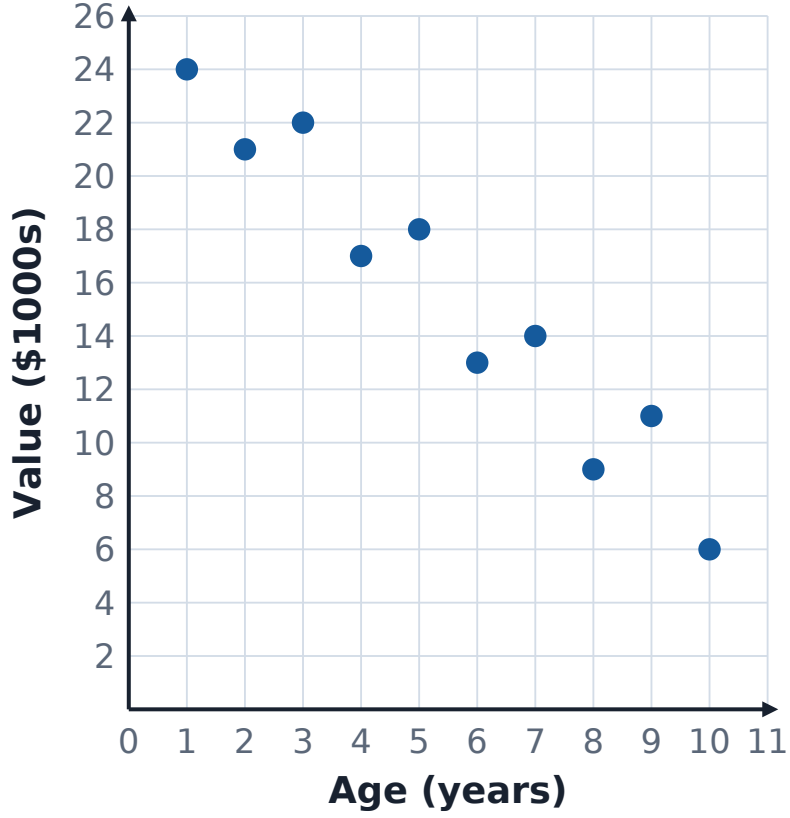
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

**Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q4/5**

In the car scatter plot, what is the resale value of the 8-year-old car, in thousands of dollars?

Value: _____




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

**Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q5/5**

SET 13 • HSS-ID.B.6

# BACKWARD 2

Describe patterns of association • Grade 8 • Statistics & Probability • 8.SP.A.1

## True or false?

When y tends to increase as x increases, the association is positive.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q1

**Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q1/5**

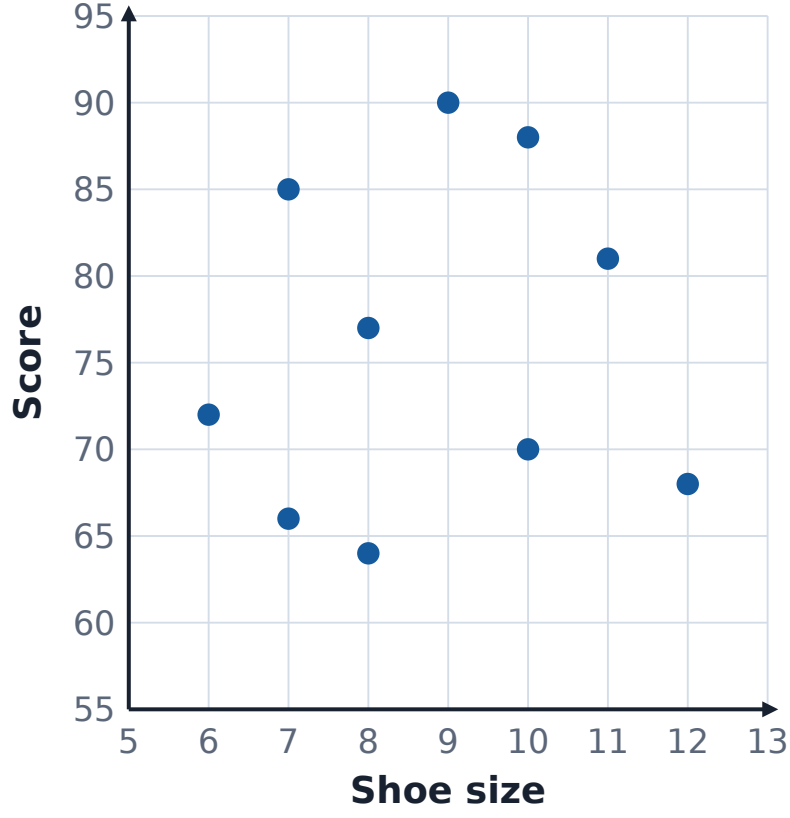
The scatter plot shows shoe size and quiz score. Which describes it?

**A.** curved

**B.** strong, positive, linear

**C.** no clear association

**D.** strong, negative, linear




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

**Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q2/5**

## True or false?

A scatter plot whose points lie close to a U-shaped curve shows a linear association.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

**Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q3/5**

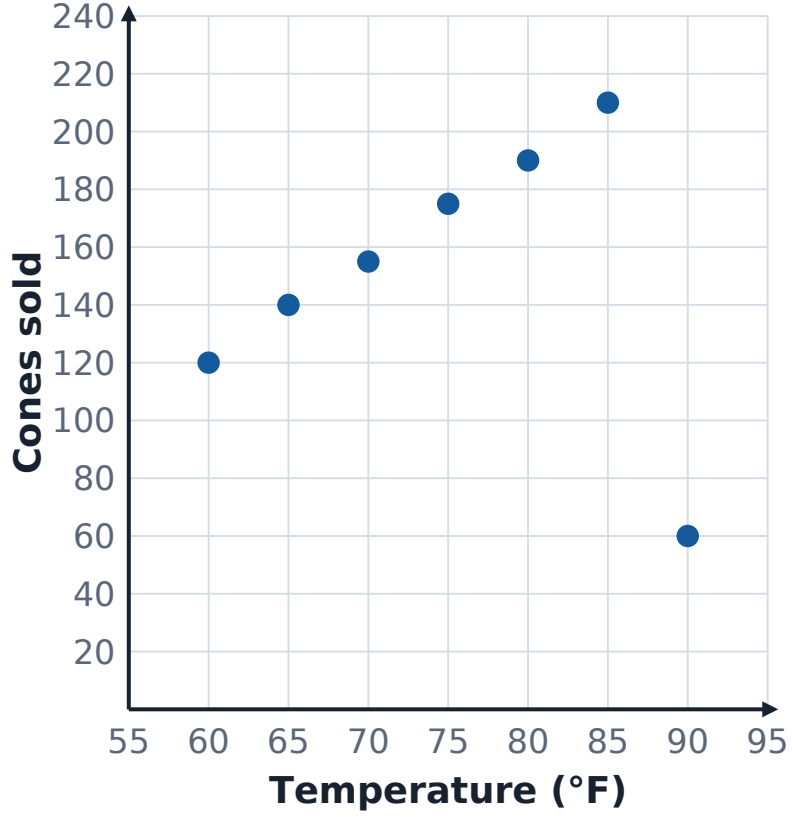
In the ice-cream scatter plot, the point (90, 60) is best described as

**A.** the trend line

**B.** an outlier

**C.** a cluster

**D.** the y-intercept



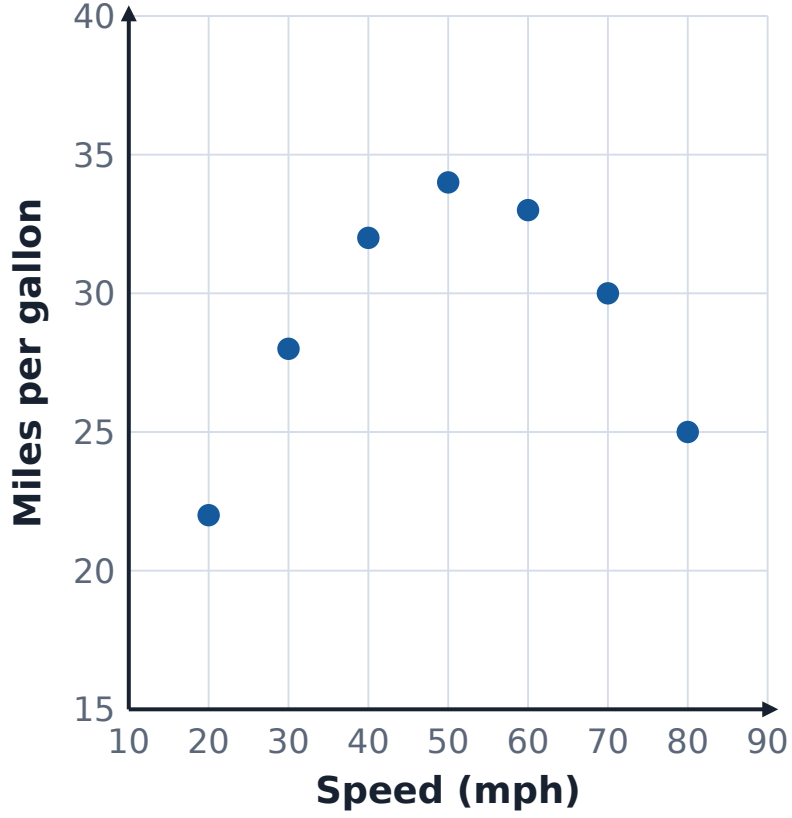

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

**Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q4/5**

In the fuel-efficiency scatter plot, at what speed is the fuel efficiency greatest?

Speed (mph): _____




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

**Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q5/5**

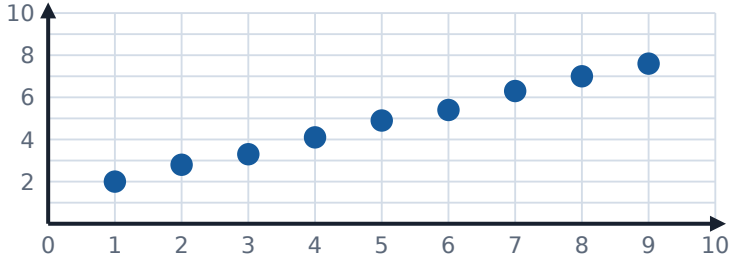
SET 13 • HSS-ID.B.6

# FORWARD 1

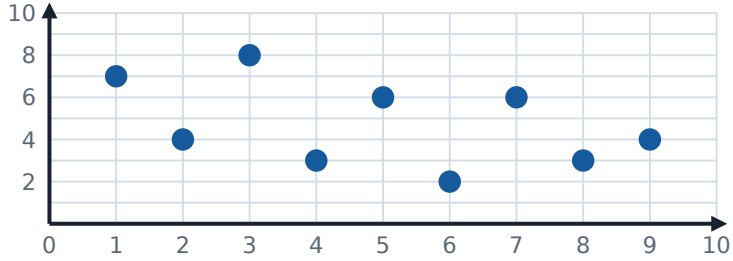
Compare the direction, form and strength of associations • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6

Which scatter plot shows a strong, negative, linear association?

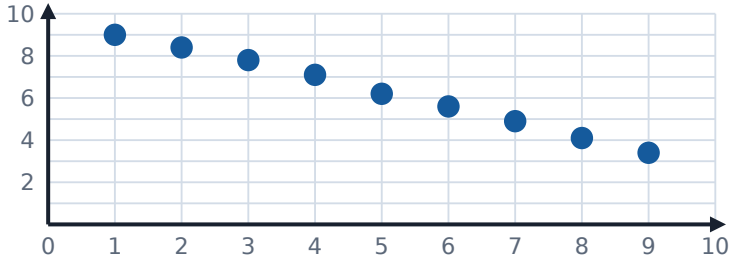
**A**



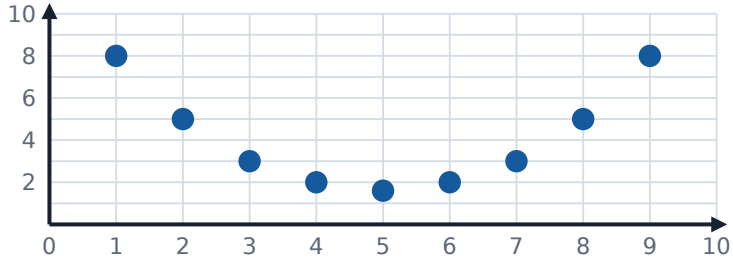

**B**




**C**




**D**




**A.** Graph A

**B.** Graph B

**C.** Graph C

**D.** Graph D

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S13-F1-Q1**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q1/5**

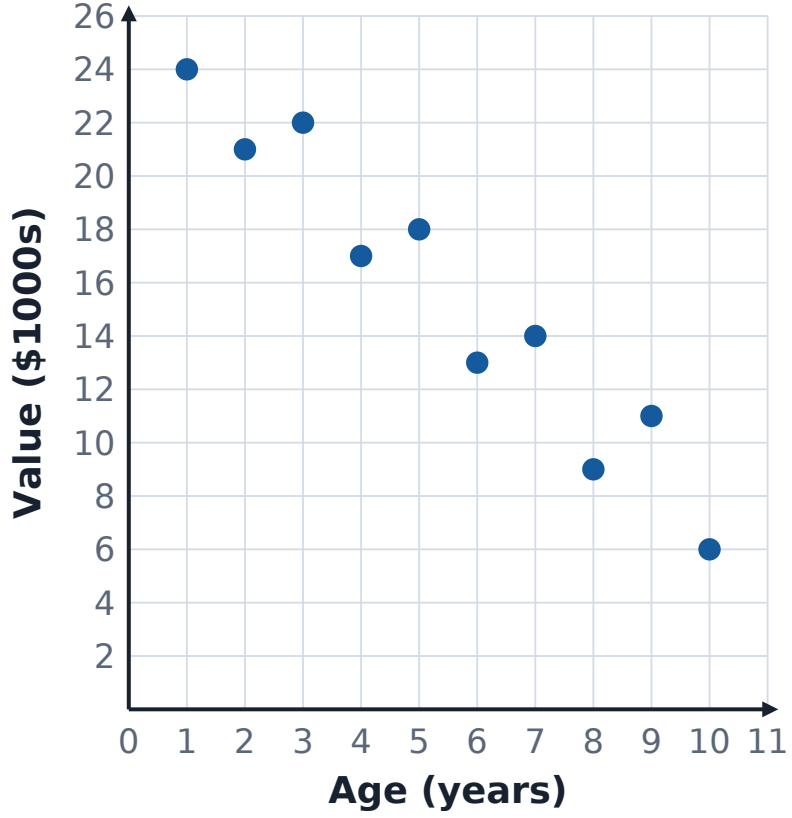
The scatter plot shows the age and resale value of cars. Which describes it?

**A.** curved, then level

**B.** strong, negative, linear

**C.** weak, positive, linear

**D.** strong, positive, linear




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q2/5

## True or false?

A pattern can show a strong association even when it is not linear.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q3/5

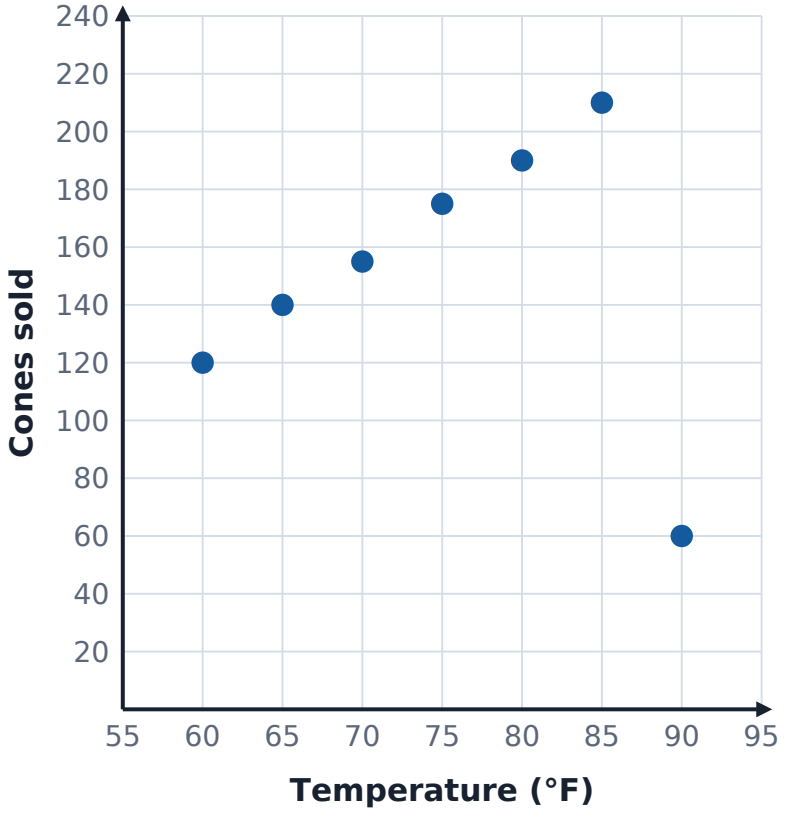
In the ice-cream scatter plot, if the outlier (90, 60) is ignored, the other six days show

**A.** a strong, positive, linear association

**B.** no association

**C.** a strong, negative, linear association

**D.** a U-shaped association




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q4/5

## True or false?

Removing an outlier can never change how strong an association looks.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q5/5

SET 13 • HSS-ID.B.6

# FORWARD 2

Distinguish association from causation • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Cities with more fire stations tend to have more fires. Which is the best explanation?

**A.** Fires cause cities to build fewer stations.

**B.** Fire stations cause fires to start.

**C.** Larger cities have both more stations and more fires.

**D.** The association must be a mistake in the data.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S13-F2-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.9 • Set 13 • FORWARD 2 • Q1/5**

## True or false?

A strong association between two variables proves that one causes the other.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 13 • FORWARD 2 • Q2/5

To show that a fertilizer causes taller plants, which study is needed?

**A.** a survey of gardeners about their plants

**B.** a scatter plot of plant heights found in gardens

**C.** a randomized experiment with fertilized and unfertilized plants

**D.** measuring only the plants that received fertilizer

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S13-F2-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.9 • Set 13 • FORWARD 2 • Q3/5**

Ice-cream sales and swimming accidents both rise during the year. What best explains the association?

**A.** The two variables are unrelated.

**B.** Swimming accidents cause ice-cream sales.

**C.** Hot weather increases both.

**D.** Eating ice cream causes swimming accidents.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S13-F2-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.9 • Set 13 • FORWARD 2 • Q4/5**

## True or false?

In an observational study, a lurking variable can create an association between two variables.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.9 • Set 13 • FORWARD 2 • Q5/5

# Use a function fitted to data to solve problems

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.B.6.a** |
| **BACKWARD 1** | Evaluate a function at an input | **HSF-IF.A.2** |
| **BACKWARD 2** | Evaluate expressions with exponents | **6.EE.A.2.c** |
| **BACKWARD 3** | Solve a linear equation for the input | **HSA-REI.B.3** |
| **FORWARD 1** | Judge whether a prediction is reasonable | **HSS-ID.B.6.a** |
| **FORWARD 2** | Compare predictions from two fitted models | **HSS-ID.B.6.a** |

SET 14 • HSS-ID.B.6.a

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

For a group of students, a line fitted to the data is $y = 2.5x + 12$, where $x$ is hours of practice and $y$ is the predicted number of free throws made out of 50. Predict $y$ for 8 hours of practice.

Predicted $y$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q1/5

A bacteria culture is modeled by $P = 200(1.15)^t$, where t is hours. Predict the population after 6 hours, to the nearest whole number. A calculator or spreadsheet may be used.

Population: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.B.6.a • Set 14 • MAIN • Q2/5

A ball's height is modeled by $h = -4.9t^2 + 20t + 1.5$ (meters, t in seconds). Predict the height at t = 2 seconds.

Height (m): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.B.6.a • Set 14 • MAIN • Q3/5

A line fitted to data is $y = 0.8x + 5$, where $x$ is the temperature (°C) and $y$ is the predicted number of visitors (hundreds). For what temperature does the model predict 29 hundred visitors?

Temperature (°C): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q4/5

A car's value is modeled by $V = 18000(0.85)^t$, where t is years. Predict the value after 4 years, to the nearest dollar. A calculator or spreadsheet may be used.

Value ($): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q5/5

SET 14 • HSS-ID.B.6.a

# BACKWARD 1

Evaluate a function at an input • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

$f(x) = 3x - 4$. Find $f(6)$.

$f(6)$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q1

**High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q1/5**

$g(t) = 100(2)^t$. Find $g(3)$.

$g(3)$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

**High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q2/5**

## True or false?

If $h(t) = -5t^2 + 10t$, then $h(1) = 5$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

**High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q3/5**

f(x) = 0.5x + 7. What is f(10)?

**A.** 7.5

**B.** 17

**C.** 57

**D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

**High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q4/5**

## True or false?

If $p(t) = 50(1.1)^t$, then $p(0) = 55$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • HSS-ID.B.6.a

# BACKWARD 2

Evaluate expressions with exponents • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate $1.5^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

**Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q1/5**

Evaluate $4 \cdot 3^2 - 6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

**Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q2/5**

## True or false?

$2 \cdot 5^2 = 100$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

**Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q3/5**

Evaluate $1000(0.9)^2$.

**A.** 1800

**B.** 900

**C.** 81

**D.** 810

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

**Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q4/5**

## True or false?

$(-3)^2 = 9$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

**Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q5/5**

SET 14 • HSS-ID.B.6.a

# BACKWARD 3

Solve a linear equation for the input • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve 0.8x + 5 = 21.

x: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q1/5

Solve 2.5x + 12 = 47.

x: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q2/5

**True or false?**

The solution of $3x - 4 = 11$ is $x = 5$.

**A.** True

**B.** False

Solve 1.2x + 6 = 30.

**A.** 28.8

**B.** 25

**C.** 20

**D.** 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q4/5

## True or false?

The solution of $0.5x + 3 = 8$ is $x = 5.5$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q5/5

SET 14 • HSS-ID.B.6.a

# FORWARD 1

Judge whether a prediction is reasonable • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

The line $y = 2.5x + 12$ was fitted to students who practiced 0 to 12 hours ($y$ = free throws made out of 50). It predicts $y = 87$ for 30 hours. Which is correct?

**A.** Reasonable, because 30 hours is a whole number of hours.

**B.** Unreasonable, because a fitted line must have a negative slope.

**C.** Unreasonable: 30 hours is far outside the data, and 87 exceeds 50.

**D.** Reasonable, because the line fits the data from 0 to 12 hours.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S14-F1-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q1/5**

A line fitted to data for x from 10 to 40 is $y = 3x - 5$. Predict y at $x = 25$, and state whether $x = 25$ is inside the data range by entering yes or no.

Predicted y: _____

Inside the range (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q2/5

## True or false?

Predicting far outside the range of the data (extrapolation) is usually less reliable than predicting inside it.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q3/5

The model $V = 18000(0.85)^t$ fits a car's value for t = 0 to 8 years. What value does it predict at t = 0?

V ($): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q4/5

## True or false?

The model $h = -4.9t^2 + 20t + 1.5$ gives sensible heights for every value of t, including t = 10.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q5/5

SET 14 • HSS-ID.B.6.a

# FORWARD 2

Compare predictions from two fitted models • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

Two models fit a town's population (thousands), t years after 2020: linear $L = 2t + 40$ and exponential $E = 40(1.05)^t$. Predict each for $t = 2$. Round E to two decimal places.

L: _____

E: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q1/5

For the models $L = 2t + 40$ and $E = 40(1.05)^t$, which statement about large t is correct?

**A.** E eventually predicts less, because 1.05 is close to 1.

**B.** The two models always predict the same population.

**C.** L always predicts more, because its slope is 2.

**D.** E eventually predicts more, because it grows by a constant percent.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S14-F2-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q2/5**

Using L = 2t + 40 (thousands), in what year t after 2020 does L predict 60 thousand?

t: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q3/5

## True or false?

If two models fit the data equally well inside the data range, they must give the same prediction far outside it.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q4/5

A residual check shows the exponential model leaves random scatter and the linear model leaves a curved pattern. Which model should be used for predictions?

**A.** the exponential model

**B.** neither, because residuals exist

**C.** the linear model

**D.** either, equally

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S14-F2-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q5/5**

# Choose and fit a linear, quadratic or exponential function

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.B.6.a** |
| **BACKWARD 1** | Tell equal differences from equal factors | **HSF-LE.A.1.a** |
| **BACKWARD 2** | Write a linear function from two points | **8.F.B.4** |
| **BACKWARD 3** | Write an exponential function from two points | **HSF-LE.A.2** |
| **FORWARD 1** | Fit a model with regression technology | **HSS-ID.B.6.a** |
| **FORWARD 2** | Check a fitted model with residuals | **HSS-ID.B.6.b** |

SET 15 • HSS-ID.B.6.a

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

The data fit an exponential function $y = a \cdot b^x$ exactly. Find a and b.

| x | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| y | 3 | 6 | 12 | 24 | 48 |

a: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q1/5

The data fit a linear function $y = mx + b$ exactly. Find $m$ and $b$.

| x | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| y | 5 | 8 | 11 | 14 | 17 |

m: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q2/5

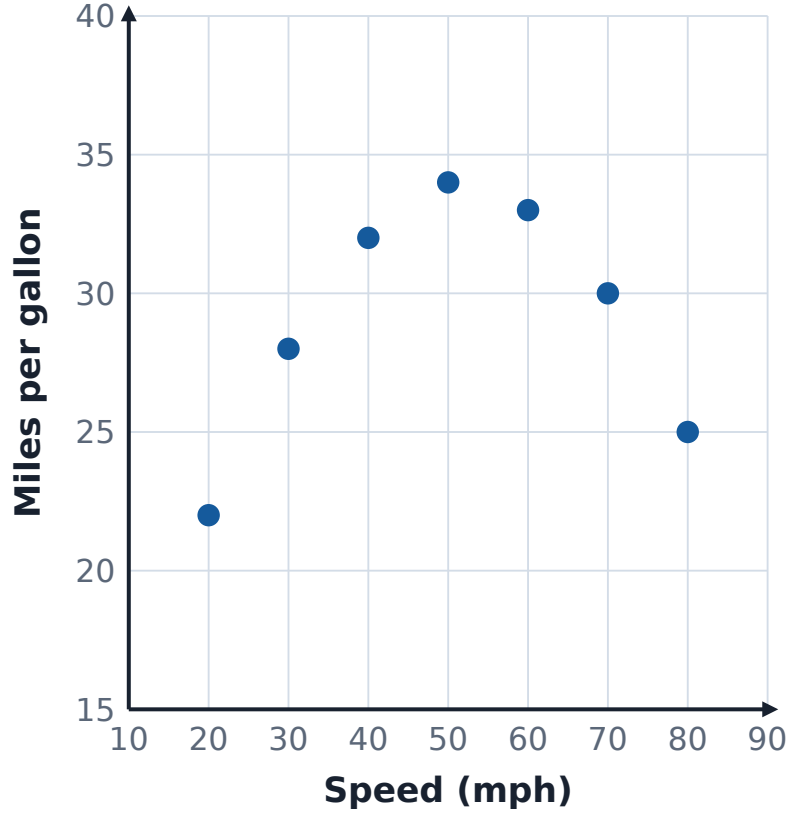
The scatter plot shows a car's speed and fuel efficiency. Which family of functions best fits the data?

**A.** quadratic

**B.** exponential

**C.** linear

**D.** constant




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.B.6.a • Set 15 • MAIN • Q3/5

A town has 12,000 people and its population grows by about 8% each year. Which model fits, with t in years?

**A.** $P = 12000(1.08)^t$

**B.** $P = 12000 + 0.08t$

**C.** $P = 12000 + 1.08t$

**D.** $P = 12000(0.08)^t$

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S15-M-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.a • Set 15 • MAIN • Q4/5**

An exponential function $y = a \cdot b^x$ passes through (0, 500) and (1, 400). Find a and b.

a: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q5/5

SET 15 • HSS-ID.B.6.a

# BACKWARD 1

Tell equal differences from equal factors • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a

## True or false?

In $y = 3x + 5$, $y$ changes by equal differences over equal intervals of $x$.

**A.** True

**B.** False

For x = 0, 1, 2, 3 the values of y are 2, 6, 18, 54. What is the pattern?

**A.** squared each step

**B.** add 3 each step

**C.** add 4 each step

**D.** multiplied by 3 each step

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q2/5

**True or false?**

The values 4, 7, 12, 19 (for x = 0 to 3) have constant first differences.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q3/5

For x = 0 to 4, y = 4, 7, 12, 19, 28. What is the constant second difference?

Second difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q4/5

Which list of y-values (for x = 0, 1, 2, 3) is linear?

**A.** 1, 2, 4, 8

**B.** 10, 7, 4, 1

**C.** 0, 1, 4, 9

**D.** 5, 5.5, 6.5, 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q5/5

SET 15 • HSS-ID.B.6.a

# BACKWARD 2

Write a linear function from two points • Grade 8 • Functions • 8.F.B.4

A line passes through (0, 5) and (2, 11). Find its slope and y-intercept.

Slope: _____

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

**Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q1/5**

Find the slope of the line through (1, 4) and (5, 12).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

**Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q2/5**

**True or false?**

The table $x = 0, 1, 2$ and $y = 7, 4, 1$ fits $y = -3x + 7$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

**Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q3/5**

Which function fits x = 0, 1, 2 and y = 2, 4.5, 7?

**A.** y = 2x + 2.5

**B.** y = 4.5x

**C.** y = 2.5x

**D.** y = 2.5x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

**Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q4/5**

**True or false?**

The line through (0, 4) and (1, 10) has slope 4.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

**Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q5/5**

SET 15 • HSS-ID.B.6.a

# BACKWARD 3

Write an exponential function from two points • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

An exponential function $y = a \cdot b^x$ passes through (0, 200) and (1, 300). Find a and b.

a: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q1/5

An exponential function $y = a \cdot b^x$ with $b > 0$ passes through (0, 80) and (2, 20). Find a and b.

a: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q2/5

## True or false?

$y = 50(1.2)^x$ passes through (1, 60).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q3/5

Which exponential function passes through (0, 4) and (1, 12)?

**A.** $y = 3(4)^x$

**B.** $y = 4(3)^x$

**C.** $y = 4 + 8x$

**D.** $y = 12(4)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q4/5

## True or false?

$y = 10(0.5)^x$ passes through (2, 5).

**A.** True

**B.** False

SET 15 • HSS-ID.B.6.a

# FORWARD 1

Fit a model with regression technology • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

Find the least-squares line $y = mx + b$ for the spring data. Round $m$ and $b$ to two decimal places. A calculator or spreadsheet may be used.

| Mass (g) | 2 | 4 | 5 | 7 | 8 | 10 |
|---|---|---|---|---|---|---|
| Length (cm) | 15 | 19 | 24 | 26 | 31 | 35 |

m: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q1/5

The points (0, 2), (1, 5), (2, 10), (3, 17) lie exactly on a quadratic $y = ax^2 + bx + c$. Find a, b and c. A calculator or spreadsheet may be used.

a: _____

b: _____

c: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q2/5

Data on y rise faster and faster as x increases, y is always positive, and each step multiplies y by about 1.3. Which regression should be used?

**A.** linear

**B.** exponential

**C.** none of these

**D.** quadratic

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S15-F1-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q3/5**

Using the fitted line y = 2.52x + 9.86, predict y when x = 6.

y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q4/5

## True or false?

Height-versus-time data for a thrown ball are better fitted by a quadratic than by a line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q5/5

SET 15 • HSS-ID.B.6.a

# FORWARD 2

Check a fitted model with residuals • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

For the model $y = 2.52x + 9.86$, find the residual (observed minus predicted) for the data point $(5, 24)$.

Residual: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q1/5

For the model $y = 2.52x + 9.86$, find the residual for the data point $(4, 19)$.

Residual: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q2/5

## True or false?

A positive residual means the data point lies above the fitted line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q3/5

A line is fitted to data that actually grow exponentially. What pattern do the residuals usually show?

**A.** every residual positive

**B.** a curved (U-shaped) pattern

**C.** random scatter around 0

**D.** all residuals equal to 0

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S15-F2-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q4/5**

## True or false?

If the residuals of a linear model are small but form a clear curve, a linear model is still the best family.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q5/5

# Assess the fit of a function with residuals

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.B.6.b** |
| **BACKWARD 1** | Subtract with signed decimals | **7.NS.A.1.c** |
| **BACKWARD 2** | Find a predicted value from a model | **HSF-IF.A.2** |
| **BACKWARD 3** | Compare an observed value with a prediction | **HSS-ID.B.6.a** |
| **FORWARD 1** | Compute least-squares residuals | **HSS-ID.B.6.b** |
| **FORWARD 2** | Compare two models with residuals | **HSS-ID.B.6.b** |

SET 16 • HSS-ID.B.6.b

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

The model $y = 2x + 1$ is used for the data in the table. Find the residuals (observed minus predicted) at $x = 1$ and at $x = 2$.

| x | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| y | 3.5 | 4.6 | 7.2 | 8.8 | 11.1 |

Residual at $x = 1$: _____

Residual at $x = 2$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.B.6.b • Set 16 • MAIN • Q1/5

A model predicts $y = 3x + 10$. Find the residuals for the observed points (4, 20) and (6, 30).

Residual for (4, 20): _____

Residual for (6, 30): _____

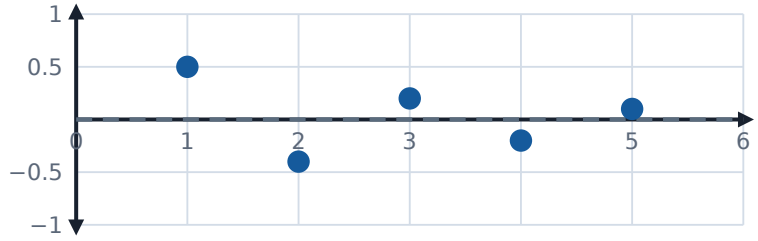
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

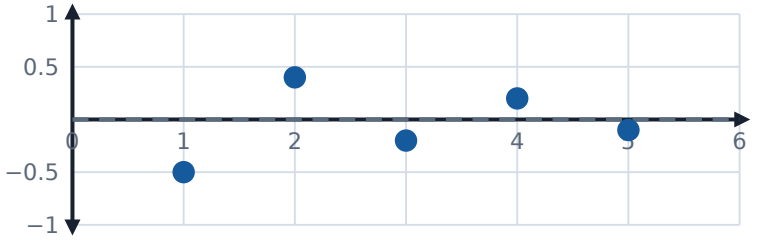
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • MAIN • Q2/5

The model y = 2x + 1 is used for the data in the table (x = 1 to 5; y = 3.5, 4.6, 7.2, 8.8, 11.1). Which plot is the residual plot (observed minus predicted)?

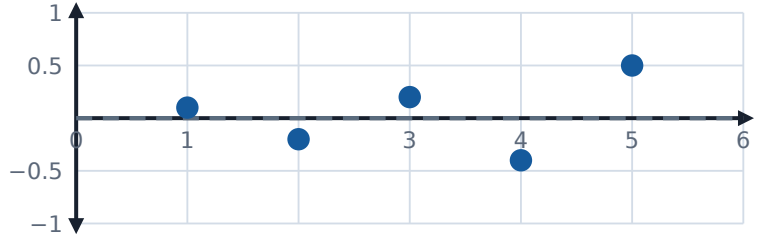
**A**



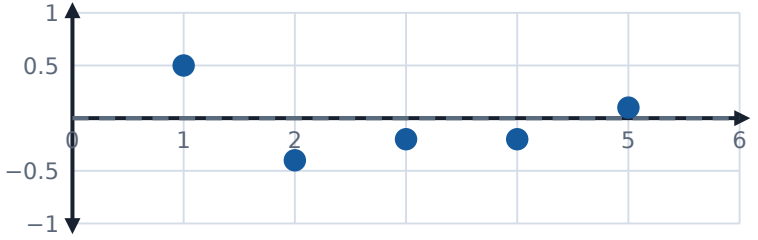

**B**




**C**




**D**




**A.** Graph A

**B.** Graph B

**C.** Graph C

**D.** Graph D

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S16-M-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.b • Set 16 • MAIN • Q3/5**

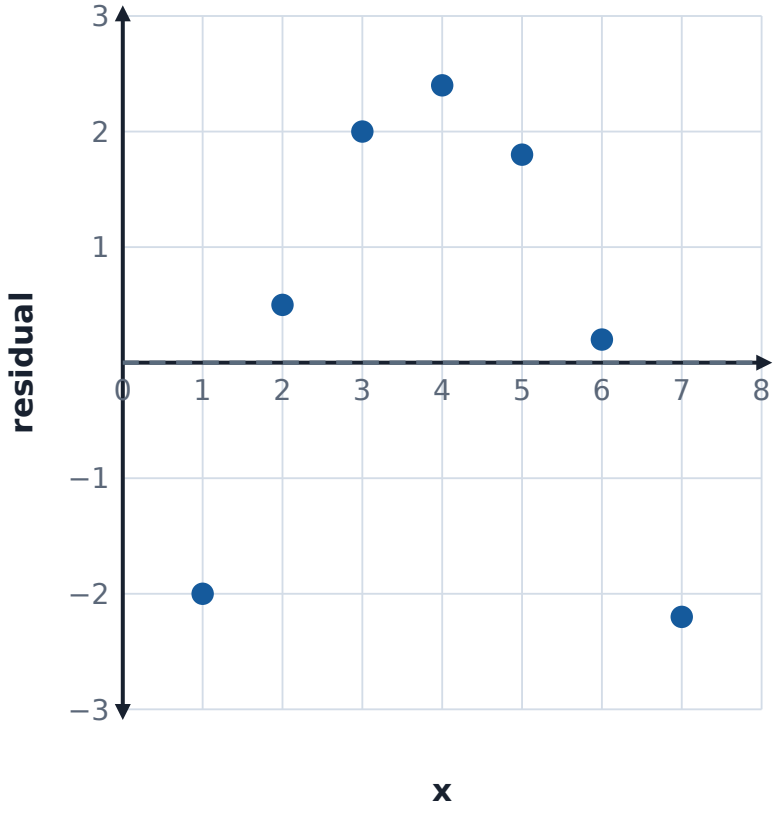
A line was fitted to a data set. The residual plot is shown. Which conclusion is correct?

**A.** The linear model is appropriate, because some residuals are near 0.

**B.** The linear model is appropriate, because residuals are both signs.

**C.** The data must contain an error, because residuals are not all 0.

**D.** A linear model is not appropriate; the curve suggests a curved model.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.B.6.b • Set 16 • MAIN • Q4/5

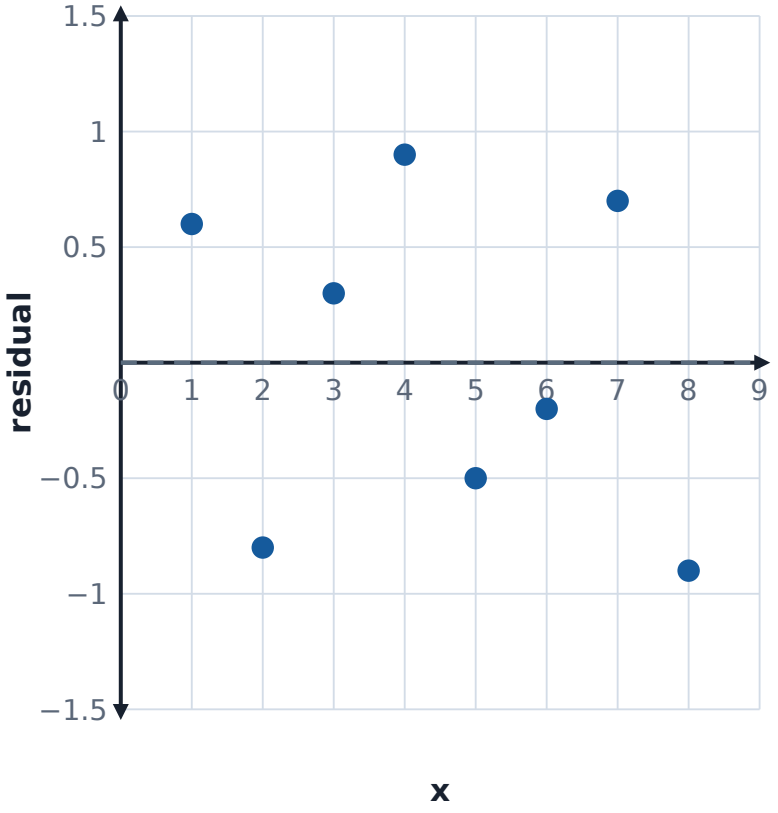
A line was fitted to a data set. The residual plot is shown. Which conclusion is correct?

**A.** The linear model is not appropriate, because residuals are not 0.

**B.** The linear model is appropriate: the residuals show no pattern.

**C.** A quadratic model is needed, because residuals change sign.

**D.** The model is perfect, because the residuals add to about 0.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.B.6.b • Set 16 • MAIN • Q5/5

SET 16 • HSS-ID.B.6.b

# BACKWARD 1

Subtract with signed decimals • Grade 7 • The Number System • 7.NS.A.1.c

Compute $3.5 - 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q1

**Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q1/5**

Compute $4.6 - 5$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

**Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q2/5**

**True or false?**

$20 - 22 = -2$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q3

**Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q3/5**

Compute $7.2 - 7.5$.

**A.** $-0.3$

**B.** $-14.7$

**C.** $14.7$

**D.** $0.3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

**Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q4/5**

## True or false?

$11 - 11.4 = 0.4$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

**Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q5/5**

SET 16 • HSS-ID.B.6.b

# BACKWARD 2

Find a predicted value from a model • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

For y = 2x + 1, find the predicted y at x = 3.

y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

**High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q1/5**

For y = 3x + 10, find the predicted y at x = 4.

y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

**High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q2/5**

## True or false?

For $y = -1.5x + 20$, the predicted y at $x = 4$ is 14.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q3/5

For y = 0.5x + 3, what is the predicted y at x = 8?

**A.** 8.5

**B.** 4

**C.** 7

**D.** 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

**High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q4/5**

## True or false?

For $y = 4x - 2$, the predicted $y$ at $x = 0$ is 2.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • HSS-ID.B.6.b

# BACKWARD 3

Compare an observed value with a prediction • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

A model predicts y = 2.5x + 12 free throws for x hours of practice. A student practiced 8 hours and made 35. What is the predicted number, and how many more did the student make than predicted?

Predicted: _____

More by: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q1/5

## True or false?

If an observed value is less than the predicted value, the point lies below the fitted line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q2/5

The observed value is 50 and the predicted value is 46. What is observed minus predicted?

**A.** 96

**B.** −4

**C.** 46

**D.** 4

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S16-B3-Q3**

**High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q3/5**

## True or false?

In this collection, a residual is defined as predicted minus observed.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q4/5

The observed value is 18 and the predicted value is 20.5. What is observed minus predicted?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q5/5

SET 16 • HSS-ID.B.6.b

# FORWARD 1

Compute least-squares residuals • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

The least-squares line for (1, 2), (2, 5), (3, 4), (4, 7) is $y = 1.4x + 1$. Find the residuals at $x = 1$ and $x = 2$.

At $x = 1$: _____

At $x = 2$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q1/5

For $y = 1.4x + 1$ and the points (1, 2), (2, 5), (3, 4), (4, 7), what is the sum of all four residuals?

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q2/5

## True or false?

For a least-squares line, the residuals always add to 0.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q3/5

For $y = 1.4x + 1$ and the points (1, 2), (2, 5), (3, 4), (4, 7), which points have the largest residuals in absolute value?

**A.** (1, 2) only

**B.** (1, 2) and (4, 7)

**C.** (4, 7) only

**D.** (2, 5) and (3, 4)

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S16-F1-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q4/5**

## True or false?

A residual of $-1.2$ means the point lies 1.2 units above the line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q5/5

SET 16 • HSS-ID.B.6.b

# FORWARD 2

Compare two models with residuals • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

Two models were fitted to the same data. Linear residuals: 3, −1, −3, −2, 1, 4. Exponential residuals: 0.2, −0.3, 0.1, 0.3, −0.2, −0.1. Which model fits better?

**A.** the exponential model

**B.** the linear model

**C.** neither model can be judged

**D.** they fit equally

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S16-F2-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q1/5**

Find the sum of squared residuals for each model: linear 3, −1, −3, −2, 1, 4 and exponential 0.2, −0.3, 0.1, 0.3, −0.2, −0.1.

Linear: _____

Exponential: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q2/5

## True or false?

Of two models fitted to the same data, the one with the smaller sum of squared residuals fits those data more closely.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q3/5

A residual plot fans out: the residuals get larger as x increases. What does this show?

**A.** the slope is negative

**B.** prediction errors grow as x increases

**C.** the model is exact

**D.** the data have no outliers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q4/5

## True or false?

A residual plot with a clear U shape shows that the model is a good fit.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q5/5

# Fit a linear function to a linear scatter plot

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.B.6.c** |
| **BACKWARD 1** | Fit a line by eye | **8.SP.A.2** |
| **BACKWARD 2** | Find slope and intercept from points | **8.F.B.4** |
| **FORWARD 1** | Use the properties of the least-squares line | **HSS-ID.B.6.c** |
| **FORWARD 2** | Interpret the fitted line's slope and intercept | **HSS-ID.C.7** |

SET 17 • HSS-ID.B.6.c

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

Find the least-squares line $y = mx + b$ for hours studied ($x$) and score ($y$). Round $m$ and $b$ to two decimal places. A calculator or spreadsheet may be used.

| Hours | 1 | 2 | 2 | 3 | 4 | 5 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|---|
| Score | 52 | 58 | 61 | 63 | 70 | 72 | 76 | 80 | 83 | 90 |

m: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.B.6.c • Set 17 • MAIN • Q1/5

Find the least-squares line y = mx + b for car age (x) and value in $1000s (y). Round m and b to two decimal places. A calculator or spreadsheet may be used.

| Age (years) | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Value ($1000s) | 24 | 21 | 22 | 17 | 18 | 13 | 14 | 9 | 11 | 6 |

m: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

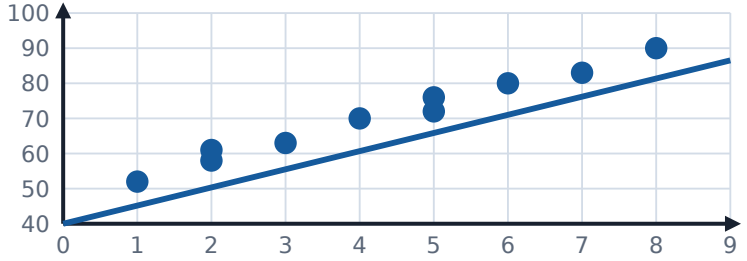
ID S17-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

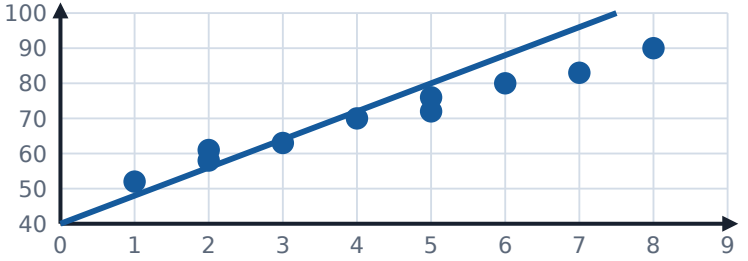
HSS-ID.B.6.c • Set 17 • MAIN • Q2/5

Which graph shows the line that best fits the study data (hours, score)?

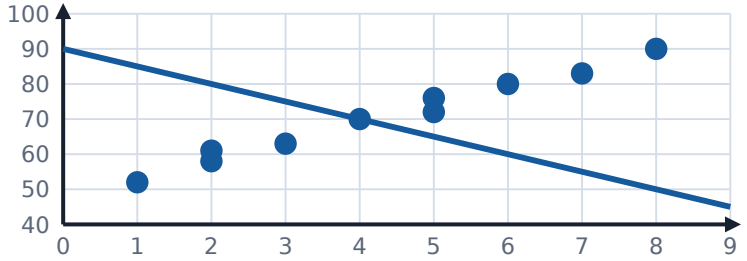
**A**



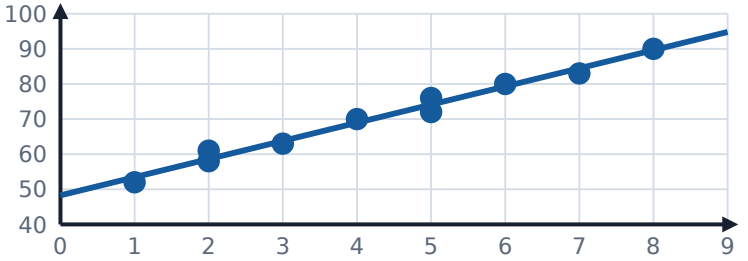

**B**




**C**




**D**




**A.** Graph A

**B.** Graph B

**C.** Graph C

**D.** Graph D

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S17-M-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.c • Set 17 • MAIN • Q3/5**

Using the least-squares line y = 5.17x + 48.28 for hours studied and score, predict the score for 3 hours.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • MAIN • Q4/5

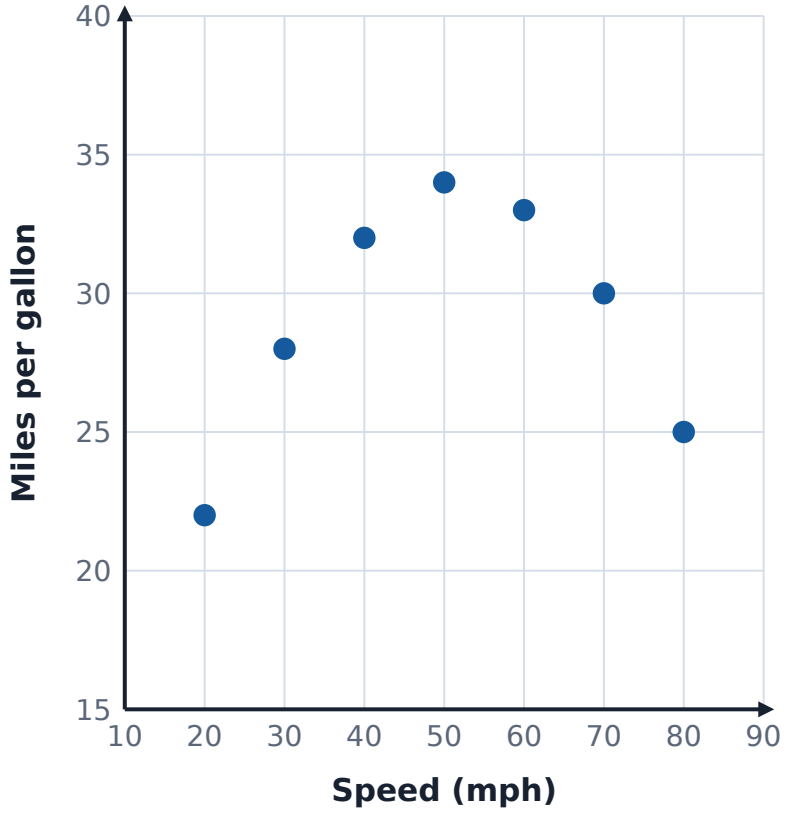
A student wants to fit a least-squares line to the data in this scatter plot. Which statement is correct?

**A.** A line is appropriate, because r can always be computed.

**B.** A line is appropriate, because the data have no outliers.

**C.** A line is not appropriate: the pattern is curved.

**D.** A line fits well, because the points rise at first.




**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S17-M-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.c • Set 17 • MAIN • Q5/5**

SET 17 • HSS-ID.B.6.c

# BACKWARD 1

Fit a line by eye • Grade 8 • Statistics & Probability • 8.SP.A.2

## True or false?

A good fitted line has about as many points above it as below it.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

**Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q1/5**

Which line passes through (0, 2), (2, 6) and (4, 10)?

**A.** $y = x + 2$

**B.** $y = 2x$

**C.** $y = 2x + 2$

**D.** $y = 4x + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

**Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q2/5**

A line drawn by eye passes through (0, 50) and (5, 75). Find its slope and y-intercept.

Slope: _____

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

**Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q3/5**

## True or false?

A fitted line must pass through every data point.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

**Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q4/5**

Use the line $y = 5x + 50$ to predict y at $x = 4$.

y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

**Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q5/5**

SET 17 • HSS-ID.B.6.c

# BACKWARD 2

Find slope and intercept from points • Grade 8 • Functions • 8.F.B.4

Find the slope of the line through (2, 58) and (6, 80).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

**Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q1/5**

A line has slope 5 and passes through (2, 60). Find its y-intercept.

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

**Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q2/5**

**True or false?**

The line through (0, 26) with slope −2 has y = 16 at x = 5.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

**Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q3/5**

Which equation is the line through (0, 10) and (4, 2)?

**A.** $y = 2x + 10$

**B.** $y = -2x + 2$

**C.** $y = -2x + 10$

**D.** $y = -0.5x + 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

**Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q4/5**

**True or false?**

The slope of the line through (1, 24) and (10, 6) is 2.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

**Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q5/5**

SET 17 • HSS-ID.B.6.c

# FORWARD 1

Use the properties of the least-squares line • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

Find the mean hours $\bar{x}$ and the mean score $\bar{y}$ for the study data.

| Hours | 1 | 2 | 2 | 3 | 4 | 5 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|---|
| Score | 52 | 58 | 61 | 63 | 70 | 72 | 76 | 80 | 83 | 90 |

$\bar{x}$: _____

$\bar{y}$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q1/5

## True or false?

The least-squares line passes through the point $(\bar{x}, \bar{y})$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q2/5

The least-squares line is the line that makes which quantity as small as possible?

**A.** the number of points off the line

**B.** the largest single residual

**C.** the sum of the squared residuals

**D.** the sum of the residuals

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S17-F1-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q3/5**

A least-squares line has slope 2 and passes through $(\bar{x}, \bar{y}) = (5, 20)$. Find its y-intercept.

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q4/5

## True or false?

Adding one point far from the others can never change the least-squares line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q5/5

SET 17 • HSS-ID.B.6.c

# FORWARD 2

Interpret the fitted line's slope and intercept • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

For hours studied (x) and score (y), the least-squares line is $y = 5.17x + 48.28$. What does 5.17 mean?

**A.** A student who studied 0 hours scored exactly 5.17.

**B.** Each extra hour goes with about 5.17 more points, on average.

**C.** The score rises 48.28 points for each hour.

**D.** Each student scored 5.17 points per test.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S17-F2-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.7 • Set 17 • FORWARD 2 • Q1/5**

Using $y = 5.17x + 48.28$, how much higher is the predicted score for a student who studies 2 more hours?

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.7 • Set 17 • FORWARD 2 • Q2/5

For car age (x, years) and value (y, $1000s), $y = -1.88x + 25.87$. What does 25.87 mean?

**A.** The oldest car is worth about $25,870.

**B.** A car is predicted to be worth $0 after 25.87 years.

**C.** A new car (age 0) has a predicted value of about $25,870.

**D.** Cars lose about $25,870 each year.

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.C.7 • Set 17 • FORWARD 2 • Q3/5

## True or false?

In $y = -1.88x + 25.87$, the slope means every car loses exactly $1,880 each year.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q4/5

Using $y = -1.88x + 25.87$, how much greater (in \$1000s) is the predicted value of a 3-year-old car than a 7-year-old car?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q5/5

# Interpret the slope and intercept of a linear model

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.C.7** |
| **BACKWARD 1** | Interpret the rate and the initial value of a linear function | **8.F.B.4** |
| **BACKWARD 2** | Use a linear model's equation to answer questions | **8.SP.A.3** |
| **FORWARD 1** | Interpret parameters with units and judge meaning | **HSS-ID.C.7** |
| **FORWARD 2** | Relate the slope to the correlation coefficient | **HSS-ID.C.8** |

SET 18 • HSS-ID.C.7

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

A model for a family's savings is $y = 85x + 1200$, where $x$ is months and $y$ is dollars. What does 85 mean?

**A.** The family started with $85 in savings.

**B.** The family saves $1200 each month.

**C.** The savings are predicted to grow by $85 each month.

**D.** The savings will reach $85 after 1200 months.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S18-M-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.7 • Set 18 • MAIN • Q1/5**

A model for weekly hours of exercise is $y = 6 - 0.2x$, where $x$ is weekly hours of TV. What does 6 mean?

**A.** A person who watches 0 hours of TV is predicted to exercise 6 hours.

**B.** Each hour of TV is predicted to add 6 hours of exercise.

**C.** Each hour of TV is predicted to cut exercise by 6 hours.

**D.** A person exercises at most 0.2 hours per week.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S18-M-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.7 • Set 18 • MAIN • Q2/5**

A tree's height is modeled by $y = 0.4x + 1.5$, with $y$ in meters and $x$ in years since planting. What is the predicted growth per year, and what is the predicted height at planting?

Growth per year (m): _____

Height at planting (m): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.C.7 • Set 18 • MAIN • Q3/5

For adults with heights from 150 to 190 cm, arm span is modeled by $y = 1.1x - 15$ (both in cm). Which statement about the intercept $-15$ is correct?

**A.** It means a person 0 cm tall has an arm span of $-15$ cm.

**B.** It means arm spans are 15 cm less than heights.

**C.** It is not meaningful: a height of 0 cm is far outside the data.

**D.** It means the slope of the model is negative.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S18-M-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.7 • Set 18 • MAIN • Q4/5**

Weekly gasoline use is modeled by $y = 0.35x + 4.2$ gallons, where $x$ is miles driven per day. By how many gallons do the predictions differ for two drivers whose daily miles differ by 20?

Gallons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.7 • Set 18 • MAIN • Q5/5

SET 18 • HSS-ID.C.7

# BACKWARD 1

Interpret the rate and the initial value of a linear function • Grade 8 • Functions • 8.F.B.4

A plumber charges $y = 50x + 80$ dollars for $x$ hours. What does 80 mean?

**A.** the charge per hour

**B.** a fixed fee charged before any hours

**C.** the number of hours

**D.** the total for 1 hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

**Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q1/5**

A phone plan costs y = 0.1x + 25 dollars for x minutes. What is the cost per minute, and what is the fixed cost?

Per minute ($): _____

Fixed ($): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

**Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q2/5**

**True or false?**

In $y = 3x + 12$, when $x$ increases by 1, $y$ increases by 12.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

**Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q3/5**

A candle's height is $h = 20 - 1.5t$ centimeters after $t$ hours. What does 1.5 mean?

**A.** the candle starts 1.5 cm tall

**B.** the candle burns 20 cm per hour

**C.** the candle burns 1.5 cm per hour

**D.** the candle lasts 1.5 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

**Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q4/5**

**True or false?**

In y = 3x + 12, the value of y when x = 0 is 12.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

**Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q5/5**

SET 18 • HSS-ID.C.7

# BACKWARD 2

Use a linear model's equation to answer questions • Grade 8 • Statistics & Probability • 8.SP.A.3

Plant height is modeled by y = 1.5x + 5 (cm after x weeks). Predict the height at 6 weeks.

Height (cm): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

**Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q1/5**

In the plant model $y = 1.5x + 5$, one more week of growth goes with

**A.** 1.5 more weeks

**B.** 5 cm more height

**C.** 1.5 cm more height

**D.** 6.5 cm more height

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

**Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q2/5**

## True or false?

In the plant model $y = 1.5x + 5$, the slope means the plants were 1.5 cm tall at the start.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

**Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q3/5**

In the plant model y = 1.5x + 5, how much greater is the predicted height at week 10 than at week 2?

Difference (cm): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

**Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q4/5**

## True or false?

A negative slope means the predicted y decreases as x increases.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

**Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q5/5**

SET 18 • HSS-ID.C.7

# FORWARD 1

Interpret parameters with units and judge meaning • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

Using y = 5.17x + 48.28 (score versus hours studied), what is the predicted score increase for 3 more hours?

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.7 • Set 18 • FORWARD 1 • Q1/5

A shipping cost is modeled by $y = 0.02x + 3.5$ dollars for $x$ grams. What does 0.02 mean?

**A.** Each package costs $0.02 no matter what it weighs.

**B.** Each additional gram adds about $0.02 to the predicted cost.

**C.** The predicted cost rises by $3.50 for each gram.

**D.** A package weighing 0 grams is predicted to cost $0.02.

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.C.7 • Set 18 • FORWARD 1 • Q2/5

## True or false?

The intercept of a linear model always has a meaningful interpretation in context.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.7 • Set 18 • FORWARD 1 • Q3/5

For ages 20 to 60, maximum heart rate is modeled by $y = 208 - 0.7x$ beats per minute, $x$ in years. Which statement about 208 is correct?

**A.** It is the prediction at age 0, outside the data: not meaningful.

**B.** It is the predicted heart rate for a 20-year-old person.

**C.** It is the decrease in maximum heart rate for each year of age.

**D.** It is the largest heart rate that any person can have.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S18-F1-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.7 • Set 18 • FORWARD 1 • Q4/5**

Using $y = 208 - 0.7x$, predict the maximum heart rate at age 40.

Beats per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.7 • Set 18 • FORWARD 1 • Q5/5

SET 18 • HSS-ID.C.7

# FORWARD 2

Relate the slope to the correlation coefficient • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

## True or false?

The sign of the correlation coefficient matches the sign of the slope of the least-squares line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q1/5

The least-squares line is $y = -1.88x + 25.87$. Which could be the correlation coefficient?

**A.** 0.97

**B.** 1.88

**C.** $-1.88$

**D.** $-0.97$

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S18-F2-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.8 • Set 18 • FORWARD 2 • Q2/5**

## True or false?

A larger slope always means a stronger correlation.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q3/5

The least-squares slope equals r·(s_y/s_x). If r = 0.8, s_y = 10 and s_x = 4, what is the slope?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q4/5

If the least-squares line has slope 0, what is r?

**A.** 0

**B.** 1

**C.** $-1$

**D.** it cannot be found

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q5/5

# Compute the correlation coefficient with technology

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.C.8** |
| **BACKWARD 1** | Find means and deviations | **6.SP.B.5.c** |
| **BACKWARD 2** | Read the direction of an association | **HSS-ID.B.6** |
| **FORWARD 1** | Compute r from deviations | **HSS-ID.C.8** |
| **FORWARD 2** | See how changes to the data affect r | **HSS-ID.C.8** |

SET 19 • HSS-ID.C.8

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

Find the correlation coefficient r for hours studied and score. Round to two decimal places. A calculator or spreadsheet may be used.

| Hours | 1 | 2 | 2 | 3 | 4 | 5 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|---|
| Score | 52 | 58 | 61 | 63 | 70 | 72 | 76 | 80 | 83 | 90 |

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.8 • Set 19 • MAIN • Q1/5

Find the correlation coefficient r for car age and value. Round to two decimal places. A calculator or spreadsheet may be used.

| Age (years) | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Value ($1000s) | 24 | 21 | 22 | 17 | 18 | 13 | 14 | 9 | 11 | 6 |

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.8 • Set 19 • MAIN • Q2/5

Find the correlation coefficient r for hours after sunrise (x) and ice thickness in cm (y). Round to two decimal places. A calculator or spreadsheet may be used.

| Hours | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Thickness (cm) | 48 | 44 | 41 | 35 | 33 | 28 |

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.8 • Set 19 • MAIN • Q3/5

Find r for the temperature and cones-sold data for all 7 days, and again without the 90-degree day. Round each to three decimal places. A calculator or spreadsheet may be used.

| Temperature (°F) | 60 | 65 | 70 | 75 | 80 | 85 | 90 |
|---|---|---|---|---|---|---|---|
| Cones sold | 120 | 140 | 155 | 175 | 190 | 210 | 60 |

All 7 days: _____

Without the 90-degree day: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.C.8 • Set 19 • MAIN • Q4/5

Find the correlation coefficient r for the spring data (mass and length). Round to two decimal places. A calculator or spreadsheet may be used.

| Mass (g) | 2 | 4 | 5 | 7 | 8 | 10 |
|---|---|---|---|---|---|---|
| Length (cm) | 15 | 19 | 24 | 26 | 31 | 35 |

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.8 • Set 19 • MAIN • Q5/5

SET 19 • HSS-ID.C.8

# BACKWARD 1

Find means and deviations • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean of 1, 2, 3, 4, 5.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q1/5**

For the values 2, 5, 4, 7 (mean 4.5), find the deviations of 2 and 7 from the mean.

2: _____

7: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q2/5**

## True or false?

The deviations of a data set from its mean always add to 0.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q3/5**

What is the mean of 48, 44, 41, 35, 33, 28, rounded to two decimal places?

**A.** 35

**B.** 38.17

**C.** 41

**D.** 38

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q4/5**

## True or false?

A value below the mean has a positive deviation.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q5/5**

SET 19 • HSS-ID.C.8

# BACKWARD 2

Read the direction of an association • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6

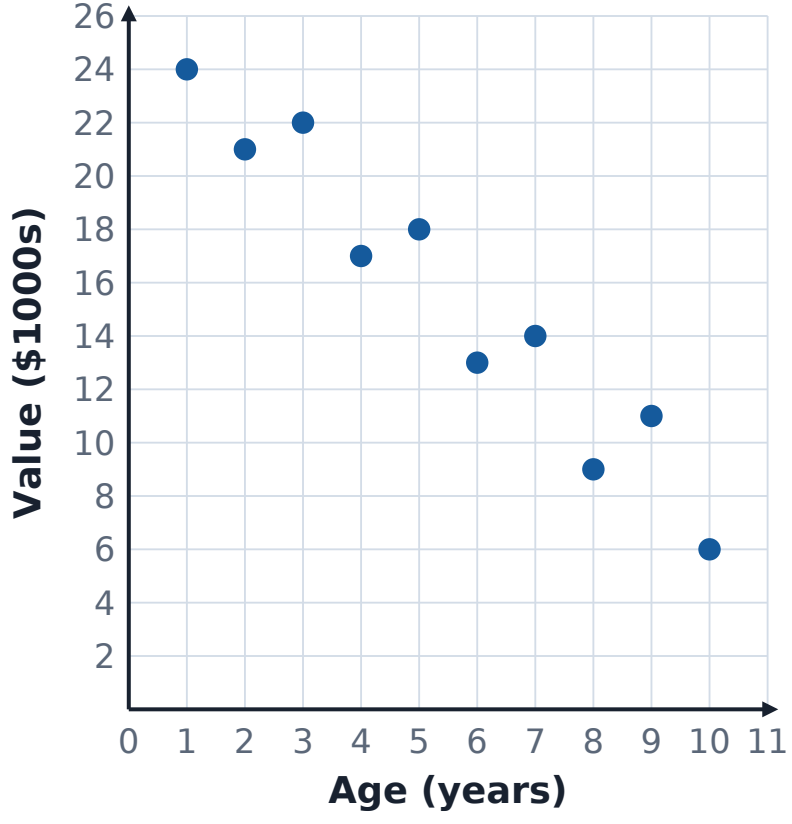
The scatter plot shows car age and value. What is the sign of r?

**A.** positive

**B.** negative

**C.** cannot be found

**D.** zero




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

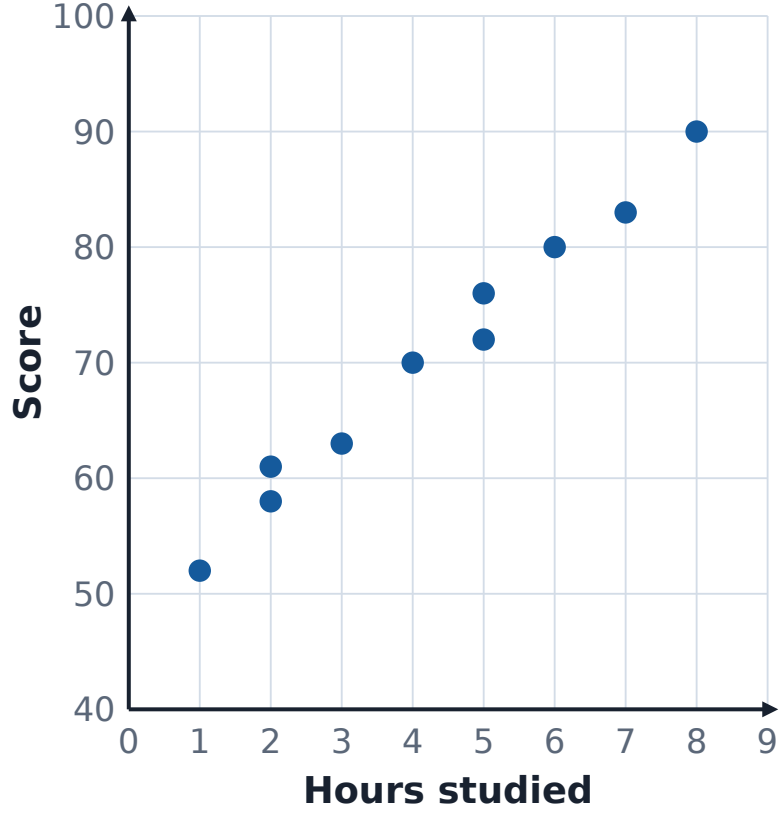
High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q1/5

## True or false?

For the hours-studied scatter plot, r is positive.

**A.** True

**B.** False




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q2/5

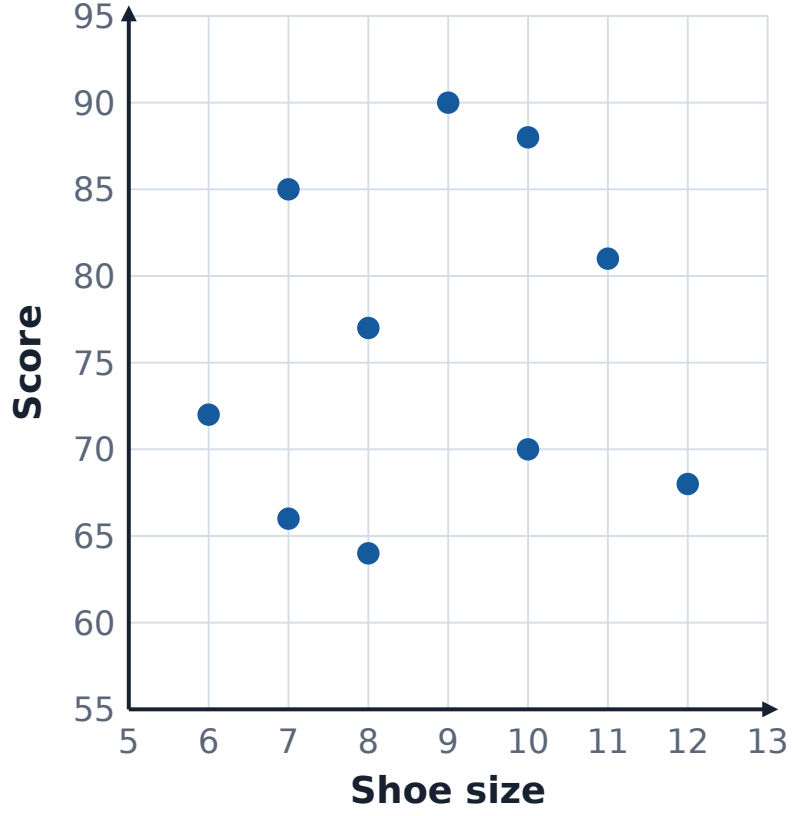
For the shoe-size scatter plot, $r$ is closest to

**A.** 1

**B.** 0

**C.** $-1$

**D.** 0.9




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q3/5

## True or false?

For a cloud of points with no pattern, r is close to 1.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q4/5

Which scatter plot would have r close to −1?

**A.** points along a U-shaped curve

**B.** a shapeless cloud

**C.** points falling closely along a line

**D.** points rising closely along a line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • HSS-ID.C.8

# FORWARD 1

Compute r from deviations • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

For (1, 2), (2, 5), (3, 4), (4, 7): $\Sigma(x - \bar{x})(y - \bar{y}) = 7$, $\Sigma(x - \bar{x})^2 = 5$ and $\Sigma(y - \bar{y})^2 = 13$. Compute $r = 7/\sqrt{(5 \cdot 13)}$, rounded to two decimal places.

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.8 • Set 19 • FORWARD 1 • Q1/5

## True or false?

If all points lie exactly on a line with positive slope, r = 1.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q2/5

For the points (1, 2), (2, 4), (3, 6), what is r?

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.8 • Set 19 • FORWARD 1 • Q3/5

A calculator reports r = 1.4. What should you conclude?

**A.** The association is very strong and positive.

**B.** The data have one outlier.

**C.** A mistake was made: r is always between −1 and 1.

**D.** The slope of the line is 1.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q4/5

## True or false?

For the points (1, 6), (2, 4), (3, 2), r = 1.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.8 • Set 19 • FORWARD 1 • Q5/5

SET 19 • HSS-ID.C.8

# FORWARD 2

See how changes to the data affect r • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

## True or false?

Changing x from hours to minutes does not change r.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q1/5

## True or false?

Swapping the roles of x and y changes the value of r.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.8 • Set 19 • FORWARD 2 • Q2/5

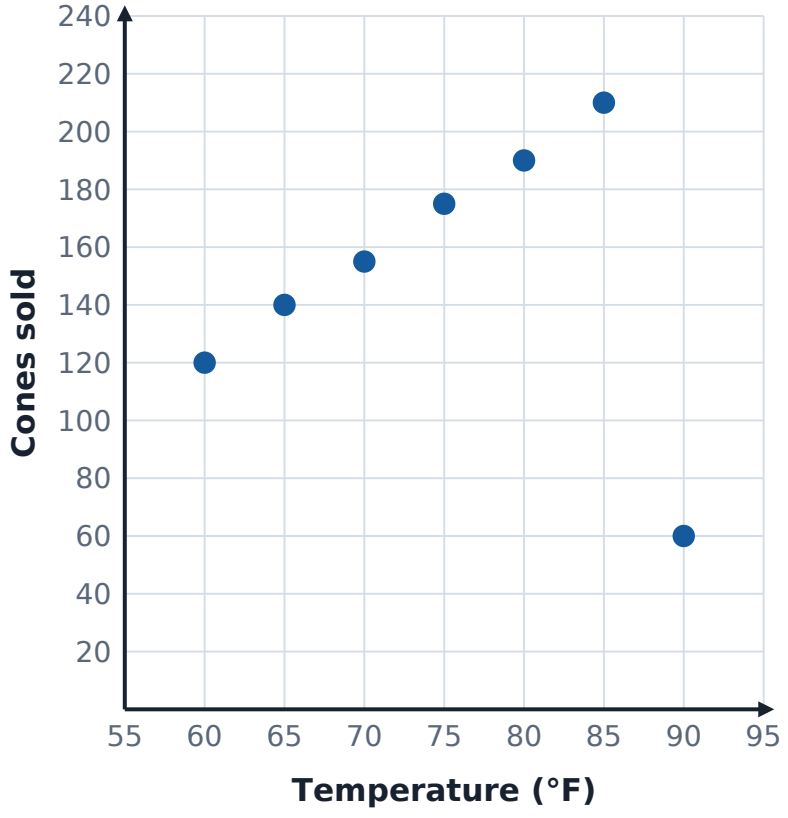
For the cones data, r = 0.999 for six days and −0.008 with the 90-degree day included. What does this show?

**A.** Temperature causes cone sales to fall.

**B.** r is not affected by outliers.

**C.** The six days have no association.

**D.** One outlier can change r dramatically.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.C.8 • Set 19 • FORWARD 2 • Q3/5

Find r for (1, 3), (2, 5), (3, 7), and for the same points with every y increased by 10. Give both.

Original: _____

After adding 10: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q4/5

Which change to a data set could change r?

**A.** adding 5 to every x-value

**B.** converting y from meters to centimeters

**C.** adding a new point far from the others

**D.** doubling every y-value

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q5/5

# Interpret the correlation coefficient

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.C.8** |
| **BACKWARD 1** | Describe strength and direction from a plot | **HSS-ID.B.6** |
| **BACKWARD 2** | Compare numbers on a scale from -1 to 1 | **6.NS.C.7.c** |
| **FORWARD 1** | Match correlation coefficients to scatter plots | **HSS-ID.C.8** |
| **FORWARD 2** | Correlation is not causation | **HSS-ID.C.9** |

SET 20 • HSS-ID.C.8

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

For car age and value, $r = -0.97$. Which interpretation is correct?

**A.** weak negative linear association

**B.** strong positive linear association

**C.** strong negative linear association

**D.** no association

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q1/5

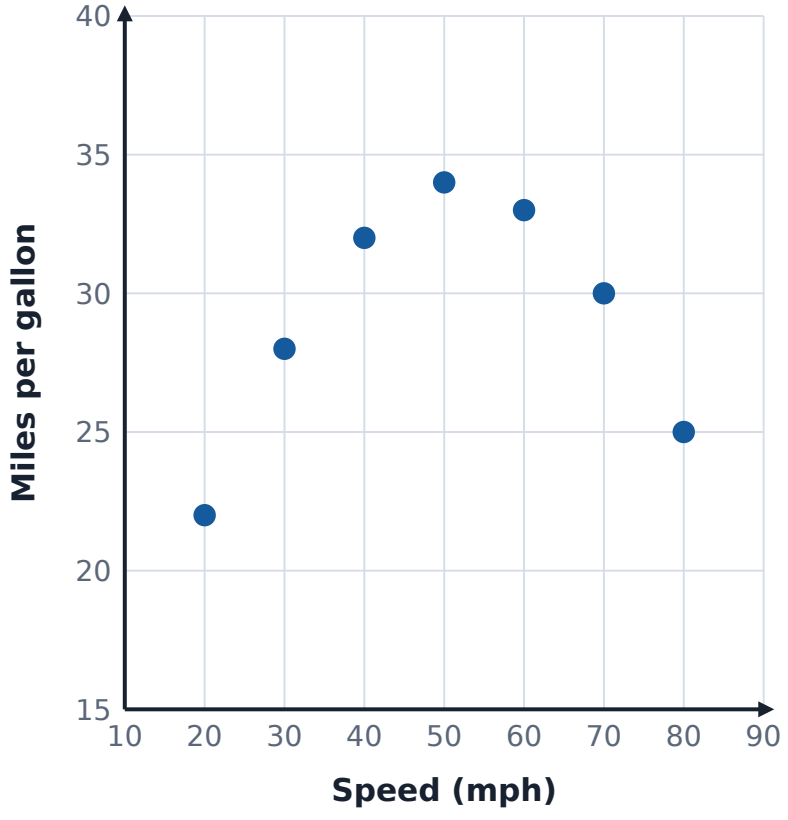
For speed and fuel efficiency, r = 0.25 but the scatter plot shows a clear curve. Which statement is correct?

**A.** r must have been computed incorrectly.

**B.** There is a weak linear relationship and nothing more.

**C.** r measures only linear association, so it misses the curve.

**D.** There is no relationship between the variables.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q2/5

Which correlation coefficient shows the strongest linear association?

**A.** $-0.91$

**B.** $0.85$

**C.** $-0.40$

**D.** $0.10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q3/5

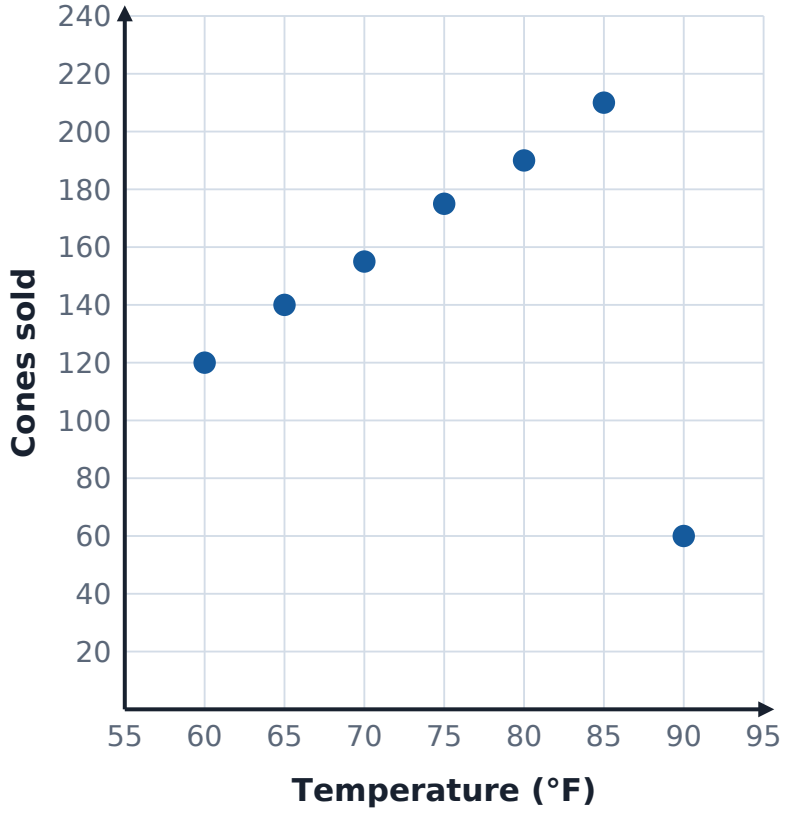
For the cones data, r = −0.008 for all 7 days but 0.999 without the 90-degree day. Which interpretation is best?

**A.** The six typical days show a strong positive linear association.

**B.** The 90-degree day is the most typical day.

**C.** Temperature and cone sales have no relationship on any day.

**D.** Hotter days have fewer cones sold.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.C.8 • Set 20 • MAIN • Q4/5

Which statement about r is correct?

**A.** r changes when x is converted from hours to minutes.

**B.** r is negative whenever the data contain an outlier.

**C.** r has no units and stays the same if the units change.

**D.** r can be as large as 5 for very strong data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q5/5

SET 20 • HSS-ID.C.8

# BACKWARD 1

Describe strength and direction from a plot • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6

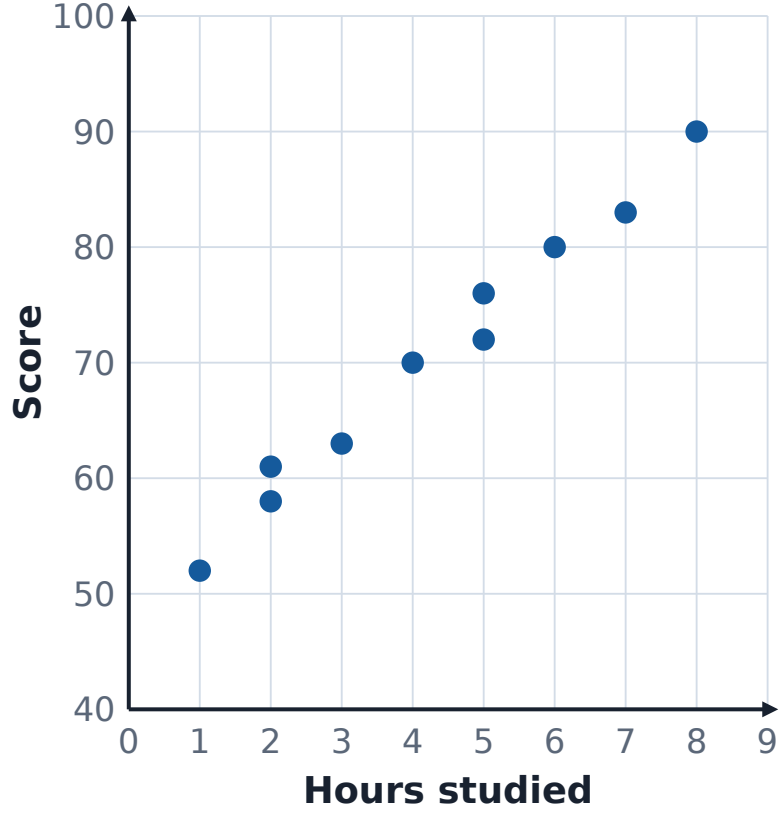
The scatter plot shows hours studied and score. How strong is the linear association?

**A.** strong

**B.** weak

**C.** cannot be judged

**D.** none




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q1/5

## True or false?

In a strong association, the points lie close to a line or curve.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q2/5

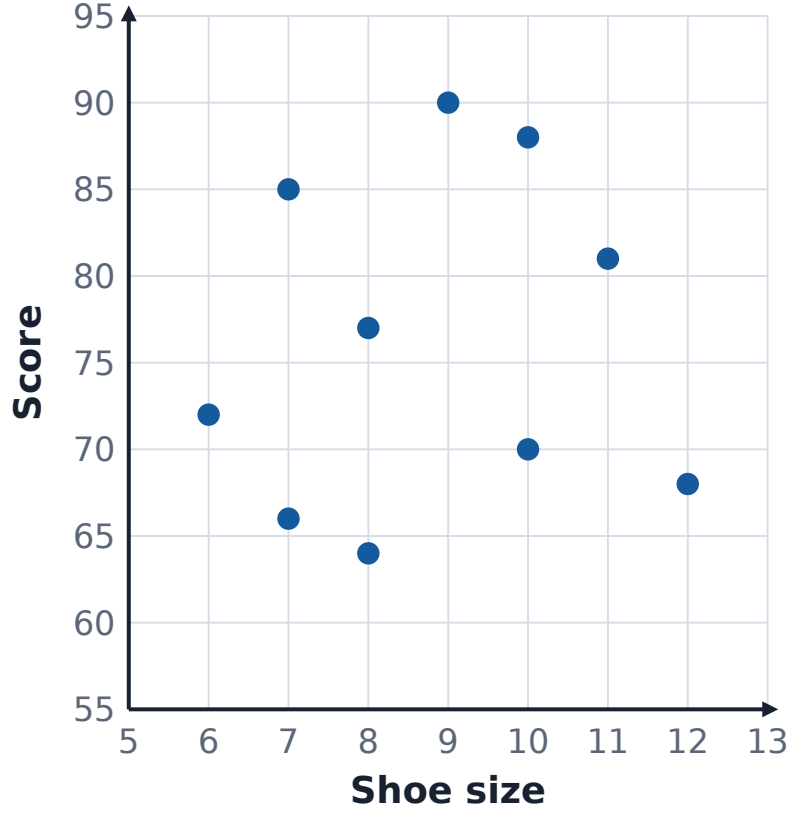
The scatter plot shows shoe size and quiz score. How strong is the association?

**A.** very weak or none

**B.** strong negative

**C.** strong positive

**D.** perfect




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q3/5

## True or false?

A scatter plot that falls from left to right shows a positive association.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q4/5

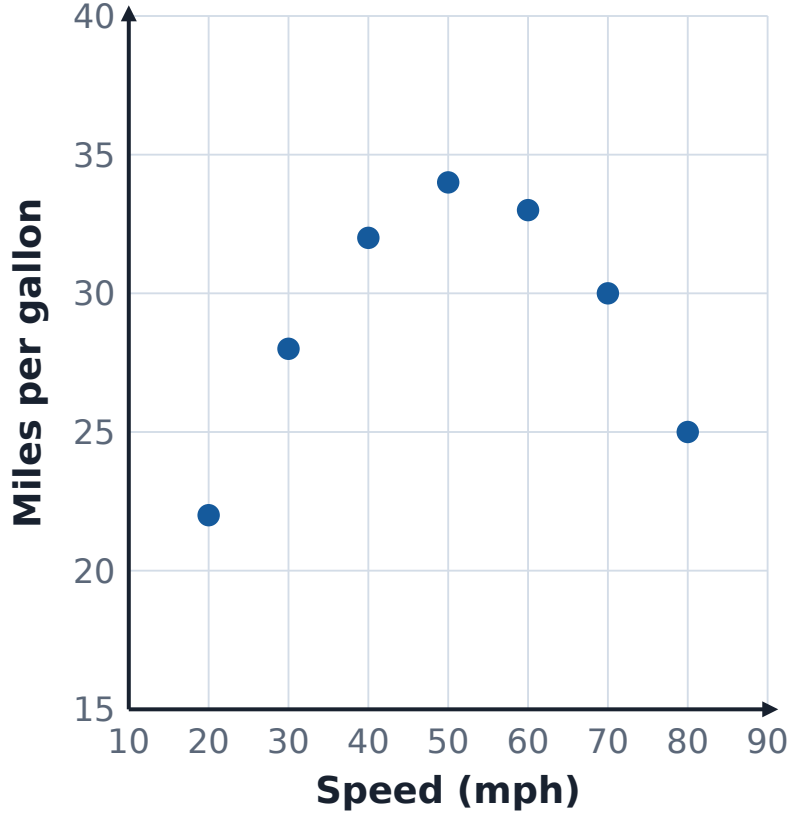
The scatter plot of speed and fuel efficiency has what form?

**A.** exponential growth

**B.** curved

**C.** linear

**D.** no form




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • HSS-ID.C.8

# BACKWARD 2

Compare numbers on a scale from −1 to 1 • Grade 6 • The Number System • 6.NS.C.7.c

What is $|-0.91|$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

**Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q1/5**

## True or false?

$|-0.8| > |0.6|$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

**Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q2/5**

Which is closest to 0?

**A.** $-0.1$

**B.** $-0.6$

**C.** 0.5

**D.** 0.9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

**Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q3/5**

**True or false?**

$-0.9$ is greater than $0.2$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

**Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q4/5**

Which has the greater absolute value, $-0.75$ or $0.7$? Enter the number.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

**Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q5/5**

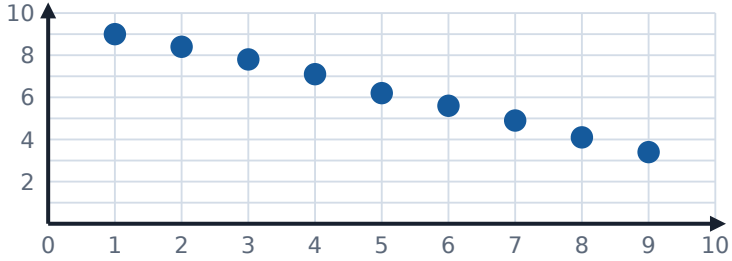
SET 20 • HSS-ID.C.8

# FORWARD 1

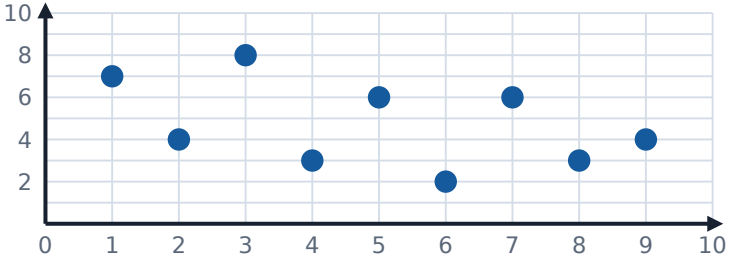
Match correlation coefficients to scatter plots • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

Which scatter plot has $r \approx -0.45$?

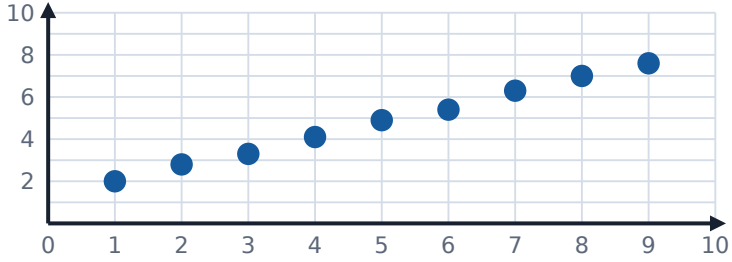
**A**



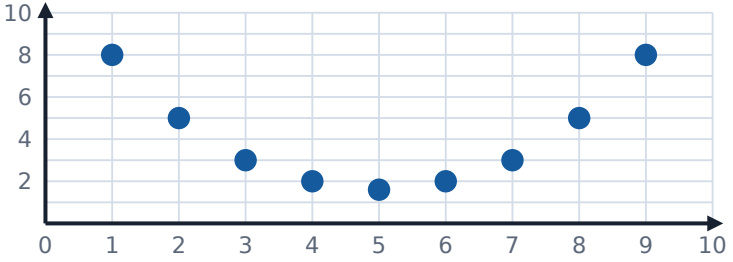

**B**




**C**




**D**




**A.** Graph A

**B.** Graph B

**C.** Graph C

**D.** Graph D

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S20-F1-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.8 • Set 20 • FORWARD 1 • Q1/5**

A U-shaped scatter plot that is perfectly symmetric has r equal to

**A.** 0.5

**B.** 0

**C.** $-1$

**D.** 1

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S20-F1-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.8 • Set 20 • FORWARD 1 • Q2/5**

## True or false?

$r = -0.45$ indicates a stronger linear association than $r = 0.30$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q3/5

Which value of r goes with points falling steadily in a narrow band?

**A.** $-0.99$

**B.** $0.99$

**C.** $0$

**D.** $-0.45$

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S20-F1-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.8 • Set 20 • FORWARD 1 • Q4/5**

## True or false?

Two data sets with the same r must have scatter plots of the same shape.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q5/5

SET 20 • HSS-ID.C.8

# FORWARD 2

Correlation is not causation • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Across towns, the number of libraries and the number of crimes have r = 0.9. Which is the best explanation?

**A.** Libraries cause crime.

**B.** r = 0.9 proves a cause in one direction.

**C.** Crime causes towns to build libraries.

**D.** Larger towns tend to have more of both.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S20-F2-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.9 • Set 20 • FORWARD 2 • Q1/5**

## True or false?

A correlation of r = 0.95 shows that changes in x cause changes in y.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q2/5

Which kind of study can show that x causes y?

**A.** a scatter plot of past data

**B.** a randomized experiment

**C.** a study with r close to 1

**D.** a survey with many people

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q3/5

## True or false?

A lurking variable can produce a strong correlation between two variables that do not affect each other.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q4/5

Students who sleep more have higher grades ($r = 0.6$) in an observational study. Which conclusion is justified?

**A.** Higher grades cause these students to sleep more.

**B.** Sleep and grades are positively associated in this group.

**C.** Sleep and grades are unrelated for these students.

**D.** More sleep causes higher grades for these students.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S20-F2-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.9 • Set 20 • FORWARD 2 • Q5/5**

# Distinguish between correlation and causation

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.C.9** |
| **BACKWARD 1** | Recognize a strong correlation | **HSS-ID.C.8** |
| **BACKWARD 2** | Describe the association in a scatter plot | **8.SP.A.1** |
| **FORWARD 1** | Tell experiments from observational studies | **HSS-IC.B.3** |
| **FORWARD 2** | Judge causal claims from experiment results | **HSS-IC.B.5** |

SET 21 • HSS-ID.C.9

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Over several years, a city's sunscreen sales and its number of pool visits are strongly correlated. What best explains this?

**A.** Swimming causes people to buy sunscreen.

**B.** Sunny, hot weather raises both.

**C.** Sunscreen causes people to swim.

**D.** The data must be wrong.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S21-M-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.9 • Set 21 • MAIN • Q1/5**

Countries with more doctors per person have longer life spans. Which conclusion is justified?

**A.** Doctors per person and life span are positively associated.

**B.** Longer life spans cause countries to train more doctors.

**C.** Adding more doctors in a country will lengthen life spans.

**D.** Doctors per person and life span are unrelated.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S21-M-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.9 • Set 21 • MAIN • Q2/5**

Researchers want to know whether a new study app raises test scores. Which design can show cause?

**A.** Find the correlation of app use and scores.

**B.** Survey students who already chose to use the app.

**C.** Compare the scores of last year's users and nonusers.

**D.** Randomly assign students to use the app or not.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S21-M-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.9 • Set 21 • MAIN • Q3/5**

Children with larger shoe sizes have higher reading scores. What is the most likely lurking variable?

**A.** age

**B.** reading-book color

**C.** school lunch menu

**D.** shoe brand

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S21-M-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.9 • Set 21 • MAIN • Q4/5**

Which statement is correct?

**A.** Association implies causation when r is above 0.9.

**B.** If two variables are not correlated, neither can cause anything.

**C.** Causation implies association, but association does not imply causation.

**D.** Causation can only be shown with a scatter plot.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S21-M-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.9 • Set 21 • MAIN • Q5/5**

SET 21 • HSS-ID.C.9

# BACKWARD 1

Recognize a strong correlation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

## True or false?

$r = -0.92$ shows a strong linear association.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q1/5

Which r shows the weakest linear association?

**A.** $-0.7$

**B.** $0.05$

**C.** $0.6$

**D.** $-0.95$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q2/5

## True or false?

A strong correlation means the two variables must be causally related.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q3/5

A data set has r = 0.88. Is the linear association positive or negative? Enter positive or negative.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q4/5

r = 0.88 for study time and grades in a survey. What does it describe?

**A.** a strong positive linear association

**B.** a curved association

**C.** a weak association

**D.** proof that study time causes grades

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • HSS-ID.C.9

# BACKWARD 2

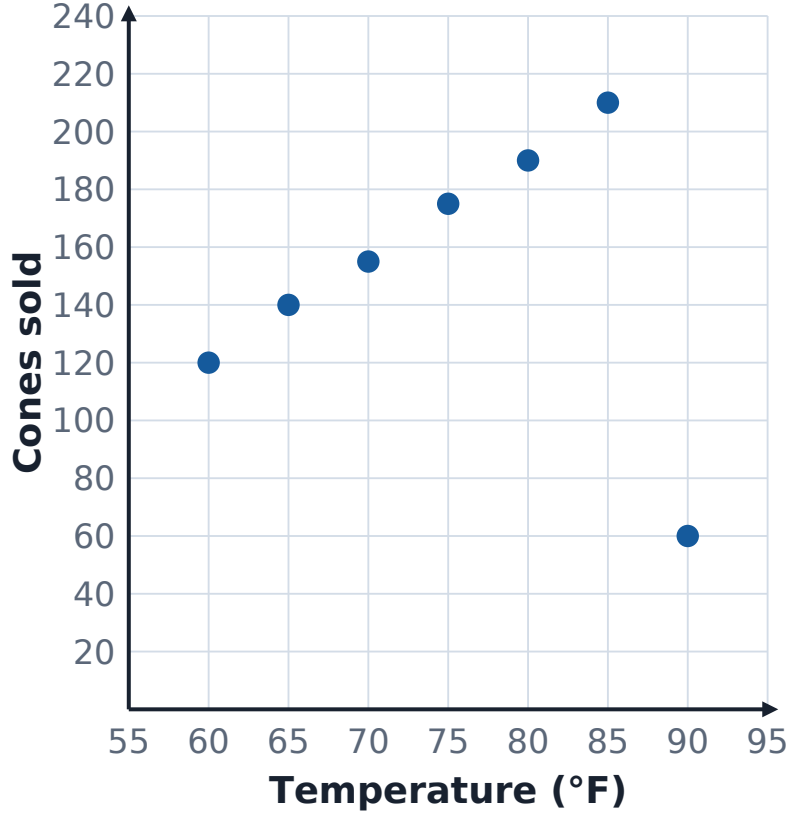
Describe the association in a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

## True or false?

In the ice-cream scatter plot, the six typical days show a positive association.

**A.** True

**B.** False




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

**Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q1/5**

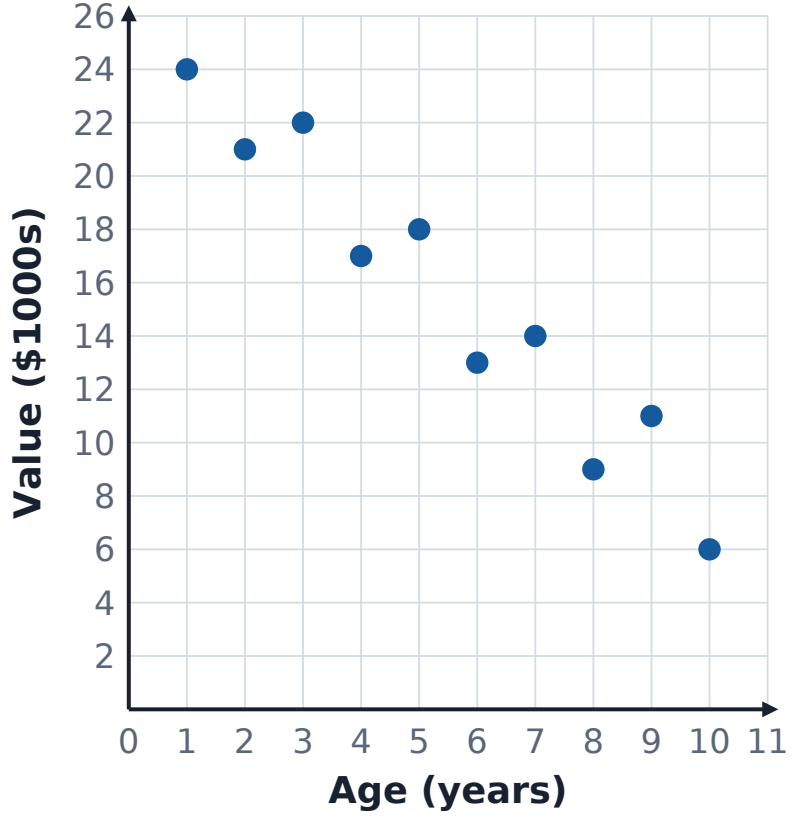
The scatter plot shows car age and value. Which describes it?

**A.** curved association

**B.** positive linear association

**C.** negative linear association

**D.** no association




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

**Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q2/5**

## True or false?

A scatter plot can show an association between two variables that do not cause each other.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

**Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q3/5**

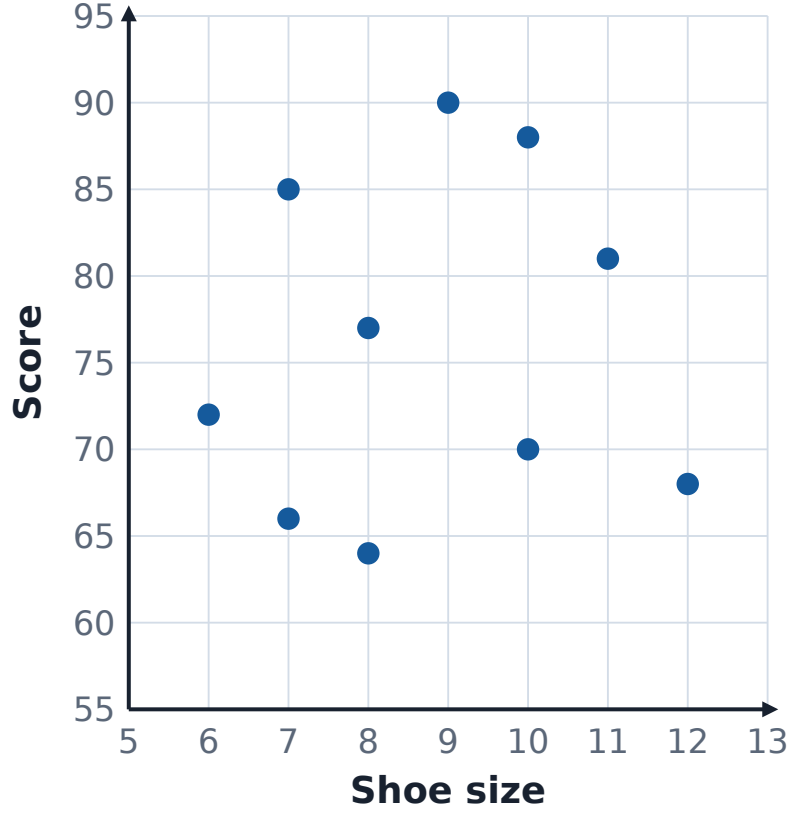
The scatter plot shows shoe size and quiz score of adults. Which describes it?

**A.** no clear association

**B.** curved

**C.** strong negative

**D.** strong positive




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

**Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q4/5**

## True or false?

An association seen in a scatter plot shows which variable is the cause.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

**Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q5/5**

SET 21 • HSS-ID.C.9

# FORWARD 1

Tell experiments from observational studies • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Researchers record the coffee intake and blood pressure of 500 adults without changing anything. What type of study is this?

**A.** randomized experiment

**B.** simulation

**C.** sample survey of opinions

**D.** observational study

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S21-F1-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q1/5**

## True or false?

In a randomized experiment, the researcher assigns the treatments by chance.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q2/5

Which study is a randomized experiment?

**A.** Researchers compare towns that chose different school lunches.

**B.** Volunteers are randomly assigned to a new diet or the usual diet.

**C.** A survey asks people which diet they prefer.

**D.** Doctors record the diets their patients already follow.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S21-F1-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q3/5**

## True or false?

An observational study can establish cause as well as a randomized experiment can.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q4/5

Why does random assignment help support causal conclusions?

**A.** It removes all variability from the measurements.

**B.** It guarantees that r will be close to 1.

**C.** It makes the sample larger than other designs do.

**D.** It tends to balance other variables between the groups.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S21-F1-Q5**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q5/5**

SET 21 • HSS-ID.C.9

# FORWARD 2

Judge causal claims from experiment results • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

In a randomized experiment, 40 plants got fertilizer and grew a mean of 18 cm; 40 got none and grew a mean of 12 cm. What is the difference in means (fertilizer minus none)?

Difference (cm): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q1/5

Re-randomizing the plants many times gave differences of 6 cm or more only 1% of the time. What can be concluded?

**A.** Fertilizer is unrelated to growth.

**B.** The fertilizer likely caused more growth.

**C.** The difference is due to chance alone.

**D.** No conclusion is possible without r.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S21-F2-Q2**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q2/5**

## True or false?

In a randomized experiment, a difference much larger than chance alone would produce supports a causal conclusion.

**A.** True

**B.** False

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S21-F2-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q3/5**

An observational study found a large difference between coffee drinkers and others. Why can't it prove cause?

**A.** Coffee drinkers may differ in other ways that matter.

**B.** Observational studies do not collect any data.

**C.** Coffee intake cannot be measured accurately.

**D.** The difference was too large to be believable.

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q4/5

## True or false?

If an experiment's difference could easily happen by chance, the experiment proves the treatment has no effect.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q5/5

# Statistics as a process of inference from samples

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-IC.A.1** |
| **BACKWARD 1** | Recognize random and representative samples | **7.SP.A.1** |
| **BACKWARD 2** | Use a sample to estimate a population count | **7.SP.A.2** |
| **FORWARD 1** | See how sample size affects sampling variability | **HSS-IC.A.1** |
| **FORWARD 2** | Attach a margin of error to an estimate | **HSS-IC.B.4** |

SET 22 • HSS-IC.A.1

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

A random sample of 200 of a school's 1,500 students found that 62 bike to school. What is the sample proportion who bike? Give a decimal.

Sample proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q1/5

A random sample of 200 of a school's 1,500 students found that 62 bike to school. What is the parameter of interest?

**A.** the 200 students who were in the sample

**B.** the proportion of all 1,500 students who bike

**C.** the sample proportion 0.31 from the survey

**D.** the 62 sampled students who said they bike

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S22-M-Q2**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q2/5**

In a random sample of 200 of a school's 1,500 students, the proportion who bike is 0.31. Use it to estimate how many of the 1,500 students bike.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q3/5

Which sample of a school's students allows an inference about all of the students?

**A.** the first 200 students to arrive at school one morning

**B.** a random sample of 200 students chosen from the school roster

**C.** all 200 students who belong to the school bike club

**D.** the 200 students who chose to answer an online poll

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S22-M-Q4**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q4/5**

Two random samples of 200 students from the same school give sample proportions 0.31 and 0.27. Why do they differ?

**A.** One of the two samples must have been chosen incorrectly.

**B.** Sample proportions can never be used to estimate a parameter.

**C.** Sample statistics vary from one random sample to another by chance.

**D.** The population proportion changed between the two samples.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S22-M-Q5**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q5/5**

SET 22 • HSS-IC.A.1

# BACKWARD 1

Recognize random and representative samples • Grade 7 • Statistics & Probability • 7.SP.A.1

## True or false?

A random sample tends to be representative of its population.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

**Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q1/5**

A town wants residents' opinion on a new park. Which method gives a random sample?

**A.** ask the first 100 people at the park

**B.** ask the members of the parks committee

**C.** post an online survey and use the first replies

**D.** pick 100 names by lottery from the voter list

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

**Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q2/5**

## True or false?

A sample of volunteers who choose to respond is a random sample.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q3

**Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q3/5**

A survey of shoppers at a sporting-goods store is used to estimate how much all town residents exercise. What is the main problem?

**A.** the sample is a census

**B.** the sample is too random

**C.** the sample has no data

**D.** the sample is not representative

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

**Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q4/5**

## True or false?

The whole group a study wants to learn about is called the population.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

**Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q5/5**

SET 22 • HSS-IC.A.1

# BACKWARD 2

Use a sample to estimate a population count • Grade 7 • Statistics & Probability • 7.SP.A.2

In a random sample of 50 students, 12 play an instrument. Estimate how many of 600 students play.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q1

**Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q1/5**

In a random sample of 40 light bulbs, 2 are defective. Estimate the number defective in 2,000 bulbs.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

**Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q2/5**

**True or false?**

If 30% of a random sample prefer pizza, about 30% of the population is estimated to prefer pizza.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

**Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q3/5**

In a random sample of 80 voters, 44 support a measure. What is the estimated percent of all voters who support it?

**A.** 36%

**B.** 44%

**C.** 80%

**D.** 55%

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

**Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q4/5**

## True or false?

Estimates from a random sample are always exactly equal to the population value.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

**Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q5/5**

SET 22 • HSS-IC.A.1

# FORWARD 1

See how sample size affects sampling variability • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

Many random samples of size 25 and many of size 400 are taken from the same population. Which is true?

**A.** Larger samples are always biased.

**B.** Proportions from samples of 25 vary less.

**C.** Both sizes vary the same amount.

**D.** Proportions from samples of 400 vary less.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S22-F1-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q1/5**

## True or false?

Increasing the size of a random sample reduces bias caused by a poor sampling method.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q2/5

A random sample of 400 voters has 228 who favor a law. What is the sample proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q3/5

Which estimate of a population proportion is likely to be closest to the true value?

**A.** one from 1,000 volunteers who called in

**B.** one from a random sample of 50

**C.** one from 50 friends of the researcher

**D.** one from a random sample of 1,000

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S22-F1-Q4**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q4/5**

## True or false?

Sample proportions from repeated random samples center around the population proportion.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q5/5

SET 22 • HSS-IC.A.1

# FORWARD 2

Attach a margin of error to an estimate • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

A random sample gives a sample proportion of 0.57 with a margin of error of 0.05. Give the interval of plausible values for the population proportion.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q1/5

A poll reports 48% with a margin of error of 3 percentage points. Which value is plausible for the population percent?

**A.** 52%

**B.** 55%

**C.** 44%

**D.** 50%

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S22-F2-Q2**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q2/5**

## True or false?

A larger random sample usually gives a smaller margin of error.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q3/5

Simulated sample proportions have standard deviation 0.025. Using margin of error = 2 × standard deviation, what is the margin of error?

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q4/5

## True or false?

A margin of error accounts for bias from a sample of volunteers.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q5/5

# Simulate a probability model with random digits

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-IC.A.2** |
| **BACKWARD 1** | Design a simulation for a compound event | **7.SP.C.8.c** |
| **BACKWARD 2** | Find a probability as a relative frequency | **7.SP.C.6** |
| **FORWARD 1** | Use two-digit groups for probabilities in hundredths | **HSS-IC.A.2** |
| **FORWARD 2** | Use simulation results to judge a model | **HSS-IC.A.2** |

SET 23 • HSS-IC.A.2

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

A player makes 70% of free throws. To simulate, digits 0-6 mean "make" and 7-9 mean "miss". Use the first 10 digits of line 1 of the table as 10 shots. How many shots are made?

| Line | Digits |
|---|---|
| 1 | 15881 39986 98797 30911 |
| 2 | 41707 53645 56815 18844 |
| 3 | 72940 55761 69871 68976 |
| 4 | 84697 88093 20819 67273 |
| 5 | 17784 86488 76634 40029 |
| 6 | 78753 47787 68045 71784 |
| 7 | 72671 47513 62701 07981 |
| 8 | 93319 08452 05997 28379 |

Shots made: _____

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S23-M-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q1/5**

A player makes 70% of free throws (digits 0-6 = make, 7-9 = miss). Each line's first 10 digits are one trial of 10 shots, giving 8 trials. In how many trials does the player make 8 or more shots?

| Line | Digits |
|---|---|
| 1 | 15881 39986 98797 30911 |
| 2 | 41707 53645 56815 18844 |
| 3 | 72940 55761 69871 68976 |
| 4 | 84697 88093 20819 67273 |
| 5 | 17784 86488 76634 40029 |
| 6 | 78753 47787 68045 71784 |
| 7 | 72671 47513 62701 07981 |
| 8 | 93319 08452 05997 28379 |

Trials: _____

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S23-M-Q2**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q2/5**

An event has probability 0.4. Which assignment of single random digits simulates it?

**A.** 4 = success; all other digits = failure

**B.** 0-3 and 9 = success; 4-8 = failure

**C.** 0-4 = success; 5-9 = failure

**D.** 0-3 = success; 4-9 = failure

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S23-M-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q3/5**

A spinner lands on red with probability 0.25. Read line 5 of the table as ten two-digit numbers; 00-24 means red. How many of the ten spins are red?

| Line | Digits |
|---|---|
| 1 | 15881 39986 98797 30911 |
| 2 | 41707 53645 56815 18844 |
| 3 | 72940 55761 69871 68976 |
| 4 | 84697 88093 20819 67273 |
| 5 | 17784 86488 76634 40029 |
| 6 | 78753 47787 68045 71784 |
| 7 | 72671 47513 62701 07981 |
| 8 | 93319 08452 05997 28379 |

Red spins: _____

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S23-M-Q4**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q4/5**

A simulation of 50 families of 4 children (each birth a girl with probability 0.5) gave the results in the table. What proportion of the simulated families had 3 or more girls? Give a decimal.

| Girls in 4 births | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Number of simulated trials | 3 | 12 | 19 | 13 | 3 |

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q5/5

SET 23 • HSS-IC.A.2

# BACKWARD 1

Design a simulation for a compound event • Grade 7 • Statistics & Probability • 7.SP.C.8.c

To simulate a fair coin with random digits, which assignment works?

**A.** 0-4 = heads; 4-9 = tails

**B.** 0 = heads; 1-9 = tails

**C.** even digits = heads; odd digits = tails

**D.** 0-6 = heads; 7-9 = tails

In the digits 4 1 7 0 7 5 3 6 4 5, digits 0-4 mean heads. How many heads?

Heads: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

**Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q2/5**

**True or false?**

A trial of a simulation for "number of heads in 10 flips" uses 10 random digits.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

**Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q3/5**

To simulate rolling a die with digits, what should be done with 0, 7, 8 and 9?

**A.** skip them and read the next digit

**B.** stop the simulation

**C.** count them as a 6

**D.** count them as a 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

**Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q4/5**

## True or false?

Running more trials of a simulation makes the estimated probability less reliable.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

**Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q5/5**

SET 23 • HSS-IC.A.2

# BACKWARD 2

Find a probability as a relative frequency • Grade 7 • Statistics & Probability • 7.SP.C.6

In 50 simulated trials, an event happened 16 times. What is its relative frequency? Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

**Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q1/5**

An event happened in 9 of 200 trials. What is its relative frequency as a percent?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

**Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q2/5**

## True or false?

A relative frequency of 0.32 means the event happened in 32% of the trials.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

**Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q3/5**

An event happened in 12 of 60 trials. What is its relative frequency?

**A.** 0.2

**B.** 0.6

**C.** 5

**D.** 0.12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

**Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q4/5**

## True or false?

The relative frequency from 10 trials is always exactly equal to the true probability.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

**Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q5/5**

SET 23 • HSS-IC.A.2

# FORWARD 1

Use two-digit groups for probabilities in hundredths • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

An event has probability 0.35. Which assignment of two-digit numbers simulates it?

**A.** 35 = success; others = failure

**B.** 00-34 = success; 35-99 = failure

**C.** 00-35 = success; 36-99 = failure

**D.** 01-34 = success; others = failure

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S23-F1-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q1/5**

Read line 3 as ten two-digit numbers; 00-34 means success. How many successes?

| Line | Digits |
|---|---|
| 1 | 15881 39986 98797 30911 |
| 2 | 41707 53645 56815 18844 |
| 3 | 72940 55761 69871 68976 |
| 4 | 84697 88093 20819 67273 |
| 5 | 17784 86488 76634 40029 |
| 6 | 78753 47787 68045 71784 |
| 7 | 72671 47513 62701 07981 |
| 8 | 93319 08452 05997 28379 |

Successes: _____

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S23-F1-Q2**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q2/5**

## True or false?

Two-digit groups 00-99 give 100 equally likely outcomes.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q3/5

A fair coin model uses 0-4 = heads. Using the first 10 digits of each of the 8 lines as 8 trials of 10 flips, in how many trials are there 3 or fewer heads?

| Line | Digits |
|---|---|
| 1 | 15881 39986 98797 30911 |
| 2 | 41707 53645 56815 18844 |
| 3 | 72940 55761 69871 68976 |
| 4 | 84697 88093 20819 67273 |
| 5 | 17784 86488 76634 40029 |
| 6 | 78753 47787 68045 71784 |
| 7 | 72671 47513 62701 07981 |
| 8 | 93319 08452 05997 28379 |

Trials: _____

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S23-F1-Q4**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q4/5**

## True or false?

To simulate a probability of 1/3 with single digits, use 1-3 = success, 4-9 = failure and skip 0.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q5/5

SET 23 • HSS-IC.A.2

# FORWARD 2

Use simulation results to judge a model • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

In 100 simulated sets of 10 fair-coin flips (table), in what proportion of the sets were there 9 or more heads? Give a decimal.

| Heads in 10 flips | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Number of simulated trials | 0 | 1 | 4 | 12 | 20 | 25 | 20 | 12 | 4 | 2 | 0 |

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q1/5

A coin claimed to be fair gave 9 heads in 10 flips. Use the table of 100 simulated sets. Decision rule: if results at least as extreme as the observed result happen in fewer than 5% of the simulated trials, the model is not consistent with the data; otherwise it is consistent. Which decision is correct?

| Heads in 10 flips | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Number of simulated trials | 0 | 1 | 4 | 12 | 20 | 25 | 20 | 12 | 4 | 2 | 0 |

**A.** Not consistent: 9 or more heads happened in only 2% of the trials.

**B.** Not consistent: a fair coin must give exactly 5 heads.

**C.** Consistent: a fair coin can land heads 9 times.

**D.** Consistent: 9 or more heads happened in 2% of the trials.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S23-F2-Q2**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q2/5**

## True or false?

A result that happens in 25% of the simulated trials is consistent with the model under the 5% rule.

**A.** True

**B.** False

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q3/5

In the table of 100 simulated sets, in what proportion were there 7 or more heads? Give a decimal.

| Heads in 10 flips | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Number of simulated trials | 0 | 1 | 4 | 12 | 20 | 25 | 20 | 12 | 4 | 2 | 0 |

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q4/5

## True or false?

If a model is consistent with the data, the simulation proves that the model is true.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q5/5

# Decide whether a model is consistent with data

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-IC.A.2** |
| **BACKWARD 1** | Read the results of a simulation | **HSS-IC.A.2** |
| **BACKWARD 2** | Compare observed frequencies with a probability model | **7.SP.C.7.b** |
| **FORWARD 1** | Decide using results extreme in either direction | **HSS-IC.A.2** |
| **FORWARD 2** | Judge a treatment difference by re-randomization | **HSS-IC.B.5** |

SET 24 • HSS-IC.A.2

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

The table shows 100 simulated sets of 10 fair-coin flips. In what proportion of the sets were there 8 or more heads? Give a decimal.

| Heads in 10 flips | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Number of simulated trials | 0 | 1 | 4 | 12 | 20 | 25 | 20 | 12 | 4 | 2 | 0 |

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q1/5

A coin claimed to be fair gave 8 heads in 10 flips. Use the table of 100 simulated sets. Decision rule: if results at least as extreme as the observed result happen in fewer than 5% of the simulated trials, the model is not consistent with the data; otherwise it is consistent. Which decision is correct?

| Heads in 10 flips | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Number of simulated trials | 0 | 1 | 4 | 12 | 20 | 25 | 20 | 12 | 4 | 2 | 0 |

**A.** Not consistent: a fair coin should give 5 heads every time.

**B.** Consistent: 8 heads happened in exactly 4% of the sets.

**C.** Consistent: 8 or more heads happened in 6% of the simulated sets.

**D.** Not consistent: 8 or more heads happened in only 6% of the sets.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S24-M-Q2**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q2/5**

A die is claimed to be fair. In 60 rolls it showed 18 sixes. In 200 simulated sets of 60 fair rolls, 18 or more sixes happened 4 times. What proportion of the simulated sets is that? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q3/5

A die claimed to be fair showed 18 sixes in 60 rolls. In 200 simulated sets of 60 fair rolls, 18 or more sixes happened 4 times. Decision rule: if results at least as extreme as the observed result happen in fewer than 5% of the simulated trials, the model is not consistent with the data; otherwise it is consistent. Which decision is correct?

**A.** Consistent: 18 sixes is possible with a fair die.

**B.** Not consistent: so many sixes happened in only 2% of simulated sets.

**C.** Not consistent: a fair die must give exactly 10 sixes.

**D.** Consistent: 4 simulated sets had 18 or more sixes.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S24-M-Q4**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q4/5**

A spinner is claimed to land on red 25% of the time. In 40 spins it landed on red 6 times. In 1,000 simulated sets of 40 spins, 6 or fewer reds happened 120 times. Decision rule: if results at least as extreme as the observed result happen in fewer than 5% of the simulated trials, the model is not consistent with the data; otherwise it is consistent. Which decision is correct?

**A.** Consistent: the spinner landed on red exactly 25% of the time.

**B.** Consistent: 6 or fewer reds happened in 12% of the sets.

**C.** Not consistent: only 120 of the sets had 6 or fewer.

**D.** Not consistent: 6 is fewer than the expected 10 reds.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S24-M-Q5**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • MAIN • Q5/5**

SET 24 • HSS-IC.A.2

# BACKWARD 1

Read the results of a simulation • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

In the table of 50 simulated families of 4 children, how many families had exactly 2 girls?

| Girls in 4 births | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Number of simulated trials | 3 | 12 | 19 | 13 | 3 |

Families: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q1/5

In the table of 50 simulated families, what proportion had 1 or fewer girls? Give a decimal.

| Girls in 4 births | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Number of simulated trials | 3 | 12 | 19 | 13 | 3 |

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q2/5

## True or false?

The table of 100 simulated coin sets shows 25 sets with exactly 5 heads.

| Heads in 10 flips | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Number of simulated trials | 0 | 1 | 4 | 12 | 20 | 25 | 20 | 12 | 4 | 2 | 0 |

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q3/5

In the table of 100 simulated coin sets, how many sets had 3 or fewer heads?

| Heads in 10 flips | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Number of simulated trials | 0 | 1 | 4 | 12 | 20 | 25 | 20 | 12 | 4 | 2 | 0 |

**A.** 12

**B.** 17

**C.** 5

**D.** 29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q4/5

## True or false?

In the table of 100 simulated coin sets, more sets had 6 heads than 4 heads.

| Heads in 10 flips | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Number of simulated trials | 0 | 1 | 4 | 12 | 20 | 25 | 20 | 12 | 4 | 2 | 0 |

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • HSS-IC.A.2

# BACKWARD 2

Compare observed frequencies with a probability model • Grade 7 • Statistics & Probability • 7.SP.C.7.b

## True or false?

A coin that lands heads 520 times in 1,000 flips is reasonably consistent with a fair coin.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

**Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q1/5**

A fair die is rolled 600 times. About how many sixes does the fair model predict?

**A.** 160

**B.** 60

**C.** 100

**D.** 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

**Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q2/5**

How many heads does a fair-coin model predict on average in 10 flips?

Heads: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

**Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q3/5**

## True or false?

A fair die rolled 60 times must show each face exactly 10 times.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

**Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q4/5**

A spinner is claimed to land on red 50% of the time. Which result in 100 spins most suggests the claim is wrong?

**A.** 50 red

**B.** 47 red

**C.** 53 red

**D.** 82 red

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

**Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 24 • BACKWARD 2 • Q5/5**

SET 24 • HSS-IC.A.2

# FORWARD 1

Decide using results extreme in either direction • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

In the table of 100 simulated sets of 10 fair flips, what proportion had 9 or more heads or 1 or fewer heads? Give a decimal.

| Heads in 10 flips | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Number of simulated trials | 0 | 1 | 4 | 12 | 20 | 25 | 20 | 12 | 4 | 2 | 0 |

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q1/5

A coin gave 1 head in 10 flips. Count results at least as far from 5 heads in either direction as 1 head (9 or more heads, or 1 or fewer), using the table. Decision rule: if results at least as extreme as the observed result happen in fewer than 5% of the simulated trials, the model is not consistent with the data; otherwise it is consistent. Which decision is correct?

| Heads in 10 flips | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Number of simulated trials | 0 | 1 | 4 | 12 | 20 | 25 | 20 | 12 | 4 | 2 | 0 |

**A.** Not consistent: a fair coin cannot give 1 head.

**B.** Not consistent: such results happened in 3% of the sets.

**C.** Consistent: 1 head happened in exactly 1 set.

**D.** Consistent: such results happened in 3% of the sets.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q2/5

## True or false?

Under the 5% rule, the smaller the proportion of simulated results as extreme as the data, the stronger the evidence against the model.

**A.** True

**B.** False

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q3/5

A coin gave 7 heads in 10 flips. In the table, what proportion of sets are at least as far from 5 heads in either direction (7 or more, or 3 or fewer)? Give a decimal.

| Heads in 10 flips | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Number of simulated trials | 0 | 1 | 4 | 12 | 20 | 25 | 20 | 12 | 4 | 2 | 0 |

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q4/5

## True or false?

If a model is judged not consistent under the 5% rule, there is no chance that the model is true.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 24 • FORWARD 1 • Q5/5

SET 24 • HSS-IC.A.2

# FORWARD 2

Judge a treatment difference by re-randomization • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

In 100 re-randomizations of an experiment's results, a difference in means of 6 or more appeared 3 times. What proportion is that? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q1/5

The observed difference was 6. In 100 re-randomizations, differences of 6 or more appeared 3 times. Decision rule: if results at least as extreme as the observed result happen in fewer than 5% of the simulated trials, the "no effect" model is not consistent with the data; otherwise it is consistent. Which decision is correct?

**A.** The difference is significant: every difference is significant.

**B.** The difference is not significant: it appeared 3 times.

**C.** The difference is significant: chance alone gave it 3% of the time.

**D.** The difference is not significant: re-randomizing changes the data.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S24-F2-Q2**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q2/5**

## True or false?

A re-randomization simulation assumes that the treatment has no effect.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q3/5

In an experiment, the treatment group's mean was 24 and the control group's mean was 19. What is the difference (treatment minus control)?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q4/5

## True or false?

If the observed difference appeared in 40% of re-randomizations, the treatment effect is significant under the 5% rule.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 24 • FORWARD 2 • Q5/5

SET 25 • HSS-IC.B.3 • MAKING INFERENCES AND JUSTIFYING CONCLUSIONS

# Sample surveys, experiments and observational studies

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-IC.B.3** |
| **BACKWARD 1** | Identify the population, sample, parameter and statistic | **HSS-IC.A.1** |
| **BACKWARD 2** | Separate association from causation | **HSS-ID.C.9** |
| **FORWARD 1** | Identify the parts of an experiment | **HSS-IC.B.3** |
| **FORWARD 2** | Choose a study design for a question | **HSS-IC.B.3** |

SET 25 • HSS-IC.B.3

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

A researcher randomly assigns 60 volunteers to a new sleep routine or their usual routine and compares their test scores. What kind of study is this?

**A.** census

**B.** sample survey

**C.** experiment

**D.** observational study

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S25-M-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q1/5**

A polling company asks a random sample of 1,000 adults which candidate they prefer. What kind of study is this?

**A.** experiment

**B.** simulation

**C.** sample survey

**D.** observational study comparing groups

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S25-M-Q2**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q2/5**

Researchers compare lung health of people who already smoke with people who do not. What kind of study is this?

**A.** experiment

**B.** census

**C.** observational study

**D.** sample survey of opinions

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S25-M-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q3/5**

Which kind of study is designed mainly to estimate a characteristic of a population?

**A.** a sample survey

**B.** an observational study of two groups

**C.** a simulation of a model

**D.** an experiment

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S25-M-Q4**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q4/5**

Why can an experiment support a cause-and-effect conclusion when an observational study cannot?

**A.** Observational studies never collect any numerical data.

**B.** Experiments do not need any comparison or control group.

**C.** Experiments always use more subjects than observational studies.

**D.** The researcher imposes treatments, so other variables can be controlled.

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • MAIN • Q5/5

SET 25 • HSS-IC.B.3

# BACKWARD 1

Identify the population, sample, parameter and statistic • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

300 randomly chosen city residents are asked how often they ride the bus. What is the population?

**A.** all city residents

**B.** the 300 residents

**C.** the city's buses

**D.** bus riders only

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S25-B1-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q1/5**

300 randomly chosen city residents are asked how often they ride the bus. What is the sample size?

Size: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q2/5

## True or false?

A statistic describes a sample.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q3/5

## True or false?

A census collects data from only part of the population.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q4/5

Which is a parameter?

**A.** the mean height of 50 sampled students

**B.** the number 50

**C.** the proportion in the sample

**D.** the mean height of all students in a state

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • HSS-IC.B.3

# BACKWARD 2

Separate association from causation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

## True or false?

An association seen without assigning treatments may be caused by a lurking variable.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q1/5

Towns with more ice-cream shops have more drownings. What is a likely lurking variable?

**A.** hot weather

**B.** the number of shops

**C.** the price of ice cream

**D.** the town's name

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q2/5

## True or false?

A correlation always implies causation when the study is large.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q3/5

Which conclusion requires an experiment?

**A.** The mean recovery time was 6 days.

**B.** Recovery time and vitamin use are associated.

**C.** Half of the patients took the vitamin.

**D.** A vitamin causes faster recovery.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q4/5

## True or false?

Random assignment helps rule out lurking variables.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • HSS-IC.B.3

# FORWARD 1

Identify the parts of an experiment • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

An experiment compares two fertilizers on 40 plots and measures crop yield. What is the response variable?

**A.** the 40 plots

**B.** the two fertilizers

**C.** crop yield

**D.** the farmer

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S25-F1-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q1/5**

An experiment tests 3 doses of a medicine and a placebo. How many treatments are there?

Treatments: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q2/5

## True or false?

A control group receives the standard treatment or a placebo so the new treatment can be compared.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q3/5

Why do experiments on medicines often give some subjects a placebo?

**A.** to increase the number of different treatments

**B.** to make the experiment cheaper to run

**C.** to account for the effect of believing one is being treated

**D.** to make sure the results come out positive

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S25-F1-Q4**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q4/5**

## True or false?

In a double-blind experiment, only the subjects do not know which treatment they receive.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 1 • Q5/5

SET 25 • HSS-IC.B.3

# FORWARD 2

Choose a study design for a question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Question: does a new math app raise test scores? Which design fits best?

**A.** a census of all students

**B.** a sample survey

**C.** an observational study of app users

**D.** a randomized experiment

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S25-F2-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q1/5**

Question: what percent of a town's residents own a bicycle? Which design fits best?

**A.** a sample survey

**B.** a randomized experiment

**C.** a simulation

**D.** an observational study of cyclists

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q2/5

Question: do people who live near highways have more asthma? Assigning where people live is not possible. Which design fits best?

**A.** an observational study

**B.** a sample survey of opinions

**C.** a simulation

**D.** a randomized experiment

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S25-F2-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q3/5**

## True or false?

It is ethical to randomly assign people to smoke in an experiment on lung disease.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q4/5

## True or false?

An observational study can be the best choice when assigning a treatment would be unethical.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 25 • FORWARD 2 • Q5/5

# How randomization relates to each kind of study

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-IC.B.3** |
| **BACKWARD 1** | Classify a study | **HSS-IC.B.3** |
| **BACKWARD 2** | Select a simple random sample | **7.SP.A.1** |
| **FORWARD 1** | Decide the scope of conclusions | **HSS-IC.B.3** |
| **FORWARD 2** | Evaluate a report's conclusion from its design | **HSS-IC.B.6** |

SET 26 • HSS-IC.B.3

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

A class has 40 students numbered 01 to 40. Read line 1 of the table as two-digit numbers, skipping 00, numbers above 40 and repeats. Which three students are chosen first?

| Line | Digits |
|---|---|
| 1 | 15881 39986 98797 30911 |
| 2 | 41707 53645 56815 18844 |
| 3 | 72940 55761 69871 68976 |
| 4 | 84697 88093 20819 67273 |
| 5 | 17784 86488 76634 40029 |
| 6 | 78753 47787 68045 71784 |
| 7 | 72671 47513 62701 07981 |
| 8 | 93319 08452 05997 28379 |

First: _____

Second: _____

Third: _____

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S26-M-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q1/5**

What is the main purpose of selecting a sample at random in a survey?

**A.** so the sample is as small as possible

**B.** so that every answer comes out the same

**C.** so the results can be generalized to the population

**D.** so a treatment can be shown to cause a change

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S26-M-Q2**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q2/5**

What is the main purpose of assigning subjects to treatments at random in an experiment?

**A.** to guarantee that the treatment will work

**B.** to make groups alike so a difference can be due to treatment

**C.** to avoid needing any control or comparison group

**D.** to let the results describe everyone in the country

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S26-M-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q3/5**

A study randomly assigned 80 volunteers to two diets and found a significant difference in weight loss. The volunteers were not a random sample. What can be concluded?

**A.** The diet causes the same difference for everyone in the country.

**B.** Nothing, because the subjects were volunteers.

**C.** The diet caused the difference for volunteers like these.

**D.** Only that the two diets are associated with weight.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S26-M-Q4**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q4/5**

Plots in a field are numbered 01 to 60. Read line 3 of the table as two-digit numbers, skipping 00, numbers above 60 and repeats. Which three plots are assigned fertilizer A first?

| Line | Digits |
|---|---|
| 1 | 15881 39986 98797 30911 |
| 2 | 41707 53645 56815 18844 |
| 3 | 72940 55761 69871 68976 |
| 4 | 84697 88093 20819 67273 |
| 5 | 17784 86488 76634 40029 |
| 6 | 78753 47787 68045 71784 |
| 7 | 72671 47513 62701 07981 |
| 8 | 93319 08452 05997 28379 |

First: _____

Second: _____

Third: _____

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S26-M-Q5**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • MAIN • Q5/5**

SET 26 • HSS-IC.B.3

# BACKWARD 1

Classify a study • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

A teacher asks a random sample of 50 students how many hours they sleep. What kind of study is this?

**A.** observational study comparing treatments

**B.** sample survey

**C.** census

**D.** experiment

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S26-B1-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q1/5**

## True or false?

In an experiment, the researcher decides which subjects get which treatment.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q2/5

Doctors compare the heart health of people who already jog with people who do not. What kind of study is this?

**A.** sample survey

**B.** simulation

**C.** observational study

**D.** experiment

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S26-B1-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q3/5**

## True or false?

A sample survey assigns treatments to people.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q4/5

Which study imposes a treatment?

**A.** asking a random sample of voters for their opinion

**B.** giving half of the volunteers a new drug, chosen by lottery

**C.** counting the cars that pass along a busy road

**D.** recording the diets that people already choose to eat

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S26-B1-Q5**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • BACKWARD 1 • Q5/5**

SET 26 • HSS-IC.B.3

# BACKWARD 2

Select a simple random sample • Grade 7 • Statistics & Probability • 7.SP.A.1

## True or false?

In a simple random sample, every group of the same size has the same chance of being chosen.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

**Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q1/5**

Which method gives a simple random sample of 5 students from a class of 30?

**A.** ask for 5 volunteers to raise their hands

**B.** choose the 5 tallest students in the class

**C.** choose the first 5 students to arrive

**D.** draw 5 names from a hat holding all 30 names

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

**Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q2/5**

Students are numbered 01 to 30. Reading the two-digit numbers 45 07 29 07 33 18, skipping numbers above 30 and repeats, which three students are chosen?

First: _____

Second: _____

Third: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

**Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q3/5**

## True or false?

Choosing every student whose name starts with A is a random sample.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

**Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q4/5**

Why are repeated numbers skipped when selecting a sample from a random-digit table?

**A.** skipping them saves time in the selection

**B.** repeated digits are always printing errors

**C.** repeated numbers are too large to use

**D.** a person cannot be chosen twice

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

**Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 26 • BACKWARD 2 • Q5/5**

SET 26 • HSS-IC.B.3

# FORWARD 1

Decide the scope of conclusions • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

A study used a random sample from a population but no random assignment. What can be concluded?

**A.** Nothing at all can be concluded from such a study.

**B.** Associations generalize to the population, but not cause.

**C.** Cause can be concluded, but only for the sample.

**D.** Cause can be concluded for the whole population.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S26-F1-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q1/5**

A study used both a random sample from a population and random assignment to treatments. What can be concluded?

**A.** Cause, but only for the sample.

**B.** Cause, generalized to the population.

**C.** Association only.

**D.** Nothing.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S26-F1-Q2**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q2/5**

## True or false?

Random assignment allows cause-and-effect conclusions.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q3/5

## True or false?

Random sampling allows cause-and-effect conclusions even without random assignment.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q4/5

A study used neither random sampling nor random assignment. What can be concluded?

**A.** Cause for the group studied.

**B.** Cause for the population.

**C.** Associations for the population.

**D.** Associations in the group studied only.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S26-F1-Q5**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 26 • FORWARD 1 • Q5/5**

SET 26 • HSS-IC.B.3

# FORWARD 2

Evaluate a report's conclusion from its design • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A news report says, "People who drink green tea live longer, so drink tea to live longer." The data came from an observational study. What is the main flaw?

**A.** The sample was too large to be trusted.

**B.** It claims cause from an observational study.

**C.** Green tea intake cannot be measured at all.

**D.** Observational studies cannot find associations.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S26-F2-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q1/5**

## True or false?

A report based on a voluntary online poll can be generalized to all adults.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q2/5

A randomized experiment on 200 college athletes found a new drink improved sprint times. To whom does the result apply most directly?

**A.** all adults

**B.** college athletes like those studied

**C.** all children

**D.** nobody

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S26-F2-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q3/5**

## True or false?

A report should state how its sample was chosen and whether treatments were randomly assigned.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q4/5

A survey of 50 gym members found 90% exercise daily, reported as "90% of Americans exercise daily." What is wrong?

**A.** Surveys can never be used to estimate percents.

**B.** The sample of 50 people is far too large.

**C.** The percent of 90 was computed incorrectly.

**D.** Gym members are not representative of all Americans.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S26-F2-Q5**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 26 • FORWARD 2 • Q5/5**

# Estimate a population proportion with a margin of error

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-IC.B.4** |
| **BACKWARD 1** | Compute a sample proportion and name it | **HSS-IC.A.1** |
| **BACKWARD 2** | Read a standard deviation as typical variation | **HSS-ID.A.2** |
| **FORWARD 1** | See how sample size changes the margin of error | **HSS-IC.B.4** |
| **FORWARD 2** | Read margins of error in poll reports | **HSS-IC.B.6** |

SET 27 • HSS-IC.B.4

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

In a random sample of 400 adults, 248 favor a new park. What is the sample proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q1/5

500 random samples of size 400 were simulated from a population with proportion 0.62. The simulated sample proportions had standard deviation 0.024. Use margin of error = 2 × (standard deviation of the simulated values). What is the margin of error?

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q2/5

A sample proportion is 0.62 and the margin of error is 0.048. Give the interval of plausible values for the population proportion.

From: _____

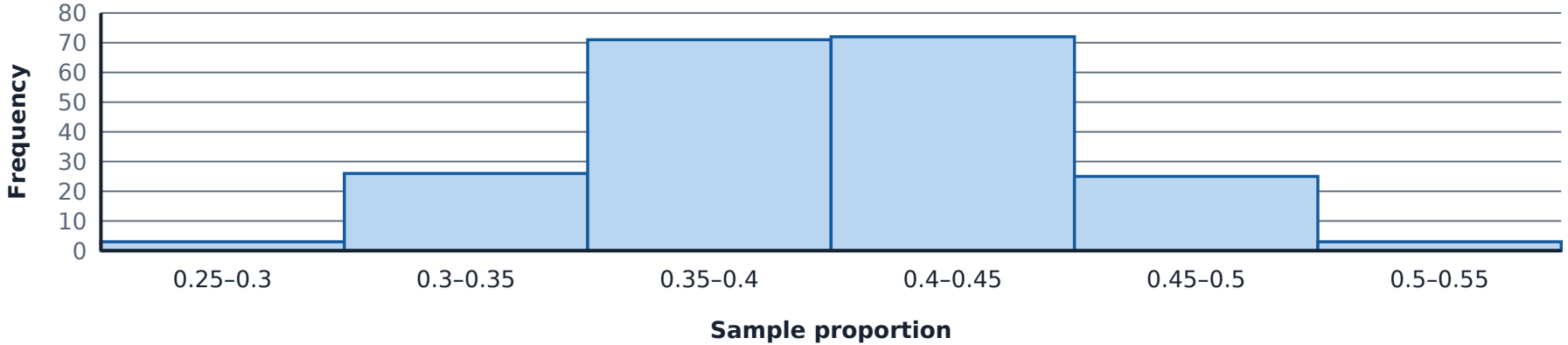
To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q3/5

The histogram shows 200 simulated sample proportions. How many of them are between 0.35 and 0.45 (bins 0.35-0.40 and 0.40-0.45)?




Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q4/5

A poll of a random sample reports that 54% of voters support a measure, with a margin of error of 3 percentage points. Which conclusion is best supported?

**A.** Exactly 54% of all voters support the measure.

**B.** A majority plausibly supports it: 51% to 57% are all above 50%.

**C.** Support could plausibly be anywhere from 0% to 100%.

**D.** The poll shows that support is below 50%.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S27-M-Q5**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • MAIN • Q5/5**

SET 27 • HSS-IC.B.4

# BACKWARD 1

Compute a sample proportion and name it • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

In a random sample of 250 students, 90 walk to school. What is the sample proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q1/5

## True or false?

A sample proportion is a statistic.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q2/5

The proportion of all voters in a state who favor a law is a

**A.** statistic

**B.** margin of error

**C.** parameter

**D.** sample

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S27-B1-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q3/5**

In a random sample of 80 phones, 6 were defective. What is the sample proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q4/5

## True or false?

Different random samples always give the same sample proportion.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • HSS-IC.B.4

# BACKWARD 2

Read a standard deviation as typical variation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

## True or false?

A standard deviation measures how far values typically are from their mean.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q1/5

Simulated proportions from samples of size 100 have SD 0.05; from samples of size 400, SD 0.025. Which vary less?

**A.** the proportions from samples of size 400

**B.** cannot tell

**C.** the proportions from samples of size 100

**D.** they vary the same

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q2/5

Compute $2 \times 0.031$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q3/5

## True or false?

A larger standard deviation means the values are more tightly clustered.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q4/5

About what percent of values in a roughly normal distribution lie within 2 SDs of the mean?

**A.** 95%

**B.** 68%

**C.** 50%

**D.** 99.7%

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 27 • BACKWARD 2 • Q5/5

SET 27 • HSS-IC.B.4

# FORWARD 1

See how sample size changes the margin of error • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

For a sample proportion near 0.5, the margin of error is about $2 \cdot \sqrt{(0.5 \cdot 0.5/n)}$. Find it for $n = 400$ and $n = 100$.

$n = 400$: _____

$n = 100$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q1/5

## True or false?

Making the sample 4 times as large cuts the margin of error about in half.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q2/5

A pollster wants a smaller margin of error. What should change?

**A.** take a larger random sample

**B.** ask only volunteers

**C.** round the results

**D.** take a smaller random sample

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S27-F1-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q3/5**

Simulated proportions from samples of size 900 have SD 0.016. Use margin of error = 2 × (standard deviation of the simulated values). What is the margin of error?

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q4/5

## True or false?

A larger sample removes the bias of a poorly worded question.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 27 • FORWARD 1 • Q5/5

SET 27 • HSS-IC.B.4

# FORWARD 2

Read margins of error in poll reports • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

Candidate A has 48% and candidate B has 45%, each with a margin of error of 4 percentage points. What is the best conclusion?

**A.** Both candidates have under 50%, so both lose.

**B.** B is certainly behind A by at least 3 points.

**C.** Too close to call: the two intervals overlap a lot.

**D.** A is certainly ahead of B by exactly 3 points.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S27-F2-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q1/5**

A report gives 62% with a margin of error of 5 percentage points. Give the interval in percents.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q2/5

## True or false?

A margin of error describes only random sampling variation, not bias from how the sample was chosen.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q3/5

A website poll of 10,000 visitors reports 70% ± 1%. Why is the margin of error misleading?

**A.** The sample is too small for a margin of error.

**B.** A margin of error must be at least 5%.

**C.** Percents cannot have a margin of error.

**D.** The visitors chose themselves, so the sample is not random.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S27-F2-Q4**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q4/5**

## True or false?

If two candidates' intervals overlap, the poll proves that they are tied.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 27 • FORWARD 2 • Q5/5

# Estimate a population mean with a margin of error

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-IC.B.4** |
| **BACKWARD 1** | Use a margin of error | **HSS-IC.B.4** |
| **BACKWARD 2** | Compute a mean | **6.SP.B.5.c** |
| **FORWARD 1** | Compare variability of sample means | **HSS-IC.B.4** |
| **FORWARD 2** | Compare two treatments' means | **HSS-IC.B.5** |

SET 28 • HSS-IC.B.4

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

A random sample of 8 students reported these hours of sleep: 7, 6.5, 8, 7.5, 6, 9, 7, 8. What is the sample mean?

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q1/5

Simulated random samples of 8 students gave sample means with standard deviation 0.15 hour. Use margin of error = 2 × (standard deviation of the simulated values). What is the margin of error?

Margin of error (hours): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q2/5

A sample mean is 7.4 hours with a margin of error of 0.3 hour. Give the interval of plausible values for the population mean.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q3/5

A study reports a mean of 7.4 hours of sleep with a margin of error of 0.3 hour, from a random sample. Which interpretation is correct?

**A.** Population means from 7.1 to 7.7 hours are plausible.

**B.** The population mean is exactly 7.4 hours.

**C.** Half of all students sleep 7.4 hours.

**D.** Every student sleeps between 7.1 and 7.7 hours.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S28-M-Q4**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q4/5**

In 1,000 simulated samples, the middle 95% of sample means ran from 41.2 to 44.8. Give the center of this interval and the margin of error (half its width).

Center: _____

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • MAIN • Q5/5

SET 28 • HSS-IC.B.4

# BACKWARD 1

Use a margin of error • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

An estimate is 0.40 with a margin of error of 0.06. Give the interval.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q1/5

## True or false?

Margin of error = 2 × 0.05 = 0.1 when the simulated values have SD 0.05.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q2/5

Which estimate has the widest interval?

**A.** $20 \pm 4$

**B.** $20 \pm 1$

**C.** $20 \pm 2$

**D.** $20 \pm 0.5$

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S28-B1-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q3/5**

An interval runs from 12 to 18. What is the margin of error?

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q4/5

## True or false?

An interval from 0.30 to 0.40 has center 0.30.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • HSS-IC.B.4

# BACKWARD 2

Compute a mean • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean of 4, 6, 8, 10.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q1/5**

Find the mean of 6.5, 7, 8.5, rounded to three decimal places.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q2/5**

**True or false?**

The mean of 2, 4, 9 is 5.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q3/5**

What is the mean of 10, 12, 14, 20?

**A.** 56

**B.** 13

**C.** 14

**D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q4/5**

**True or false?**

The mean of a data set must be one of its values.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 28 • BACKWARD 2 • Q5/5**

SET 28 • HSS-IC.B.4

# FORWARD 1

Compare variability of sample means • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

Sample means from samples of size 9 have SD 2; from samples of size 36, SD 1. Which statement is correct?

**A.** Means of larger samples vary more, so the margin is larger.

**B.** Both sample sizes give the same margin of error.

**C.** Sample size has no effect at all on sample means.

**D.** Means of larger samples vary less, so the margin of error is smaller.

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q1/5

Simulated sample means from samples of size 36 have SD 1. Use margin of error = 2 × (standard deviation of the simulated values). What is the margin of error?

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q2/5

## True or false?

Individual values vary more than means of samples of those values.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q3/5

An estimate of 52 has a margin of error of 2. Is 55 a plausible population mean? Enter yes or no.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q4/5

## True or false?

A margin of error for a mean can be computed without any random sample.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 28 • FORWARD 1 • Q5/5

SET 28 • HSS-IC.B.4

# FORWARD 2

Compare two treatments' means • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

Plant food A: mean growth 13 cm; plant food B: mean growth 10 cm. What is the difference (A minus B)?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q1/5

An experiment's difference in means could be due to the treatments or to chance. What tool checks chance?

**A.** a census

**B.** a larger margin of error

**C.** a re-randomization simulation

**D.** a scatter plot

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S28-F2-Q2**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q2/5**

## True or false?

A difference in means from a randomized experiment is always caused by the treatment.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q3/5

In 200 re-randomizations, a difference of 3 or more occurred 4 times. What proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q4/5

## True or false?

A difference that chance alone produces in only 2% of re-randomizations is significant under a 5% rule.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 28 • FORWARD 2 • Q5/5

# Compare two treatments with a randomization simulation

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-IC.B.5** |
| **BACKWARD 1** | Decide whether a model is consistent with data | **HSS-IC.A.2** |
| **BACKWARD 2** | Explain random assignment | **HSS-IC.B.3** |
| **BACKWARD 3** | Compute means of small groups | **6.SP.B.5.c** |
| **FORWARD 1** | Use differences in either direction | **HSS-IC.B.5** |
| **FORWARD 2** | Evaluate a report of an experiment | **HSS-IC.B.6** |

SET 29 • HSS-IC.B.5

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

Ten plants were randomly assigned to plant food A or B. Find the mean growth for each group and the difference (A minus B).

| Group | Growth (cm) | Mean |
|---|---|---|
| A | 14, 11, 13, 12, 15 | 13 |
| B | 10, 12, 9, 11, 8 | 10 |

Mean A: _____

Mean B: _____

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q1/5

The 10 results were re-randomized into two groups of 5 a total of 200 times. A difference of 3 or more occurred 4 times. What proportion of the re-randomizations is that? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q2/5

In 200 re-randomizations, a difference of 3 or more (the observed difference) occurred 4 times. Decision rule: if results at least as extreme as the observed result happen in fewer than 5% of the simulated trials, the "no difference" model is not consistent with the data; otherwise it is consistent. Which decision is correct?

**A.** The difference is significant: any difference is significant.

**B.** The difference is not significant: it happened 4 times by chance.

**C.** The difference is significant: chance gave it only 2% of the time.

**D.** The difference is not significant: the groups had only 5 plants.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S29-M-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q3/5**

What does one re-randomization do?

**A.** Repeats the whole experiment with many more plants.

**B.** Swaps the means of the two original groups of plants.

**C.** Reshuffles the 10 results into groups of 5, ignoring food.

**D.** Grows 10 new plants, again using the two plant foods.

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q4/5

Another experiment found a difference of 1.2. In 200 re-randomizations, differences of 1.2 or more occurred 62 times. Decision rule: if results at least as extreme as the observed result happen in fewer than 5% of the simulated trials, the "no difference" model is not consistent with the data; otherwise it is consistent. Which decision is correct?

**A.** Not significant: re-randomizations can never be significant.

**B.** Not significant: chance gave such a difference 31% of the time.

**C.** Significant: 62 is a large number of re-randomizations.

**D.** Significant: the difference 1.2 is greater than 0.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S29-M-Q5**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • MAIN • Q5/5**

SET 29 • HSS-IC.B.5

# BACKWARD 1

Decide whether a model is consistent with data • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

## True or false?

A result that would happen rarely under a model is evidence against the model.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q1/5

A result at least as extreme happened in 3 of 150 simulated trials. What proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q2/5

Under the 5% rule, a result as extreme as the data occurred in 12% of trials. The decision is

**A.** the model is proven

**B.** not consistent with the model

**C.** no decision

**D.** consistent with the model

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S29-B1-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q3/5**

## True or false?

Under the 5% rule, 4% of trials means the model is consistent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q4/5

Which proportion of simulated results gives the strongest evidence against a model?

**A.** 0.2

**B.** 0.04

**C.** 0.001

**D.** 0.5

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S29-B1-Q5**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 29 • BACKWARD 1 • Q5/5**

SET 29 • HSS-IC.B.5

# BACKWARD 2

Explain random assignment • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

## True or false?

Random assignment makes the treatment groups similar before treatment.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q1/5

Why were the plants assigned to foods at random?

**A.** so that more of the plants grow well

**B.** so that no plant needs to be measured

**C.** so growth differences can be credited to food

**D.** so the two group means come out equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q2/5

## True or false?

In an experiment, subjects choose which treatment they receive.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q3/5

Ten plants are numbered 1-10. Which assignment is random?

**A.** draw 5 numbers from a hat for food A

**B.** put the 5 tallest plants on food A

**C.** let the gardener choose

**D.** put plants 1-5 on food A

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S29-B2-Q4**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q4/5**

## True or false?

Random assignment is what allows an experiment to support a conclusion about cause.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • HSS-IC.B.5

# BACKWARD 3

Compute means of small groups • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean of 14, 11, 13, 12, 15.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q1/5**

Find the mean of 10, 12, 9, 11, 8.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q2/5**

**True or false?**

The difference of the means 13 and 10 is 3.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q3/5**

Find the mean of 2.5, 3, 3.5, 5.

**A.** 3

**B.** 3.5

**C.** 3.25

**D.** 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q4/5**

**True or false?**

The mean of 8, 9, 10, 11, 12 is 11.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 29 • BACKWARD 3 • Q5/5**

SET 29 • HSS-IC.B.5

# FORWARD 1

Use differences in either direction • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

In 200 re-randomizations, differences of 3 or more occurred 4 times and differences of −3 or less occurred 5 times. What proportion is at least 3 away from 0 in either direction? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q1/5

## True or false?

Under a 5% rule, a proportion of 0.045 means the difference is significant.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q2/5

With more plants per group, re-randomized differences tend to be

**A.** always positive

**B.** closer to 0

**C.** farther from 0

**D.** exactly 0

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S29-F1-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q3/5**

Treatment mean 31, control mean 28. What is the difference (treatment minus control)?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q4/5

## True or false?

A significant difference proves that the treatment helps every individual.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 29 • FORWARD 1 • Q5/5

SET 29 • HSS-IC.B.5

# FORWARD 2

Evaluate a report of an experiment • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A company reports: "Our drink improved scores in a study." The study gave the drink to 10 volunteers and compared them with 10 who chose not to drink it. What is the main flaw?

**A.** Drinks cannot be studied in an experiment.

**B.** Twenty subjects is too many for a study.

**C.** Scores of two groups cannot be compared.

**D.** Subjects chose their own group: no random assignment.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S29-F2-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q1/5**

## True or false?

A report should say whether a difference could plausibly be due to chance.

**A.** True

**B.** False

A report says a new method raised mean scores from 70 to 71, "a significant improvement." Re-randomization shows such differences 40% of the time. What is the best response?

**A.** The new method actually lowered scores.

**B.** Proven true: the study was an experiment.

**C.** Supported: 71 is higher than 70.

**D.** Not supported: chance often gives such differences.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S29-F2-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q3/5**

## True or false?

A randomized experiment on adults automatically applies to children.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q4/5

Which detail most strengthens a report's causal claim?

**A.** subjects were randomly assigned to treatments

**B.** the study had a long and technical name

**C.** the researchers felt confident in the result

**D.** the report used colorful, eye-catching graphs

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S29-F2-Q5**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 29 • FORWARD 2 • Q5/5**

# Evaluate the conclusions of reports based on data

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-IC.B.6** |
| **BACKWARD 1** | Know what each design can support | **HSS-IC.B.3** |
| **BACKWARD 2** | Recognize lurking variables | **HSS-ID.C.9** |
| **FORWARD 1** | Identify the type of bias in a report | **HSS-IC.B.6** |
| **FORWARD 2** | Weigh several flaws and decide what a report shows | **HSS-IC.B.6** |

SET 30 • HSS-IC.B.6

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A report says: "Teens who eat breakfast get better grades, so eating breakfast raises grades." The data came from a survey of 2,000 randomly chosen teens. What is the main problem?

**A.** A random sample of 2,000 teens is far too small for any survey.

**B.** Grades cannot be compared between different groups of teens.

**C.** Random samples cannot be used to describe a population of teens.

**D.** A survey shows association, not that breakfast causes better grades.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S30-M-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q1/5**

A magazine asked readers to mail in whether they liked their jobs; 80% of the 3,000 replies said no. The magazine reports that "80% of workers dislike their jobs." What is the main problem?

**A.** Three thousand replies is too few to report.

**B.** The percent 80% was computed incorrectly.

**C.** Readers chose to reply, so the sample is likely biased.

**D.** Job satisfaction cannot be surveyed.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S30-M-Q2**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q2/5**

A survey asked, "Don't you agree that the unfair new tax should be repealed?" What is the problem?

**A.** The wording pushes people toward one answer.

**B.** The question is too short to be answered.

**C.** The sample must have been too large.

**D.** Tax questions cannot appear in surveys.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S30-M-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q3/5**

A randomized experiment found that a reading program raised scores of 3rd graders in one district. Which conclusion is best supported?

**A.** The program will raise scores for students of every age everywhere.

**B.** The program caused higher scores for students like those in the study.

**C.** The program is associated with scores but did not cause any change.

**D.** Nothing can be concluded, because only one district was studied.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S30-M-Q4**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q4/5**

A report on 12 patients says a new treatment "works for everyone," because 9 of the 12 improved. Which criticism is most important?

**A.** Twelve patients is too many to study at once.

**B.** The fraction 9/12 should have been written 12/9.

**C.** Improvement cannot be measured in patients.

**D.** No comparison group, and 3 of the 12 did not improve.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S30-M-Q5**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • MAIN • Q5/5**

SET 30 • HSS-IC.B.6

# BACKWARD 1

Know what each design can support • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

## True or false?

Only a randomized experiment can support a cause-and-effect conclusion.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q1/5

Which design lets results be generalized to a population?

**A.** a group of volunteers

**B.** a convenience sample

**C.** a random sample from that population

**D.** an experiment on friends

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S30-B1-Q2**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q2/5**

## True or false?

An observational study can prove cause if it is large enough.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q3/5

A study used volunteers and random assignment. Its conclusions can include

**A.** nothing at all

**B.** cause, for subjects like the volunteers

**C.** cause, for the whole population

**D.** association only

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S30-B1-Q4**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q4/5**

## True or false?

Random sampling and random assignment serve different purposes.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • HSS-IC.B.6

# BACKWARD 2

Recognize lurking variables • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Cities with more police officers have more crimes. What is a likely lurking variable?

**A.** city population

**B.** the number of officers

**C.** crime reports

**D.** police uniforms

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q1/5

## True or false?

A lurking variable affects both variables in an association.

**A.** True

**B.** False

Students who take music lessons score higher in math. Which is a possible lurking variable?

**A.** the music teacher's name

**B.** the number of notes played

**C.** family income

**D.** math test length

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q3/5

## True or false?

If an association is strong, there can be no lurking variable.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q4/5

## True or false?

Random assignment protects an experiment from lurking variables.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • HSS-IC.B.6

# FORWARD 1

Identify the type of bias in a report • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A phone survey calls only landlines during work hours. Which bias is most likely?

**A.** undercoverage of people away or with only cell phones

**B.** no bias, because calls are made at random

**C.** response bias from the wording of questions

**D.** voluntary response bias from self-selection

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S30-F1-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q1/5**

Many people chosen for a survey refuse to answer. Which bias is this?

**A.** measurement error

**B.** nonresponse bias

**C.** wording bias

**D.** voluntary response bias

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S30-F1-Q2**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q2/5**

## True or false?

A larger sample removes bias caused by a poor sampling method.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q3/5

A survey of mall shoppers is used to describe all residents' shopping habits. Which bias is this?

**A.** a convenience sample that is not representative

**B.** random sampling error only, with no bias

**C.** wording bias from a leading question

**D.** nonresponse bias from refusals to answer

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S30-F1-Q4**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q4/5**

## True or false?

People may give socially acceptable answers instead of true ones when asked sensitive questions.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 1 • Q5/5

SET 30 • HSS-IC.B.6

# FORWARD 2

Weigh several flaws and decide what a report shows • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

An app maker reports: "Users who used our app for 30 days lost 8 pounds on average." The users were volunteers and there was no comparison group. Which conclusion is justified?

**A.** The app causes weight loss in everyone who uses it.

**B.** Only that these volunteers lost weight; the app's role is unclear.

**C.** Every single user of the app lost exactly 8 pounds.

**D.** The study has shown that the app has no effect on weight.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S30-F2-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q1/5**

## True or false?

A report that gives no information about how its sample was chosen should be treated with caution.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q2/5

Which change would most improve the app study?

**A.** ask the users whether they liked the app

**B.** randomly assign volunteers to use the app or not

**C.** extend the study from 30 days to 31 days

**D.** report the weight loss as a percent instead

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S30-F2-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q3/5**

## True or false?

A study funded by a company is always wrong about that company's product.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q4/5

Two reports disagree. Report 1 is a randomized experiment with 400 subjects; report 2 is a voluntary online poll of 5,000. Which deserves more weight for a causal claim?

**A.** report 1

**B.** equal weight

**C.** report 2

**D.** neither

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 30 • FORWARD 2 • Q5/5

SET 31 • HSS-IC.B.6 • MAKING INFERENCES AND JUSTIFYING CONCLUSIONS

# Evaluate the statistics and displays in reports

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-IC.B.6** |
| **BACKWARD 1** | Use margins of error to compare results | **HSS-IC.B.4** |
| **BACKWARD 2** | Compute percent change | **7.RP.A.3** |
| **FORWARD 1** | Spot misleading features of displays | **HSS-IC.B.6** |
| **FORWARD 2** | Reason about conditional rates in reports | **HSS-CP.A.5** |

SET 31 • HSS-IC.B.6

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A bar graph's vertical axis starts at 50. One bar shows 52 and another shows 55. How tall is the second bar compared with the first (ratio of drawn heights), and what is the ratio of the actual values? Round the second answer to two decimal places.

Ratio of drawn heights: _____

Ratio of values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q1/5

An ad says a risk "increased 80%," from 1 in 10,000 to 1.8 in 10,000. Which statement is correct?

**A.** Most people will now be affected by it.

**B.** The risk is now 80 out of every 100 people.

**C.** The absolute risk rose by only 0.8 in 10,000.

**D.** The risk went from 1% of people to 80%.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S31-M-Q2**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q2/5**

A poll shows candidate A at 51% and B at 49%, with a margin of error of 3 percentage points. A headline says "A leads!" What is the best evaluation?

**A.** Correct: A has a majority of all the voters.

**B.** Misleading: B is certainly ahead of A.

**C.** Correct: 51% is more than 49%, so A leads.

**D.** Misleading: the gap is well within the margin of error.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S31-M-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q3/5**

A report gives the mean income in a town as $95,000, but most households earn under $50,000 and a few earn millions. What is the best evaluation?

**A.** The report must have added incomes incorrectly.

**B.** The median would describe a typical household better.

**C.** The mean is the best summary for skewed data.

**D.** The range is the best measure of center.

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q4/5

A pictograph shows sales doubling by drawing a square icon twice as tall and twice as wide. How many times as large is the new icon's area?

Times as large: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • MAIN • Q5/5

SET 31 • HSS-IC.B.6

# BACKWARD 1

Use margins of error to compare results • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

An estimate is 51% with a margin of error of 3 points. Give the interval.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q1/5

## True or false?

If two intervals overlap a lot, the data do not clearly show which value is larger.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q2/5

Which pair of estimates (each ± 2 points) clearly differs?

**A.** 48% and 50%

**B.** 49% and 51%

**C.** 45% and 47%

**D.** 40% and 50%

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S31-B1-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q3/5**

## True or false?

A margin of error of 3 points means the estimate is off by exactly 3 points.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q4/5

An interval is 46% to 52%. What is the estimate at its center?

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • HSS-IC.B.6

# BACKWARD 2

Compute percent change • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A value rises from 1 to 1.8. What is the percent increase?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

**Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q1/5**

A price rises from $40 to $50. What is the percent increase?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

**Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q2/5**

## True or false?

A rise from 2 to 3 is a 50% increase.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

**Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q3/5**

A count falls from 200 to 150. What is the percent decrease?

**A.** 33%

**B.** 25%

**C.** 75%

**D.** 50%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

**Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q4/5**

**True or false?**

A rise from 50 to 55 is a 5% increase.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

**Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 31 • BACKWARD 2 • Q5/5**

SET 31 • HSS-IC.B.6

# FORWARD 1

Spot misleading features of displays • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A line graph of monthly sales has a vertical axis from 980 to 1,000. What effect does this have?

**A.** Large changes look very small.

**B.** It hides the months.

**C.** Small changes look very large.

**D.** It has no effect.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S31-F1-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q1/5**

## True or false?

A pie chart whose percents add to 130% is misleading or in error.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q2/5

A graph's vertical axis starts at 90. Bars show 92 and 98. What is the ratio of the drawn heights?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q3/5

## True or false?

Using a 3-D effect on a pie chart never changes how large slices appear.

**A.** True

**B.** False

Which display choice is most honest for comparing two counts of 52 and 55?

**A.** a bar graph with the axis starting at 51

**B.** a 3-D pie chart

**C.** icons of different widths

**D.** a bar graph with the axis starting at 0

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S31-F1-Q5**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 31 • FORWARD 1 • Q5/5**

SET 31 • HSS-IC.B.6

# FORWARD 2

Reason about conditional rates in reports • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

A report says "most car accidents happen within 5 miles of home, so driving near home is dangerous." What is the main flaw?

**A.** Accidents cannot be counted.

**B.** Most driving happens near home.

**C.** Five miles is too short.

**D.** Cars are safer far away.

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q1/5

## True or false?

Knowing that 90% of people in a hospital are sick does not tell you the chance that a sick person is in the hospital.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q2/5

A test catches 95% of people who have a rare disease. A report concludes that 95% of people who test positive have it. What is the error?

**A.** Rare diseases cannot be tested for at all.

**B.** It confuses P(positive given disease) with P(disease given positive).

**C.** A test must catch 100% of cases before it is reported.

**D.** The 95% should have been written as a fraction.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S31-F2-Q3**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q3/5**

## True or false?

A higher number of accidents involving red cars proves that red cars are more dangerous.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q4/5

To compare the safety of two car colors fairly, which statistic is needed?

**A.** the number of different car colors sold

**B.** the average price of the cars of each color

**C.** the total number of accidents for each color

**D.** accidents per car (or per mile) for each color

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S31-F2-Q5**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 31 • FORWARD 2 • Q5/5**

# Describe sample spaces and events

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-CP.A.1** |
| **BACKWARD 1** | List a sample space with a list, table or tree | **7.SP.C.8.b** |
| **BACKWARD 2** | Find probabilities in a uniform model | **7.SP.C.7.a** |
| **FORWARD 1** | Describe events in multi-stage sample spaces | **HSS-CP.A.1** |
| **FORWARD 2** | Combine events with "and", "or" and "not" | **HSS-CP.A.1** |

SET 32 • HSS-CP.A.1

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

A fair coin is flipped and a fair six-sided die is rolled. How many outcomes are in the sample space?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q1/5

Two fair coins are flipped; the sample space is HH, HT, TH, TT. How many outcomes are in the event "at least one head"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q2/5

A coin is flipped twice and the sequence of results is recorded. Which is the sample space?

**A.** {H, T}

**B.** {HH, HT, TH, TT}

**C.** {HH, HT, TT}

**D.** {HH, TT}

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S32-M-Q3**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q3/5**

Two fair dice are rolled. How many of the 36 equally likely outcomes are in the event "the sum is 7"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q4/5

A spinner has 8 equal sections numbered 1 to 8. Event A is "an even number greater than 3". How many outcomes are in A, and what is P(A)? Give a fraction in lowest terms.

Outcomes in A: _____

P(A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • MAIN • Q5/5

SET 32 • HSS-CP.A.1

# BACKWARD 1

List a sample space with a list, table or tree • Grade 7 • Statistics & Probability • 7.SP.C.8.b

A bag has a red and a blue marble; a coin is flipped. How many outcomes (color, coin) are there?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q1

**Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q1/5**

**True or false?**

A tree diagram for flipping a coin 3 times has 8 final branches.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q2

**Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q2/5**

A sandwich shop offers 3 breads and 4 fillings. How many bread-filling choices are there?

**A.** 81

**B.** 7

**C.** 12

**D.** 34

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q3

**Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q3/5**

**True or false?**

The sample space for rolling two dice has 12 outcomes.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

**Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q4/5**

A spinner with 3 equal colors is spun twice. How many outcomes are in the sample space?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

**Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 32 • BACKWARD 1 • Q5/5**

SET 32 • HSS-CP.A.1

# BACKWARD 2

Find probabilities in a uniform model • Grade 7 • Statistics & Probability • 7.SP.C.7.a

A fair die is rolled. What is P(a number greater than 4)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

**Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q1/5**

## True or false?

In a uniform model, every outcome has the same probability.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q2

**Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q2/5**

A letter is chosen at random from the word MATH. What is P(vowel)?

**A.** 1/2

**B.** 1/3

**C.** 1/4

**D.** 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

**Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q3/5**

A number is chosen at random from 1 to 20. What is P(multiple of 5)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

**Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q4/5**

**True or false?**

A spinner with sections of different sizes is a uniform model.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

**Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 32 • BACKWARD 2 • Q5/5**

SET 32 • HSS-CP.A.1

# FORWARD 1

Describe events in multi-stage sample spaces • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

Three fair coins are flipped. How many outcomes are in the sample space, and how many have exactly 2 heads?

Outcomes: _____

Exactly 2 heads: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q1/5

Three fair coins are flipped. What is P(exactly 2 heads)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q2/5

## True or false?

For two dice, the event "doubles" contains 6 outcomes.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q3/5

For two dice, which event contains exactly 4 outcomes?

**A.** the sum is 7

**B.** the sum is 12

**C.** the sum is 5

**D.** doubles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q4/5

Two dice are rolled. How many outcomes have a sum of 10 or more?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 1 • Q5/5

SET 32 • HSS-CP.A.1

# FORWARD 2

Combine events with "and", "or" and "not" • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

A die is rolled. A = "even", B = "greater than 3". How many outcomes are in "A and B"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q1/5

A die is rolled. A = "even", B = "greater than 3". How many outcomes are in "A or B"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q2/5

## True or false?

The event "not A" contains every outcome of the sample space that is not in A.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q3/5

A die is rolled. A = "even". Which set is "not A"?

**A.** {2, 4, 6}

**B.** {1, 3, 5}

**C.** {1, 2, 3}

**D.** { }

## True or false?

The event "A and B" always contains at least as many outcomes as "A or B".

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 32 • FORWARD 2 • Q5/5

# Unions, intersections and complements of events

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-CP.A.1** |
| **BACKWARD 1** | Count the outcomes in an event | **HSS-CP.A.1** |
| **BACKWARD 2** | Find a probability by counting | **7.SP.C.8.a** |
| **FORWARD 1** | Use a Venn diagram of counts | **HSS-CP.A.1** |
| **FORWARD 2** | Apply the Addition Rule | **HSS-CP.B.7** |

SET 33 • HSS-CP.A.1

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

A fair die is rolled. A = "even" and B = "greater than 3". How many outcomes are in A and B, and what is P(A and B)? Give a fraction in lowest terms.

Outcomes: _____

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q1/5

A fair die is rolled. A = "even" and B = "greater than 3". How many outcomes are in A or B, and what is P(A or B)? Give a fraction in lowest terms.

Outcomes: _____

P(A or B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q2/5

A fair die is rolled. A = "even". What is P(not A)? Give a fraction in lowest terms.

P(not A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q3/5

A fair die is rolled. A = "even" and B = "greater than 3". Which describes the event {1, 3}?

**A.** A or B

**B.** A and B

**C.** not A

**D.** not (A or B)

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q4/5

A card is drawn from a standard 52-card deck. H = "heart", F = "face card (jack, queen or king)". Find P(H and F) and P(H or F). Give a fraction in lowest terms.

P(H and F): _____

P(H or F): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • MAIN • Q5/5

SET 33 • HSS-CP.A.1

# BACKWARD 1

Count the outcomes in an event • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

A number from 1 to 10 is chosen. How many outcomes are in "multiple of 3"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q1/5

## True or false?

For one roll of a die, the event "less than 3" is {1, 2}.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q2/5

A letter is chosen from the word PROBABILITY. How many outcomes (letter positions) are in "the letter B"?

**A.** 11

**B.** 1

**C.** 2

**D.** 3

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S33-B1-Q3**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q3/5**

How many cards in a standard deck are face cards?

Cards: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q4/5

## True or false?

A standard deck has 4 aces and 13 hearts, so it has 17 cards that are aces or hearts.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • HSS-CP.A.1

# BACKWARD 2

Find a probability by counting • Grade 7 • Statistics & Probability • 7.SP.C.8.a

Two coins are flipped. What is P(two heads)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

**Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q1/5**

A die is rolled. What is P(1 or 2)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

**Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q2/5**

## True or false?

The probability of an event plus the probability of its complement is 1.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

**Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q3/5**

A card is drawn from a standard deck. What is P(king)?

**A.** 1/13

**B.** 1/4

**C.** 1/52

**D.** 4/13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

**Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q4/5**

## True or false?

If P(A) = 0.3, then P(not A) = 0.3.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

**Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 33 • BACKWARD 2 • Q5/5**

SET 33 • HSS-CP.A.1

# FORWARD 1

Use a Venn diagram of counts • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

Of 30 students, 18 play soccer, 12 play basketball and 5 play both. How many play soccer only, and how many play neither?

Soccer only: _____

Neither: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q1/5

Of 30 students, 18 play soccer, 12 play basketball and 5 play both. A student is chosen at random. What is P(soccer or basketball)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q2/5

## True or false?

In a Venn diagram, the overlap of two circles shows "A and B".

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q3/5

Of 30 students, 18 play soccer, 12 play basketball and 5 play both. What is P(not soccer)?

**A.** 3/5

**B.** 7/30

**C.** 1/6

**D.** 2/5

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S33-F1-Q4**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q4/5**

## True or false?

Adding 18 and 12 gives the number who play soccer or basketball.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 33 • FORWARD 1 • Q5/5

SET 33 • HSS-CP.A.1

# FORWARD 2

Apply the Addition Rule • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7

P(A) = 0.5, P(B) = 0.4 and P(A and B) = 0.2. Find P(A or B).

P(A or B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q1/5

A card is drawn. Find P(king or heart). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q2/5

## True or false?

If A and B cannot happen together, P(A or B) = P(A) + P(B).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q3/5

P(A) = 0.6, P(B) = 0.3, P(A or B) = 0.7. What is P(A and B)?

**A.** 0.9

**B.** 0.2

**C.** 0.1

**D.** 0.18

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S33-F2-Q4**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 33 • FORWARD 2 • Q4/5**

## True or false?

P(A or B) can be greater than 1 if A and B are both likely.

**A.** True

**B.** False

# Decide independence with the product rule

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-CP.A.2** |
| **BACKWARD 1** | Multiply probabilities | **5.NF.B.4** |
| **BACKWARD 2** | Find P(A and B) by counting | **HSS-CP.A.1** |
| **FORWARD 1** | Use independence for several events | **HSS-CP.A.2** |
| **FORWARD 2** | Interpret independence with conditional probability | **HSS-CP.A.3** |

SET 34 • HSS-CP.A.2

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

$P(A) = 0.4$, $P(B) = 0.5$ and $P(A \text{ and } B) = 0.2$. Find $P(A)\cdot P(B)$, and decide whether A and B are independent (enter yes or no).

$P(A)\cdot P(B)$: _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q1/5

P(A) = 0.3, P(B) = 0.6 and P(A and B) = 0.25. Find P(A)·P(B), and decide whether A and B are independent (enter yes or no).

P(A)·P(B): _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q2/5

A and B are independent with P(A) = 0.7 and P(B) = 0.2. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q3/5

A die is rolled. A = "even" and B = "1 or 2". Are A and B independent?

**A.** No: A and B share the outcome 2.

**B.** No: P(A and B) = 1/6, not 1/2.

**C.** Yes: P(A and B) = 1/6 = (1/2)(1/3).

**D.** Yes: A and B have no outcomes in common.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S34-M-Q4**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q4/5**

In the survey table, a person is chosen at random. M = "man" and Y = "said yes". Find P(M and Y), and decide whether M and Y are independent (enter yes or no).

| | Yes | No | Total |
|---|---|---|---|
| Men | 24 | 16 | 40 |
| Women | 36 | 24 | 60 |
| Total | 60 | 40 | 100 |

P(M and Y): _____

Independent (yes or no): _____

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S34-M-Q5**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • MAIN • Q5/5**

SET 34 • HSS-CP.A.2

# BACKWARD 1

Multiply probabilities • Grade 5 • Number & Operations–Fractions • 5.NF.B.4

Compute (1/2)(1/3). Give a fraction in lowest terms.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q1

**Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q1/5**

Compute 0.4 · 0.5.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q2

**Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q2/5**

**True or false?**

$(2/3)(3/4) = 1/2$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

**Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q3/5**

Compute 0.3 · 0.6.

**A.** 0.18

**B.** 1.8

**C.** 0.36

**D.** 0.9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

**Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q4/5**

## True or false?

$0.7 \cdot 0.2 = 1.4$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

**Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 34 • BACKWARD 1 • Q5/5**

SET 34 • HSS-CP.A.2

# BACKWARD 2

Find P(A and B) by counting • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

A die is rolled. What is P(even and greater than 3)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q1/5

A coin is flipped and a die is rolled. What is P(heads and 6)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q2/5

## True or false?

For one die, P(odd and even) = 0.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q3/5

Two coins are flipped. What is P(first is heads and second is tails)?

**A.** 3/4

**B.** 1/4

**C.** 1

**D.** 1/2

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S34-B2-Q4**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q4/5**

## True or false?

For one die, P(less than 3 and odd) = 1/3.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • HSS-CP.A.2

# FORWARD 1

Use independence for several events • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

A fair coin is flipped 3 times. What is P(all three heads)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q1/5

Each of 3 independent lights works with probability 0.9. What is P(all 3 work), and P(at least one fails)?

All work: _____

At least one fails: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q2/5

## True or false?

For independent events, P(A and B and C) = P(A)·P(B)·P(C).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q3/5

A fair die is rolled twice. What is P(two sixes)?

**A.** 1/36

**B.** 1/6

**C.** 2/6

**D.** 1/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q4/5

## True or false?

If P(A and B) = P(A)·P(B), then A and B cannot happen together.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 34 • FORWARD 1 • Q5/5

SET 34 • HSS-CP.A.2

# FORWARD 2

Interpret independence with conditional probability • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

P(A and B) = 0.2 and P(B) = 0.5. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q1/5

## True or false?

If A and B are independent, P(A given B) = P(A).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q2/5

P(A) = 0.4 and P(A given B) = 0.4. What can be concluded?

**A.** A and B are independent.

**B.** P(B) = 0.4.

**C.** B always happens.

**D.** A and B are mutually exclusive.

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q3/5

P(A) = 0.3, P(B) = 0.6 and P(A and B) = 0.18. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q4/5

## True or false?

If P(A given B) is greater than P(A), then A and B are independent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 34 • FORWARD 2 • Q5/5

# Compute conditional probability as P(A and B)/P(B)

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-CP.A.3** |
| **BACKWARD 1** | Divide decimals and fractions | **6.NS.B.3** |
| **BACKWARD 2** | Count outcomes in a restricted sample space | **HSS-CP.A.1** |
| **FORWARD 1** | Compute conditional probability in both directions | **HSS-CP.A.3** |
| **FORWARD 2** | Test independence with conditional probability | **HSS-CP.A.3** |

SET 35 • HSS-CP.A.3

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

P(A and B) = 0.12 and P(B) = 0.4. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q1/5

A fair die is rolled. What is P(6 given the roll is even)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q2/5

Two fair dice are rolled. What is P(doubles given the sum is 8)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q3/5

On a given day, P(rain) = 0.3 and P(rain and wind) = 0.12. What is P(wind given rain)?

**A.** 0.4

**B.** 2.5

**C.** 0.12

**D.** 0.036

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q4/5

P(B) = 0.25 and P(A given B) = 0.6. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • MAIN • Q5/5

SET 35 • HSS-CP.A.3

# BACKWARD 1

Divide decimals and fractions • Grade 6 • The Number System • 6.NS.B.3

Compute 0.12 ÷ 0.4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

**Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q1/5**

Compute 0.18 ÷ 0.6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

**Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q2/5**

## True or false?

$0.2 \div 0.5 = 0.4$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

**Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q3/5**

Compute 0.05 ÷ 0.2.

**A.** 0.01

**B.** 2.5

**C.** 0.25

**D.** 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

**Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q4/5**

**True or false?**

$0.3 \div 0.6 = 2.$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

**Grade 6 • The Number System • 6.NS.B.3 • Set 35 • BACKWARD 1 • Q5/5**

SET 35 • HSS-CP.A.3

# BACKWARD 2

Count outcomes in a restricted sample space • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

A die is rolled and shows an even number. How many outcomes remain possible?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q1/5

Two dice are rolled and the sum is 8. How many outcomes give a sum of 8?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q2/5

## True or false?

Knowing that a card is a heart leaves 13 possible cards.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q3/5

A die shows a number greater than 2. How many of the remaining outcomes are odd?

**A.** 3

**B.** 4

**C.** 2

**D.** 1

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S35-B2-Q4**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q4/5**

## True or false?

Knowing a card is a face card leaves 13 possible cards.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 35 • BACKWARD 2 • Q5/5

SET 35 • HSS-CP.A.3

# FORWARD 1

Compute conditional probability in both directions • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

A card is drawn. Find P(king given face card) and P(face card given king). Give a fraction in lowest terms.

P(king given face card): _____

P(face card given king): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q1/5

## True or false?

P(A given B) and P(B given A) are always equal.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q2/5

P(A) = 0.5, P(B) = 0.2 and P(A and B) = 0.1. Find P(A given B) and P(B given A).

P(A given B): _____

P(B given A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q3/5

Which formula gives P(B given A)?

**A.** P(A)/P(B)

**B.** P(A)·P(B)

**C.** P(A and B)/P(B)

**D.** P(A and B)/P(A)

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S35-F1-Q4**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 1 • Q4/5**

## True or false?

P(A given B) = P(B given A) exactly when P(A) = P(B), provided P(A and B) > 0.

**A.** True

**B.** False

SET 35 • HSS-CP.A.3

# FORWARD 2

Test independence with conditional probability • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

P(A) = 0.6, P(B) = 0.5, P(A and B) = 0.3. Find P(A given B), and decide whether A and B are independent (enter yes or no).

P(A given B): _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q1/5

## True or false?

If P(A given B) = P(A), then also P(B given A) = P(B), provided P(A) > 0.

**A.** True

**B.** False

Two cards are drawn without replacement. Are "first card is an ace" and "second card is an ace" independent?

**A.** No: two aces can never be drawn one after another.

**B.** Yes: the deck always contains exactly 4 aces.

**C.** No: the first draw changes the chance for the second.

**D.** Yes: each card in the deck is equally likely to be drawn.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S35-F2-Q3**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q3/5**

P(A) = 0.4, P(B) = 0.5, P(A and B) = 0.1. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q4/5

## True or false?

If P(A given B) = 0.2 and P(A) = 0.4, then A and B are independent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 35 • FORWARD 2 • Q5/5

# Interpret independence through conditional probability

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-CP.A.3** |
| **BACKWARD 1** | Compute a conditional probability | **HSS-CP.A.3** |
| **BACKWARD 2** | Test independence with the product rule | **HSS-CP.A.2** |
| **FORWARD 1** | Check both directions of independence | **HSS-CP.A.3** |
| **FORWARD 2** | Approximate conditional probabilities from data | **HSS-CP.A.4** |

SET 36 • HSS-CP.A.3

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

$P(A) = 0.5$, $P(B) = 0.4$ and $P(A \text{ and } B) = 0.2$. Find $P(A \text{ given } B)$, and decide whether A and B are independent (enter yes or no).

$P(A \text{ given } B)$: _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q1/5

$P(A) = 0.3$, $P(B) = 0.5$ and $P(A \text{ and } B) = 0.1$. Find $P(A \text{ given } B)$, and decide whether A and B are independent (enter yes or no).

$P(A \text{ given } B)$: _____

Independent (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q2/5

A and B are independent. Which statement is correct?

**A.** Whenever B happens, then A must also happen.

**B.** P(A) and P(B) must be equal to each other.

**C.** A and B can never happen at the same time as each other.

**D.** Knowing that B happened does not change the probability of A.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S36-M-Q3**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q3/5**

Two cards are drawn from a standard deck. Find P(second is an ace given the first is an ace) when the first card is replaced, and when it is not replaced. Give a fraction in lowest terms.

With replacement: _____

Without replacement: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q4/5

Which pair of events is independent?

**A.** a die shows 6; a separate coin shows heads

**B.** a student studies; the student passes the test

**C.** first card drawn is an ace; second card (not replaced) is an ace

**D.** it rains today; the ground is wet today

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S36-M-Q5**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • MAIN • Q5/5**

SET 36 • HSS-CP.A.3

# BACKWARD 1

Compute a conditional probability • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

P(A and B) = 0.06 and P(B) = 0.3. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q1/5

P(A and B) = 0.25 and P(A) = 0.5. Find P(B given A).

P(B given A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q2/5

## True or false?

P(A given B) = P(A and B)/P(B).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q3/5

P(A and B) = 0.1, P(B) = 0.4. Find P(A given B).

**A.** 0.4

**B.** 4

**C.** 0.25

**D.** 0.04

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q4/5

## True or false?

P(A given B) = P(A and B)·P(B).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • BACKWARD 1 • Q5/5

SET 36 • HSS-CP.A.3

# BACKWARD 2

Test independence with the product rule • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

## True or false?

If P(A) = 0.5, P(B) = 0.4 and P(A and B) = 0.2, A and B are independent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q1/5

P(A) = 0.2 and P(B) = 0.5. If A and B are independent, what is P(A and B)?

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q2/5

$P(A) = 0.3$, $P(B) = 0.4$, $P(A \text{ and } B) = 0.2$. Are A and B independent?

**A.** no, because $0.2 > 0$

**B.** yes, because $0.3 < 0.4$

**C.** yes, because $0.2 < 1$

**D.** no, because $0.3 \cdot 0.4 = 0.12$

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S36-B2-Q3**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q3/5**

## True or false?

Mutually exclusive events with positive probabilities are independent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q4/5

P(A) = 0.6 and P(B) = 0.5 for independent events. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • HSS-CP.A.3

# FORWARD 1

Check both directions of independence • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

P(A) = 0.6, P(B) = 0.25 and P(A and B) = 0.15. Find P(A given B) and P(B given A).

P(A given B): _____

P(B given A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q1/5

## True or false?

If P(A given B) = P(A), then A and B are independent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q2/5

P(A) = 0.5, P(A given B) = 0.5, and P(B) = 0.3. What is P(B given A)?

**A.** 0.15

**B.** 0.3

**C.** 0.6

**D.** 0.5

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S36-F1-Q3**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q3/5**

A and B are independent, P(A) = 0.7 and P(B) = 0.4. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q4/5

## True or false?

If P(B given A) = 0.25 and P(B) = 0.5, then A and B are independent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 36 • FORWARD 1 • Q5/5

SET 36 • HSS-CP.A.3

# FORWARD 2

Approximate conditional probabilities from data • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

In the survey table, a person is chosen at random. Find P(yes given man) and P(yes).

| | Yes | No | Total |
|---|---|---|---|
| Men | 24 | 16 | 40 |
| Women | 36 | 24 | 60 |
| Total | 60 | 40 | 100 |

P(yes given man): _____

P(yes): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q1/5

## True or false?

In the survey table, gender and answer are independent.

| | Yes | No | Total |
|---|---|---|---|
| Men | 24 | 16 | 40 |
| Women | 36 | 24 | 60 |
| Total | 60 | 40 | 100 |

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q2/5

In the driving table, find P(drives given junior) and P(drives).

| | Drives | Does not drive | Total |
|---|---|---|---|
| Juniors | 60 | 60 | 120 |
| Seniors | 30 | 50 | 80 |
| Total | 90 | 110 | 200 |

P(drives given junior): _____

P(drives): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q3/5

## True or false?

In the driving table, grade and driving are independent.

| | Drives | Does not drive | Total |
|---|---|---|---|
| Juniors | 60 | 60 | 120 |
| Seniors | 30 | 50 | 80 |
| Total | 90 | 110 | 200 |

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q4/5

In the driving table, which is P(junior given drives)?

| | Drives | Does not drive | Total |
|---|---|---|---|
| Juniors | 60 | 60 | 120 |
| Seniors | 30 | 50 | 80 |
| Total | 90 | 110 | 200 |

**A.** 1/2 **B.** 3/10 **C.** 2/3 **D.** 9/20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 36 • FORWARD 2 • Q5/5

# Construct two-way frequency tables

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-CP.A.4** |
| **BACKWARD 1** | Fill in a two-way frequency table | **8.SP.A.4** |
| **BACKWARD 2** | Read joint and marginal frequencies | **HSS-ID.B.5** |
| **FORWARD 1** | Check statements against a built table | **HSS-CP.A.4** |
| **FORWARD 2** | Use the table as a sample space | **HSS-CP.A.4** |

SET 37 • HSS-CP.A.4

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

Of 200 students, 120 are juniors and 80 are seniors. 90 students drive to school, including 60 juniors. Complete the two-way table: juniors who do not drive, seniors who drive, seniors who do not drive.

Juniors, do not drive: _____

Seniors, drive: _____

Seniors, do not drive: _____

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S37-M-Q1**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q1/5**

Of 150 adults, 90 own a car, 40 own a bike, and 25 own both. In a two-way table (car yes/no by bike yes/no), how many own a car but no bike, and how many own neither?

Car, no bike: _____

Neither: _____

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q2/5

Of 200 students, 120 are juniors, and 60 of the juniors drive. What percent of juniors drive?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q3/5

80 patients took part in a study. Which description matches the table?

| | Improved | Did not | Total |
|---|---|---|---|
| Drug | 35 | 15 | 50 |
| Placebo | 12 | 18 | 30 |
| Total | 47 | 33 | 80 |

**A.** 50 got the drug (35 improved); 12 of the placebo patients improved.

**B.** 50 patients got the placebo and the other 30 got the drug.

**C.** 50 patients improved, and 35 of those improved patients got the drug.

**D.** 35 patients got the drug and 12 patients got a placebo instead.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S37-M-Q4**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q4/5**

80 patients: 50 got a drug and 30 a placebo; 35 drug patients and 12 placebo patients improved. How many drug patients and how many placebo patients did not improve?

Drug, did not: _____

Placebo, did not: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • MAIN • Q5/5

SET 37 • HSS-CP.A.4

# BACKWARD 1

Fill in a two-way frequency table • Grade 8 • Statistics & Probability • 8.SP.A.4

A table has row totals 40 and 60. One cell in the first row is 25. What is the other cell in that row?

Cell: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q1

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q1/5**

## True or false?

In a two-way table, the column totals add to the grand total.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q2

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q2/5**

Of 50 people, 30 are adults. 12 of the adults wear glasses. How many adults do not?

**A.** 12

**B.** 18

**C.** 38

**D.** 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q3/5**

A two-way table with two columns classifies 100 objects. One column total is 45. What is the other column total?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q4/5**

## True or false?

A cell can be found from its row total and the other cells in that row.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 37 • BACKWARD 1 • Q5/5**

SET 37 • HSS-CP.A.4

# BACKWARD 2

Read joint and marginal frequencies • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

In the drug table, how many patients improved in total?

| | Improved | Did not | Total |
|---|---|---|---|
| Drug | 35 | 15 | 50 |
| Placebo | 12 | 18 | 30 |
| Total | 47 | 33 | 80 |

Improved: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q1/5

In the drug table, what fraction of all patients got the drug and improved? Give lowest terms.

| | Improved | Did not | Total |
|---|---|---|---|
| Drug | 35 | 15 | 50 |
| Placebo | 12 | 18 | 30 |
| Total | 47 | 33 | 80 |

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q2/5

## True or false?

In the drug table, 30 patients got the placebo.

| | Improved | Did not | Total |
|---|---|---|---|
| Drug | 35 | 15 | 50 |
| Placebo | 12 | 18 | 30 |
| Total | 47 | 33 | 80 |

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q3/5

In the drug table, which number is a marginal total?

| | Improved | Did not | Total |
|---|---|---|---|
| Drug | 35 | 15 | 50 |
| Placebo | 12 | 18 | 30 |
| Total | 47 | 33 | 80 |

**A.** 12 **B.** 35 **C.** 50 **D.** 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q4/5

## True or false?

In the drug table, 35/47 is the fraction of drug patients who improved.

| | Improved | Did not | Total |
|---|---|---|---|
| Drug | 35 | 15 | 50 |
| Placebo | 12 | 18 | 30 |
| Total | 47 | 33 | 80 |

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 37 • BACKWARD 2 • Q5/5

SET 37 • HSS-CP.A.4

# FORWARD 1

Check statements against a built table • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

## True or false?

In the drug table, a greater percent of drug patients than placebo patients improved.

| | Improved | Did not | Total |
|---|---|---|---|
| Drug | 35 | 15 | 50 |
| Placebo | 12 | 18 | 30 |
| Total | 47 | 33 | 80 |

**A.** True

**B.** False

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S37-F1-Q1**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q1/5**

In the drug table, what percent of drug patients improved, and what percent of placebo patients improved?

| | Improved | Did not | Total |
|---|---|---|---|
| Drug | 35 | 15 | 50 |
| Placebo | 12 | 18 | 30 |
| Total | 47 | 33 | 80 |

Drug: _____

Placebo: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q2/5

A report says "most patients who improved got the drug." Using the drug table, is this correct?

| | Improved | Did not | Total |
|---|---|---|---|
| Drug | 35 | 15 | 50 |
| Placebo | 12 | 18 | 30 |
| Total | 47 | 33 | 80 |

**A.** No: only 35 patients got the drug.

**B.** Yes: 35 of the 47 who improved got the drug.

**C.** Yes: 70% of all patients improved.

**D.** No: 50 patients got the drug.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S37-F1-Q3**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q3/5**

Of 120 voters, 70 are under 40. 45 of all voters support a plan, including 30 of those under 40. How many voters 40 or older do not support it?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q4/5

## True or false?

A two-way table can be built when each object is classified by only one category.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 1 • Q5/5

SET 37 • HSS-CP.A.4

# FORWARD 2

Use the table as a sample space • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

A patient is chosen at random from the drug table. What is P(improved)? Give a fraction in lowest terms.

| | Improved | Did not | Total |
|---|---|---|---|
| Drug | 35 | 15 | 50 |
| Placebo | 12 | 18 | 30 |
| Total | 47 | 33 | 80 |

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q1/5

A patient is chosen at random from the drug table. What is P(improved given drug)? Give a fraction in lowest terms.

| | Improved | Did not | Total |
|---|---|---|---|
| Drug | 35 | 15 | 50 |
| Placebo | 12 | 18 | 30 |
| Total | 47 | 33 | 80 |

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q2/5

## True or false?

In the drug table, improving and treatment are independent.

| | Improved | Did not | Total |
|---|---|---|---|
| Drug | 35 | 15 | 50 |
| Placebo | 12 | 18 | 30 |
| Total | 47 | 33 | 80 |

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q3/5

A patient is chosen at random from the drug table. What is P(drug given improved)?

| | Improved | Did not | Total |
|---|---|---|---|
| Drug | 35 | 15 | 50 |
| Placebo | 12 | 18 | 30 |
| Total | 47 | 33 | 80 |

**A.** 35/47 **B.** 47/80 **C.** 7/16 **D.** 35/50

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S37-F2-Q4**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q4/5**

A patient is chosen at random. What is P(placebo and did not improve)? Give a fraction in lowest terms.

| | Improved | Did not | Total |
|---|---|---|---|
| Drug | 35 | 15 | 50 |
| Placebo | 12 | 18 | 30 |
| Total | 47 | 33 | 80 |

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 37 • FORWARD 2 • Q5/5

# Use two-way tables to find conditional probabilities and judge independence

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-CP.A.4** |
| **BACKWARD 1** | Compute conditional relative frequencies | **HSS-ID.B.5** |
| **BACKWARD 2** | Compare P(A given B) with P(A) | **HSS-CP.A.3** |
| **FORWARD 1** | Use larger two-way tables | **HSS-CP.A.4** |
| **FORWARD 2** | Explain table results in everyday language | **HSS-CP.A.5** |

SET 38 • HSS-CP.A.4

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

A student is chosen at random from the driving table. Find P(drives given junior) and P(drives given senior). Give decimals.

| | Drives | Does not drive | Total |
|---|---|---|---|
| Juniors | 60 | 60 | 120 |
| Seniors | 30 | 50 | 80 |
| Total | 90 | 110 | 200 |

Given junior: _____

Given senior: _____

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S38-M-Q1**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q1/5**

In the driving table, are "drives" and "junior" independent?

| | Drives | Does not drive | Total |
|---|---|---|---|
| Juniors | 60 | 60 | 120 |
| Seniors | 30 | 50 | 80 |
| Total | 90 | 110 | 200 |

**A.** No: P(drives given junior) ≠ P(drives given senior).

**B.** Yes: every student is a junior or a senior.

**C.** Yes: 60 juniors drive and 60 do not.

**D.** No: more juniors than seniors were surveyed.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S38-M-Q2**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q2/5**

A student is chosen at random from the sports table. Find P(plays a sport) and P(plays a sport given 9th grade). Give decimals.

| | Plays a sport | Does not | Total |
|---|---|---|---|
| 9th grade | 30 | 20 | 50 |
| 10th grade | 42 | 28 | 70 |
| Total | 72 | 48 | 120 |

P(sport): _____

P(sport given 9th): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q3/5

In the sports table, are "plays a sport" and "9th grade" independent?

| | Plays a sport | Does not | Total |
|---|---|---|---|
| 9th grade | 30 | 20 | 50 |
| 10th grade | 42 | 28 | 70 |
| Total | 72 | 48 | 120 |

**A.** No: 30 is not equal to 42.

**B.** Yes: P(sport given 9th) = P(sport) = 0.6.

**C.** No: the grades have different sizes.

**D.** Yes: more students play than do not.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S38-M-Q4**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q4/5**

A student who drives is chosen at random from the driving table. What is P(junior given drives)? Give a fraction in lowest terms.

| | Drives | Does not drive | Total |
|---|---|---|---|
| Juniors | 60 | 60 | 120 |
| Seniors | 30 | 50 | 80 |
| Total | 90 | 110 | 200 |

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • MAIN • Q5/5

SET 38 • HSS-CP.A.4

# BACKWARD 1

Compute conditional relative frequencies • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

In the driving table, what fraction of seniors drive? Give lowest terms.

| | Drives | Does not drive | Total |
|---|---|---|---|
| Juniors | 60 | 60 | 120 |
| Seniors | 30 | 50 | 80 |
| Total | 90 | 110 | 200 |

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q1/5

## True or false?

In the sports table, 42/70 of 10th graders play a sport.

| | Plays a sport | Does not | Total |
|---|---|---|---|
| 9th grade | 30 | 20 | 50 |
| 10th grade | 42 | 28 | 70 |
| Total | 72 | 48 | 120 |

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q2/5

In the sports table, what fraction of students who play a sport are in 10th grade?

| | Plays a sport | Does not | Total |
|---|---|---|---|
| 9th grade | 30 | 20 | 50 |
| 10th grade | 42 | 28 | 70 |
| Total | 72 | 48 | 120 |

**A.** 5/12 **B.** 7/20 **C.** 7/12 **D.** 3/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q3/5

In the driving table, what percent of all students drive?

| | Drives | Does not drive | Total |
|---|---|---|---|
| Juniors | 60 | 60 | 120 |
| Seniors | 30 | 50 | 80 |
| Total | 90 | 110 | 200 |

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q4/5

## True or false?

A row conditional frequency divides by the grand total.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • HSS-CP.A.4

# BACKWARD 2

Compare P(A given B) with P(A) • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

## True or false?

If P(A given B) = P(A), A and B are independent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q1/5

P(A) = 0.45 and P(A given B) = 0.5. Are A and B independent?

**A.** no

**B.** only if P(B) = 0.5

**C.** yes

**D.** cannot tell

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S38-B2-Q2**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q2/5**

P(A and B) = 0.3 and P(B) = 0.5. Find P(A given B).

P(A given B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q3/5

## True or false?

P(A given B) is always less than P(A).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q4/5

P(A and B) = 0.18 and P(B) = 0.3. If P(A) = 0.6, are A and B independent?

**A.** yes

**B.** only if P(A) = 0.3

**C.** cannot tell

**D.** no

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 38 • BACKWARD 2 • Q5/5

SET 38 • HSS-CP.A.4

# FORWARD 1

Use larger two-way tables • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

A student is chosen at random from the music table. Find P(rock) and P(rock given high school). Give decimals.

| | Pop | Rock | Other | Total |
|---|---|---|---|---|
| Middle school | 16 | 24 | 10 | 50 |
| High school | 24 | 36 | 15 | 75 |
| Total | 40 | 60 | 25 | 125 |

P(rock): _____

P(rock given high school): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q1/5

## True or false?

In the music table, school level and favorite type of music are independent.

| | Pop | Rock | Other | Total |
|---|---|---|---|---|
| Middle school | 16 | 24 | 10 | 50 |
| High school | 24 | 36 | 15 | 75 |
| Total | 40 | 60 | 25 | 125 |

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q2/5

In the music table, what is P(middle school given pop)? Give a fraction in lowest terms.

| | Pop | Rock | Other | Total |
|---|---|---|---|---|
| Middle school | 16 | 24 | 10 | 50 |
| High school | 24 | 36 | 15 | 75 |
| Total | 40 | 60 | 25 | 125 |

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q3/5

In the music table, what is P(other given middle school)?

| | Pop | Rock | Other | Total |
|---|---|---|---|---|
| Middle school | 16 | 24 | 10 | 50 |
| High school | 24 | 36 | 15 | 75 |
| Total | 40 | 60 | 25 | 125 |

**A.** 2/25 **B.** 1/5 **C.** 2/5 **D.** 1/3

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S38-F1-Q4**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q4/5**

## True or false?

In the music table, P(pop given middle school) is greater than P(pop given high school).

| | Pop | Rock | Other | Total |
|---|---|---|---|---|
| Middle school | 16 | 24 | 10 | 50 |
| High school | 24 | 36 | 15 | 75 |
| Total | 40 | 60 | 25 | 125 |

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 38 • FORWARD 1 • Q5/5

SET 38 • HSS-CP.A.4

# FORWARD 2

Explain table results in everyday language • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

In the driving table, P(drives given junior) = 0.5 and P(drives given senior) = 0.375. Which sentence says this in everyday language?

| | Drives | Does not drive | Total |
|---|---|---|---|
| Juniors | 60 | 60 | 120 |
| Seniors | 30 | 50 | 80 |
| Total | 90 | 110 | 200 |

**A.** Most of the students who drive to school are seniors.

**B.** Juniors and seniors are equally likely to drive.

**C.** Half of all the students who drive are juniors.

**D.** Half of the juniors drive, but fewer than 4 in 10 seniors drive.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S38-F2-Q1**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q1/5**

## True or false?

"Among students who drive, two-thirds are juniors" describes P(junior given drives).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q2/5

Which statement describes independence in the sports table?

| | Plays a sport | Does not | Total |
|---|---|---|---|
| 9th grade | 30 | 20 | 50 |
| 10th grade | 42 | 28 | 70 |
| Total | 72 | 48 | 120 |

**A.** A student's grade does not change the chance of playing a sport.

**B.** Every student in the table plays exactly one sport.

**C.** The two grades have the same number of students.

**D.** Students in 10th grade play more sports than 9th graders.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S38-F2-Q3**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q3/5**

## True or false?

"Most juniors drive" and "most drivers are juniors" mean the same thing.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 38 • FORWARD 2 • Q4/5

"Among seniors, 3 in 8 drive." Which probability is this?

**A.** P(senior given drives)

**B.** P(drives given senior)

**C.** P(senior and drives)

**D.** P(senior)

# Conditional probability and independence in everyday language

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-CP.A.5** |
| **BACKWARD 1** | Compute and read conditional probabilities | **HSS-CP.A.3** |
| **BACKWARD 2** | Read conditional rates in a table | **HSS-CP.A.4** |
| **FORWARD 1** | Translate statements and compute | **HSS-CP.A.5** |
| **FORWARD 2** | Find a conditional probability as a fraction of outcomes | **HSS-CP.B.6** |

SET 39 • HSS-CP.A.5

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

Which statement describes a conditional probability?

**A.** Among smokers, the chance of lung disease is higher.

**B.** One in four cards in a deck is a heart.

**C.** A fair coin lands heads half of the time.

**D.** The chance of rain tomorrow is 30 percent.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S39-M-Q1**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q1/5**

"Most people with lung disease were smokers." Which probability does this sentence say is high?

**A.** P(lung disease)

**B.** P(smoker given lung disease)

**C.** P(smoker and lung disease)

**D.** P(lung disease given smoker)

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S39-M-Q2**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q2/5**

Which pair of events is most likely to be independent?

**A.** heavy rain today; a baseball game being canceled

**B.** a student's shirt color; the school bus being late

**C.** studying hard for a test; passing that test

**D.** a person's height; the person's shoe size

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S39-M-Q3**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q3/5**

A fair coin has landed heads 5 times in a row. What is the probability of heads on the next flip?

**A.** 1/32

**B.** 1/2

**C.** less than 1/2

**D.** more than 1/2

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S39-M-Q4**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q4/5**

In a town, 40% of adults own a dog, and among dog owners 70% walk daily. What percent of all adults are dog owners who walk daily?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • MAIN • Q5/5

SET 39 • HSS-CP.A.5

# BACKWARD 1

Compute and read conditional probabilities • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

P(A and B) = 0.28 and P(A) = 0.4. Find P(B given A).

P(B given A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q1/5

## True or false?

"The probability of B given A" means the chance of B when A is known to happen.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q2/5

P(passing given studying) is the probability that

**A.** a student passes

**B.** a student studies, among students who pass

**C.** a student both studies and passes

**D.** a student passes, among students who study

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S39-B1-Q3**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q3/5**

P(B) = 0.5 and P(A given B) = 0.3. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 39 • BACKWARD 1 • Q4/5

## True or false?

P(rain given clouds) is the same as P(clouds given rain).

**A.** True

**B.** False

SET 39 • HSS-CP.A.5

# BACKWARD 2

Read conditional rates in a table • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

In the driving table, what fraction of juniors drive? Give lowest terms.

| | Drives | Does not drive | Total |
|---|---|---|---|
| Juniors | 60 | 60 | 120 |
| Seniors | 30 | 50 | 80 |
| Total | 90 | 110 | 200 |

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q1/5

**True or false?**

In the driving table, "among drivers, 2 in 3 are juniors" is correct.

| | Drives | Does not drive | Total |
|---|---|---|---|
| Juniors | 60 | 60 | 120 |
| Seniors | 30 | 50 | 80 |
| Total | 90 | 110 | 200 |

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q2/5

In the driving table, which sentence is correct?

| | Drives | Does not drive | Total |
|---|---|---|---|
| Juniors | 60 | 60 | 120 |
| Seniors | 30 | 50 | 80 |
| Total | 90 | 110 | 200 |

**A.** Among seniors, 1 in 2 drive.

**B.** Among juniors, 3 in 8 drive.

**C.** Among drivers, 3 in 8 are seniors.

**D.** Among seniors, 3 in 8 drive.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S39-B2-Q3**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q3/5**

In the driving table, what fraction of drivers are seniors? Give lowest terms.

| | Drives | Does not drive | Total |
|---|---|---|---|
| Juniors | 60 | 60 | 120 |
| Seniors | 30 | 50 | 80 |
| Total | 90 | 110 | 200 |

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q4/5

**True or false?**

In the driving table, "among seniors, most drive" is correct.

| | Drives | Does not drive | Total |
|---|---|---|---|
| Juniors | 60 | 60 | 120 |
| Seniors | 30 | 50 | 80 |
| Total | 90 | 110 | 200 |

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 39 • BACKWARD 2 • Q5/5

SET 39 • HSS-CP.A.5

# FORWARD 1

Translate statements and compute • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5

"If it is a school day, there is an 80% chance that Sam takes the bus." Which notation matches?

**A.** P(bus) = 0.8

**B.** P(school day given bus) = 0.8

**C.** P(bus given school day) = 0.8

**D.** P(bus and school day) = 0.8

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S39-F1-Q1**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q1/5**

5 of 7 days are school days, and on a school day Sam takes the bus with probability 0.8. What is P(school day and bus)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q2/5

## True or false?

"Knowing it is a school day changes the chance that Sam takes the bus" means the events are not independent.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q3/5

A test is positive for 95% of people with a disease. Which sentence states this correctly?

**A.** 95% of all people have the disease.

**B.** Among people with the disease, 95% test positive.

**C.** 95% of all people test positive.

**D.** Among people who test positive, 95% have the disease.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S39-F1-Q4**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q4/5**

## True or false?

"Among people who test positive, 95% have the disease" follows from "95% of people with the disease test positive".

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.5 • Set 39 • FORWARD 1 • Q5/5

SET 39 • HSS-CP.A.5

# FORWARD 2

Find a conditional probability as a fraction of outcomes • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6

A die is rolled. Given the roll is odd, what is P(the roll is 5)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q1/5

A card is red. What is P(it is a king)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q2/5

## True or false?

P(A given B) can be found as (outcomes in both A and B)/(outcomes in B) in a uniform model.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q3/5

A number from 1 to 10 is even. What is P(it is greater than 6)?

**A.** 1/2

**B.** 2/5

**C.** 2/3

**D.** 1/5

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S39-F2-Q4**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q4/5**

## True or false?

Given the roll of a die is even, P(roll is 3) = 1/6.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 39 • FORWARD 2 • Q5/5

# Find conditional probability as a fraction of outcomes

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-CP.B.6** |
| **BACKWARD 1** | Count outcomes in events | **HSS-CP.A.1** |
| **BACKWARD 2** | Use the definition of conditional probability | **HSS-CP.A.3** |
| **FORWARD 1** | Condition on "at least one" | **HSS-CP.B.6** |
| **FORWARD 2** | Multiply along a chain of conditions | **HSS-CP.B.8** |

SET 40 • HSS-CP.B.6

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6

Two fair dice are rolled. The first die shows 5. What is P(the sum is 10 or more)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q1/5

A card drawn from a standard deck is red. What is P(it is an ace)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q2/5

A patient from the drug table is chosen at random and got the placebo. What is P(improved)? Give a fraction in lowest terms.

| | Improved | Did not | Total |
|---|---|---|---|
| Drug | 35 | 15 | 50 |
| Placebo | 12 | 18 | 30 |
| Total | 47 | 33 | 80 |

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q3/5

A number from 1 to 10 is chosen at random and is odd. What is P(it is prime)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q4/5

From the drug table, P(improved given placebo) = 2/5. What does this mean in the model?

| | Improved | Did not | Total |
|---|---|---|---|
| Drug | 35 | 15 | 50 |
| Placebo | 12 | 18 | 30 |
| Total | 47 | 33 | 80 |

**A.** Of the patients who improved, 2 in 5 got the placebo.

**B.** 2 in 5 of all patients got the placebo and improved.

**C.** Of the patients given the placebo, 2 in 5 improved.

**D.** 2 in 5 of all patients got the placebo.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S40-M-Q5**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • MAIN • Q5/5**

SET 40 • HSS-CP.B.6

# BACKWARD 1

Count outcomes in events • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

How many red cards are in a standard deck?

Cards: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q1/5

Two dice: how many outcomes have the first die equal to 5?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q2/5

## True or false?

There are 2 red aces.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q3/5

How many numbers from 1 to 10 are odd and prime?

**A.** 2

**B.** 5

**C.** 3

**D.** 4

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S40-B1-Q4**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 40 • BACKWARD 1 • Q4/5**

## True or false?

There are 4 red kings.

**A.** True

**B.** False

SET 40 • HSS-CP.B.6

# BACKWARD 2

Use the definition of conditional probability • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

P(A and B) = 1/12 and P(B) = 1/4. Find P(A given B). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q1/5

## True or false?

In a uniform model, P(A and B)/P(B) equals (count of A and B)/(count of B).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q2/5

P(A and B) = 0.1 and P(B) = 0.5. Find P(A given B).

**A.** 5

**B.** 0.5

**C.** 0.05

**D.** 0.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q3/5

P(A and B) = 2/36 and P(B) = 6/36. Find P(A given B). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 40 • BACKWARD 2 • Q4/5

## True or false?

P(A given B) = P(A and B)/P(A).

**A.** True

**B.** False

SET 40 • HSS-CP.B.6

# FORWARD 1

Condition on "at least one" • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6

Two fair dice are rolled and at least one shows 6. What is P(the sum is 7)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q1/5

A family has two children; each is equally likely to be a girl or a boy. Given at least one is a girl, what is P(both are girls)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q2/5

## True or false?

Given at least one of two coins shows heads, P(both heads) = 1/2.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q3/5

Two dice are rolled and the sum is at least 11. What is P(doubles)?

**A.** 1/36

**B.** 1/6

**C.** 1/3

**D.** 1/2

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S40-F1-Q4**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q4/5**

## True or false?

Conditioning on an event removes the outcomes outside it from the sample space.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 40 • FORWARD 1 • Q5/5

SET 40 • HSS-CP.B.6

# FORWARD 2

Multiply along a chain of conditions • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.8

A bag has 5 red and 3 blue marbles. Two are drawn without replacement. What is P(both red)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q1/5

## True or false?

P(A and B) = P(A)·P(B given A).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q2/5

P(A) = 0.6 and P(B given A) = 0.5. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q3/5

Two cards are drawn without replacement. What is P(second is a heart given the first is a heart)?

**A.** 1/4

**B.** 13/52

**C.** 13/51

**D.** 12/51

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S40-F2-Q4**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability**
**HSS-CP.B.8 • Set 40 • FORWARD 2 • Q4/5**

## True or false?

For draws without replacement, P(second red given first red) equals P(first red).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 40 • FORWARD 2 • Q5/5

# Apply the Addition Rule

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-CP.B.7** |
| **BACKWARD 1** | Count a union without double counting | **HSS-CP.A.1** |
| **BACKWARD 2** | Add and subtract fractions | **5.NF.A.1** |
| **FORWARD 1** | Use the rule in reverse and with exclusive events | **HSS-CP.B.7** |
| **FORWARD 2** | Combine the Addition Rule with complements | **HSS-CP.B.7** |

SET 41 • HSS-CP.B.7

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7

P(A) = 0.45, P(B) = 0.35 and P(A and B) = 0.15. Find P(A or B).

P(A or B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q1/5

A card is drawn from a standard deck. Find P(red or king). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q2/5

Two fair dice are rolled. Find P(doubles or sum of 8). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q3/5

Of 200 students, 120 take Spanish, 70 take art, and 40 take both. A student is chosen at random. What is P(Spanish or art)?

**A.** 3/4

**B.** 7/10

**C.** 19/20

**D.** 1/5

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S41-M-Q4**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q4/5**

For a randomly chosen shopper, P(buys bread or milk) = 0.65. What does this mean?

**A.** About 65% of shoppers buy both bread and milk.

**B.** About 65% of shoppers buy neither one.

**C.** About 65% of shoppers buy bread, milk, or both.

**D.** About 65% of shoppers buy exactly one.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S41-M-Q5**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • MAIN • Q5/5**

SET 41 • HSS-CP.B.7

# BACKWARD 1

Count a union without double counting • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

A die is rolled. How many outcomes are "even or less than 3"?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q1/5

## True or false?

Hearts or face cards: $13 + 12 - 3 = 22$ cards.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q2/5

How many cards are aces or spades?

**A.** 4

**B.** 16

**C.** 13

**D.** 17

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S41-B1-Q3**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q3/5**

How many numbers from 1 to 20 are multiples of 2 or multiples of 5?

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q4/5

## True or false?

The number of outcomes in "A or B" is always the count of A plus the count of B.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • HSS-CP.B.7

# BACKWARD 2

Add and subtract fractions • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Compute 1/2 + 1/13 − 1/26. Give a fraction in lowest terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q1

**Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q1/5**

Compute 1/6 + 5/36 − 1/36. Give a fraction in lowest terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q2

**Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q2/5**

**True or false?**

1/4 + 1/3 = 7/12.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q3

**Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q3/5**

Compute 3/5 + 7/20 − 1/5.

**A.** 17/20

**B.** 1

**C.** 3/4

**D.** 9/20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

**Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q4/5**

**True or false?**

1/2 + 1/3 = 2/5.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

**Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 41 • BACKWARD 2 • Q5/5**

SET 41 • HSS-CP.B.7

# FORWARD 1

Use the rule in reverse and with exclusive events • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7

P(A) = 0.5, P(B) = 0.4 and P(A or B) = 0.7. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q1/5

## True or false?

If A and B are mutually exclusive, P(A or B) = P(A) + P(B).

**A.** True

**B.** False

A die is rolled. Find P(1 or 6). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q3/5

P(A) = 0.3, P(B) = 0.5, and A and B are independent. What is P(A or B)?

**A.** 0.65

**B.** 0.15

**C.** 0.5

**D.** 0.8

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S41-F1-Q4**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q4/5**

## True or false?

If P(A) = 0.6 and P(B) = 0.7, then A and B could be mutually exclusive.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 1 • Q5/5

SET 41 • HSS-CP.B.7

# FORWARD 2

Combine the Addition Rule with complements • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7

P(A or B) = 0.65. What is P(neither A nor B)?

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q1/5

Of 200 students, 120 take Spanish, 70 take art and 40 take both. What is P(neither)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q2/5

## True or false?

P(not (A or B)) = 1 − P(A or B).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q3/5

A card is drawn. What is P(neither red nor a king)?

**A.** 1/2

**B.** 6/13

**C.** 7/13

**D.** 11/26

## True or false?

P(neither A nor B) = 1 − P(A) − P(B) in every case.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 41 • FORWARD 2 • Q5/5

# Apply the general Multiplication Rule (+)

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-CP.B.8** |
| **BACKWARD 1** | Rearrange the conditional probability formula | **HSS-CP.A.3** |
| **BACKWARD 2** | Multiply fractions | **5.NF.B.4** |
| **BACKWARD 3** | Multiply for independent events | **HSS-CP.A.2** |
| **FORWARD 1** | Combine orders and longer chains | **HSS-CP.B.8** |
| **FORWARD 2** | Count arrangements to find probabilities | **HSS-CP.B.9** |

SET 42 • HSS-CP.B.8

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.8

A bag has 5 red and 3 blue marbles. Two are drawn without replacement. What is P(both red)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q1/5

A bag has 5 red and 3 blue marbles. Two are drawn without replacement. What is P(first red and second blue)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q2/5

Two cards are drawn without replacement. What is P(both aces)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q3/5

P(A) = 0.6 and P(B given A) = 0.5. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q4/5

A committee has 3 boys and 2 girls. Two members are chosen at random, one after the other. What is P(both girls)?

**A.** 2/5

**B.** 4/25

**C.** 1/5

**D.** 1/10

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • MAIN • Q5/5

SET 42 • HSS-CP.B.8

# BACKWARD 1

Rearrange the conditional probability formula • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

## True or false?

From P(B given A) = P(A and B)/P(A), it follows that P(A and B) = P(A)·P(B given A).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q1/5

P(A) = 0.4 and P(B given A) = 0.25. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q2/5

P(B) = 0.5 and P(A given B) = 0.6. What is P(A and B)?

**A.** 0.1

**B.** 1.1

**C.** 0.83

**D.** 0.3

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S42-B1-Q3**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q3/5**

P(A and B) = 0.12 and P(A) = 0.3. Find P(B given A).

P(B given A): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q4/5

## True or false?

P(A and B) = P(A) + P(B given A).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • HSS-CP.B.8

# BACKWARD 2

Multiply fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4

Compute (5/8)(4/7). Give a fraction in lowest terms.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q1

**Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q1/5**

Compute (4/52)(3/51). Give a fraction in lowest terms.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

**Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q2/5**

**True or false?**

(2/5)(1/4) = 1/10.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q3

**Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q3/5**

Compute (3/8)(2/7).

**A.** 6/15

**B.** 3/28

**C.** 3/56

**D.** 5/15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

**Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q4/5**

**True or false?**

(1/2)(1/3) = 2/5.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q5

**Grade 5 • Number & Operations–Fractions • 5.NF.B.4 • Set 42 • BACKWARD 2 • Q5/5**

SET 42 • HSS-CP.B.8

# BACKWARD 3

Multiply for independent events • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

A die is rolled twice. What is P(both 6)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q1/5

## True or false?

With replacement, P(red then red) from 5 red and 3 blue is (5/8)(5/8).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q2/5

A coin is flipped and a die is rolled. P(tails and even)?

**A.** 2/3

**B.** 1/12

**C.** 1/2

**D.** 1/4

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S42-B3-Q3**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q3/5**

Independent events have P(A) = 0.5 and P(B) = 0.3. Find P(A and B).

P(A and B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q4/5

## True or false?

Without replacement, draws are independent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 42 • BACKWARD 3 • Q5/5

SET 42 • HSS-CP.B.8

# FORWARD 1

Combine orders and longer chains • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.8

A bag has 5 red and 3 blue marbles. Two are drawn without replacement. What is P(one of each color)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q1/5

Three cards are drawn without replacement. What is P(all three are hearts)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q2/5

## True or false?

P(A and B) = P(B)·P(A given B) gives the same answer as P(A)·P(B given A).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q3/5

A bag has 4 green and 6 yellow marbles. Two are drawn without replacement. What is P(both yellow)?

**A.** 9/25

**B.** 2/5

**C.** 3/5

**D.** 1/3

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S42-F1-Q4**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability**
**HSS-CP.B.8 • Set 42 • FORWARD 1 • Q4/5**

## True or false?

P(first red and second blue) equals P(first blue and second red) for 5 red and 3 blue drawn without replacement.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 42 • FORWARD 1 • Q5/5

SET 42 • HSS-CP.B.8

# FORWARD 2

Count arrangements to find probabilities • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

How many ways can 2 marbles be chosen from 8 (order does not matter)?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q1/5

How many ways can 2 red marbles be chosen from 5 red?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q2/5

## True or false?

P(both red) = 10/28 agrees with (5/8)(4/7).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q3/5

How many ordered ways can 2 of 8 marbles be drawn one after the other?

**A.** 56

**B.** 16

**C.** 28

**D.** 64

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S42-F2-Q4**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability**
**HSS-CP.B.9 • Set 42 • FORWARD 2 • Q4/5**

## True or false?

Counting with combinations always gives a different probability than the Multiplication Rule.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 42 • FORWARD 2 • Q5/5

# Count ordered arrangements with permutations (+)

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-CP.B.9** |
| **BACKWARD 1** | Count outcomes with a tree or the counting principle | **7.SP.C.8.b** |
| **BACKWARD 2** | Multiply several whole numbers | **5.NBT.B.5** |
| **FORWARD 1** | Count arrangements with conditions | **HSS-CP.B.9** |
| **FORWARD 2** | Count selections where order does not matter | **HSS-CP.B.9** |

SET 43 • HSS-CP.B.9

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

Five runners race. In how many ways can first, second and third place be awarded?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q1/5

In how many ways can 6 different books be arranged on a shelf?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q2/5

A 4-digit code uses the digits 0-9 with no digit repeated. How many codes are possible?

Codes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q3/5

What is the value of 8P2 (the number of ordered arrangements of 2 objects from 8)?

**A.** 56

**B.** 28

**C.** 16

**D.** 64

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S43-M-Q4**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability**
**HSS-CP.B.9 • Set 43 • MAIN • Q4/5**

A club of 10 members chooses a president, a vice president and a treasurer. How many ways are possible?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • MAIN • Q5/5

SET 43 • HSS-CP.B.9

# BACKWARD 1

Count outcomes with a tree or the counting principle • Grade 7 • Statistics & Probability • 7.SP.C.8.b

An outfit is a shirt (4 choices) and pants (3 choices). How many outfits?

Outfits: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q1

**Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q1/5**

## True or false?

A tree for 3 stages with 2 choices each has 8 paths.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

**Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q2/5**

A meal is a main (5 choices), a side (3) and a drink (2). How many meals?

**A.** 60

**B.** 15

**C.** 10

**D.** 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

**Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q3/5**

A license code is a letter (26 choices) followed by a digit (10 choices). How many codes?

Codes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

**Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q4/5**

**True or false?**

With 3 shirts and 2 pants there are 5 outfits.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

**Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 43 • BACKWARD 1 • Q5/5**

SET 43 • HSS-CP.B.9

# BACKWARD 2

Multiply several whole numbers • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5

Compute $10 \cdot 9 \cdot 8$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q1

**Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q1/5**

Compute $6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

**Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q2/5**

## True or false?

$5 \cdot 4 \cdot 3 = 60$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q3

**Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q3/5**

Compute $7 \cdot 6 \cdot 5$.

**A.** 343

**B.** 180

**C.** 42

**D.** 210

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

**Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q4/5**

**True or false?**

$4 \cdot 3 \cdot 2 \cdot 1 = 10$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

**Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 43 • BACKWARD 2 • Q5/5**

SET 43 • HSS-CP.B.9

# FORWARD 1

Count arrangements with conditions • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

In how many ways can the letters of MATH be arranged if M must be first?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q1/5

In how many ways can 5 people stand in a line if Ana and Ben must be at the two ends?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q2/5

## True or false?

$nPr = n!/(n - r)!$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q3/5

How many 3-letter arrangements (no repeats) can be made from the letters A, B, C, D, E if the first letter is a vowel?

**A.** 24

**B.** 60

**C.** 12

**D.** 48

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q4/5

## True or false?

The number of ways to arrange 4 people in a line is $4 + 3 + 2 + 1$.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 1 • Q5/5

SET 43 • HSS-CP.B.9

# FORWARD 2

Count selections where order does not matter • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

How many ways can 3 of 5 students be chosen for a team (order does not matter)?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q1/5

## True or false?

Each unordered group of 3 people can be ordered in 6 ways.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q2/5

Which situation is counted with combinations rather than permutations?

**A.** choosing 3 pizza toppings

**B.** awarding gold, silver and bronze

**C.** making a 4-digit code

**D.** arranging books on a shelf

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S43-F2-Q3**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability**
**HSS-CP.B.9 • Set 43 • FORWARD 2 • Q3/5**

A permutation count is 120 for choosing 3 of 6 in order. How many unordered groups is that?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q4/5

## True or false?

There are more combinations than permutations of 3 objects from 6.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 43 • FORWARD 2 • Q5/5

# Count unordered selections with combinations (+)

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-CP.B.9** |
| **BACKWARD 1** | Count ordered arrangements | **HSS-CP.B.9** |
| **BACKWARD 2** | Represent and count outcomes of compound events | **7.SP.C.8.b** |
| **FORWARD 1** | Count selections from separate groups | **HSS-CP.B.9** |
| **FORWARD 2** | Turn counts into probabilities | **HSS-CP.B.9** |

SET 44 • HSS-CP.B.9

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

How many ways can 3 students be chosen from 5 (order does not matter)?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q1/5

A pizza shop has 8 toppings. How many 3-topping pizzas are possible (no repeated toppings)?

Pizzas: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q2/5

How many 4-person committees can be chosen from 10 people?

Committees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q3/5

What is the value of 6C2?

**A.** 15

**B.** 12

**C.** 36

**D.** 30

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S44-M-Q4**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability**
**HSS-CP.B.9 • Set 44 • MAIN • Q4/5**

Seven people each shake hands once with every other person. How many handshakes are there?

Handshakes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • MAIN • Q5/5

SET 44 • HSS-CP.B.9

# BACKWARD 1

Count ordered arrangements • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

How many ordered ways can 3 of 5 students be lined up?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q1/5

## True or false?

$5P3 = 60$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q2/5

How many ordered pairs of different people can be formed from 7 people?

**A.** 21

**B.** 42

**C.** 49

**D.** 14

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S44-B1-Q3**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability**
**HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q3/5**

How many ways can 3 people be arranged in order?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q4/5

## True or false?

$10P4 = 210$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • HSS-CP.B.9

# BACKWARD 2

Represent and count outcomes of compound events • Grade 7 • Statistics & Probability • 7.SP.C.8.b

List the 2-letter groups (order does not matter) from A, B, C. How many are there?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

**Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q1/5**

**True or false?**

From A, B, C, D, the groups AB and BA are the same group.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

**Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q2/5**

How many 2-letter groups (order does not matter) can be chosen from A, B, C, D?

**A.** 8

**B.** 4

**C.** 12

**D.** 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q3

**Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q3/5**

How many ordered pairs of different letters can be made from A, B, C, D?

Pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

**Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q4/5**

**True or false?**

An organized list of groups should include both AB and BA when order does not matter.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

**Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 44 • BACKWARD 2 • Q5/5**

SET 44 • HSS-CP.B.9

# FORWARD 1

Count selections from separate groups • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

A committee needs 3 of 5 boys and 2 of 4 girls. How many committees are possible?

Committees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q1/5

How many ways can 2 hearts be chosen from the 13 hearts in a deck?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q2/5

## True or false?

$nCr = n!/(r!(n - r)!)$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q3/5

How many 5-card hands can be chosen from 52 cards?

**A.** 2,598,960

**B.** 380,204,032

**C.** 260

**D.** 311,875,200

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S44-F1-Q4**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability**
**HSS-CP.B.9 • Set 44 • FORWARD 1 • Q4/5**

## True or false?

Choosing 2 from 4 girls and 3 from 5 boys gives 6 + 10 = 16 committees.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 1 • Q5/5

SET 44 • HSS-CP.B.9

# FORWARD 2

Turn counts into probabilities • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

3 people are chosen at random from 5 boys and 4 girls. How many groups are possible in all?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q1/5

3 people are chosen at random from 5 boys and 4 girls. What is P(all three are girls)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q2/5

## True or false?

In a uniform model, P(event) = (number of outcomes in the event)/(number of outcomes in all).

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q3/5

2 cards are drawn from a deck. What is P(both hearts)?

**A.** 1/16

**B.** 1/4

**C.** 1/17

**D.** 13/52

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S44-F2-Q4**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability**
**HSS-CP.B.9 • Set 44 • FORWARD 2 • Q4/5**

## True or false?

P(all three are boys) from 5 boys and 4 girls is 3/9.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 44 • FORWARD 2 • Q5/5

# Compute probabilities with permutations and combinations (+)

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-CP.B.9** |
| **BACKWARD 1** | Count with permutations and combinations | **HSS-CP.B.9** |
| **BACKWARD 2** | Multiply conditional probabilities along draws | **HSS-CP.B.8** |
| **FORWARD 1** | Use complements with counting | **HSS-CP.B.9** |
| **FORWARD 2** | Define a random variable from a counting model | **HSS-MD.A.1** |

SET 45 • HSS-CP.B.9

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

A lottery draws 3 different numbers from 1 to 10 (order does not matter). What is the probability that one ticket with 3 numbers wins? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q1/5

A 3-person group is chosen at random from 5 boys and 4 girls. What is P(all girls)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q2/5

Five people line up in random order. What is P(Ann is first and Bo is second)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q3/5

Two cards are dealt from a standard deck. What is P(both are hearts)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • MAIN • Q4/5

A 3-person group is chosen at random from 5 boys and 4 girls. What is P(exactly one girl)?

**A.** 1/3

**B.** 10/21

**C.** 5/21

**D.** 4/9

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S45-M-Q5**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability**
**HSS-CP.B.9 • Set 45 • MAIN • Q5/5**

SET 45 • HSS-CP.B.9

# BACKWARD 1

Count with permutations and combinations • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

Compute 10C3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q1/5

Compute 5!.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q2/5

## True or false?

$9C3 = 84$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q3/5

Compute 52C2.

**A.** 676

**B.** 2,652

**C.** 104

**D.** 1,326

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q4/5

## True or false?

$5C2 = 20$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • HSS-CP.B.9

# BACKWARD 2

Multiply conditional probabilities along draws • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.8

Using the Multiplication Rule, find P(both hearts) for two cards dealt without replacement. Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q1/5

## True or false?

$(4/9)(3/8)(2/7) = 1/21$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q2/5

From 5 boys and 4 girls, two are chosen in order without replacement. P(both boys)?

**A.** 25/81

**B.** 1/4

**C.** 5/9

**D.** 5/18

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S45-B2-Q3**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability**
**HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q3/5**

Five people line up at random. P(Ann first)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q4/5

## True or false?

Drawing without replacement, P(second girl given first girl) = 4/9.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.8 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • HSS-CP.B.9

# FORWARD 1

Use complements with counting • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

A 3-person group is chosen at random from 5 boys and 4 girls. What is P(no girls)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q1/5

A 3-person group is chosen at random from 5 boys and 4 girls. What is P(at least one girl)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q2/5

## True or false?

P(at least one) = 1 − P(none).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q3/5

Four cards are dealt. What is P(no aces)? (48C4 = 194,580 and 52C4 = 270,725.)

**A.** 38,916/54,145

**B.** 194,580/52

**C.** 4/52

**D.** 1/13

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q4/5

## True or false?

P(at least one girl) = 3 · (4/9).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 45 • FORWARD 1 • Q5/5

SET 45 • HSS-CP.B.9

# FORWARD 2

Define a random variable from a counting model • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.1

X = number of girls in a random group of 3 from 5 boys and 4 girls. What values can X take? Give the smallest and the largest.

Smallest: _____

Largest: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q1/5

For X = number of girls in the group, find P(X = 0) and P(X = 3). Give a fraction in lowest terms.

P(X = 0): _____

P(X = 3): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q2/5

## True or false?

The probabilities $P(X = 0)$, $P(X = 1)$, $P(X = 2)$, $P(X = 3)$ add to 1.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q3/5

For X = number of girls in the group, what is P(X = 2)?

**A.** 2/9

**B.** 1/21

**C.** 10/21

**D.** 5/14

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q4/5

## True or false?

A random variable assigns a probability, not a number, to each outcome.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 45 • FORWARD 2 • Q5/5

# Define a random variable and graph its distribution (+)

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-MD.A.1** |
| **BACKWARD 1** | Describe outcomes in a sample space | **HSS-CP.A.1** |
| **BACKWARD 2** | Read and draw a distribution display | **HSS-ID.A.1** |
| **FORWARD 1** | Find probabilities of ranges of values | **HSS-MD.A.1** |
| **FORWARD 2** | Compute the expected value of a random variable | **HSS-MD.A.2** |

SET 46 • HSS-MD.A.1

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.1

Three fair coins are flipped. X = number of heads. Find P(X = 0), P(X = 1), P(X = 2) and P(X = 3). Give a fraction in lowest terms.

P(X = 0): _____

P(X = 1): _____

P(X = 2): _____

P(X = 3): _____

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q1/5

Two fair dice are rolled and X = the sum. How many different values can X take, and what is P(X = 7)? Give a fraction in lowest terms.

Number of values: _____

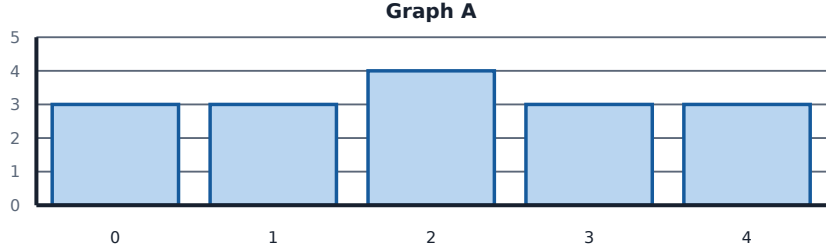
P(X = 7): _____

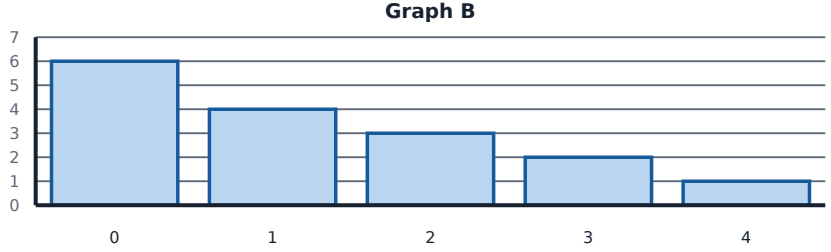
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

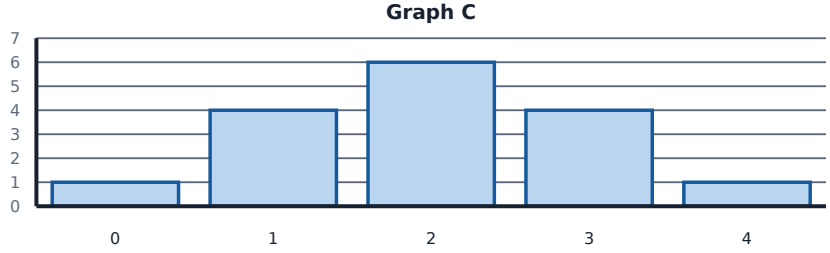
ID S46-M-Q2

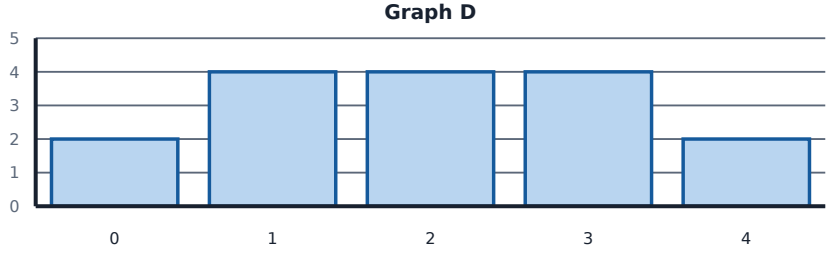
High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q2/5

Four fair coins are flipped and X = number of heads. Each bar graph shows, for X = 0 to 4, how many of the 16 equally likely outcomes give that value. Which graph shows the distribution of X? In each histogram, the horizontal axis shows number of heads and the vertical axis shows frequency.




**A.** Graph A

**B.** Graph B

**C.** Graph C

**D.** Graph D

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S46-M-Q3**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions**
**HSS-MD.A.1 • Set 46 • MAIN • Q3/5**

Which list could be a probability distribution for X = 1, 2, 3?

**A.** 0.2, 0.5, 0.3

**B.** 0.3, 0.4, 0.4

**C.** 0.2, 0.2, 0.2

**D.** 0.5, −0.1, 0.6

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q4/5

X takes the values 0, 1, 2, 3 with P(X = 0) = 0.1, P(X = 1) = 0.35 and P(X = 3) = 0.15. Find P(X = 2).

P(X = 2): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • MAIN • Q5/5

SET 46 • HSS-MD.A.1

# BACKWARD 1

Describe outcomes in a sample space • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

Three coins are flipped. How many outcomes have exactly 1 head?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q1/5

## True or false?

Three coin flips have 8 equally likely outcomes.

**A.** True

**B.** False

Two dice are rolled. How many outcomes have a sum of 4?

**A.** 2

**B.** 6

**C.** 4

**D.** 3

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S46-B1-Q3**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q3/5**

Four coins are flipped. How many outcomes are there?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q4/5

## True or false?

For two dice, the sums 2 and 12 each come from 2 outcomes.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • HSS-MD.A.1

# BACKWARD 2

Read and draw a distribution display • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

## True or false?

In a bar graph of a distribution, taller bars show more likely values.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q1/5

A bar graph shows counts 1, 4, 6, 4, 1 for X = 0 to 4. What is its shape?

**A.** skewed right

**B.** symmetric

**C.** uniform

**D.** skewed left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q2/5

Counts 1, 4, 6, 4, 1 are shown for X = 0 to 4. What is the total count?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q3/5

## True or false?

A bar graph with counts 6, 4, 3, 2, 1 is symmetric.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q4/5

Counts are 1, 4, 6, 4, 1 for X = 0 to 4. Which value of X is most likely?

**A.** 1

**B.** 0

**C.** 4

**D.** 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • HSS-MD.A.1

# FORWARD 1

Find probabilities of ranges of values • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.1

For X = number of heads in 3 flips, find P(X ≥ 2). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q1/5

For X = sum of two dice, find P(X ≤ 4). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q2/5

## True or false?

The probabilities of all values of a random variable add to 1.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q3/5

X takes the values 0 to 3 with probabilities 0.1, 0.35, 0.4, 0.15. What is P(X > 1)?

**A.** 0.9

**B.** 0.45

**C.** 0.4

**D.** 0.55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q4/5

## True or false?

$P(X \geq 1) = 1 - P(X = 1)$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 46 • FORWARD 1 • Q5/5

SET 46 • HSS-MD.A.1

# FORWARD 2

Compute the expected value of a random variable • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

X = number of heads in 3 flips. Find the expected value of X.

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q1/5

## True or false?

The expected value is the probability-weighted average of the values.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q2/5

X takes 0, 1, 2, 3 with probabilities 0.1, 0.35, 0.4, 0.15. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q3/5

X = result of rolling one fair die. What is E(X)?

**A.** 3

**B.** 6

**C.** 3.5

**D.** 21

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S46-F2-Q4**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions**
**HSS-MD.A.2 • Set 46 • FORWARD 2 • Q4/5**

## True or false?

The expected value must be one of the values X can take.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 46 • FORWARD 2 • Q5/5

# Calculate and interpret expected value (+)

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-MD.A.2** |
| **BACKWARD 1** | Compute a mean | **6.SP.B.5.c** |
| **BACKWARD 2** | Read a probability distribution | **HSS-MD.A.1** |
| **BACKWARD 3** | Multiply and add decimals | **6.NS.B.3** |
| **FORWARD 1** | Expected value with gains and losses | **HSS-MD.A.2** |
| **FORWARD 2** | Find expected values of theoretical distributions | **HSS-MD.A.3** |

SET 47 • HSS-MD.A.2

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

X = number of heads when 3 fair coins are flipped (table). Find E(X).

| x | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| P(X = x) | 1/8 | 3/8 | 3/8 | 1/8 |

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q1/5

Find E(X) for the distribution in the table.

| x | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| P(X = x) | 0.1 | 0.35 | 0.4 | 0.15 |

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q2/5

X = the number rolled on a fair six-sided die. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q3/5

For the distribution in the table, E(X) = 1.6. What does this mean?

| x | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| P(X = x) | 0.1 | 0.35 | 0.4 | 0.15 |

**A.** The single most likely value of X is 1.6.

**B.** X will equal exactly 1.6 on most of the trials.

**C.** X is 1.6 or less with a probability of 1.6.

**D.** Over many trials, the average value of X is about 1.6.

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q4/5

A game pays $10 with probability 0.2, $2 with probability 0.3, and $0 otherwise. Find the expected payout.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • MAIN • Q5/5

SET 47 • HSS-MD.A.2

# BACKWARD 1

Compute a mean • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean of 1, 2, 3, 4, 5, 6.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q1/5**

Find the mean of 0, 1, 1, 1, 2, 2, 2, 3.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q2/5**

**True or false?**

A value listed 3 times counts 3 times in the mean.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q3/5**

Find the mean of 2, 2, 2, 8.

**A.** 14

**B.** 5

**C.** 2

**D.** 3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q4/5**

## True or false?

The mean of 0, 0, 0, 4 is 2.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 47 • BACKWARD 1 • Q5/5**

SET 47 • HSS-MD.A.2

# BACKWARD 2

Read a probability distribution • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.1

In the distribution table with values 0 to 3, what is P(X = 2)?

| x | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| P(X = x) | 0.1 | 0.35 | 0.4 | 0.15 |

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q1/5

## True or false?

A probability distribution lists every value of X with its probability.

**A.** True

**B.** False

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q2/5

In the coin table, which values of X are equally likely?

| x | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| P(X = x) | 1/8 | 3/8 | 3/8 | 1/8 |

**A.** 2 and 3

**B.** 0 and 1

**C.** all four

**D.** 1 and 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q3/5

A distribution has $P(X = 5) = 0.3$ and $P(X = 10) = 0.7$. What is $P(X = 5) + P(X = 10)$?

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q4/5

## True or false?

A distribution may have probabilities adding to 1.2.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.1 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • HSS-MD.A.2

# BACKWARD 3

Multiply and add decimals • Grade 6 • The Number System • 6.NS.B.3

Compute 2(0.4) + 3(0.15).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q1

**Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q1/5**

Compute 10(0.2) + 2(0.3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q2

**Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q2/5**

**True or false?**

$1(0.35) + 2(0.4) = 1.15.$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q3

**Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q3/5**

Compute 5(0.3) + 10(0.7).

**A.** 7

**B.** 1.5

**C.** 15

**D.** 8.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q4

**Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q4/5**

**True or false?**

$4(0.25) + 2(0.5) = 1.5$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q5

**Grade 6 • The Number System • 6.NS.B.3 • Set 47 • BACKWARD 3 • Q5/5**

SET 47 • HSS-MD.A.2

# FORWARD 1

Expected value with gains and losses • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

A game: win $5 with probability 0.2, lose $2 with probability 0.8. Find the expected gain.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q1/5

## True or false?

A negative expected gain means a player loses money on average over many plays.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q2/5

X has values −3, 0, 6 with probabilities 0.5, 0.2, 0.3. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q3/5

A game has expected gain $-\$0.60$. About how much does a player expect to lose in 100 plays?

**A.** $60

**B.** $600

**C.** $6

**D.** $0.60

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q4/5

## True or false?

If E(X) = 0, the player breaks even on every single play.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 47 • FORWARD 1 • Q5/5

SET 47 • HSS-MD.A.2

# FORWARD 2

Find expected values of theoretical distributions • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.3

A student guesses on 4 true-false questions.
X = number correct. Find P(X = 2). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q1/5

A student guesses on 4 true-false questions. Find the expected number correct.

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q2/5

## True or false?

For n independent trials with success probability p, the expected number of successes is np.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q3/5

A student guesses on 10 four-choice questions. What is the expected number correct?

**A.** 2.5

**B.** 0.25

**C.** 10

**D.** 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q4/5

## True or false?

Guessing on 4 true-false questions, getting all 4 right has probability 1/4.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 47 • FORWARD 2 • Q5/5

# Develop a theoretical probability distribution (+)

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-MD.A.3** |
| **BACKWARD 1** | Count arrangements of successes | **HSS-CP.B.9** |
| **BACKWARD 2** | Multiply probabilities of independent trials | **HSS-CP.A.2** |
| **FORWARD 1** | Build the whole theoretical distribution | **HSS-MD.A.3** |
| **FORWARD 2** | Compare a theoretical distribution with data | **HSS-MD.A.4** |

SET 48 • HSS-MD.A.3

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.3

A student guesses on 4 true-false questions.
X = number correct. Find P(X = 2). Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q1/5

A student guesses on 4 true-false questions.
X = number correct. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q2/5

A student guesses on 5 questions, each with 4 choices. Find P(all 5 correct) and the expected number correct. Give a fraction in lowest terms.

P(all correct): _____

Expected number correct: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q3/5

A student guesses on 3 four-choice questions. What is P(exactly 1 correct)?

**A.** 27/64

**B.** 9/64

**C.** 1/4

**D.** 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q4/5

A student guesses on 3 four-choice questions. What is P(at least 1 correct)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • MAIN • Q5/5

SET 48 • HSS-MD.A.3

# BACKWARD 1

Count arrangements of successes • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

In how many ways can exactly 2 of 4 answers be correct?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q1/5

In how many ways can exactly 1 of 3 answers be correct?

Ways: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q2/5

## True or false?

$5C5 = 1$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q3/5

In how many ways can exactly 3 of 5 answers be correct?

**A.** 60

**B.** 15

**C.** 10

**D.** 5

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S48-B1-Q4**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability**
**HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q4/5**

## True or false?

$4C2 = 8$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • HSS-MD.A.3

# BACKWARD 2

Multiply probabilities of independent trials • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

Compute $(1/2)^4$. Give a fraction in lowest terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q1/5

Compute (1/4)(3/4)(3/4). Give a fraction in lowest terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q2/5

## True or false?

Guesses on different questions are independent.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q3/5

P(correct) = 1/4 on each of 2 independent guesses. P(both correct)?

**A.** 1/16

**B.** 1/8

**C.** 1/2

**D.** 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 48 • BACKWARD 2 • Q4/5

## True or false?

$(3/4)^3 = 9/12$.

**A.** True

**B.** False

SET 48 • HSS-MD.A.3

# FORWARD 1

Build the whole theoretical distribution • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.3

A student guesses on 3 four-choice questions; X = number correct. Find P(X = 0), P(X = 1), P(X = 2), P(X = 3). Give a fraction in lowest terms.

P(X = 0): _____

P(X = 1): _____

P(X = 2): _____

P(X = 3): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q1/5

For X = number correct out of 3 four-choice guesses, find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q2/5

## True or false?

The four probabilities 27/64, 27/64, 9/64, 1/64 add to 1.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q3/5

For 3 four-choice guesses, which value of X is least likely?

**A.** 2

**B.** 0

**C.** 3

**D.** 1

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S48-F1-Q4**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions**
**HSS-MD.A.3 • Set 48 • FORWARD 1 • Q4/5**

## True or false?

For 3 four-choice guesses, $X = 1$ is more likely than $X = 0$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.3 • Set 48 • FORWARD 1 • Q5/5

SET 48 • HSS-MD.A.3

# FORWARD 2

Compare a theoretical distribution with data • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.4

A die is rolled 60 times. How many sixes does the theoretical distribution predict?

Sixes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q1/5

In 60 rolls, a die showed 6 sixes. What is the empirical probability of a six? Give a decimal.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q2/5

## True or false?

An empirical probability is found from observed data.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q3/5

In 100 households, the numbers of TVs gave an average of 1.9. This average is

**A.** the expected value of the empirical distribution

**B.** a theoretical probability from a model

**C.** always equal to 2 for any survey

**D.** a margin of error for the survey

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S48-F2-Q4**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions**
**HSS-MD.A.4 • Set 48 • FORWARD 2 • Q4/5**

## True or false?

Empirical and theoretical probabilities for the same event are always equal.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 48 • FORWARD 2 • Q5/5

# Develop an empirical probability distribution (+)

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-MD.A.4** |
| **BACKWARD 1** | Find relative frequencies | **7.SP.C.6** |
| **BACKWARD 2** | Compute an expected value | **HSS-MD.A.2** |
| **FORWARD 1** | Compare empirical and theoretical results | **HSS-MD.A.4** |
| **FORWARD 2** | Find the expected payoff of a game of chance | **HSS-MD.B.5.a** |

SET 49 • HSS-MD.A.4

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.4

100 households were surveyed (table). X = number of TVs in a randomly chosen household. Find P(X = 2) and P(X ≥ 3). Give decimals.

| TVs in household | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Households | 5 | 30 | 40 | 20 | 5 |

P(X = 2): _____

P(X ≥ 3): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q1/5

Using the TV survey, find the expected number of TVs in a household.

| TVs in household | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Households | 5 | 30 | 40 | 20 | 5 |

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q2/5

Using the expected value 1.9 TVs per household, about how many TVs are expected in 500 households?

TVs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q3/5

The TV survey gives E(X) = 1.9. Which interpretation is correct?

| TVs in household | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Households | 5 | 30 | 40 | 20 | 5 |

**A.** Most of the households have exactly 1.9 TVs.

**B.** Households in this population average about 1.9 TVs.

**C.** Every household surveyed has 1 or 2 TVs.

**D.** There is a 1.9% chance of owning a TV.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S49-M-Q4**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions**
**HSS-MD.A.4 • Set 49 • MAIN • Q4/5**

A store recorded complaints for 50 days (table). Find the expected number of complaints in a day.

| Complaints in a day | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| Days | 20 | 15 | 10 | 5 |

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • MAIN • Q5/5

SET 49 • HSS-MD.A.4

# BACKWARD 1

Find relative frequencies • Grade 7 • Statistics & Probability • 7.SP.C.6

In 100 households, 40 have 2 TVs. What is the relative frequency? Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

**Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 49 • BACKWARD 1 • Q1/5**

On 20 of 50 days there were no complaints. What is the relative frequency? Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

**Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 49 • BACKWARD 1 • Q2/5**

## True or false?

Relative frequencies of all categories add to 1.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

**Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 49 • BACKWARD 1 • Q3/5**

On 5 of 50 days there were 3 complaints. Relative frequency?

**A.** 5

**B.** 0.1

**C.** 0.05

**D.** 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

**Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 49 • BACKWARD 1 • Q4/5**

## True or false?

A relative frequency of 0.25 means 25 observations.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q5

**Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 49 • BACKWARD 1 • Q5/5**

SET 49 • HSS-MD.A.4

# BACKWARD 2

Compute an expected value • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

X takes 0, 1, 2 with probabilities 0.5, 0.3, 0.2. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 49 • BACKWARD 2 • Q1/5

## True or false?

$$E(X) = \Sigma\, x \cdot P(X = x).$$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 49 • BACKWARD 2 • Q2/5

X takes 1 and 3 with probabilities 0.5 each. E(X)?

**A.** 4

**B.** 2

**C.** 1.5

**D.** 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 49 • BACKWARD 2 • Q3/5

X takes 0, 2, 4 with probabilities 0.25, 0.5, 0.25. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 49 • BACKWARD 2 • Q4/5

## True or false?

E(X) is found by adding the probabilities.

**A.** True

**B.** False

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • HSS-MD.A.4

# FORWARD 1

Compare empirical and theoretical results • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.4

A die was rolled 60 times: the faces 1-6 appeared 7, 13, 9, 11, 14, 6 times. Find the empirical P(6) and the theoretical P(6). Give decimals rounded to two places.

Empirical: _____

Theoretical: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • FORWARD 1 • Q1/5

Using the 60 rolls (faces 1-6 appeared 7, 13, 9, 11, 14, 6 times), find the empirical expected value of a roll.

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • FORWARD 1 • Q2/5

## True or false?

With many more rolls, empirical probabilities tend to get closer to the theoretical ones.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • FORWARD 1 • Q3/5

Which result would most suggest the die is unfair?

**A.** In 600 rolls, a 6 appeared 104 times.

**B.** In 60 rolls, a 6 appeared 6 times.

**C.** In 6 rolls, no 6 appeared.

**D.** In 6,000 rolls, a 6 appeared 1,800 times.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • FORWARD 1 • Q4/5

## True or false?

An empirical distribution from 10 observations is as reliable as one from 10,000.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.4 • Set 49 • FORWARD 1 • Q5/5

SET 49 • HSS-MD.A.4

# FORWARD 2

Find the expected payoff of a game of chance • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.5.a

A $2 ticket wins $500 with probability 1/1000. Find the expected net gain per ticket.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 49 • FORWARD 2 • Q1/5

## True or false?

A game with negative expected net gain loses money for the player in the long run.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 49 • FORWARD 2 • Q2/5

A spinner: win $8 with probability 1/4, lose $2 otherwise. Find the expected gain.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 49 • FORWARD 2 • Q3/5

A game's expected net gain is −$1.50. Which interpretation is correct?

**A.** The player wins $1.50 on about half the plays.

**B.** Every player loses exactly $1.50 on each play.

**C.** Over many plays, a player loses about $1.50 per play.

**D.** The game is fair, since some plays win.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S49-F2-Q4**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions**
**HSS-MD.B.5.a • Set 49 • FORWARD 2 • Q4/5**

## True or false?

A game with expected net gain $0 is called fair.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 49 • FORWARD 2 • Q5/5

# Weigh the payoffs of a game of chance (+)

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-MD.B.5.a** |
| **BACKWARD 1** | Compute an expected value | **HSS-MD.A.2** |
| **BACKWARD 2** | Add and subtract signed numbers | **7.NS.A.3** |
| **FORWARD 1** | Set a price that makes a game fair | **HSS-MD.B.5.a** |
| **FORWARD 2** | Compare decisions by expected value | **HSS-MD.B.5.b** |

SET 50 • HSS-MD.B.5.a

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.5.a

A lottery ticket costs $2 and wins $500 with probability 1/1000; otherwise it wins nothing. Find the expected net gain per ticket.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • MAIN • Q1/5

A game: roll a fair die. If it shows 6, you win $5; otherwise you lose $1. Find the expected gain per play.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • MAIN • Q2/5

A raffle sells 500 tickets at $5 each for one $1,000 prize. Find the expected net gain per ticket.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • MAIN • Q3/5

A lottery ticket has expected net gain −$1.50. Which interpretation is correct?

**A.** The lottery is fair, because some tickets win a prize.

**B.** Over many tickets, a buyer loses about $1.50 per ticket on average.

**C.** Every ticket loses exactly $1.50 for the person who buys it.

**D.** Buyers win $1.50 on about half of the tickets they buy.

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • MAIN • Q4/5

A spinner wins $8 with probability 1/4 and loses $2 otherwise. Find the expected gain per spin.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • MAIN • Q5/5

SET 50 • HSS-MD.B.5.a

# BACKWARD 1

Compute an expected value • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

X = 10 with probability 0.1 and X = 0 otherwise. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 50 • BACKWARD 1 • Q1/5

## True or false?

$$E(X) = \Sigma\, x \cdot P(X = x).$$

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 50 • BACKWARD 1 • Q2/5

$X = 4$ with probability 0.25 and $X = -1$ otherwise. $E(X)$?

**A.** 1

**B.** $-0.75$

**C.** 0.25

**D.** 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 50 • BACKWARD 1 • Q3/5

X = 100 with probability 0.01 and 0 otherwise. Find E(X).

E(X): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 50 • BACKWARD 1 • Q4/5

## True or false?

$E(X) = 5$ when $X = 10$ with probability 0.2 and $X = 0$ otherwise.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • HSS-MD.B.5.a

# BACKWARD 2

Add and subtract signed numbers • Grade 7 • The Number System • 7.NS.A.3

Compute $0.5 - 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q1

**Grade 7 • The Number System • 7.NS.A.3 • Set 50 • BACKWARD 2 • Q1/5**

Compute 5(1/6) − 1(5/6).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q2

**Grade 7 • The Number System • 7.NS.A.3 • Set 50 • BACKWARD 2 • Q2/5**

**True or false?**

$2 - 1.5 = 0.5$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q3

**Grade 7 • The Number System • 7.NS.A.3 • Set 50 • BACKWARD 2 • Q3/5**

Compute $2 - 5$.

**A.** 3

**B.** $-7$

**C.** 7

**D.** $-3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

**Grade 7 • The Number System • 7.NS.A.3 • Set 50 • BACKWARD 2 • Q4/5**

**True or false?**

$-1.5 + 2 = -3.5$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

**Grade 7 • The Number System • 7.NS.A.3 • Set 50 • BACKWARD 2 • Q5/5**

SET 50 • HSS-MD.B.5.a

# FORWARD 1

Set a price that makes a game fair • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.5.a

A game pays $12 if a die shows 6 and nothing otherwise. What price makes the game fair?

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • FORWARD 1 • Q1/5

A raffle has one $1,000 prize and 400 tickets. What ticket price makes the raffle fair?

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • FORWARD 1 • Q2/5

## True or false?

A game is fair when the price equals the expected payout.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • FORWARD 1 • Q3/5

A game costs $3 and pays $10 with probability 0.25. What is the expected net gain over 100 plays?

**A.** −$0.50

**B.** −$50

**C.** $50

**D.** $250

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • FORWARD 1 • Q4/5

## True or false?

A raffle with 400 tickets and a $1,000 prize is fair at $5 per ticket.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 50 • FORWARD 1 • Q5/5

SET 50 • HSS-MD.B.5.a

# FORWARD 2

Compare decisions by expected value • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.5.b

Option X gains $1,000 with probability 0.6 and loses $500 with probability 0.4. Option Y gains $300 for sure. Find the expected gain of each.

X: _____

Y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 50 • FORWARD 2 • Q1/5

Option X: expected gain $400 but a 40% chance of losing $500. Option Y: $300 for sure. Which statement is correct?

**A.** X is certain to gain $400 for whoever chooses it.

**B.** X and Y have exactly the same expected value.

**C.** Y has the higher expected value of the two.

**D.** X has the higher expected value, but Y has no risk of loss.

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 50 • FORWARD 2 • Q2/5

## True or false?

Expected value describes the long-run average, not what happens in one decision.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 50 • FORWARD 2 • Q3/5

Insurance costs $150. Without it, a $800 repair is needed with probability 0.1. What is the expected repair cost without insurance?

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 50 • FORWARD 2 • Q4/5

## True or false?

Buying insurance always has a greater expected value than not buying it.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 50 • FORWARD 2 • Q5/5

# Evaluate and compare strategies by expected value (+)

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-MD.B.5.b** |
| **BACKWARD 1** | Find the expected payoff of one choice | **HSS-MD.B.5.a** |
| **BACKWARD 2** | Read a distribution and its mean | **HSS-MD.A.2** |
| **FORWARD 1** | Find break-even probabilities | **HSS-MD.B.5.b** |
| **FORWARD 2** | Make fair decisions with probability | **HSS-MD.B.6** |

SET 51 • HSS-MD.B.5.b

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.5.b

Plan A: premium $1,200, deductible $500. Plan B: premium $900, deductible $1,000. A claim larger than $1,000 occurs with probability 0.2 (at most one claim). Find the expected yearly cost of each plan (premium plus expected deductible).

Plan A: _____

Plan B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • MAIN • Q1/5

Plan A has expected cost $1,300 and plan B $1,100 per year. Which plan is better on the basis of expected cost?

**A.** Plan B

**B.** It cannot be decided

**C.** Plan A

**D.** They are equal

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions**
**HSS-MD.B.5.b • Set 51 • MAIN • Q2/5**

Option X gains $2,000 with probability 0.3 and loses $400 with probability 0.7. Option Y gains $250 for sure. Find the expected gain of each.

X: _____

Y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • MAIN • Q3/5

X has expected gain $320 with a 70% chance of losing $400; Y gains $250 for sure. Which statement is correct?

**A.** Y is better on expected value, so everyone should prefer Y.

**B.** X is certain to gain $320 for the person who chooses it.

**C.** X and Y have the same expected value, so they are equal.

**D.** X is better on average, but someone who cannot risk a loss may prefer Y.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S51-M-Q4**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions**
**HSS-MD.B.5.b • Set 51 • MAIN • Q4/5**

A phone warranty costs $150. Without it, a repair costing $800 is needed with probability 0.1. What is the expected repair cost without the warranty, and how much more is the warranty than that?

Expected cost: _____

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • MAIN • Q5/5

SET 51 • HSS-MD.B.5.b

# BACKWARD 1

Find the expected payoff of one choice • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.5.a

A choice gains $200 with probability 0.5 and $0 otherwise. Find the expected gain.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 51 • BACKWARD 1 • Q1/5

A choice loses $1,000 with probability 0.2 and $0 otherwise. Find the expected loss.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 51 • BACKWARD 1 • Q2/5

## True or false?

A sure gain of $300 has expected value $300.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 51 • BACKWARD 1 • Q3/5

A bet wins $50 with probability 0.3 and loses $20 otherwise. Expected gain?

**A.** $30

**B.** −$14

**C.** $1

**D.** $15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 51 • BACKWARD 1 • Q4/5

## True or false?

A choice with expected gain $0 can never make money.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.a • Set 51 • BACKWARD 1 • Q5/5

SET 51 • HSS-MD.B.5.b

# BACKWARD 2

Read a distribution and its mean • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

## True or false?

The option with the larger expected value is better on average.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 51 • BACKWARD 2 • Q1/5

$X = 1000$ with probability 0.6 and $-500$ with probability 0.4. Find $E(X)$.

$E(X)$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 51 • BACKWARD 2 • Q2/5

Which has the larger expected value: $50 for sure, or $120 with probability 0.5?

**A.** $120 with probability 0.5

**B.** cannot tell

**C.** they are equal

**D.** $50 for sure

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 51 • BACKWARD 2 • Q3/5

Cost = 500 with probability 0.2 and 0 otherwise. Find the expected cost.

Dollars: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 51 • BACKWARD 2 • Q4/5

## True or false?

Expected cost = cost × (1 − probability).

**A.** True

**B.** False

SET 51 • HSS-MD.B.5.b

# FORWARD 1

Find break-even probabilities • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.5.b

A $150 warranty covers an $800 repair. For what repair probability is the expected repair cost equal to $150? Give a decimal.

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • FORWARD 1 • Q1/5

## True or false?

If the repair probability is below the break-even value, skipping the warranty has the lower expected cost.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • FORWARD 1 • Q2/5

Plan A costs $1,200 plus a $500 deductible; plan B costs $900 plus a $1,000 deductible. For what claim probability p are the expected costs equal?

p: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • FORWARD 1 • Q3/5

With claim probability 0.8, which plan has the lower expected cost (A: 1200 + 500p; B: 900 + 1000p)?

**A.** Plan A

**B.** Plan B

**C.** They are equal

**D.** Neither

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S51-F1-Q4**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions**
**HSS-MD.B.5.b • Set 51 • FORWARD 1 • Q4/5**

## True or false?

The better plan never depends on the claim probability.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.5.b • Set 51 • FORWARD 1 • Q5/5

SET 51 • HSS-MD.B.5.b

# FORWARD 2

Make fair decisions with probability • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

Three friends want to decide fairly who gets one ticket. Which method is fair?

**A.** Roll a die: 1-2 Ana, 3-4 Ben, 5-6 Cy.

**B.** Flip a coin: heads Ana, tails Ben.

**C.** Roll a die: 1 Ana, 2 Ben, 3-6 Cy.

**D.** Choose the person who asks first.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S51-F2-Q1**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions**
**HSS-MD.B.6 • Set 51 • FORWARD 2 • Q1/5**

## True or false?

A method is fair if every person has the same probability of being chosen.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 51 • FORWARD 2 • Q2/5

Ana has 3 tickets, Ben 5 and Cy 2 in a drawing of 10 tickets. What is P(Ben wins)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 51 • FORWARD 2 • Q3/5

In a two-dice game, A wins if the sum is 6, 7 or 8; otherwise B wins. Is it fair?

**A.** No: A wins with probability 3/11.

**B.** No: A wins with probability 4/9.

**C.** Yes: A has 3 sums and B has 8.

**D.** Yes: each player has 1/2.

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 51 • FORWARD 2 • Q4/5

## True or false?

Letting the oldest person choose is a fair random method.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 51 • FORWARD 2 • Q5/5

# Use probabilities to make fair decisions (+)

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-MD.B.6** |
| **BACKWARD 1** | Simulate with random digits | **HSS-IC.A.2** |
| **BACKWARD 2** | Find probabilities in a uniform model | **7.SP.C.7.a** |
| **FORWARD 1** | Design fair methods for awkward numbers | **HSS-MD.B.6** |
| **FORWARD 2** | Analyze a decision with a medical test | **HSS-MD.B.7** |

SET 52 • HSS-MD.B.6

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

Three friends want to decide fairly who gets the last ticket. Which method is fair?

**A.** Roll a die: 1 Ana, 2 Ben, 3-6 Cy.

**B.** Flip a coin: heads Ana, tails Ben; Cy only on an edge.

**C.** Ana chooses, since she asked first.

**D.** Roll a die: 1-2 Ana, 3-4 Ben, 5-6 Cy.

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • MAIN • Q1/5

Four people A, B, C, D are chosen from with random digits: 0-1 = A, 2-3 = B, 4-5 = C, 6-7 = D, and 8-9 are skipped. Reading line 2 of the table from the left, who is chosen? Enter A, B, C or D.

| Line | Digits |
|---|---|
| 1 | 15881 39986 98797 30911 |
| 2 | 41707 53645 56815 18844 |
| 3 | 72940 55761 69871 68976 |
| 4 | 84697 88093 20819 67273 |
| 5 | 17784 86488 76634 40029 |
| 6 | 78753 47787 68045 71784 |
| 7 | 72671 47513 62701 07981 |
| 8 | 93319 08452 05997 28379 |

Person: _____

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S52-M-Q2**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions**
**HSS-MD.B.6 • Set 52 • MAIN • Q2/5**

Two players each flip a coin. Ana wins if the coins match; Ben wins if they differ. What is P(Ana wins)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • MAIN • Q3/5

Two dice are rolled. Player A wins if the sum is 6, 7 or 8; otherwise B wins. Is the game fair?

**A.** Yes: the sums are split fairly.

**B.** Yes: each player wins with probability 1/2.

**C.** No: A wins with probability 3/11.

**D.** No: A wins with probability 4/9.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S52-M-Q4**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions**
**HSS-MD.B.6 • Set 52 • MAIN • Q4/5**

In a drawing, Ana has 3 tickets, Ben 5 and Cy 2, out of 10. What is P(Ben wins)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • MAIN • Q5/5

SET 52 • HSS-MD.B.6

# BACKWARD 1

Simulate with random digits • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

## True or false?

Each digit 0-9 in a random-digit table is equally likely.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 52 • BACKWARD 1 • Q1/5

To choose among 4 people with single digits, which assignment is fair?

**A.** 0-1, 2-3, 4-5, 6-7 and skip 8-9

**B.** 0-2, 3-5, 6-8 and 9

**C.** 0-1, 2-3, 4-5, 6-9

**D.** 0, 1, 2, 3 and stop

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S52-B1-Q2**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 52 • BACKWARD 1 • Q2/5**

Using 0-1 = A, 2-3 = B, 4-5 = C, 6-7 = D, skip 8-9, what does the digit 9 mean?

Answer (enter skip or a letter): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 52 • BACKWARD 1 • Q3/5

## True or false?

Skipping digits 8 and 9 makes the assignment unfair.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 52 • BACKWARD 1 • Q4/5

Using the same assignment, the digits read 8, 9, 3. Who is chosen?

**A.** A

**B.** no one

**C.** D

**D.** B

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S52-B1-Q5**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 52 • BACKWARD 1 • Q5/5**

SET 52 • HSS-MD.B.6

# BACKWARD 2

Find probabilities in a uniform model • Grade 7 • Statistics & Probability • 7.SP.C.7.a

A die is rolled. What is P(1 or 2)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

**Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 52 • BACKWARD 2 • Q1/5**

## True or false?

With a fair coin, heads and tails each have probability 1/2.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

**Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 52 • BACKWARD 2 • Q2/5**

Names of 5 people are in a hat. P(a particular person is drawn)?

**A.** 1/2

**B.** 1/5

**C.** 1/4

**D.** 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

**Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 52 • BACKWARD 2 • Q3/5**

Two coins are flipped. What is P(they match)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

**Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 52 • BACKWARD 2 • Q4/5**

## True or false?

A spinner with one half red and two quarters blue and green is fair among the three colors.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

**Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 52 • BACKWARD 2 • Q5/5**

SET 52 • HSS-MD.B.6

# FORWARD 1

Design fair methods for awkward numbers • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.6

Five people must be chosen from fairly with one die. Which method works?

**A.** Assign 1-5 to the people and give 6 to the first person.

**B.** Assign 1-5 to the people and reroll a 6.

**C.** Assign 1-6 and let the sixth face mean "everyone".

**D.** Roll and divide the result by 5.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S52-F1-Q1**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions**
**HSS-MD.B.6 • Set 52 • FORWARD 1 • Q1/5**

With the "reroll a 6" method for 5 people, what is each person's probability of being chosen? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • FORWARD 1 • Q2/5

## True or false?

A method that gives one person 2 of 6 die faces and the others 1 each is unfair.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • FORWARD 1 • Q3/5

Using two-digit numbers 00-99, how many numbers should each of 3 people get for a fair choice, if the rest are skipped?

Numbers each: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • FORWARD 1 • Q4/5

## True or false?

Drawing names one at a time from a hat without replacement is unfair to the last name.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.6 • Set 52 • FORWARD 1 • Q5/5

SET 52 • HSS-MD.B.6

# FORWARD 2

Analyze a decision with a medical test • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

Of 1,000 people, 2% have a disease. How many have it?

People: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 52 • FORWARD 2 • Q1/5

A test is positive for 90% of the 20 people with the disease. How many test positive?

People: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 52 • FORWARD 2 • Q2/5

## True or false?

A test can give positive results for people who do not have the disease.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 52 • FORWARD 2 • Q3/5

Of the 980 healthy people, 5% test positive. How many false positives?

**A.** 49

**B.** 5

**C.** 18

**D.** 931

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 52 • FORWARD 2 • Q4/5

## True or false?

If a test is positive for 90% of sick people, then 90% of positives are sick.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 52 • FORWARD 2 • Q5/5

# Analyze decisions about medical testing (+)

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-MD.B.7** |
| **BACKWARD 1** | Read a two-way table as a sample space | **HSS-CP.A.4** |
| **BACKWARD 2** | Find a conditional probability as a fraction of outcomes | **HSS-CP.B.6** |
| **FORWARD 1** | See how prevalence changes the answer | **HSS-MD.B.7** |
| **FORWARD 2** | Analyze strategies in games and quality control | **HSS-MD.B.7** |

SET 53 • HSS-MD.B.7

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

1,000 people are tested. 2% have a disease; the test is positive for 90% of people with the disease. How many have the disease, and how many of them test positive?

Have the disease: _____

Test positive: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • MAIN • Q1/5

Of the 980 people without the disease, 5% test positive. How many false positives are there?

People: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • MAIN • Q2/5

Use the table for 1,000 people. A person tests positive. What is P(has disease given positive)? Give a fraction in lowest terms.

| | Tests positive | Tests negative | Total |
|---|---|---|---|
| Has disease | 18 | 2 | 20 |
| Does not | 49 | 931 | 980 |
| Total | 67 | 933 | 1000 |

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • MAIN • Q3/5

In the table, P(disease given positive) = 18/67, about 27%. What does this show?

| | Tests positive | Tests negative | Total |
|---|---|---|---|
| Has disease | 18 | 2 | 20 |
| Does not | 49 | 931 | 980 |
| Total | 67 | 933 | 1000 |

**A.** Most people who test positive do not have the disease.

**B.** Most people who have the disease test negative.

**C.** The test is wrong 73% of the time for everyone.

**D.** A positive test means a 90% chance of disease.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S53-M-Q4**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions**
**HSS-MD.B.7 • Set 53 • MAIN • Q4/5**

Use the table. A person tests negative. What is P(does not have the disease given negative)? Give a fraction in lowest terms.

| | Tests positive | Tests negative | Total |
|---|---|---|---|
| Has disease | 18 | 2 | 20 |
| Does not | 49 | 931 | 980 |
| Total | 67 | 933 | 1000 |

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • MAIN • Q5/5

SET 53 • HSS-MD.B.7

# BACKWARD 1

Read a two-way table as a sample space • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

In the test table, how many people test positive in all?

| | Tests positive | Tests negative | Total |
|---|---|---|---|
| Has disease | 18 | 2 | 20 |
| Does not | 49 | 931 | 980 |
| Total | 67 | 933 | 1000 |

People: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 53 • BACKWARD 1 • Q1/5

In the test table, how many people test negative in all?

| | Tests positive | Tests negative | Total |
|---|---|---|---|
| Has disease | 18 | 2 | 20 |
| Does not | 49 | 931 | 980 |
| Total | 67 | 933 | 1000 |

People: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 53 • BACKWARD 1 • Q2/5

## True or false?

In the test table, 20 people have the disease.

| | Tests positive | Tests negative | Total |
|---|---|---|---|
| Has disease | 18 | 2 | 20 |
| Does not | 49 | 931 | 980 |
| Total | 67 | 933 | 1000 |

**A.** True

**B.** False

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S53-B1-Q3**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 53 • BACKWARD 1 • Q3/5**

In the test table, which cell counts false positives?

| | Tests positive | Tests negative | Total |
|---|---|---|---|
| Has disease | 18 | 2 | 20 |
| Does not | 49 | 931 | 980 |
| Total | 67 | 933 | 1000 |

**A.** does not have the disease, tests negative

**B.** does not have the disease, tests positive

**C.** has the disease, tests negative

**D.** has the disease, tests positive

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 53 • BACKWARD 1 • Q4/5

## True or false?

In the test table, false negatives number 18.

| | Tests positive | Tests negative | Total |
|---|---|---|---|
| Has disease | 18 | 2 | 20 |
| Does not | 49 | 931 | 980 |
| Total | 67 | 933 | 1000 |

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 53 • BACKWARD 1 • Q5/5

SET 53 • HSS-MD.B.7

# BACKWARD 2

Find a conditional probability as a fraction of outcomes • High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6

Of 67 people who test positive, 18 have the disease. What fraction is that?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q1

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 53 • BACKWARD 2 • Q1/5

## True or false?

P(disease given positive) uses the positive column as the sample space.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 53 • BACKWARD 2 • Q2/5

Of 20 people with the disease, 18 test positive. P(positive given disease)?

**A.** 18/67

**B.** 2/20

**C.** 1/50

**D.** 9/10

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S53-B2-Q3**

**High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 53 • BACKWARD 2 • Q3/5**

Of 980 healthy people, 49 test positive. What is P(positive given healthy)? Give a decimal.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.6 • Set 53 • BACKWARD 2 • Q4/5

## True or false?

P(positive given disease) = P(disease given positive).

**A.** True

**B.** False

SET 53 • HSS-MD.B.7

# FORWARD 1

See how prevalence changes the answer • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

Now 10% of 1,000 people have the disease; the test is positive for 90% of them and for 5% of the healthy. How many true positives and how many false positives?

True positives: _____

False positives: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 1 • Q1/5

With 90 true positives and 45 false positives, what is P(disease given positive)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 1 • Q2/5

## True or false?

The same test gives a higher P(disease given positive) when the disease is more common.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 1 • Q3/5

Why do doctors often order a second test after a positive result for a rare disease?

**A.** Negative results are never correct.

**B.** Many first positives are false positives.

**C.** The first test is always wrong.

**D.** A second test lowers the prevalence.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S53-F1-Q4**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions**
**HSS-MD.B.7 • Set 53 • FORWARD 1 • Q4/5**

## True or false?

For a rare disease, a positive result makes having the disease certain.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 1 • Q5/5

SET 53 • HSS-MD.B.7

# FORWARD 2

Analyze strategies in games and quality control • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

A box of 10 parts has 2 defective. An inspector checks 2 parts at random. What is P(no defective part is found)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 2 • Q1/5

## True or false?

Checking more parts raises the chance of finding a defective part.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 2 • Q2/5

A test question has 4 choices; a correct answer earns 1 point and a wrong answer loses 1/3 point. What is the expected score of a random guess, and of a guess after one wrong choice is eliminated? Give a fraction in lowest terms.

Random guess: _____

After eliminating one: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 2 • Q3/5

Using those expected scores, when is guessing worthwhile?

**A.** after eliminating at least one wrong choice

**B.** always, even with no information

**C.** only on the last question of the test

**D.** never, under any circumstances

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 2 • Q4/5

## True or false?

A strategy with a higher expected value always wins in a single try.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 53 • FORWARD 2 • Q5/5

# Analyze strategies and decisions with probability (+)

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-MD.B.7** |
| **BACKWARD 1** | Compare expected values of choices | **HSS-MD.A.2** |
| **BACKWARD 2** | Count selections to find probabilities | **HSS-CP.B.9** |
| **FORWARD 1** | Weigh probabilities against consequences | **HSS-MD.B.7** |
| **FORWARD 2** | Combine several probability tools in one decision | **HSS-MD.B.7** |

SET 54 • HSS-MD.B.7

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

A hockey team trails by 1 goal with 2 minutes left. Keeping the goalie gives a 0.08 chance of tying; pulling the goalie for an extra skater gives a 0.20 chance of tying (but more chance of allowing another goal, which also loses). Give both chances of tying.

Keep goalie: _____

Pull goalie: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • MAIN • Q1/5

The team loses unless it ties. Keeping the goalie gives a 0.08 chance of tying; pulling gives 0.20. Which strategy is better?

**A.** Pull the goalie: it gives the better chance of avoiding a loss.

**B.** Keep the goalie: allowing another goal would be worse.

**C.** Keep the goalie: 0.08 is the safer probability.

**D.** Either one: both chances are less than one-half.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S54-M-Q2**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions**
**HSS-MD.B.7 • Set 54 • MAIN • Q2/5**

A box of 12 parts has 3 defective. An inspector checks 2 parts at random. What is P(no defective part is found)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • MAIN • Q3/5

A test question has 5 choices; a correct answer earns 1 point and a wrong answer loses 1/4 point. Find the expected score of a random guess, and of a guess after eliminating one wrong choice. Give a fraction in lowest terms.

Random guess: _____

After eliminating one: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • MAIN • Q4/5

A company can test 2 or 5 parts from each box of 10 (2 defective). With 2 tested, P(missing both defectives) = 28/45; with 5 tested, it is 2/9. Which is correct?

**A.** Testing 5 parts always finds both defects.

**B.** Testing 2 parts finds the defects more often.

**C.** Testing 5 parts lowers the chance of missing the defects.

**D.** The number tested does not change anything.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S54-M-Q5**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions**
**HSS-MD.B.7 • Set 54 • MAIN • Q5/5**

SET 54 • HSS-MD.B.7

# BACKWARD 1

Compare expected values of choices • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.A.2

Strategy A scores 3 with probability 0.5 and 0 otherwise. Find its expected score.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 54 • BACKWARD 1 • Q1/5

## True or false?

Between two strategies, the one with the larger expected value is better on average.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 54 • BACKWARD 1 • Q2/5

Strategy B scores 2 for sure; strategy A scores 3 with probability 0.5. Which is better on average?

**A.** cannot tell

**B.** A

**C.** B

**D.** equal

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 54 • BACKWARD 1 • Q3/5

A guess earns 1 with probability 1/3 and loses 1/3 otherwise. Expected score? Give a fraction in lowest terms.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 54 • BACKWARD 1 • Q4/5

## True or false?

An expected score of 0 means the strategy always scores 0.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.A.2 • Set 54 • BACKWARD 1 • Q5/5

SET 54 • HSS-MD.B.7

# BACKWARD 2

Count selections to find probabilities • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

Compute 10C2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 54 • BACKWARD 2 • Q1/5

Compute 8C2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 54 • BACKWARD 2 • Q2/5

## True or false?

$8C5 = 56$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 54 • BACKWARD 2 • Q3/5

Compute 10C5.

**A.** 30240

**B.** 252

**C.** 120

**D.** 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 54 • BACKWARD 2 • Q4/5

## True or false?

P(no defectives in 2 of 10 with 2 defective) = 8/10.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • HSS-MD.B.7

# FORWARD 1

Weigh probabilities against consequences • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

A product test costs $50 per box and finds the defects with probability 0.8. An undetected defective box costs $400. For a defective box, what is the expected cost of shipping without testing, and with testing (test cost plus expected cost of a missed defect)?

Without testing: _____

With testing: _____

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 1 • Q1/5

## True or false?

A strategy can have a better expected value even if it sometimes produces the worst outcome.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 1 • Q2/5

A student must decide whether to guess on 10 four-choice questions with a 1/3-point penalty. Which strategy has the highest expected score?

**A.** Guess only where at least one choice can be eliminated.

**B.** Leave every question blank.

**C.** Guess on every question, even with no information.

**D.** Never guess at all.

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 1 • Q3/5

With two choices eliminated (2 left), what is the expected score of a guess (1 point right, −1/3 wrong)? Give a fraction in lowest terms.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q4

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 1 • Q4/5

## True or false?

Probability analysis can tell exactly what will happen in one game.

**A.** True

**B.** False

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 1 • Q5/5

SET 54 • HSS-MD.B.7

# FORWARD 2

Combine several probability tools in one decision • High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

A quiz has 5 four-choice questions. A student guesses on all. What is the expected number correct, and P(all wrong)? Give a fraction in lowest terms.

Expected correct: _____

P(all wrong): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q1

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 2 • Q1/5

## True or false?

P(at least one correct) = 1 − P(all wrong).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q2

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 2 • Q2/5

A quiz has 5 four-choice questions and a student guesses on all. What is P(at least one correct)? Give a fraction in lowest terms.

P: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q3

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 2 • Q3/5

A team must choose between a play that gains 2 points with probability 0.5 and one that gains 3 points with probability 0.3. Which has the higher expected gain?

**A.** equal

**B.** cannot tell

**C.** the 2-point play

**D.** the 3-point play

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S54-F2-Q4**

**High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions**
**HSS-MD.B.7 • Set 54 • FORWARD 2 • Q4/5**

## True or false?

If the team needs 3 points to win, the 2-point play is still the better choice.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q5

High School — Probability & Statistics, Advanced Extension (+) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 54 • FORWARD 2 • Q5/5